Guide to Bridge Technology Part 7

Maintenance and Management of Existing Bridges

Austroads

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Austroads

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Abstract

Austroads Guide to Bridge Technology provides bridge owners and agencies with advice on bridge ownership, design procurement, vehicle and pedestrian accessibility, and bridge maintenance and management practices. The Guide has eight parts.

Part 7 discusses the structural management of existing bridges in practical technical terms. It documents practices relevant to forward works programming, inspection, recording, reporting, evaluation of bridge condition and fitness for purpose. Bridge testing and structural monitoring, such as available monitoring technologies and instrumentation, are discussed. Technical rehabilitation and strengthening treatments are also included.

Keywords

Maintenance, programming, economic evaluation, defects, maintenance issues, deterioration, management, bridges, structural inspection, load testing, non-destructive test, monitoring, instrumentation, inspector training and certification, load-carrying capacity, bearings, deck joints, masonry, timber, steel, wrought iron, rehabilitation, strengthening treatments, foundations, heritage bridges

Edition 2.1 published March 2024

Edition 2.1 updates Section 3: Inspection, Testing, Monitoring and Reporting.

Edition 2.0 published February 2018

Edition 2.0 provides updated information in various sections, and information contained in other Guides has been removed. Major changes include:

- Information on the operational aspects of bridge management practices, e.g. routine bridge inspection and maintenance activities has been removed. Refer to Austroads Guide to Asset Management for further reading on network level bridge management.
- Information has been removed which overlapped with other Parts of the Guide to Bridge Technology
- Information on the inspection, maintenance and replacement issues of bridge bearings and bridge expansion joints has been updated, based on the findings of *Design Rules for Bridge Bearings and Expansion Joints* (Austroads 2012)

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About Austroads

Austroads is the association of Australasian road transport and traffic agencies.

Austroads' purpose is to support our member organisations to deliver an improved Australasian road transport network. To succeed in this task, we undertake leading-edge road and transport research which underpins our input to policy development and published guidance on the design, construction and management of the road network and its associated infrastructure.

Austroads provides a collective approach that delivers value for money, encourages shared knowledge and drives consistency for road users.

Austroads is governed by a Board consisting of senior executive representatives from each of its eleven member organisations:

- Transport for NSW
- Department of Transport and Planning Victoria
- Queensland Department of Transport and Main Roads
- Main Roads Western Australia
- Department for Infrastructure and Transport South Australia
- Department of State Growth Tasmania
- Department of Infrastructure, Planning and Logistics Northern Territory
- Transport Canberra and City Services Directorate, Australian Capital Territory
- Department of Infrastructure, Transport, Regional Development, Communications and the Arts
- Australian Local Government Association
- NZ Transport Agency Waka Kotahi.

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1. Introduction

1.1 Scope

Part 7 of the Austroads *Guide to Bridge Technology (AGBT)* provides guidance on the maintenance and management of existing bridges and addresses topics such as:

- bridge inspection
- load rating and monitoring
- maintenance, repair and strengthening
- bridge preservation.

Part 7 provides documentation of the technical issues and practices relevant to the above topics and is intended to present good practice in the technical aspects of bridge management consistent with *Integrated Asset Management Guidelines for Road Networks* (Austroads 2002). *Part 7* can be used to complement *Guide to Asset Management Technical Information Part 13: Structures Asset Management* (Austroads 2018), which provides guidance on the establishment and maintenance of structure asset inventories and the monitoring of asset performance.

The focus of *Part 7* is on the practical and technical procedures and strategies available for managing the existing bridges in a network and maintaining bridges as fit for purpose. Included is the aim to provide information to assist management of the existing bridge asset to achieve the optimum mobility of freight and exceptionally heavy loads for the benefit of the community, while ensuring preservation of the bridge assets and safety of road users. It is based on practices adopted by Austroads members and therefore reflects the combined experience of the Australasian road agencies.

The target audience includes all those involved with the management of the bridge asset, industry, and students seeking to learn more about the management of existing bridges and appropriate maintenance practices and procedures.

The bridge manager is faced with many choices in the allocation of scarce resources to meet competing priorities. Systematic bridge management aims to ensure the existing bridge asset is fit for purpose and maintained within available resources in an economic and cost-effective manner.

It is accepted that bridges represent a significant but distinctive part of the road infrastructure requiring the development and application of specific requirements and practices to facilitate sound bridge asset management.

The management of existing bridges involves dealing with the numerous bridges within the road agencies' networks including individual bridges that are complex structures. The management of existing bridges therefore involves procedures and practices tailored to the requirements of bridge network management (bridges as a group) and project management (bridges as individual structures).

The Australian experience is of a diverse history and practice from:

- 1800s to 1950s a diverse approach regarding; economy connection (A & B Class), local materials, and a railway legacy
- 1950s to 1970s practice based on American Association of State Highway and Transportation Officials (AASHTO)
- 1976 practice based on Austroads Bridge Design and T44 loading
- 2007 practice based on Standards Australia Bridge Design, AS 5100 (revised in 2017).

The New Zealand experience is one of continuous development of design practice:

- New Zealand's early bridges based on traction engine loadings with a variation of a double traction engine to cover two traffic lanes
- from the early 1940s loadings based on AASHTO H20-S16-44
- 1961 – H20-S16-T16 was introduced taking into account changes in truck configurations
- 1972 National Roads Board adopted a code for bridge design that incorporated HN-HO-72 design loading.

The scope of *Part 7* includes coverage of:

- timber bridges
- steel girder bridges and reinforced concrete deck
- reinforced concrete bridges.

Excluded from scope are bridges with spans greater than 100 m such as:

- steel truss bridges
- long span box girders
- cable-stayed or suspension bridges.

While this document is primarily aimed at the management of existing bridges, the same principles may be applied to the management of other highway structures (e.g. major culverts, retaining walls, tunnels, overhead gantries, large traffic management signs etc.).

In preparing this publication extensive use has been made of a series of reports published by Austroads as part of an ongoing research effort to advance the state-of-the-art of bridge management.

In particular, it is acknowledged that significant material was derived from the following documents that contributed to and is contained in this document:

- Guidelines for Bridge Management: Structure Information (Austroads 2004)
- Bridge Inspection and Maintenance Manual (NZ Transport Agency (NZTA) 2001).

1.2 Guide Structure

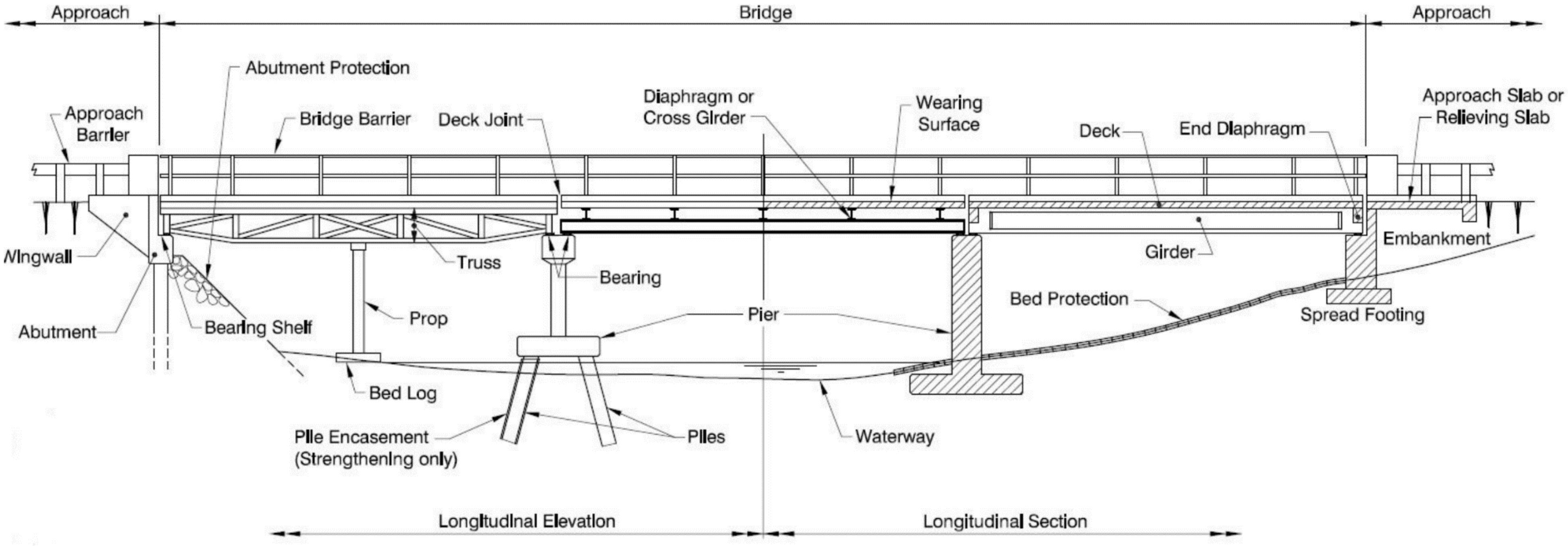
The *AGBT* is published in eight parts and addresses a range of bridge technology issues, each of which is summarised in Table 1.1.

Table 1.1: Parts of the Guide to Bridge Technology

Part	Title	Content
Part 1	Introduction and Bridge Performance	• Scope of the *Guide to Bridge Technology* and its relationship to the bridge design standards. • Factors affecting bridge performance and technical and non-technical design influences. • Evolution of bridges, bridge construction methods and equipment and bridge loadings. • Specifications and quality assurance in bridge construction.
Part 2	Materials	• The full range of bridge building materials including concrete, steel, timber and non-metallic components. • Material characteristics including individual stress mechanisms.

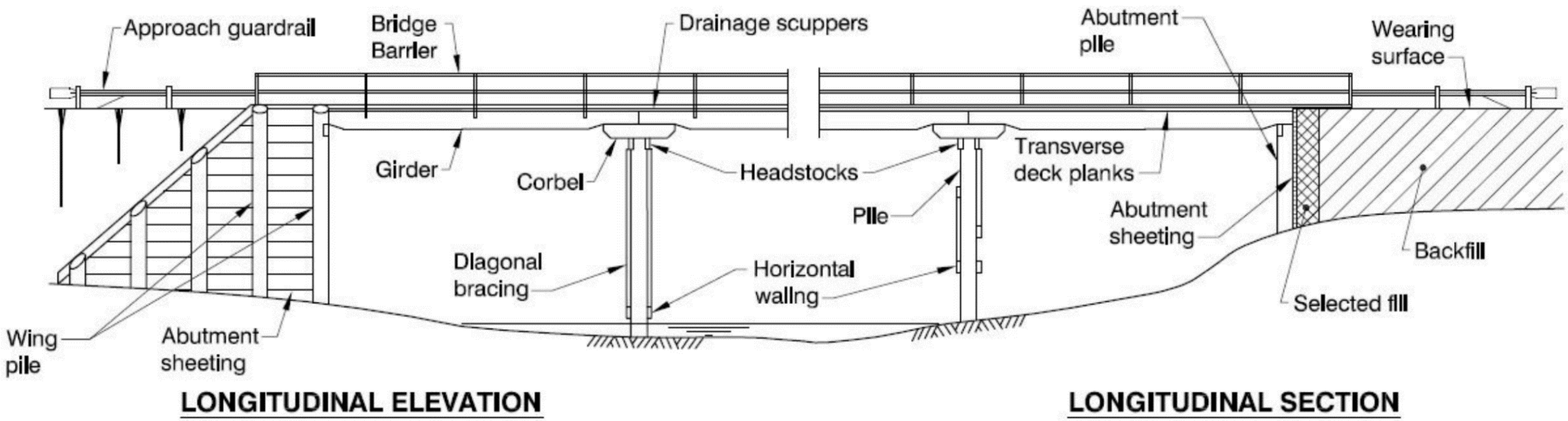
Part	Title	Content
Part 3	Typical Bridge Superstructures, Substructures and Components	• Superstructure and substructure components – namely timber, steel, wrought iron, reinforced and pre-stressed concrete. • Typical bridge types such as suspension, cable stayed and arched types. • Bridge foundations.
Part 4	Design Procurement and Concept Design	• Bridge design process procurement models, specification requirements, design and delivery management processes, design checking and review concepts, the use of standardised components, aesthetics/architectural requirements, standard presentation of drawings and reports, designing for constructability and maintenance. • Service life of the structure and components, mining and subsidence, flood plains, bridge loadings, and geotechnical and environmental considerations.
Part 5	Structural Drafting	• Detailed drawing aspects required to clearly convey to the consultant/construction contractor the specifics of the project. • Standards including details required for cost estimating and material quantities. • Reinforcement identification details.
Part 6	Bridge Construction	• Guidance to the bridge owner's representative on site. • Focuses on bridge technology, high-risk construction processes e.g. piling, pre-stressing, and the relevant technical surveillance requirements during the construction phase. • Bridge geometry, the management of existing road traffic and temporary works.
Part 7	Maintenance and Management of Existing Bridges	• Maintenance issues for timber, reinforced and pre-stressed concrete, steel, wrought and cast iron bridges. • Maintenance of bridge components including bridge bearings and deck joints. • Monitoring, inspection and management of bridge conditions.
Part 8	Hydraulic Design of Waterway Structures	• Waterway design of bridge structures • Design flood standards and estimation methods, general considerations in waterway design and design considerations of waterway structures. • Design of new bridges for scour, as well as monitoring and evaluation of scour at existing bridge sites.

Figure 1.1: General terminology for bridges



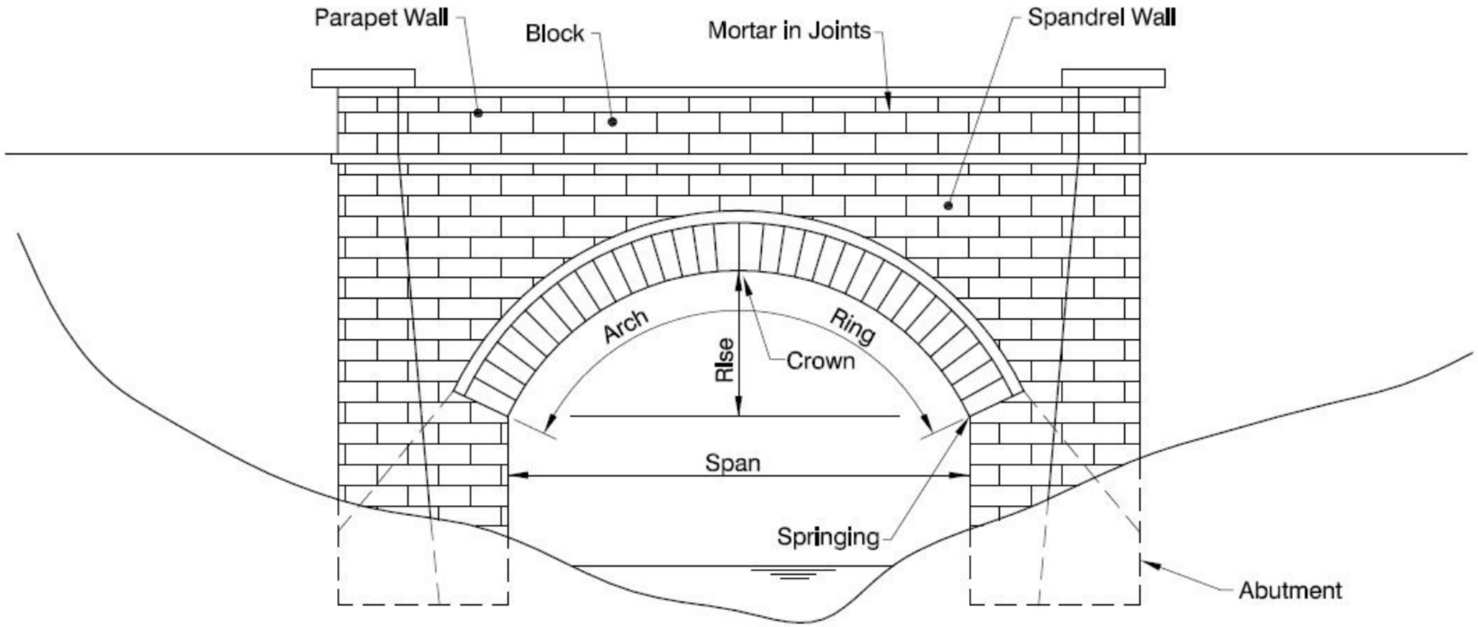
Source: Queensland Department of Transport and Main Roads (TMR) (2016).

Figure 1.2: Special terminology for timber bridges



Source: TMR (2016).

Figure 1.3: Special terminology for masonry bridges



Source: TMR (2016).

2. Maintenance and Management of Existing Bridges

2.1 Introduction

Bridges are key elements in any road network and represent a major investment of community resources which has been built up progressively over time. Because of their strategic location over natural or other obstacles, any failure of a bridge or reduction in level of service may limit or severely restrict road traffic over a large part of the road network with consequent inconvenience and economic loss to the community. While there will always be a focus on new bridges to service increased growth/expectations it must be recognised that the existing bridge stock must be managed through maintenance and renewal.

Bridges are valuable assets that require ongoing attentive maintenance. Asset management practices need to be implemented by road agencies to achieve and maintain target levels of service in the most cost-effective manner. Appropriate maintenance of existing bridges is necessary and fundamental to achieving this aim.

This is achieved through understanding:

- the performance of bridges
- the consequences of not carrying out maintenance, strengthening and upgrading
- how maintenance, strengthening, and upgrading will influence their performance
- lifecycle cost minimisation and improving the performance of the asset are key considerations.

The asset manager necessarily has a focus on the 'network level' budget and economies along with responsibilities for the decision on the allocation of funds and risks. Engineering management of existing bridges focuses on the 'project level' aspects of asset management. It deals with particular bridges and bridges as a group requiring technical engineering issues to be addressed rather than overall network level management.

2.2 General Administration and Management Systems

Effective and efficient decision-making regarding the management of highway structures requires reliable data and information on a range of structural aspects. Key information includes:

- condition assessment
- load rating, predicted deterioration
- expected remaining service life
- major risks
- historical maintenance costs and predicted forward works investment.

All of this information needs to be captured and processed in a fit-for-purpose asset management system, able to produce data for the asset manager to answer key questions.

The ISO 55000 suite of Asset Management Standards published in 2014 provides an overview of asset management practices as well as specifying requirements for the establishment, implementation, maintenance and improvement of an asset management system. These standards seek to promote global harmonisation and are increasingly referenced by Australian and New Zealand jurisdictions as the benchmark for their management systems. The ISO 55000 suite superseded the widely used and publicly available standard PAS 55 *Specification for the optimised management of physical assets*.

Another widely referenced document is the *International Infrastructure Management Manual* (Institute of Public Works Engineering Australasia 2015) which expands on the framework provided by ISO 55000 and promotes international best management practice for infrastructure assets.

The inspection and maintenance program allows the bridge stock condition to be better understood so that the overall maintenance and development strategies can be implemented.

Defects requiring attention will develop during a bridge's life. It is important that these defects are identified at an early stage to:

- ensure public safety
- protect the investment by extending the life of the structure, in a manner consistent with the individual bridge strategy
- minimise the cost of repairs.

Efficient and effective procedures are required to ensure:

- consistency and uniformity in inspections so that remedial work can be correctly prioritised
- use of sound assessment techniques
- use of effective repair methods
- feedback is provided to designers.

When a large bridge stock is to be managed and a number of inspection teams are used it is more difficult to ensure consistency of inspection and prioritisation of repair work. In these cases the bridge inspection engineer or consultant commissioned by the road jurisdiction needs to undertake regular auditing and review, and, if necessary, implement training to ensure consistency is achieved. Particular care is needed when inspection and maintenance services are contracted out.

2.3 Record Systems and Reporting

The success of any bridge inspection and maintenance program depends on its reporting system, as this is the means by which works items identified through inspection are included in budgets and repair work is undertaken.

Accurate record-keeping enables the bridge inspection engineer to:

- program maintenance work
- check effects of changes in vehicle loadings and traffic volumes
- assess structural adequacy and provide information for assessment of load carrying capacity, which ensures that posting and overweight permit structural details are accurate and up-to-date
- monitor the progress of structural changes
- establish a performance history
- provide feedback to designers.

The form of recording system used needs to be fit for purpose and designed to suit local conditions. There is no benefit in wasting time and money having one that is too elaborate and provides data that is not going to be used. Typically, the recording system will take the form of, or be a constituent part of an asset information management system (bridge information system). For structures of sufficient complexity, this may be supplemented by the use of Building Information Modelling where warranted.

The objective of a 'bridge information system' is to ensure, among other objectives relative to the 'network level' of asset management, the acquisition and documentation of structural bridge information. This is aimed to facilitate 'project level' management of specific structures and groups of structures within the road jurisdiction's organisation. The intent is to ensure that necessary and sufficient relevant information for every bridge is adequately recorded to enable effective inspection, maintenance, rehabilitation, strengthening and replacement programs to be planned. It is extremely important that this documentation is preserved for use throughout the life of the bridge.

Information required for an effective structural assessment

This information includes network data such as:

- ownership
- administration and management
- structure location and identification
- structure significance e.g. strategic importance or heritage value
- route significance
- traffic demand.

Structure details include:

- drawing registers, and detailed drawings
- design criteria
- as constructed information – construction, maintenance, rehabilitation and strengthening details including any subsequent alterations
- construction diaries, photographs, material test reports and completion of construction report
- services carried
- physical attributes including structure type, design capacity and clearances
- material specifications and materials used, e.g. details of concrete mixes
- value, historic cost and replacement costs.

The Austroads *Guide to Asset Management* provides details of typical data that could be recorded.

Records captured in the system should consist of all of the following in whatever form is most convenient:

- bridge inventory
- as-built drawings and photos
- inspection forms, reports, photos and diagrams (condition history)
- remedial work records, photos and costs (maintenance history).

A computer-based system is typically the most efficient way to store, update and manipulate data, particularly for a large number of structures.

Bridge documentation should provide the facility to monitor the progress of the recommended rehabilitation measures and to ensure that the work is undertaken satisfactorily within the specified time.

To ensure consistency in reporting, standard inspection procedures and forms should be used, as this provides a check list when collecting data on site. It is also important to use a reference system that allows defects to be located in the future for monitoring or repair.

Each state/territory road jurisdiction and the NZ Transport Agency (NZTA) has their own inspection policy, procedures and inspection forms, usually published in a bridge or structures inspection manual and available for download from their respective websites.

2.4 Regulatory Issues

All staff involved in inspecting and specifying, managing and carrying out maintenance and repair activities need to be aware of their obligations and liabilities under the statutory legislation applicable to their circumstances, e.g. the NZ *Resource Management Act 1991*, the *Health and Safety in Employment Act 1992*, and the *Building Act 1991*. Australian legal obligations are laid down in various Acts such as State Occupational Health and Safety Acts and Codes such as AS 1742.3.

Applicable legislation, codes of practice, standards and internal policies/manuals should be listed in each jurisdiction's inspection manual/policy.

2.5 Financial Control

2.5.1 Preparation of Funding Request

It is important to identify and work to priorities, particularly where there is a shortage of funds.

For inspection and routine maintenance activities (such as cleaning drains and joints, and painting handrails etc.), it is usual practice to base the funding request on historical expenditure.

For structural maintenance work (such as joint and bearing repairs or replacement and major repaints etc.) it is usual practice to prioritise the work as high, medium or low priority on a job-by-job basis and to seek funding for all high-priority jobs and a proportion of medium-priority jobs.

2.5.2 Monitoring of Expenditure

Once the budget has been set it is important to monitor expenditure against budget. This can be done by preparing a monthly expenditure forecast and comparing monthly:

- actual costs against forecast
- forecast expenditure against budget.

Comparisons can be by spreadsheet or by graphical means.

At the beginning of the financial year the forecast expenditure would equal the budgeted amount. However, if during the year the forecast expenditure is predicted to exceed the budget, additional funds should be sought or lower priority work deferred. If the forecast expenditure is predicted to be less than the budget, lower priority work can be advanced or the surplus declared.

2.6 The Inspection Process

The general inspection procedures, personnel, equipment and techniques are covered in jurisdiction specific manuals developed to meet the specific needs of each road jurisdiction. Example manuals include:

- *NSW: Bridge Inventory, Inspection and Condition Rating: Policy PN 158* (Roads and Maritime Services 2011)
- *NZ: Bridges and Other Significant Highway Structures Inspection Policy NZTA S6* (NZTA 2015)
- *QLD: Structures Inspection Manual* (TMR 2016)
- *VIC: Road Structures Inspection Manual* (VicRoads 2014).

Refer also to Section 3 of this publication and the further reading list included in References.

2.7 Feedback to Designers

Often designers are unaware of problems in the field, so it is important that there is a transfer of information between inspectors, designers, and owners to ensure that sound techniques are promoted and poor experiences highlighted so that problems are not perpetuated.

Identifying areas for better design and detailing should result in better performance and durability of future bridge structures and replacement components. Thus the bridge inspection engineer should ensure that inspection reports include comments on design performance of structures and their constituent components where relevant and should pass these comments to designers.

2.8 Economic Evaluation

2.8.1 General

Economic evaluation techniques can assist decision making on many aspects of bridge maintenance and rehabilitation and ensure the most cost-effective management strategy is adopted. For instance, economic evaluation enables management to make more rational decisions about when to replace an old structure rather than persist with high-cost maintenance.

It enables the comparison of a number of rehabilitation options that restore the bridge to its original level of service and extend its life. Economic evaluation also provides a means for assessing the total benefit to the land transport system (including benefits to road users) of upgrading weak links such as weight-restricted bridges.

Not all aspects are covered here and some are described only briefly. More detail on economic evaluation procedures can be found in documents such as:

- *NZ: Economic Evaluation Manual* (NZTA 2016a)
- *QLD: Cost-benefit Analysis Manual: Road Projects* (TMR 2011).

Other Australian road agencies have similar manuals. Also see Appendix A for an example of an economic evaluation procedure.

2.8.2 Determining Priorities for the Bridge Maintenance Program

Each item in the bridge maintenance program should be the most cost-effective response to the maintenance need identified in the inspection. Items should only be included in the program if they will give a future saving that exceeds the cost of the item. The method for determining this is summarised in Section 2.8.3. It is fundamental to understand why the defect has developed (outside cause, inherent design or construction problem or general deterioration) and anticipate what future changes may be likely and how quickly they might develop. Only then can repair options be identified and their effectiveness assessed. Maintenance funds should not be spent for cosmetic reasons.

When the budget for road works is restricted it is necessary to take the evaluation of the bridge maintenance program one step further and determine priorities. Some items that the analysis has shown will give future savings in excess of cost must be deferred until the following year. To prioritise the bridge maintenance items, the following factors should be considered:

- risk to the public
- importance of the component
- importance of the bridge
- condition of the component
- cost consequences of delaying the maintenance.

Priorities must be set consistently. Although a subjective decision can be made for small maintenance items, an empirical procedure based on the above factors should be implemented for larger maintenance and rehabilitation items. NZTA has implemented a specific prioritisation procedure for structural maintenance items. It will be found that some items can be deferred for a year with little impact on future maintenance or rehabilitation costs. Other items, however, may need to be undertaken promptly in order to avoid much larger costs in future years.

2.8.3 Economics of Maintenance Items

Savings resulting from maintenance items should be calculated using standard road jurisdiction evaluation parameters.

For example, NZTA specifies:

- a discount rate of 6%
- an evaluation period of 40 years.

Evaluation is done in constant dollars (i.e. no allowance is made for inflation).

All costs expected over the next 40 years, including that of the maintenance item itself, are listed at current prices along with the year in which they are likely to be incurred. These are obtained from historical records for the bridge concerned, and judgement about future needs, based on knowledge of the performance of other similar structures.

Every cost is then discounted using the appropriate discount rate defined by the road jurisdiction, e.g. say, a 6% discount rate.

All discounted costs are then added together to give a net present value of cost for that maintenance item. This is done for each possible maintenance option (including the ‘do nothing’ option). This process is then repeated, but with the assumption that the proposed maintenance item be carried out in future years. Only routine maintenance costs are allowed for this year. If the proposed maintenance item is essential, leaving it out of the current year’s program will have consequential changes on costs in future years.

The maintenance option with the lowest net present cost is the most economically favourable option.

2.8.4 Economics of Continued Maintenance Compared with Replacement

The calculation method described above for maintenance items can also be used to compare other options such as more substantial rehabilitation or even replacement. This section considers rehabilitation or replacement options on the existing alignment that reduce future maintenance costs but offer no other benefits to road users.

The procedure and parameters to be used are the same as in Section 2.8.3. For the bridge replacement option the future costs include the cost of the new bridge in the first year, along with the greatly reduced maintenance costs that will apply over the next 40 years.

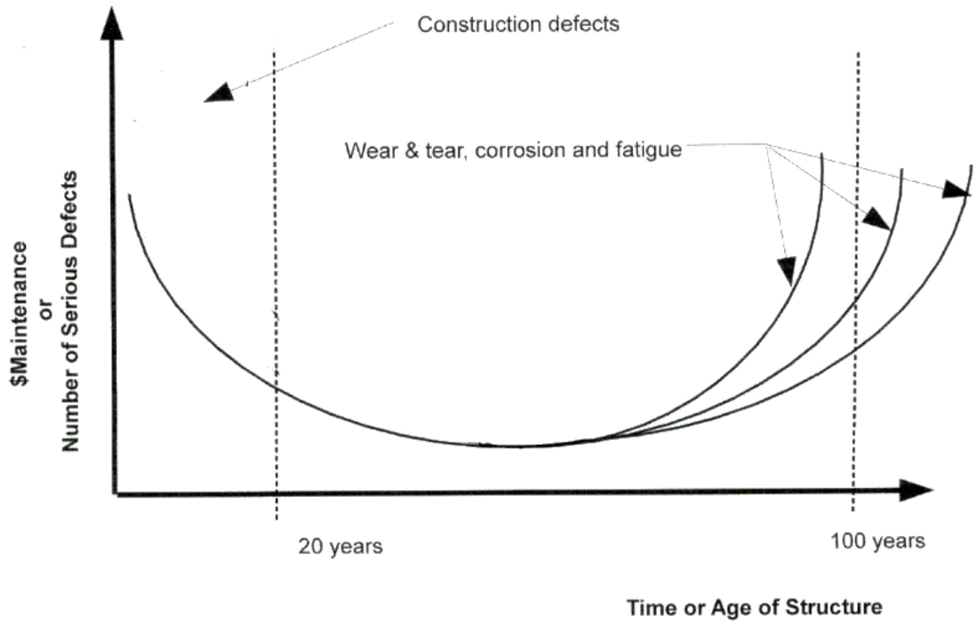
Replacement should also be compared with other options such as a minor or major rehabilitation. For a rehabilitation option, the future costs might include a substantial cost in the first year to prolong the life of the structure by a few years and reduced maintenance costs for the next few years that then steadily increase again until the bridge is replaced. The replacement cost is included in the year in which it is projected to be necessary and is followed for the remainder of the 40 year evaluation period by the much lower routine maintenance cost expected for the replacement bridge. If the rehabilitation will prolong the bridge life by more than 40 years then the replacement cost does not need to be considered.

Agencies are required to apply a specific legislated discount rate. For example, the 6% discount rate required by the NZTA *Economic Evaluation Manual* (NZTA 2016a) often has the effect of making rehabilitation the most economically attractive option, even if the work is only expected to extend the life of a bridge by 10 or 15 years.

Whichever option has the lowest net present value of costs is the most economically favourable option. However, other funding criteria for the particular agency may dictate the final outcome.

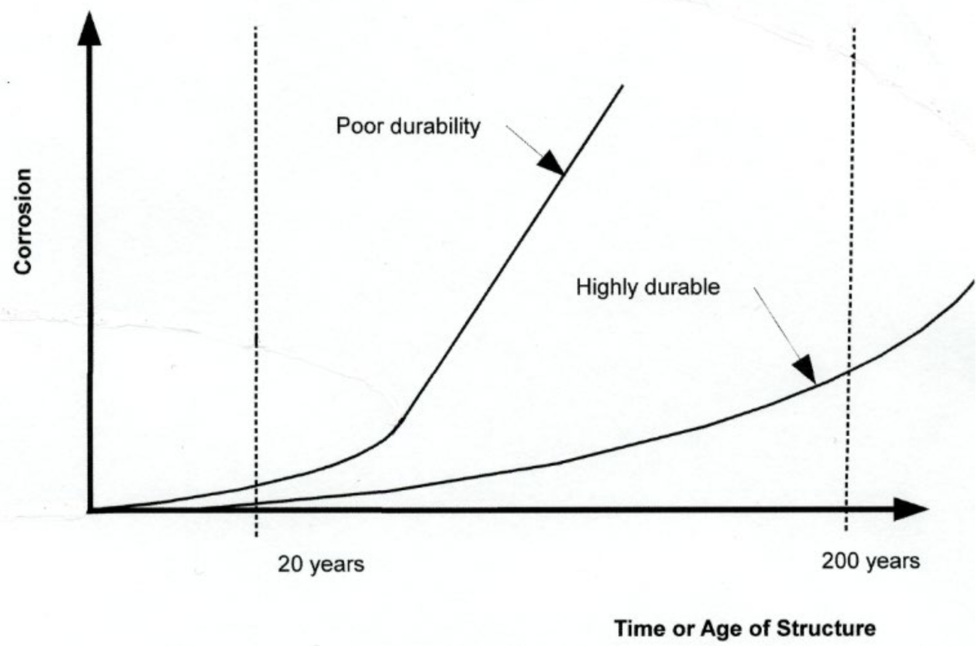
It is useful to consider whether defects may be attributed to construction defects or fair wear and tear, corrosion and fatigue. The occurrence of failures should be related to the lifecycle of the bridge. Figure 2.1 shows the relationship between time/age of the structure against the number of serious defects or cost of maintenance. Figure 2.2 shows the general relationship between poor durability and highly durable structures based on amount of corrosion and time or age of the structure.

Figure 2.1: Cost of maintenance and number of serious defects vs time or age of structure



Source: F McGuire (c2009).

Figure 2.2: Poor and high durability as related to the degree of corrosion vs time



Source: F McGuire (c2009).

2.8.5 Economics of Road Benefits

When a bridge is rehabilitated or replaced there are often benefits to road users in addition to the future maintenance savings for the road agency. These benefits can tip the scales in favour of such options in preference to continued maintenance of the existing situation, even though the continued maintenance has the lowest present value of cost.

Strengthening, for instance, can enable a weight restriction to be lifted. This would permit more efficient freight transport on that route and could also allow heavy vehicles to transfer from other longer routes, thereby saving on vehicle operating costs and travel time costs. Rehabilitation might also reduce accident costs, for instance, by widening the deck of a narrow two-lane bridge.

With replacement, further benefits are possible, as the new bridge can be built on a different alignment that allows travel time savings and possibly reduces the accident rate.

Methods for calculating these benefits are contained in the various road jurisdiction manuals, e.g. NZTA *Economic Evaluation Manual* (NZTA 2016a). The benefits are discounted in the same way as the costs, and given as a present value of benefits for the option. The present value of benefits is divided by the present value of costs for the option less the present value of costs for the 'do-minimum', as calculated earlier, to obtain the benefit/cost ratio for the proposed improvement. This is used to rank the proposed rehabilitation or replacement against other road improvement proposals.

2.8.6 Risk Assessment

In calculating costs for various options it is sometimes not possible to pinpoint exactly when a particular item of expenditure such as a bridge replacement might be necessary. For instance analysis of the structure could reveal that it is likely to fail in a certain size flood or earthquake. However, it is impossible to predict when such an event will occur. The solution is to estimate probabilities of occurrence in each year and then carry out a risk assessment.

2.9 Health and Safety

It is of paramount importance that health and safety have a high priority at all times during all field operations. Rules and regulations for the health and safety of personnel and safety of traffic must always be adhered to.

All inspection and maintenance activities should comply with the specific requirements of the relevant jurisdiction along with any applicable legislation, codes of practice and standards.

While structure inspections, detailed investigations and maintenance activities may not be categorised as a 'specified work' category, documented safe work procedures (for example, Safe Work Method Statements) should be prepared in order to:

- reduce the risk to staff undertaking field work
- provide documentary evidence of fulfilling obligations under the Act.

The documented procedures should address all aspects of the work including access requirements, and a copy should be maintained on site at all times.

Consideration should be given to specific issues such as:

- temporary traffic control
- work site safety
- use of hazardous materials and need for appropriate equipment
- presence or suspected presence of asbestos containing materials within the bridge with the potential to be disturbed during the planned activities (e.g. permanent formwork, service pits etc.)
- site characteristics, height of structures
- access to inspect and make repair works
- environment hazards arising from flood, scour, seismic events, fauna, confined spaces etc.

Each road jurisdiction will have specific requirements stipulated in either their manuals or contract requirements and specifications.

Where inspections are to be carried out on structures located over or under the assets of other Agencies, the relevant Regulations and Codes of Practice relating to work on or close to their assets must also be adhered to and, where necessary, referred to in the safe work procedures developed for the inspection.

Appropriate first aid training should be provided to site operatives. The level of first aid training qualification required and numbers of trained staff required to be present on site should take into account the location and type of activity being undertaken.

Because inspection staff often work in remote areas they must have a communication system that operates throughout the inspection area. They must also be familiar with the physical hazards and traffic control required or off-road parking available at each site. If staff are in the field for an extended period then additional communication protocols, such as a daily reporting should be implemented. There is increasing recognition of the hazards associated with bridge inspection and jurisdictions are now commonly specifying a minimum of two operatives in the field for inspection activities.

3. Inspection, Testing, Monitoring and Reporting

3.1 Introduction

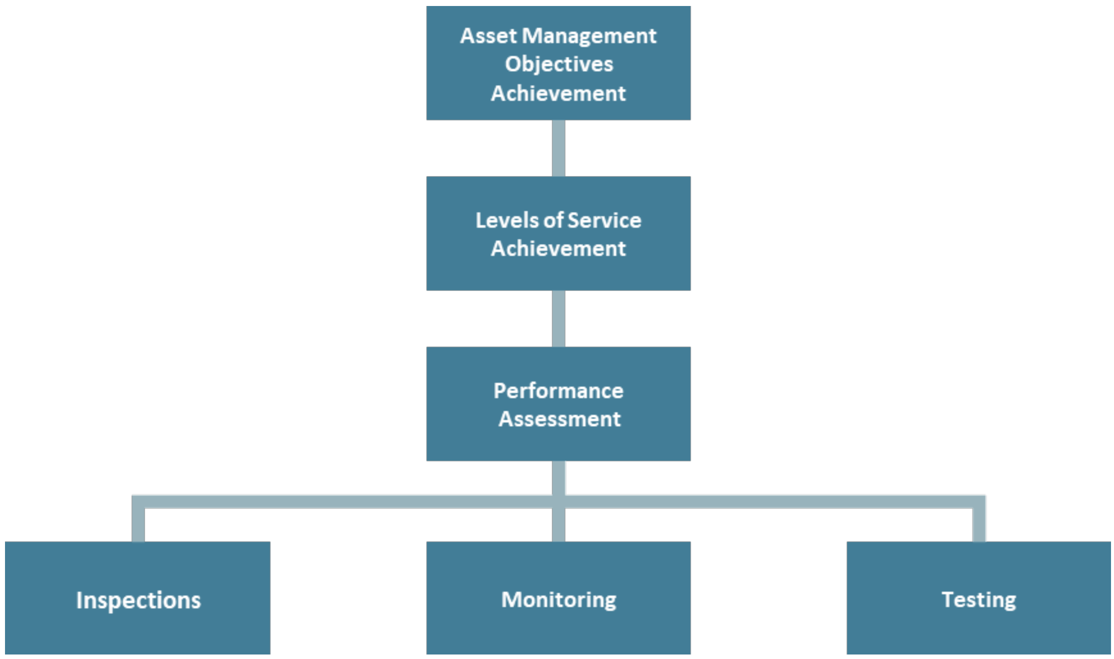
3.1.1 Purpose

The overall objective for the inspection, testing, monitoring, and reporting on bridge assets is to ensure bridges are maintained to achieve lowest whole of life cost and provide the appropriate levels of service as outlined as outlined in *Engineering Guideline to Bridge Asset Management* (Austroads 2021). They include:

- Live load access – through understanding current defects and their potential impact on bridge capacity and understanding the impact of changing vehicle loadings and volumes.
- Resilience – Inspections after events documenting any damage caused and identifying remedial actions (if any) to return the bridge to a serviceable state.
- Aesthetics – through the identification of visual defects such surface rust on steel components and graffiti.
- Freight access and efficiency – through maximising the availability of the structure and monitoring the performance over time of material types and details to enable accurate modelling of when intervention criteria will be met.
- Traffic safety impacted by shoulder widths, geometry, barriers, signage, surfacing, etc.
- Multi-modal access – through ensuring availability of non-vehicular users.

The inspection, testing and monitoring of bridges provide an integral part of the wider asset management system, and a key factor in providing data to determine the achievement of levels of service and the asset management objectives as shown in Figure 3.1.

Figure 3.1: Inspections and monitoring within the Asset Management System



To ensure bridge inspections are effective and efficient, it is essential to have:

- appropriate timings for different inspection types
- proper equipment and techniques
- consistent and effective inspection procedures
- consistency in condition and inventory data management
- appropriately trained and experienced personnel
- good preparation
- effective quality control processes in place.

3.1.2 Scope

This document provides a framework that can be implemented by jurisdictions with included flexibility where appropriate to allow for varying degrees of maturity.

The guideline is specifically focused on the inspection regime, and the following seven aspects:

- bridge and large culvert terminology
- inspection type classification
- inspection frequencies
- inspector qualification
- inspection auditing
- quality control
- reporting.

The framework is intended to define the outcomes - with jurisdictions defining their own policies, processes and procedures that will deliver those outcomes.

3.1.3 Framework Level 2 Inspection Approach

In relation to Level 2 inspections, this document outlines two approaches.

- Condition assessment framework. This framework rates the condition of the components on a 1 to 4 scale and is the most widely used method within Australasia.

  The advantages of this inspection type include:

  - cost effective, consistent, and sufficiently detailed inspection
  - commonly used throughout Australia and New Zealand
  - quick identification of component replacement, or rehabilitation.

- Defect Mapping framework. This framework includes the initial mapping of defects on components, then monitoring the changes in the defects over time.

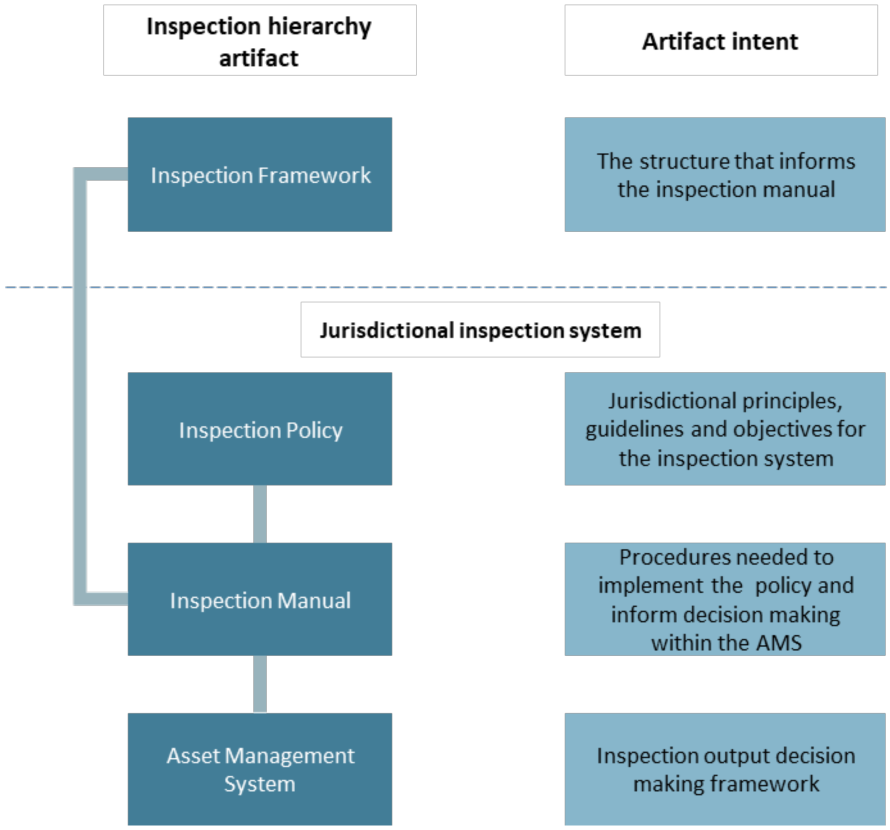
  The advantages of this inspection type include:

  - allows for changes in defects to be monitored
  - interventions are more targeted to specific faults
  - deterioration rates of faults easily monitored over time.

3.1.4 Documentation Hierarchy

This framework is envisioned to inform the inspection manuals of a road controlling authority. This will assist the inspection manuals in delivering the objectives of the inspections contained within the policy and ensure the outputs derived from the inspection system inform the decision making and performance measuring within the asset management system. How this framework sits within a documentation hierarchy is shown in Figure 3.2.

Figure 3.2: Inspections and monitoring framework within the documentation hierarchy



3.1.5 Methodology

The information included within this report was obtained from a literature review of the current Bridge Inspection Policies, Manuals, and stakeholder interviews of each of the Austroads jurisdictions to understand current Australian and New Zealand jurisdictional practice. The literature review included:

- Roads and Traffic Authority 2007 *Bridge inspection procedure manual,* Sydney, Roads and Traffic Authority of New South Wales
- Roads and Traffic Authority 2011 *Policy Number PN158; Bridge inventory, inspection and condition rating policy,* Sydney, Roads and Traffic Authority of New South Wales
- New Zealand Transport Agency, Waka Kotahi 2019 *Policy Number S6; Bridges, geotechnical structures, and other significant highways structures inspection policy,* Wellington, New Zealand Transport Agency, Waka Kotahi
- Department of Transport 2014, *Policy, Routine maintenance bridge and major culvert inspections, Level 1,* Darwin, Northern Territory Government.
- Department of Infrastructure 2013, *Northern Territory Structure Inspection Manual Draft,* Darwin, Northern Territory Government.
- Transit New Zealand 2001, *Bridge inspection manual*, Wellington, Transit New Zealand

- Department of Transport and Main Roads 2016, *Structures inspection manual,* Brisbane, Queensland Government
- Department of Planning, Transport and Infrastructure, South Australia 2020*, Road structures inspection manual*, Adelaide, Government of South Australia
- Vicroads 2018, *Road structures inspection manual*, Melbourne, Vicroads
- Main Roads Western Australia 2013, *Structures inspection and information management policy,* Perth, Main Roads Western Australia
- Main Roads Western Australia 2013, *Routine visual bridge inspection guidelines (Level 1 inspections for bridges),* Perth, Main Roads Western Australia
- Main Roads Western Australia 2017, *Detailed visual bridge inspection guidelines for concrete and steel bridges, (Level 2 inspections),* Perth, Main Roads Western Australia
- Main Roads Western Australia 2014, *Detailed visual bridge inspection guidelines for timber bridges, (Level 2 inspections),* Perth, Main Roads Western Australia
- Main Roads Western Australia 2012, *Detailed non-destructive bridge inspection guidelines, concrete and steel bridges, (Level 3 inspections),* Perth, Main Roads Western Australia
- Western Australia Local Government Association 2020, *Level 1 Bridge inspection framework*, Perth, Western Australia Local Government Association.

International practice was considered through the review of the following manuals:

- Federal Highway Administration 2012, *Bridge inspectors reference manual*, Washington, U.S. Department of Transportation
- Highways England: *CS450, Inspection of highway structures*, Birmingham, Highways England.

3.2 Terminology

3.2.1 Bridge and Large Culvert Definitions

To ensure a consistent approach to the management of bridge assets this section outlines standard definition that can be adopted. A bridge and culvert are defined as follows:

- a bridge is a structure that carries a road and/or path over an obstacle. A bridge has a clear span of greater than 3m, a culvert with a diameter ≥1.8m or sectional area greater than $3m^2$
- a culvert has the primary purpose of carrying water or is an underpass under a road for livestock or pedestrians. A culvert has a waterway/underpass sectional area of less than $3m^2$ or a diameter of ≤1.8m.

At present there is a range of documented terminology in use across jurisdictions as outlined in the Table 3.1.

Table 3.1: Jurisdictional bridge and culvert definitions

Jurisdiction	Definitions
Transport for NSW	• Bridges and culverts which carry State Roads over depressions and obstructions such as waterways, roadways, or railways. They must have an opening of six metres or more when measured along the road centre line, or spring lines of arches, or extreme ends of openings for multiple cells. • State Assets that are defined as bridges controlled by the TfNSW and located on Regional or Local Roads including border bridges. • Special structures under State Roads, e.g. pedestrian subways, utility tunnels and wildlife and stock access, regardless of the six metre length limit. • Other structures as determined by IAM.
NZ Transport Agency Waka Kotahi	• Bridge shall include all structures which directly support vehicle traffic, including culverts and multiple culverts with a total waterway area greater than $3.4m^2$, critical small culverts with a total waterway area less than or equal to $3.4m^2$ and all stock underpasses and pedestrian subways.
Department of Infrastructure, Planning, and Logistics, Northern Territory	• Major culverts (culverts with a span or diameter greater than 1.8 m or cross-sectional waterway area greater than 3 m^2).

Department of Transport and Main Roads, Queensland	• Major Culverts: All major culverts meeting the following criteria:
	- metal culverts (steel and aluminium):at least one barrel (cell) with span, height or diameter ≥1.2 m, or
	- all other culverts:
	  - pipes with at least one barrel (cell) with diameter ≥ 1.8 m, or
	  - rectangular/oval/arch culverts at least one barrel (cell) with span > 1.8 and height > 1.5 m.
	- stock and pedestrian underpasses.

Department of Infrastructure and Transport, South Australia	• Bridges: A structure with a minimum span or diameter ≥ 1.8 m for the primary purpose of carrying a road or path over an obstacle. • Major Culvert: A structure with a minimum span or diameter ≥ 1.8 m (head wall, kerb or more) with a height ≥ 0.9 m and for the primary purpose of carrying water.
Department of Transport Planning, Victoria	• Bridges: A structure with a minimum span or diameter ≥ 1.8 m or a waterway area ≥ 3 m^2 for the primary purpose of carrying a road or path over an obstacle. • Major culverts: A structure with a minimum span or diameter ≥ 1.8 m or a waterway area ≥3 m^2 for the primary purpose of carrying water.
Main Roads Western Australia	• Bridge: a structure (except for sign gantries) having a clear opening in any span of greater than 3 metres measured between the faces of piers and/or abutments or structures of a lesser span with a deck supported on timber stringers. Note: A traffic-supporting structure that is a combination of bridge and culvert openings is managed as a 'Bridge'. • Precast Box Unit Bridge: a specific superstructure bridge type consisting of precast box units of clear opening in any span of greater than 3 metres. Referred to as a Bridge • Culvert: a structure under a road having only clear openings of less than or equal to 3 metres measured between the faces of piers and/or abutments or a pipe shaped structure of any diameter.

3.2.2 Bridge Components

Terminology for bridge components varies across jurisdictions. The tables below highlight changes in terminology used between jurisdictions (except for NZ Transport Agency Waka Kotahi where standard component names are not documented). The component names are based on those used in Figure 1.1, Figure 1.2, and Figure 1.3 provided by Department of Transport and Main Roads, Queensland.

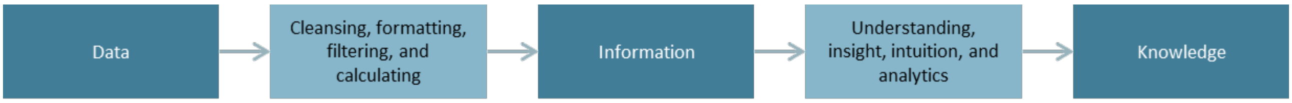
Table 3.2: Jurisdictional component naming

Component	Jurisdictional component names					
	Transport for NSW	Department of Infrastructure, Planning, and Logistics, Northern Territory	Department of Infrastructure and Transport, South Australia	Department of State Growth, Tasmania	Department of Transport Planning, Victoria	Main Roads Western Australia
Concrete and steel bridge terminology (refer Figure 1.1)						
Piles/ Columns	Piers	Piers	Piers	Piers/ Columns	Piers/ Columns	Piers/ Columns
Abutment Protection	Scour Protection	Abutment protection	Scour Protection	Scour Protection	Scour Protection	Batter Protection
Spread Footing	Footing	Footing	Spread Footing	Footing	Footing	Footing
Pile Cap	Pile Cap	Headstock	Pier Cap	Cross head	Cross head	Cap Beam
Cross Girder/ Diaphragm	Cross Girder / Diaphragm	Cross Girder / Diaphragm / Bracing	Cross Beam/ Cross Girder	Diaphragm / Bracing	Diaphragm / Bracing	Diaphragm / Bracing
End Diaphragm	End Beam	Not stated	End Beam	Diaphragm	Diaphragm	Diaphragm
Wearing Surface	Wearing Surface	Wearing Surface	Running Surface	Wearing Surface	Wearing Surface	Surface
Bridge Barrier	Bridge Barrier	Barrier/ Parapet	Bridge Barrier / Parapet	Bridge Railing / Parapet	Bridge Railing / Parapet	Bridge Guardrailing
Timber Bridges (refer Figure 1.2)						
Horizontal Waling	Wale	Not stated	Horizontal Waling	Waler	Waler	Waler
Diagonal Bracing	Brace	Not stated	Diagonal Bracing	Cross Braces	Cross Braces	Bracing
Headstock	Capwale / Headstock / Sill	Not stated	Half Caps	Cross Head	Cross Head	Halfcap
Girder	Beam	Not stated	Stringer	Stringer	Stringer	Stringer
Transverse Deck Planks	Transverse Deck Planks	Deck Planks	Transverse Deck Planks	Deck Planks	Deck Planks	Deck Planks
Masonry Bridges (refer Figure 1.3)						
Springing	Not stated	Spring line	Springing	Not stated	Not stated	Not stated
Arch Ring	Not stated	Arch	Arch Ring	Not stated	Not stated	Not stated
Rise	Not stated	Not documented	Rise	Not stated	Not stated	Not stated
Crown	Not stated	Not documented	Crown	Not stated	Not stated	Not stated
Spandrel Wall	Not stated	Wall	Spandrel Wall	Side Wall	Side Wall	Head Wall
Block	Not stated	Block	Block	Block	Block	Not Stated

3.3 Data Collection

In order to broaden knowledge on a structure, it starts with ensuring the right data is collected. The data is a series of facts collected at a point in time or from continual monitoring. This is converted into information through cleansing, formatting, filtering, and calculating, depending on the information required. The information is converted to knowledge through understanding, insight, intuition and analytics.

Figure 3.3: Converting data to knowledge



The uses of the data collected from the inspection processes, and in particular Level 2 inspections, needs to be understood to ensure the right data is collected on site and that it is catalogued appropriately. To whom and how the data will be shared across the organisation should be understood before data is collected to avoid additional site visits to capture missing information. The users of this data may be from within the bridge asset management team or outside of it (example of uses are shown in Table 3.3).

Table 3.3: Example bridge inventory, condition and defect data uses

Data use	Data/information required
Level 2 inspection request	Level 1 inspection report including details on specific issues requiring further investigation. Details of previous repairs/construction undertaken since last Level 2 Inspection.
Level 3 inspection request	Level 2 inspection report including details on specific issues requiring further investigation.
Works prioritisation	Condition or defect data to enable works scopes to be developed and prioritised. This can be sourced from any level of inspection or design report.
Maintenance requests	Observed defect or condition data on issues requiring rectification from maintenance budget allocations. This can be sourced from any level of inspection or design report.
Capital works funding requests	Observed defect or condition data on issues requiring rectification from capital budget allocations. This can be sourced from any level of inspection or design report.
Third party data requests	External to the organisation request can be made for inspection or inventory data.
Over mass/over dimension permit system	Inventory data, condition/defect data that can impact the load carrying capacity of the bridge.
Asset valuations	Inventory data to enable a modern equivalent replacement cost to be determined.
Asset depreciation	Condition or defect data to enable the remaining useful life to be determined.
Impairment assessments	Post event inspection data either condition or defect based to understand event impact on the asset.
Performance reporting	Condition or defect data that is used in calculating performance indexes or trend analysis.

3.3.1 Current Practice

The inspection process will generate data that needs to be stored within the asset management information system. The data structures will be organisational specific and largely governed by the system the data is to be stored within.

The *Austroads Data Standard – Priority Data Sets and Metrics* (Austroads 2020) provides the minimum requirements for Road Controlling Authorities. A more extensive data set is included in the *Data Standard for Road Management and Investment in Australia and New Zealand Version 4* (Austroads 2022). Actual data collected and stored will be specific to the organisation’s requirements.

The current assessment forms (manual or electronic) used by organisations for routine and condition inspections collect the data required by that organisation. This is the data the organisation deems necessary to manage its bridge stock and aid in the wider asset management decision making processes. In time it is likely the data requirements will change as organisational maturity increases, and in these instances the data previously collected should not be discarded.

Table 3.4 is a summary of current data types collected during routine inspections (Level1), condition inspections (Level 2) and special inspections (Level 3 and 4).

Table 3.4: Jurisdictional bridge inspection data collection

Jurisdiction	Inspection level	Data types collected
Transport for NSW	1	Data is collected on visible accident damage or other deformations in the superstructure such as trusses and information regarding the status and performance of ancillary elements such as barriers, deck scuppers, and waterways.
	2	General information data, condition rating of components and quantities at each condition state, maintenance requirements, and inventory verification, recommended changes to next inspection date. Underwater inspections include defect quantification, the photos and videos taken as part of the inspection
	3	Specialist structural inspection including obtaining data needed to assess issues within a defined scope.
	4	Data required for load rating assessment due to proposed changes in legal loadings, new vehicle types, or required to determine the structural capacity of the bridges.
NZ Transport Agency Waka Kotahi	1	Capture of routine maintenance and any other obvious structural defects.
	2 & 3	Verification of structural inventory, defect identification and documentation, remedial works required.
	4	Specialist structural inspections/investigations are undertaken as required to obtain more detailed information on a specific defect.
Department of Infrastructure, Planning, and Logistics, Northern Territory	1	Identification of maintenance required identification of the need for a Level 2 inspection to assess specific defects.
	2	Verification of structure inventory, component and overall structure condition rating, identification and prioritisation of repairs, identification of components needing monitoring, identification of more detailed inspections and testing requirements.
	3	Undertaken on an as needs basis to collect data to assess the structural condition and capacity of structures which have been identified as potential candidates for rehabilitation, strengthening, widening or replacement.
Department of Transport and Main Roads, Queensland	1	Defects/distressed components, maintenance requirements, inventory data verification, monitoring requirements and higher-level inspections required.
	2	Quantifying structural defects, identification of maintenance requirements, condition rating components and the structure, identification of components requiring further investigation/monitoring, and inventory verification.
	3	Inventory data collection, design verification data, post flooding data, specialist inspection/testing results.
Department of Infrastructure and Transport, South Australia	1	Capture of defects that may impact safety, routine maintenance, and issues requiring further identification.
	2	Verification of structure inventory, component and overall structure condition rating, identification and prioritisation of repairs, identification of components needing monitoring, identification of more detailed inspections and testing requirements.
	3	Undertaken on an as needs basis to collect data to assess the structural condition and capacity of structures which have been identified as potential candidates for rehabilitation, strengthening, widening or replacement.

Jurisdiction	Inspection level	Data types collected
Department of State Growth, Tasmania	1	Identification of damage to components or significant changes in component condition, identifying when higher order inspection is required.
	2	Identification of maintenance needs, performance of previous maintenance activities, rating condition of components and structure, identification of components requiring detailed inspection, and components requiring monitoring.
	3	Undertaken on an as needs basis to collect data to assess the structural condition and capacity of structures.
Department of Transport Planning, Victoria	1	Identification of damage to components or significant changes in component condition, identifying when higher order inspection is required.
	2	Identification of maintenance needs, performance of previous maintenance activities, rating condition of components and structure, identification of components requiring detailed inspection, and components requiring monitoring.
	3	Undertaken on an as needs basis to collect data to assess the structural condition and capacity of structures.
Main Roads Western Australia	1	Check overall safety and performance of the structure. Identification of major accident damage or obvious failure of structural components.
	2	Record physical and functional condition, assess performance of previous repairs, record inventory changes, and identify maintenance needs.
	3	Undertaken on an as needs basis to collect data to assess the structural condition and capacity of structures which have been identified as potential candidates for rehabilitation, strengthening, widening or replacement.

3.3.2 Data Collection Framework

Level 1 Data Framework

The data collected by each of the jurisdictions shows strong alignment. The framework for data collection for Level 1 inspections includes:

- capture defects that impact on functionality and safety that can be either routine maintenance or capital works
- capture maintenance items to be completed
- capture the need for items that have deteriorated quickly and require an out of cycle Level 2 inspection
- capture data on components inspected
- capture data on defects affecting asset user safety.

Level 2 Component Condition Rating Data Framework

The data collected by each of the jurisdictions shows good alignment for the condition assessment framework. The data collected for these inspection types includes:

- validation/collection of the structural inventory
- capture maintenance items to be completed
- condition rating of all the bridge components and structure overall (if used in decision making processes)
- capture the need for items that have deteriorated quickly and require monitoring and/or a higher-level inspection.

Level 2 Defect Mapping Data Framework

The defect mapping framework will collect much of the same data as the condition inspection framework, except for the condition rating data. In place of this new defect or change in defect data is collected.

Level 3 Specialist Inspections Data Framework

Specialist inspection include all inspection types other than Levels 1 and 2. This is a broad range of inspections typically undertaken on an as needed basis. These inspections are specifically scoped and can included data capture on:

- as constructed drawing verification/structural inventory data
- capture of data to allow post inspection analysis on specific issues
- material testing and durability assessments
- specific data captured that cannot typically be collected through Level 2 inspections due to inaccessibility.

3.4 Component Condition Rating

The overall rating of a bridge and components can be undertaken using different techniques. The methods currently used are typically a visual inspection as part of a Level 2 inspections. This data can be used to determine the remaining life of a component or bridge to trigger a renewal or rehabilitation of that component or bridge.

3.4.1 Component Condition Rating Framework

Table 3.5 shows a generic component rating framework. Most jurisdictions have adapted similar rating criteria that are specific to different components and construction types to suit their requirements in using the data.

Table 3.5: Component condition rating framework

Condition State	Description
Concrete components	
1	• The components have minor fine cracking due to corroding reinforcement but there should be no shear cracking or spalling of the concrete. • Hair line cracking may exist at the built-in supports or fine vertical shrinkage cracks may appear in the beams due to the locked-up movements of the structure. • The component shows only minor scaling or efflorescence having no effect on strength.
2	• The components may have fine flexural and/or shear cracking. • Vertical shrinkage cracks and cracking at built-in supports may be fine. • Longitudinal cracking along the bottom of the beams due to reinforcement corrosion may be of fine size with minor spalling. • Construction joints opening. • Scaling of the concrete surface may be in larger patches with an increase in white efflorescence powder on the surface. • Minor impact damage of no consequence is evident.
3	• Flexural cracking and shear cracking may be medium sized. • Vertical shrinkage cracks and cracking at built-in supports may be medium in size. • Longitudinal cracking may be medium along the bottom of the beams due to reinforcement corrosion and there may be large spalls with delaminated cover concrete. • Exposed reinforcement may have heavy corrosion with section loss up to 20% in areas. The components may have medium cracking in the bearing areas at the ends of the beams. • Scaling and efflorescence are prevalent. • Impact damage evident with minor effect on component strength or structure

Condition State	Description
4	• Flexural and shear cracking may be heavy with medium cracking along joints. • Vertical shrinkage cracks and cracking at built-in supports may be heavy in size. • Severe spalling or delamination of the of the beams components is occurring, with heavy corrosion of the reinforcement with section loss more than 20% in areas. • Beams have heavy cracking in the bearing area with severe loss of bearing support. • The component may be beginning to lose shape due to movement of a footing, or there may be cracking due to slight differential movement of the foundations. • The arch may have lost shape due to movement of a footing, or there may be heavy cracking due to differential settlement of the foundations. • Impact damage may be severe and having a definite effect on the component or structure.
Concrete cracking definitions	• Hairline =<0.1mm • Fine >0.1mm to <=0.3mm • Medium >0.3mm to <=0.7mm • Heavy >0.7mm
Steel components	
1	• The protective coating system is sound with only minor chalking, peeling, or curling of paint. There are no exposed metal surfaces. • All weld, bolts, or rivets are in good condition, with no signs of weld cracks, or movement of plates or sections of the component evident.
2	• Rust spotting of the protective coating system to 5% of the component surface area is occurring and the protective coating system is no longer effective. No corrosion or section loss has occurred. • All welds or bolts are in good condition with no cracking, corrosion, or loose bolts. • Minor impact damage of no consequence is evident.
3	• Some surface pitting is present and active corrosion is occurring in isolated areas, but no loss of a whole area is occurring to affect the strength of the component. • Protective coating system has broken down with surface rusting to 10% of component area and pitting present in several locations. • Nuts and bolts may be corroding but are still tight. No cracking of welds has occurred. • Impact damage to component is evident and has minor effect on the component’s stiffness or alignment.
4	• Corrosion is well advanced, and some loss of section has occurred which may have a detrimental effect on the strength of the member, i.e., a flange badly corroded over much of its length. • Welds cracks are visible. Nuts or bolts are severely corroded, and possibly no longer functioning to full capacity. • Impact damage to components is evident and has major effect on strength of the component.
Timber components	
1	• The component is in good condition with little or no pipe rot or decay. • There may be minor splits or checks having no effect on component strength. • Glued laminated beams have no separation of the laminations and no splits or checks within the laminations.
2	• Components are in good condition and may have up to 30 % of the diameter pipe rot or decay. They may also have minor decay, splitting, checking, or crushing but not of sufficient magnitude to affect the strength of the component. • Glue laminated beams have no separation of the laminations but may have minor splits or checks within the laminations but not of sufficient size or magnitude to affect component strength.
3	• Component has a reasonable amount of pipe rot or decay, up to 50% of the diameter. They may have large splits or checks, which may have a reduction in strength of the component. • Splits may be separating under load causing crushing of the component, or crushing may be due to water ingress softening the load bearing areas of the timber. • Glued laminated beams may have fine splitting along laminated joints and the outer fibres may be peeling or a fine moment crack may exist in the outer lamination. There may also be fine splitting within the laminations due to tensile stresses between the fibres.

Condition State	Description
4	• The component may have excessive pipe rot or decay, up to 70 % of the diameter, accompanied by severe splitting or crushing. • Strength of the member has been severely affected and failure may be imminent. • Glued laminated beams have extensive splitting along the lamination joint. • Medium moment cracking of the laminations has also occurred with the outer fibre splitting apart due to tensile stresses, and the beam has partially failed.
Masonry components	
1	• The component shows little or no deterioration with no cracking of mortar or loss of mortar between the blocks. • Small, localised areas of dampness and efflorescence may be present.
2	• Minor cracking of the mortar or minor loss of mortar between the blocks is visible, but not sufficient to affect the strength of the component. • The shape of the component is still good and there is no cracking or bulging evident. • Large areas of dampness and efflorescence may be evident.
3	• Cracking or loss of mortar between blocks which has a minor effect on the strength of the component. • Some blocks may have slipped/moved slightly due to the loss of mortar. • Minor settlements, movements, loss of component shape, or cracking has occurred, but not of sufficient magnitude to cause concern.
4	• There is severe cracking or loss of mortar between blocks which has a significant effect on the strength of the component. Some soffit blocks may have slipped significantly, and some blocks may have cracked through, or edges broken off. • Abutments or piers may have settled or moved significantly causing a loss of shape of the component. • Differential settlement of the foundations may have also caused heavy cracking along the component soffit. • Earth pressure on a sidewall may have caused heavy cracking, movement, or large bulging of the blocks to occur.

3.4.2 Defect Mapping Framework

The framework alternative does not consider or collect condition ratings on the components or structure. Refer section 3.6.3.

3.5 Bridge Indices

Condition data can be used to calculate a range of indices post inspection and care should be taken when defining condition ratings to ensure they are compatible with the models it is used within. Such models used by jurisdictions in New Zealand and Australia include:

- Bridge health index. This index is used in assessing the structural or functional health of a bridge. There are many models in-use that fit within the following four generic types (Federal Highway Administration 2016):
  - Ratio-based methods assign a bridge condition index or bridge condition number based on the ratio of the current condition to the condition of the structure when it was new.
  - The weighted averaging approach is suitable for planning bridge maintenance and rehabilitation activities. The approach estimates the condition of the whole structure by combining condition ratings of all individual bridge elements weighted by their significance or contribution to the structural integrity of the bridge.
  - The worst-conditioned component approach is common in systems that carry out inspections on key bridge components. This method is used to extract the critical defects in bridge components.
  - Qualitative methods do not report the condition of the bridge on a numerical scale. They describe a structure as either "Poor," "Fair," or "Good," based on the condition state and importance of the elements under investigation.

- An overall bridge condition index (BCI) can be calculated from the component ratings. The BCI gives an overall mean condition to enable the bridge to be assessed against other bridges across the network. Typically, the BCI is not calculated by the inspector and is undertaken in post processing of condition data.

  While the BCI gives an indication of components in distress compared to other bridges, if used for prioritisation additional measures should be applied considering risks associated with meeting levels of service. These would include items such as:

  – structural sufficiency, capacity and performance.

  – potential for live load reduction if repairs are not undertaken

  – AADT

  – road hierarchy

  – impact on non-vehicular users of the bridge

  – impact on long term resilience

  – aesthetic impacts.

3.6 Maintenance Intervention Criteria

Intervention criteria can trigger either maintenance activities to repair minor defects, or larger capital works such as reinstallation of the protective coating system, replacement of components, insect infestation treatments or bridge renewals.

3.6.1 Level 1 Inspections

Level 1 inspections are limited to visible components visible/inspectable from the ground. These inspections do not require access equipment or lane closures. Typically these inspections identify minor defects that cyclic maintenance can address. There are no set intervention criteria except for determining if action is required by the inspector. Minor maintenance activities can be completed in conjunction with the inspection when the inspector is part of the maintenance crew. Additional focus to ensure the safety of the asset user is part of this level of inspection.

Defects to be identified (and rectified) include:

- Approach surfacing and pavement defects.
- Deck wearing course.
- Bridge footpath defects.
- Signage, delineation completeness, damage, cleanliness, orientation, visibility to road user, and loose and missing bolts.
- Barrier alignment and damaged posts and rails.
- Barrier with loose and/or missing bolts and/or clamps, missing and damaged spacer blocks, corrosion, correct rail height and alignment. The approach barrier and bridge barrier connection. Barrier expansion mechanism working properly.
- Deck joint leakage and loose, missing, and damaged bolts, seals, and other components; dirt or objects impeding free movement and proper functioning.
- Accumulation of debris on the deck, in gutters, scuppers and drains which may obstruct free drainage and cause ponding.
- Accumulations of debris and growth inside drains, channels, inlet and outlet pits and sumps which may obstruct free drainage and cause ponding.
- Weepholes in abutments and retaining walls blocked or partially blocked prevent free drainage.
- Vent holes in superstructures which prevent flotation of structures are blocked or partially blocked.

- Scour occurring, especially at outlets to culverts and off deck drainage channels.
- Erosion and scour of embankments.
- Vegetation within 2.0m of abutments and wingwalls.
- Slope protection damage and undermining by scour or embankment movement.
- Bridge bearings, pads, and pedestals are free of debris, dirt, rust vegetation. Are well maintained and functioning properly with no obstruction or seizing of mechanism.

3.6.2 Level 2 Component Condition Rating Framework

Table 3.6 is a review of jurisdictional practice in relation to assigning maintenance intervention criteria.

Table 3.6: Current maintenance intervention criteria

Jurisdiction	Inspection level	Data types collected
Transport for NSW	Any	Maintenance actions are captured in the Level 2 inspection report. It is the Inspector who assigns a risk rating which is used by the Bridge Maintenance Planner to determine the invention criteria
NZ Transport Agency Waka Kotahi	Any	Maintenance requirements are documented and prioritised by the inspector, against the risk-based intervention criteria. These are validated by a Senior Bridge Inspection Engineer.
Department of Infrastructure, Planning, and Logistics, Northern Territory	Any	Specific intervention criteria are not attached to a condition state. As part of the inspection report and maintenance schedule is developed which defines the intervention period for all defects on the following basis: • Urgent - within 6 months • High - within 12 months • Medium - within 2 years • Low - Either 5 years (or as resources allow) • Monitor – (monitor for further deterioration)
Department of Transport and Main Roads, Queensland	3	Under this condition state intervention (which is not exclusively related to physical intervention at the site) is considered required, however the manual does not give specific time frames. ‘Defects affecting the durability/serviceability which may require monitoring and/or remedial action or inspection by a structural engineer. Component or element shows marked and advancing deterioration including loss of protective coating and minor loss of section from the parent material is evident intervention is normally required’
	4	Under this condition state immediate intervention (which is not exclusively related to physical intervention at the site) is required as outlined in the inspection manual. ‘Defects affecting the performance and structural integrity which require immediate intervention including an inspection by a structural engineer, if principal components are affected. Component or element shows advanced deterioration, loss of section from the parent material, signs of overstressing or evidence that it is acting differently to its intended design mode or function’
Department of Infrastructure and Transport, South Australia	Any	Maintenance actions are defined, quantified, and prioritised by the inspector.
Department of State Growth, Tasmania	Any	A deterministic algorithm is used to assess all maintenance activities where the condition state is >=3. The algorithm considers condition, load capacity, road status, remaining life, heavy vehicle usage and determines a risk score. The prioritised list is reviewed, and works are programmed up until estimated costs meet available funding.

Jurisdiction	Inspection level	Data types collected
Department of Transport Planning, Victoria	Any	Maintenance requirements are not assigned based on condition. Maintenance is prioritised by the inspector and assigned one of the following priorities. • M1 (1st priority - Maintenance works required). M1 - structures that require maintenance works within 2 years because structural and non-structural components have advanced deterioration and/or reduced durability. Failure to address these issues may lead to safety, serviceability, or financial risks in the short term. • M2 (2nd priority - Maintenance works required). M2 - structures that require maintenance works within the next 2 to 4 years because structural and non-structural components are beginning to show isolated or not so advanced signs of deterioration and/or reduced durability. Failure to address these issues may lead to safety, serviceability, or financial risks in the medium term. • M3 (3rd priority - Maintenance works required). M3 - structures that require desirable maintenance works because structural and non-structural components are showing minor signs of deterioration and/or reduced durability. Failure to address these issues may lead to safety, serviceability, or financial risks in the long term.
Main Roads Western Australia	3 and 4	Maintenance requirements are required to be assigned for elements with condition 3 or 4 components and captured through the work item numbers for periodic, routine, and specific works. Maintenance is prioritised by the inspector and assigned one of the following priorities. • 0 – Critical. Immediate works required to be completed within 6 months. • 1 – High Priority. Works to be completed within 3 years. • 2 – Medium Priority. Works to be completed within 4-6 years. • 3 – Low Priority (monitor). Assess again at next Detailed (Level 2) Inspection (5 years for timber bridges, 7 years for non-timber bridges).

Assigning intervention criteria is varied across the jurisdictions with only two ensuring that maintenance interventions are identified for condition states 3 and 4 components. By definition of the condition states included in Section 3.4 it can be expected that maintenance will be required. By setting intervention criteria based on the Western Australia criteria these can be grouped against condition states and provide guidance to inspectors. It is anticipated that while these provide a framework for defining maintenance interventions there will be times when these cannot be applied.

The proposed intervention criteria are defined below and summarised in Table 3.7:

- High Priority – Non-routine maintenance required to be completed within 0 to 12 months due to severe deterioration and required to preserve structure load carrying capacity and component durability. Proposed timing should initially be assigned by the inspector and reviewed by the bridge engineer.
- Medium Priority – Non-routine maintenance to be completed within 1 to 2 years due to structural and non-structural components have advanced deterioration and/or reduced durability.
- Low Priority - Non-routine maintenance to be completed within 4 years due to structural and non-structural components are beginning to show isolated or not so advanced signs of deterioration and/or reduced durability.
- Monitor – Deterioration is evident in isolated areas, however there is no short-term foreseen impact on component durability.

Table 3.7: Condition based expected maintenance interventions

Condition rating	Intervention criteria
1	• Monitor
2	• Low Priority • Monitor
3	• High Priority • Medium Priority • Low Priority
4	• Urgent • High Priority

3.6.3 Defect Mapping Framework

Under the defect mapping framework where defects are all monitored and mapped rather than a condition being assigned to the components priority for maintenance are assigned on the visible defects. The following definitions can be used.

- Urgent – Non-routine maintenance required in the short term to prevent further degradation of components that could impact load carrying capacity and/or user safety.
- High Priority – Non-routine maintenance required to be completed within 12 months due to severe deterioration and required to preserve structure load carrying capacity and component durability.
- Medium Priority – Non-routine maintenance to be completed within 2 years due to structural and non-structural components showing advanced deterioration and/or reduced durability.
- Low Priority - Non-routine maintenance to be completed within 4 years due to structural and non-structural components are beginning to show isolated or not so advanced change in the rate of deterioration and/or reduced durability.
- Monitor – Change in the rate of deterioration is minimal and in isolated areas, however there is no short-term foreseen impact on component durability.

3.6.4 Risk Based Intervention Approach

For either the component condition rating framework or the defect mapping framework a risk-based approach can be used to prioritise works across the portfolio, considering differing categories and priorities. For example, the defect intervention criteria can be assessed against a range of categories derived from the Levels of Service such as:

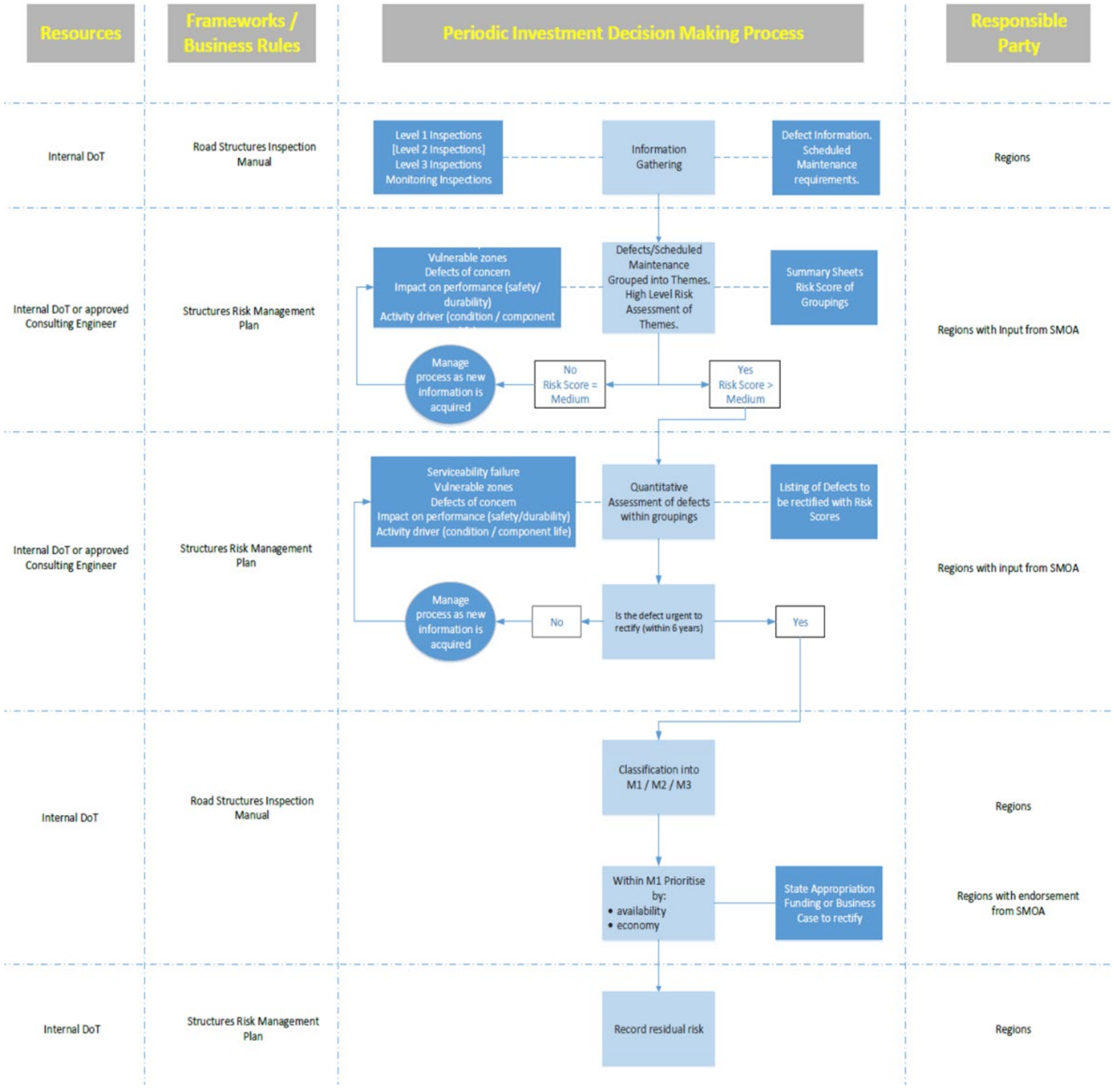
- impact on bridge capacity or component durability
- impact on structure resilience
- impact on aesthetics
- impact on freight access and efficiency
- impact on traffic safety
- impact on non-vehicular users.

Additional categories can also be used such as

- legal implications
- environmental implications
- heritage implications.

By assigning values in a matrix and weighting the categories based on jurisdictional importance an overall ranking can be achieved to prioritise maintenance activities across the portfolio. An example of the decision logic used by the Department of Transport and Planning Victoria is shown in Figure 3.4.

Figure 3.4: Department of Transport and Planning Victoria decision making logic



3.7 Inspection Levels

Inspections are the primary mechanism that allows asset managers and owners to understand if the bridges are providing the required level of service. Different inspection types will deliver data against different levels of service and/or provide varying insight into the level of service being provided.

At present there is variation in inspection types and definitions across jurisdictions as shown in Table 3.8.

Table 3.8: Current inspection types

Jurisdiction	Inspection level	Inspection type	Inspection scope
Transport for NSW	1	Drive by inspection	An as required inspection of specific components.
	2	Condition rating inspection	• A detailed visual assessment of element condition reported in accordance with parameters defined within the Bridge Inspection Procedure • Complex bridges are assessed in accordance with the inspection plan for the bridge, an Engineer is required on this inspection to assess specific components/known issues. Due to the nature of the inspection this sits between a level 2 and level 3 inspection. • Under water/scour inspections are utilised to determine defect in the chosen sample parts of the underwater elements. Due to the nature of the inspection this sits between a level 2 and level 3 inspection.
	3	Structural engineering inspection	• Structural inspection of the complete bridge except where the RBMP deems a partial inspection is adequate. Partial structural inspections are allowed for inspections relating to bridge emergencies, monitoring or follow-up inspections within two years after a complete Level 3 inspection. • The scope of the inspection shall be defined by the Regional Bridge Maintenance Planner and/or the Senior Bridge Engineer (A&E). The scope of the inspection will cover the full bridge unless otherwise specified.
	4	Load capacity assessment	These are an advanced Level 3 inspection and involve load assessment due to proposed changes in legal loading, new vehicle types, or the need to confirm the structural capacity of a bridge.
NZ Transport Agency Waka Kotahi	1	Routine surveillance inspection	The inspections identify any obvious defect which may affect the safety of highway users or anything else needing urgent attention
	2	General inspection	• A general inspection shall comprise a visual inspection of all parts of the structure that can be inspected without the need for additional access equipment, using safe, ground level viewing positions around the structure. • The purpose of a general inspection is to provide information on the physical condition of all visible elements of a highway structure.
	3	Principal inspection	• A principal inspection shall comprise a close examination, within touching distance, of all accessible parts of a structure. • The purpose of a principal inspection is to provide information on the physical condition of all accessible parts of a highway structure. • A principal inspection is more comprehensive and provides more detailed information than a general inspection.
	4	Special inspection	Special inspections involve particular types of structure or particular circumstances. The Structure Inspection Engineer shall identify structures requiring special inspections, document them as such in Highway Structures Information Management System and maintain a schedule of structures requiring regular special inspections, which define the specific inspection requirements including frequency and access requirements.
Department of Infrastructure, Planning, and Logistics,	1	Routine maintenance inspection	Usually carried out in conjunction with routine pavement inspections to check the general serviceability of the structure for road users.
	2	Condition inspection	Assess the condition of each structure and its components

Jurisdiction	Inspection level	Inspection type	Inspection scope
Northern Territory	3	Detailed engineering inspections	Undertaken on a needs basis to assess the structural condition and capacity of structures which have been identified as potential candidates for rehabilitation, strengthening, widening or replacement.
Department of Transport and Main Roads, Queensland	1	Routine maintenance inspection	A visual inspection to check the general serviceability of the structure, particularly for the safety of road users, and to identify any emerging problems.
	2	Condition rating inspection	An inspection to assess and rate the condition of a structure (as a basis for assessing the effectiveness of past maintenance treatments, identifying current maintenance needs, modelling, and forecasting future changes in condition and estimating future budget requirements).
	3	Special Inspection	An inspection to provide improved knowledge of the condition, load capacity, in-service performance, or any other characteristic beyond the scope of other types of inspection. Special inspections may be used to inform/develop the scope of other types of inspection. Level 3 inspection categories include: • Structural engineering • ACM identification • ACM verification • Underwater access • Fracture critical/low redundancy • Sub-standard load rating • Complex/unique structures • Known/suspected deficiencies • Confined space inspection.
Department of Infrastructure and Transport, South Australia	1	Routine maintenance inspection	Routine visual inspections which are used to check the general serviceability of a structure particularly for the safety of road-users, and to identify any emerging problems.
	2	Detailed visual condition inspections	Inspections are detailed visual condition inspections which are used to assess and rate the condition of structures and their components.
	3	Special inspections and investigations	Inspections are special inspections which may be instigated by request for a specific reason. They are not scheduled but may be required due to concerns over the structure’s safety, condition, load capacity or for structures subject to complex associated repair, strengthening or widening works. They may also result from scheduled follow-up material surveys or be required to inspect those components that are not accessible for a normal Level 2 inspection.
Department of State Growth, Tasmania	1	Routine visual inspection	Check for visible defects which might affect the safety of road users and/or the serviceability of a structure. Identify items that may require routine maintenance and/or urgent attention/further investigation.
	2	Condition inspection	Inspections are visual and are intended to measure and rate the condition of structures to: • Identify and prioritise maintenance needs. • Assess the effectiveness of past maintenance treatments. • Estimate future requirements for maintenance budgets.
	3	Detailed engineering investigation	Level 3 inspection may arise from reacting to a significant defect identified during a level 1 or 2 inspection and will usually include detailed engineering investigation comprising of field investigation and theoretical analysis. It may target a specific issue relevant to an individual structure or a class of structures. The investigation is usually comprehensive and may include behaviour of structural components and testing of material. The result of the detailed investigation is used as a basis for options on structure management and access management.

Jurisdiction	Inspection level	Inspection type	Inspection scope
Department of Transport Planning, Victoria	1	Routine visual inspection	Check for visible defects which might affect the safety of road users and/or the serviceability of a structure. Identify items that may require routine maintenance and/or urgent attention/further investigation.
	2	Condition inspection	Identify and prioritise maintenance needs. Assess the effectiveness of past maintenance treatments. Estimate future requirements for maintenance budgets.
	3	Detailed engineering investigation	Level 3 investigations are detailed engineering investigations that generally include a combination of field investigation and theoretical analysis which target a specific issue relevant either to an individual structure or to a class of structures. Level 3 reports and data are used as a basis for the assessment of structural management options.
Main Roads Western Australia	1	Routine visual inspection	They are visual in nature. They are intended to check on the overall safety and performance of the structure and the identification of any major accident damage or incident and any obvious failure of structural components. They are also important in ensuring maintenance works are being carried out.
	2	Detailed visual inspection and condition assessment	Quantitative data on structural components are collected for use in engineering analyses and deterioration models, and every structural aspect is reported. Level 2 inspection is a detailed visual inspection where all components are inspected closely (within 1.0 m). Also includes requirements for "complex bridges" and "under water / scour inspections".
	3	Special inspections and investigations	This is a special inspection which may be instigated by request for a specific reason. They are not scheduled but may be required due to concerns over the structure's safety, condition, and load capacity or for structures subject to complex associated repair, strengthening or widening works. They may also result from scheduled follow-up material surveys or be required to inspect those components that are not accessible for a normal Level 2 inspection.

3.7.1 Level 1 Inspections

There is strong consistency across all jurisdictions on the intent of Level 1 inspections. The inspection intent and definition remain the same for both the component condition rating framework and the defect mapping framework.

Level 1 inspections are typically undertaken by network maintenance personnel in conjunction with cyclic maintenance activities. A process needs to be established within the jurisdiction to ensure that the structures asset management team have visibility of Level 1 reports, and maintenance programming of identified works to ensure minor issues are treated when required.

Level 1 inspections are routine/cyclic inspection. These inspections are undertaken to check for:

- defects that can impact on user safety
- identify accident damage
- identify emerging issues and obvious failure of components (structural and non-structural)
- identify routine maintenance activities that need to be undertaken
- recommend when Level 2 inspections are to be undertaken out of cycle
- while cyclic in nature, unscheduled Level 1 inspections can/should be undertaken when events occur (ie weather, traffic accidents, etc) that may have changed an asset's condition to determine a need for a higher level inspection.

The outputs from these inspections are primarily used to:

- program maintenance activities
- program out of cycle Level 2 inspections
- program monitoring inspections.

3.7.2 Level 2 Inspection

Level 2 inspections provide the most value in terms of managing the bridges and the bridge portfolio. These are the primary means for the identification of defects and other issues.

These inspections are undertaken by specifically trained personnel on a cyclic basis.

Component Condition Rating Framework

Level 2 inspections for the component conditioning framework are detailed visual inspections of the bridge. These inspections include the following:

- compilation of, or confirmation of the inventory at a component level
- inspection and reporting on the components and an assessment of condition using the jurisdictional specific condition rating criteria for the components
- reporting on the condition of the principal structure components and condition rating for the structure as a whole
- identifying, and prioritising defect treatment requirements
- collect a photographic record of the structure
- recommending requirements for next inspection (if any) and nominating components for closer monitoring as appropriate
- requesting detailed engineering investigations of the structure if warranted because of incident damage or apparent rapid changes in structural condition and/or deterioration
- recommending supplementary testing as appropriate.

The outputs from these inspections are used to:

- program maintenance activities
- program additional monitoring
- program Level 3 investigations
- track changing condition of the components over time to enable appropriate timing of interventions
- identify the need for capital works.

Advantages of the component condition rating:

- cost effective inspection
- commonly used throughout Australia and New Zealand
- quick identification of needed component replacement or rehabilitation.

Disadvantages of component condition rating:

- repeatability of inspections can be an issue due to subjective nature, subject to the extent of guidance provided
- experience of inspector can impact results when assessing condition percentages in a component.

Defect Mapping Framework

The defect mapping framework Level 2 inspection focuses on the same activities as the component condition rating framework except for rating the components. In place of this the existing mapped defects are updated, and new defects documented.

The outputs from these inspections are used to:

- program maintenance activities
- program additional monitoring
- program Level 3 investigations
- track changing condition of the components over time to enable appropriate timing of interventions
- identify the need for capital works.

Advantages of the defect mapping:

- allows for changes in defects to be monitored
- interventions are more targeted to specific faults
- deterioration rates of faults easily monitored over time.

Disadvantages of defect mapping:

- time consuming in terms of mapping defects initially
- additional training of inspectors required.

3.7.3 Level 3 Specialist Inspections

Level 3 inspections typically require inspection/supervision by engineers qualified and experienced in the appropriate discipline relevant to the area of concern.

The frequency of Level 3 inspections is determined on a case-by-case basis and may be one-off events or form part of an ongoing management plan for structure(s) of concern.

Characteristics of these inspections include:

- specialist access outside the scope of Level 2 inspections (e.g., confined space entry, underwater inspection etc.)
- site testing and/or laboratory testing maybe required outside of routine testing undertaken as part of Level 2 inspections
- monitoring in-service performance of structures with theoretical or known deficiencies
- complex or unique structures requiring specialist knowledge or equipment to accurately determine condition/performance e.g., suspension/cable stayed bridges, heritage structures etc.

Level 3 inspections include the following inspection types:

Detailed Engineering Inspection

This is a specific, detailed investigation of a structure, the scope of which is typically developed to address concerns about the individual structure or class of structures. They are typically undertaken to provide improved knowledge of the condition, load carrying capacity, in-service performance or other characteristics that are beyond the scope of Level 1 and Level 2 inspections.

As a minimum, the field inspection component will comprise a visual examination of all components of the structure that need inspection to cover the scope of the brief, supplemented, where necessary, by examinations, testing, or analyses.

Asbestos Containing Material Identification Inspection

Periodic inspection of structures that are known to contain asbestos is required to ensure the asbestos register is kept up to date as required by Safe Work Australia. Once asbestos is removed and the register updated to detail this, periodic inspections are no longer required. Where asbestos is suspected, these bridges also should have asbestos inspections undertaken to confirm if asbestos is present. The hazards associated with exposure to airborne asbestos fibres are well documented and there are numerous documents available relating to the management of asbestos.

Asbestos may be present in bridges constructed before 2003. Possible locations of ACM include:

- permanent/sacrificial formwork between girders on concrete and steel girder constructed bridges
- deck unit constructed bridges with cast in situ bridge decks (that is, no post-tensioning)
- external and internal service mains pipelines such as stormwater and sewer pipes
- service pits
- internal service conduits
- half pipes for drainage channels
- drainage systems on the bridge or near the bridge
- asbestos bonded buried corrugated metal culverts.

It is almost certain that any compressed fibre products used on structures constructed and completed prior to 1985 will contain asbestos.

Complex Structures

Complex Structures are:

- individual or classes of bridges for which the standard Level 2 inspection does not provide sufficient information to assess the condition of bridge components
- bridges that require a bridge-specific inspection and management plan.

These bridges should be identified within the bridge asset management information system.

Structures may be included in the complex list for several reasons:

- structural form e.g., cable, suspension, truss, lifting bridge, high skew angle
- the size of the structure e.g., its length, height, or number of spans
- special knowledge or training beyond that of a Level 2 inspector is required to conduct an inspection e.g., metallurgical knowledge, fibre reinforced polymer strengthening
- the structure includes fracture-critical or fatigue prone components e.g., steel components subjected to cyclic loading, welds, gusset-plates
- bridges for which special access provisions are necessary e.g., elevated work platforms, scaffolding, barges, diving gear, or similar and for investigations on railway property, or within confined spaces.

Examples of complex structure types:

- cable-stayed and suspension bridges
- prestressed concrete segmental bridges
- steel and concrete box girder bridges
- moveable bridges such as bascule, vertical lift span and swing bridges
- through and half-through trusses and girders
- bridges with half joints and drop-in-spans
- riveted bridges
- high bridges
- bridges over large expanses of water
- other structure types with limited or no structural redundancy.

Underwater Inspections

Relate to the need for divers to undertake below water level inspections of components. These inspections can be programmed typically on large critical bridges or undertaken on an as needed basis if there are concerns raised in Level 2 inspections. Construction diving (including inspections) requirements vary between jurisdictions and must be followed.

Remotely operated underwater vehicles (ROV) can be used for underwater inspections to reduce risk. The use of these vehicles will depend on the intent of the inspection such as visual only, or if testing is also required. The capabilities in terms of visual inspections have improved with the use of both video enhancing in low visibility areas, and sonar mapping in zero visibility areas.

Difficult Access Inspections

Difficult access inspections are those that require specialist access equipment outside of what would normally be considered acceptable for Level 2 inspections. This includes utilising access equipment that requires the inspector or operator to have specific training in the use of, and not specifically covered as part of other inspection types outlined above, such as:

- abseiling equipment
- using an elevated work platform, underbridge inspection unit, or cherry picker
- powered and unpowered watercraft
- movable suspended scaffold platforms, underslung gantries, climbing stages, and movable underbridge platforms.

Specialist access equipment is generally not required to undertake Level 1 and 2 inspections; however, they may be needed to complete Level 3 inspections due to the specialist nature of the inspections.

Monitoring Inspections

Monitoring is the periodic or continuous measurement of structural behaviour by visual or electronic means, e.g., deflections, strains, and crack sizes. There are many instances where measurements can be taken periodically, or in rare circumstances, continuously, so that condition and performance can be monitored over time.

Monitoring inspections may be required for.

- bridges that have serious deficiencies and are programmed for repair or replacement but require monitoring until refurbishment has been completed
- bridges with performance issues that are difficult to analyse or resolve analytically
- bridges employing new materials, design and/or construction characteristics where in-service performance requires validation
- bridges that fail load rating assessment but show little or no physical signs of distress
- bridges where external events/factors may impact on performance/integrity (e.g. flooding, scour or vehicle impact).

The benefits of monitoring include:

- decreased ongoing inspection and maintenance costs
- increased structural safety
- improved understanding of behaviour and durability of monitored structures
- improved understanding of in situ structural behaviour
- early damage detection
- assurances of the structure's strength and serviceability
- reduction in downtime due to assessment or maintenance procedures
- improved maintenance and management strategies for better allocation of resources
- enabling/encouraging the use of innovative materials.

Visual Monitoring

Visual monitoring includes the periodic checking or known specific defects to determine changes in condition. Any equipment installed on the structure is read manually during the inspection process. Issues that can be visually monitored include:

- abutment movement
- differential settlement
- scour
- flexural cracking
- weld cracking.

Electronic Monitoring

Electronic monitoring includes the deploying of sensors on a structure to take continuous measurements. It is vital to establish a complete understanding of why the structure is being monitored. Without this understanding it will be difficult to design an appropriate monitoring system that provides the desired data. Care must be taken to understand a range of factors that will impact the success of an electronic monitoring program such as:

- identify the mechanism/event of interest and categorise the influence of the mechanism on the physical response of the structure or its principal components
- establish the characteristic response level of key parameters (e.g., strain, vibration, displacement, etc.)
- selecting the appropriate monitoring devices and technologies

- define the site characteristics in terms of environment, security, access requirements, traffic control, turning points (for test vehicles), ongoing equipment maintenance requirements
- develop an unambiguous specification of the system being developed, it is essential that all key decisions/criteria are captured in a systematic and consistent manner.

3.7.4 Other Inspection Types

Other inspections are those which are adhoc, or do not require engineering experience to be undertaken. These inspection types may include other specific skill sets to be completed.

Heritage and Historic Bridges

Heritage bridges should be identified within the bridge asset management information system. The identification should include details if the bridge is historically significant in its entirety or only specific elements. While these will be inspected as part of the Level 2 inspection process, special inspections maybe required by heritage specialists to develop conservation plans, assist in scoping activities.

Conservation management plans are required in all states and territories for heritage bridges. These plans should be developed by specialist heritage architects. In general, the plan should:

- identify the heritage importance of the structure
- define the heritage features and components to be inspected and reported on
- describe the current condition of the structure, outstanding maintenance and resulting threats to the structure
- outline the frequency for regular inspections, provide guidance on mechanisms of deterioration and signs of distress in components
- provide guidance for the maintenance including appropriate materials, surface treatments and application methods for the various components.

Disused Bridges

Jurisdictions have a responsibility for the management of disused and other potentially hazardous structures within the road reservation. A road bridge which is no longer in use by vehicular traffic (but which may be in use by pedestrians) or a closed pedestrian bridge is considered a disused bridge.

Disused structures not open to vehicles and pedestrians often have a lower maintenance priority to other structures. As a result, a disused structure could deteriorate to a point where it is unstable or has partially collapsed and, provided that the structure is not subject to statutory or other protection, partial or total dismantling may be necessary to properly control the hazard.

Disused structures are typically included in the normal bridge inspection program for Level 1 and 2 inspections even if they are not used by traffic.

Emergency Inspections

Initial Post Flooding Inspection and Actions

Structures over severely flood-affected watercourses and those with known vulnerability to flood-damage are to be given priority after a flood event. An inspection is to be conducted as soon as safe access to the structure is possible. The inspector should perform an initial visual inspection from a safe position. If the structure is under water and invisible, this must be reported and the structure, together with its approach roads, should be closed.

Other than monitoring, no further action should be taken until the water-level has subsided and the whole of the structure i.e., the superstructure and substructure is visible. If the structure is accessible, the inspector should carefully assess the immediate approaches to the structure and the visible parts of the bridge for following irregularities:

- bridge approaches (voids in surface, settlement, slippage of embankment)
- barriers (missing, misaligned, foundation washed away)
- abutments (displaced, damaged, unsupported, voids)
- deck/beams (holes, missing beams, misalignment)
- kerbs (misalignment, settlement)
- movement joints (missing, displaced)
- crossheads (displaced, damaged, unsupported)
- piles (missing, displaced, damaged)
- pile-caps (displaced, damaged, unsupported) changes in riverbed (alignment, depth, profile) and visible scouring under bridge foundations
- debris (accumulation on superstructure, lodged in substructure)
- if there is any doubt regarding the stability of the structure or the adjacent road embankments, the road must be closed immediately.

If there are any concerns a Detailed Engineering inspection should be recommended.

Bridge Strikes

Following a bridge strike the immediate response is a Level 1 inspection generally undertaken by the maintenance contractor. Depending on severity of the strike a Detailed Engineering inspection maybe requested by the maintenance contractor or emergency services on-site controller.

When attending bridge strikes the on-site incident controller should be the first point of contact to ensure that your activities do not impact on emergency services operations.

Bridge strike assessments should identify visible damage to:

- barriers (missing, misaligned)
- abutments (displaced, damaged)
- deck/beams (deformation, misalignment, damaged)
- crossheads (displaced, misalignment)
- piles (displaced, deformed, damaged)
- debris (impact debris, cargo disbursed)
- if there is any doubt regarding the stability of the structure, the road must be closed immediately.

Other Emergency Inspections

Other emergency inspections include:

- post-earthquake inspections
- chemical and hazardous goods spills
- post fire inspections.

Unmanned Aerial Vehicle Inspections

Prior to undertaking a UAV (drone) inspection all CASA (Australia) and CAA (New Zealand) requirements must be followed. Operators should be certified and have all appropriate insurances.

A UAV is an alternative solution when there is difficult access into certain areas of a bridge, including confined spaces within a bridge. However, if the bridge is within a no-fly zone for UAV's, they cannot be used.

A UAV use could reduce inspection costs through eliminating the need of elevated work platforms, vessels, and scaffolding. However, requirements for traffic control may still exist due to the regulatory body, and ability to access all areas of interest may be limited by regulations (visual line of sight) or by site conditions (weather).

UAVs at present can only undertake visual inspections through the capture of images which require post inspection interpretation.

Environmental Conditions and Hazard Identification.

Specialist inspections maybe undertaken to assess specific environmental risks, conditions, and hazards. These inspections are likely to increase going forward to determine the resilience of road networks with regards to the impacts of changing climatic conditions. The inspections are likely to be a part of a wider study into the asset's resilience.

These inspections are likely to include:

- assessments of the suitability of the bridge and its condition
- assess the surrounding environment for risks that could impact the resilience of the structure
- validation of desk top assessments findings and risk mitigation measures.

Construction Handover or Major Works Handover Inspection

Upon completion of construction, a detailed inspection 'completion of construction' report should be prepared, and the design plans should be amended to record the as-constructed structure. Other relevant documentation, such as correspondence and photos, should also be preserved and filed for use during the bridge's life. These records can be useful in assessing when damage has occurred (during or after construction) and what the probable causes may be.

Similarly, upon completion of rehabilitation and strengthening work, a detailed inspection should be undertaken, and a report should be prepared so that the bridge inventory can be updated to record the 'as-rehabilitated/strengthened' structure. This report is similar in nature to a 'completion of construction' report detailing the 'as-constructed' structure.

3.8 Access Requirements

Access should be considered in two distinct areas:

- accessing the site
- accessing the bridge to inspect.

Provision needed to access the site and to access the bridge should be captured in the bridge asset management information system. This will simplify the planning process prior to undertaking the inspections.

3.8.1 Accessing the Site

Accessing the site can apply to all levels of inspections. To obtain access to the bridge site the planning process will need to consider the following:

- timing of inspections, particularly in remote areas, to account for the seasonal conditions and the impact of these on road access to the site
- the need to obtain approvals to enter indigenous land and townships, including the need for inspectors to undertake appropriate training
- obtaining approvals where relevant to enter the rail corridor and ensuring the inspectors have undertaken the appropriate training/certification with the rail authority
- obtain approvals from private landowners if required.

3.8.2 Accessing the Bridge

The need for specialist access equipment is subject to specific jurisdictional requirements, however the practice is generally confined to Level 2 and Level 3 inspections. This equipment is typically needed for:

- work site safe access and risk mitigation while undertaking the inspection
- working at heights
- working on water or under water
- confined space works.

Work Site Safe Access and Risk Mitigation

Any activities which impact the carriageway should have a traffic management plan developed and approved by the road controlling authority. The traffic management plan requirements and approval processes are jurisdictionally based.

Working at Heights Plan

To access components from above or below deck level, or from the ground a working at heights plan will be required. Below are different access methods that can be considered. With all options below a safe work method statement must be developed and an on-site risk assessment should also be undertaken prior to works commencing, and local jurisdictional legislation complied with.

Ladders: While one of the cheapest access solutions they are dangerous and should only be used as a last resort. They can be used to access areas at relatively low heights, and care must be undertaken to not over-reach, the ladder is founded on flat ground, the slope angle is correct, and a second person is securing the base of the ladder.

Cherry pickers and elevated work platforms: Both cherry pickers and elevated work platforms are limited in terms of heights they can reach. Elevated work platforms do provide a stable platform for inspection activities but have lower reaches than cherry pickers. Operation of these vehicles requires specific jurisdictional based training. Both vehicles are generally used on relatively flat ground or from the carriageway.

Under Bridge inspection unit*:* These are purpose-built access vehicles for inspection under deck components of the bridge while parked on the deck. Reach lengths and bucket sizes will vary between units, when hiring these units care should be taken to ensure both are adequate for the intended use.

Permanent inspection structure: On larger bridge permanent inspection equipment can be installed such as catwalks, handrails, and movable gentries. Catwalks are typically used for maintenance access but can be used for inspecting components accessible from them, where justified particularly in relation to monitoring sites, permanent catwalks can be constructed. Handrails are fixed to the structure and allow the inspector to walk along sufficiently wide enough bridge component such as box girders. Movable gantries running in a transverse manner to the main bridge beams allow for a far wider inspection are and catwalks or handrails.

Rigging: Comprising of cables, platforms or stages, rigging can be used in areas where access by other means is not feasible or where special inspection procedures are required. Rigging can be used where lane closures are not tolerable, however care needs to be taken to ensure there is sufficient clearance for vehicles passing under the platform of stage.

Scaffolding: Typically used where only small specific areas require inspection, due to the time to setup and dismantle. Scaffolds can be founded on the ground or suspended in a fixed location from a bridge.

Abseiling: Without or without a Bosun chair abseiling can be used to access components of bridges. Typically used in below deck applications where under bridge units cannot reach such as pier inspections. To undertake this inspection type specialist training is required. In addition, a second person is required who is trained in at height rescues.

Working on or Under Water

When working on water a vessel such as a barge, boat, or other type of watercraft may be required. The vessel operator must be licenced to operate the vessel within the jurisdiction, and specific approvals maybe required for operating the vessel in designated waterways.

Barges can be used in conjunction with other equipment to provide floating platforms for the inspections including elevated work platforms or cherry picker.

Specialist divers will be required to undertake below water level inspections of components. These inspections can be programmed typically on large critical bridges or undertaken on an as needed basis if there are concerns raised in Level 2 inspections. Construction diving (including inspections) requirements vary between jurisdictions and must be followed. The use of divers is particularly useful in situations where a clear view of the components is obstructed by site elements such as marine growth which they can remove or turbidity, where they can use tactile techniques and clear boxes.

Remotely operated underwater vehicles (ROV) can be used for underwater inspections to reduce risk. The use of these vehicles will depend on the intent of the inspection such as visual only, or if testing is also required. The capabilities in terms of visual inspections have improved with the use of both video enhancing in low visibility areas, and sonar mapping in zero visibility areas. Typically, current light duty ROVs do not have capability to address items such as marine growth or turbidity and can be challenging to use in tidal or fast flowing waters.

Confined Space Works

Confined spaces relate to internal inspections of box girders, culverts, and other difficult places to access. Each jurisdiction has specific requirements when working in these spaces which must be followed. This includes specific training of the inspectors, and PPE that must be on site.

Depending on the reason for accessing the confined space the risk can be eliminated using an unmanned aerial vehicle, or unmanned ground vehicles that can incorporate robotics.

3.9 Testing

Testing comprises a range of activities that provide information on the condition of a structure, its behaviour or material characteristics. These activities are almost exclusively the domain of the Level 3 inspection, however some tests identified below can be undertaken as part of the Level 2 inspection process. Testing is categorised as non-destructive or destructive.

Non-destructive testing (NDT) such as electrode potential measurements or ultrasonic inspection, to assist in the detection of defects that may be difficult to detect visually, such as cracks in welded joints or those hidden within the structure.

Destructive testing can be subdivided into:

- **Material sampling** methods for taking samples of materials from the structure to determine composition and properties of the material or the presence of deleterious substances such as chlorides in concrete.
- **Intrusive testing** such as drilling holes, to determine the condition inside the structure that is not revealed by normal visual inspection, e.g., the condition of post-tensioning tendons or the interior of box girder sections.
- **Testing to destruction** such as testing replaced components, or entire disused structures.

Table 3.9 summarises some of the more commonly used testing techniques:

Table 3.9: Commonly used testing techniques

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Integrity and Structural Performance					
Visual Inspection	Observe, classify, and document the appearance of distress and defects on exposed surfaces of the structure. Map distress and defects.	Surface defects such as cracking, spalling, leaching, erosion, or construction defects.	Simplest and least expensive; extensive information can be gathered from visual inspection to give a preliminary indication of the condition of the structure and allow formulation of a subsequent testing program. Does not cover areas not visible.	✓	NDT
Delamination Survey	Tap the concrete surface using a light hammer to identify delaminated concrete through a 'hollow' impact sound.	Assessment and location and extent of discontinuity in the cover concrete, which is substantially separated, but not completely detached, from the concrete.	Low cost; quick; no instrumentation needed; easy; portable; can measure large areas; can identify a variety of additional information (hardness, voids, peeling etc.). Indicative only; need access to surface; depths of around 100 mm; non-specific; inconsistent results; requires further tests to confirm results; not good for thin components; cannot define deep voids.	✓	NDT

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Impact Echo	Receiver adjacent to impact point monitors arrival of stress waves as they undergo multiple reflections between surface and opposite side of plate-like member or from internal defects. Frequency analysis permits determination of distance to reflector if wave speed is known.	Locate a variety of defects within concrete components such as delamination, voids, honeycombing, or measure component thickness.	Relatively low cost; easy; portable; direct results; surface roughness does not affect results. Experience for testing and interpretation; requires surface preparation; background noise. Reinforcement clouds results.		NDT
Ground Penetrating Radar (GPR)	Radio frequency waves from radar transmitter are directed into the material. The waves propagate through the material until a boundary of different electrical characteristic is encountered. Then part of the incident energy is reflected and the remainder travels across the boundary at a new velocity. The reflected (echo) wave is picked up by a receiver. The transducer is drawn over a surface and forms a continuous profile of the substrate below.	It can detect several parameters within concrete structures such as the location of reinforcement, the depth of cover, the location of voids, location of cracks, in situ density and moisture content variations.	Good identification of reinforcing bars, prestressing strands, cable ducts, zones of varying moisture content and thickness of slabs, and a fair assessment of delamination and large voids in concrete. Quick; non-disruptive; non-destructive; good coverage. Expensive; cannot see through areas with heavily congested steel; does not differentiate between defect types: reliant upon operator judgement; complicated equipment setup; specialist experience in data interpretation. Additional tests needed to confirm results.		NDT
Concrete Properties Affecting Durability and Deterioration					
Cement Content and Type	Cement content by calcium oxide – analysis for hardened concrete.	Assess concrete quality.	Quick; low cost; reliable. Reliability is affected by knowledge of cement chemistry and aggregates related to the structure. Experience and correlation with other test data is needed for interpretation.		Intrusive
Chloride and Sulfate Content	The sample is dissolved in hot nitric acid to provide a solution from which aliquots may be tested for chloride or sulfate content.	Assess susceptibility of concrete to sulfate attack. Provide input data for chloride induced corrosion service life modelling.	Low cost; quick; direct results, reliable; accurate. Interpretation requires experience. Need to drill holes or collect core samples and repair.		Material sampling

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Ultrasonic Transmission Velocity (Ultrasonic Pulse Velocity)	This technique measures the transit time (in microseconds) of ultrasonic waves passing from an emitter transducer through a concrete sample to a receiver transducer.	Determination of the variability and quality of concrete by measuring pulse velocity. Using transmission method, the extent of such defects such as voids, honeycombing, cracks, and segregation may be determined. This technique is useful when examining fire damaged concrete.	Easy; portable; relatively quick; relatively low cost; measures from one side only; excellent for determining the quality and uniformity of concrete; path lengths of 10 m to 15 m can be inspected with suitable equipment. Data can only be usefully interpreted where the distance between the transducers is accurately known (generally better than ±2%). Indirect surface testing; difficult to use on rough surface (PUNDIT or AU2000); variations with concrete quality; no information about defect depth; affected by many factors including type of concrete, aggregate, temperature, humidity, roughness, high density of reinforcement etc.; calibration to estimate concrete strength is required; expertise is needed to interpret the results; very time consuming as it takes only point readings.		NDT
Identification of Presence of Alkali Silica Reaction (ASR)	Application of acidic uranyl acetate solution forming a complex with components of ASR gel that fluoresces under UV-C light.	Identification of likely presence of ASR.	Quick and relatively inexpensive. Requires experience in interpretation, especially in structures exposed to marine or other high salinity conditions.		NDT
Petrographic Examination	Microscopic visual examination of polished thin sections under polarised and unpolarised light.	Identification of the presence of ASR susceptible aggregates and diagnosis of the presence of ASR.	Accurate determination of cause(s) of distress; degree of damage; quality of concrete when originally cast and current. Expensive, needs to be performed by a person experienced with concrete and specifically with diagnosis of ASR. Additionally, can provide information on: Cement content and type; aggregate type; cement replacement materials; water cement ratio; air void content including entrained air and entrapped air; deterioration mechanisms such as sulfate, acid sulfate, leaching, fire damage, delayed ettringite formation; aggregate or cement paste shrinkage; carbonation; detection of unsound contaminants.		Material sampling

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Apparent Volume of Permeable Voids (AVPV)	The volume of interconnected void space of a concrete specimen which is emptied during the specified oven drying and filled with water during the subsequent immersion and boiling as a result of capillary suction, expressed as a percentage.	Determine the water porosity or permeability of the concrete microstructure.	General indication of quality of the concrete Can be used on a regular basis as a diagnostic tool as a part of condition surveys of existing concrete structures. As it requires the extraction of concrete cores, it nevertheless provides a very cheap and a non specialised way of establishing the quality and potential long-term performance of concrete, although is not intended as an absolute measure of durability.		Material sampling
Location of Reinforcement					
GPR	As above.	As above.	As above.		
Cover Meter	A low frequency magnetic field is applied on the surface of the structure; the presence of embedded reinforcement alters this field, and a measurement of this change provides information on the reinforcement.	Locate embedded reinforcement, measure depth of cover, and estimate approximate diameter of reinforcement.	Cost effective; quick; portable; easy; direct results; can measure large areas; reasonably accurate; readily available. Difficult to identify separate bars in heavily reinforced areas; need access to the surface; depth range 30-180mm; can only detect the first layer of reinforcement; accuracy depends on correct calibration; high voltage cables can disturb readings; presence of perpendicular reinforcement and iron inclusions in the concrete can alter readings; cannot detect stainless steel; must make numerous readings.	✓	NDT
Corrosion of Embedded Steel					
Carbonation Depth	Apply phenolphthalein solution uniformly to freshly exposed concrete surface and measure the depth of carbonation (absence of colour).	Assess corrosion protection value of concrete with depth and susceptibility of steel reinforcement to corrosion due to carbonation. Carbonation depth is used to assess whether the reinforcement is likely to have de-passivated leading to corrosion. The results can be used to model concrete carbonation rates to estimate remaining service life.	Low cost; quick; direct results, easy; evaluation; reliable; accurate. Need trained people; need to drill holes or collect core samples and repair; only freshly exposed surface can be tested; powder contamination can affect the results; one location only; not valid for concrete with large aggregate unless a representative area of surface can be tested.		Intrusive

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Chloride Profile	Core samples or powder samples are collected for laboratory analysis. The sample is dissolved in hot nitric acid to provide a solution from which aliquots may be tested for chloride or sulfate content.	Assess risk of steel reinforcement to corrosion due to chloride ingress. Chloride profile will show if chloride has reached the corrosion activation threshold concentration at the steel reinforcement. The chloride profile also can be used to model future deterioration and remaining service life.	Direct results; easy; rapid results for estimate of chloride content; low cost compared to other methods; accurate, variation only 4-5%; quick; strong indicator of corrosion potential; determine areas to be rehabilitated; helps to determine type of repair needed. Need trained people; need access to surface; repair of drill or core holes required; specialist evaluation, can be time consuming; need several samples to draw reliable conclusions. Interpretation of concentration profiles and service life modelling requires experience.		Material sampling
Corrosion Potential (Electrochemical, Half-cell)	Measure the potential difference (voltage) between the steel reinforcement and a standard reference electrode; the measured voltage provides an indication of the likelihood that corrosion is occurring in the reinforcement.	Identify region or regions in a reinforced concrete structure where there is a high probability that corrosion is occurring at the time of the measurement.	Quick; easy; objective data; relatively low cost; large areas can be measured; strong indicator of corrosion potential; can result in timely intervention; helps to determine type of repair needed; allows quantification of area likely to be corroding. Need trained people; specialist interpretation required; requires complementary testing to verify results; calibration is required before using the results; can only measure potentials on first layer of reinforcement; needs connection to reinforcement, with subsequent repairs required; influenced by humidity in the concrete; cannot be used when it is cold, <5°C; coatings may need to be removed; readings overhead could be misleading; requires numerous readings; results can be inconsistent; experience in corrosion required for assessment.		Intrusive

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Concrete Resistivity	Resistivity is measured by inserting 4 electrodes into small holes on the surface and passing a current between the outer electrodes and measuring the voltage between the inner contacts. This potential difference and measured current provide an estimate of resistance which can be converted into resistivity.	It is used for measuring the ability of the concrete to conduct the corrosion current. It gives an indication of the rate of corrosion which may occur if corrosion of the reinforcement commences.	Low cost; objective data; quick; safe to use; indication of corrosion rate; large areas can be measured; low operative expertise is sufficient; very useful when used in conjunction with other methods of testing, e.g., half-cell potential. Requires complementary testing (e.g., use of cover meter) to obtain best results; coatings may need to be removed locally; concrete must have some moisture content; precise results are not usually obtained; interpretation by experienced personnel required.		Intrusive
Corrosion Rate Measurement - Linear Polarisation	Measure the current required to change by a fixed amount the potential difference between the reinforcement and a standard reference electrode; the measured current and voltage allow determination of the polarisation resistance, which is related to the rate of corrosion.	Determine the instantaneous corrosion rate of the reinforcement located below that test point.	Objective data; possible to estimate residual load capacity of the bridge; monitoring corrosion over time. Very expensive and time consuming to perform on more than a limited number of representative locations; special equipment; need trained people and effective teamwork; specialised interpretation required; need complimentary testing for verification; measures localised corrosion at a single point in time that cannot be extrapolated over seasonal variations; difficult to obtain reliable data in solutions of high linear resistance; need to expose reinforcement. To provide an overall assessment of a structure, repetitive measurements at many locations are required over a representative annual cycle.		Intrusive
Concrete Breakout	To remove (by saw cutting or coring) small, isolated areas of concrete covering the reinforcement to enable inspection and measurement of the exposed reinforcement.	Determine reinforcement details (e.g., bar size, type, orientation and taped cover) and its condition (e.g., corrosion state and loss of cross-sectional area).	Direct results: low cost compared to other methods; accurate, quick; strong indicator of corrosion; provide connection for half-cell potential measurements and corrosion rate measurements; opportunity to collect the breakout sample for laboratory testing. Need access to surface; repair of breakouts required; may need several breakouts to draw reliable conclusions.		Intrusive

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Concrete Strength					
Pullout	Measurement of the force required to pull an embedded metal insert and the attached concrete fragment from a concrete test specimen or structure. With the help of calibration charts the maximum force gives an indication of the strength of concrete. The insert can be either cast into fresh concrete or installed in hardened concrete.	It provides an estimation of the compressive and tensile strengths of hardened concrete; comparison of strength in different locations.	User expertise is low and can be used in the field; in-place strength of concrete can be measured quickly and appears to give good prediction of concrete strength. Small areas of concrete are removed, necessitating minor repairs; only tests a limited depth of material.	✓	Intrusive
Compressive Strength	Cores are extracted from hardened concrete by using a core drill (AS 1012.14), then trimmed, capped, and tested for compressive strength (AS 1012.9).	Strength of in-place concrete; comparison of strength in different locations.	Fairly direct results and accurate; easy; rapid results for overall strength assessment; indirect assessment of concrete durability; can be utilised for structural analysis. Destructive testing: repair of core holes required; care should be taken as core holes may cause damage to the member from which the core is taken.		Material sampling
Tensile Strength (Indirect)	Cores are extracted from hardened concrete by using a core drill (AS 1012.14), then prepared, and tested (AS 1012.10) by compression forces to determine the indirect tensile strength.	Estimation of tensile strength of in-place concrete; comparison of strength in different locations.	Fairly direct results and accurate; easy; rapid results for overall strength assessment; can be utilised for structural analysis. Destructive testing: repair of sample core holes required; care should be taken as core holes may cause damage to the member from which the core is taken.		Material sampling
Rebound Hammer	The test is based on the principle of the elastic rebound of a spring-driven mass running down a central guide bar onto a plunger pressed firmly against the surface of the concrete.	Provides a measure of the local surface ‘hardness’ of the concrete and under laboratory conditions the resulting rebound number has been empirically related to compressive strength of concrete.	Low cost; quick; easy; portable; direct results; gives accurate assessment of the strength of the surface layer of material; the entire structure can be tested in its 'as-built' condition. It is an imprecise test and does not provide a reliable prediction of the strength of concrete; indicative only (+/-25%), other tests needed to confirm results; can be affected by smoothness of the concrete surface, moisture content of the		NDT

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
			concrete, type, size and location of coarse aggregate, shape, and rigidity of the component and carbonation of the concrete surface; affected by direction of application.		
Steel Structures Deterioration					
Visual Inspection	Observe, classify, and document the appearance of distress and defects on exposed surfaces of the component, including welds.	Identifies defects on the surface only.	Simplest and least expensive method. Detects only surface defects; need well trained inspector; need full access to surface.	✓	NDT
Weld Inspection – Dye Penetrant Testing	A developer is applied to test surface to reveal locations where the fluorescent or visible dye has penetrated.	Identify the location and extent of weld discontinuities open to the surface e.g., cracks, porosity, seams and surface defects such as fatigue cracks.	Portable, low cost; high accuracy; expedient results; very small surface cracks with a minimum depth of 3 times surface roughness can be detected, if the surface preparation is diligent; personnel are easy to train. Results are easy to interpret. Surface films such as coatings, scale, and smeared metal may hide defects; surface must be cleaned and protected after evaluation; surface roughness can give rise to spurious indications.	✓	NDT
Weld Inspection – Magnetic Particle	A magnetic field is induced in a ferromagnetic material and then the surface is dusted with iron particles (either dry or suspended in liquid). Surface and near-surface imperfections distort the magnetic field and concentrate iron particles near imperfections, previewing a visual indication of the flaw.	Surface and near surface (up to 2 mm below the surface) cracks in ferromagnetic materials.	Low cost; expedient; personnel are easy to train; exact results are obtained for locally limited area. No limit to the size or shape which can be tested; very small surface cracks on accessible surfaces up to a width of 0.2 µm and length of 0.5–2 mm with use of reference samples (BS EN ISO 9934-2, Non-destructive testing – Magnetic particle test); convenient for inspection of target oriented small areas. Can inspect through thin coatings. Can only detect surface or near surface defects; should not be applied during direct sun exposure; not usable for non- ferromagnetic material; photographic documentation is to be made without flash. Surface must be protected after testing; structures painted with aluminum paint can provide poor results.	✓	NDT

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Ultrasonic Testing	Very high frequency sound waves are passed through the metal structure under test. The waves are reflected by either a defect or from the far surface of the member.	Ultrasonic flaw detection methods can detect voids and defects within a metal section and are best viewed as being complementary methods. Identifies most weld discontinuities including cracks, slag, lack of fusion; accurate metal thickness measurements possible.	Most sensitive to planar type defects; immediate results; portable; provides relatively rapid, and cost effective, defect detection; detect defects that are too small, or incorrectly oriented, for detection by radiography; requires access from one surface only; when correctly calibrated and employed, permits the detection of extremely small defects. Irregular, rough, non-homogeneous, very small or thin components are difficult to test; surface condition should be suitable for coupling of transducer; requires highly skilled operators to use the equipment and to interpret the results.		NDT
Radiographic Testing (Gamma)	Penetration of electromagnetic radiation through the body under the test to produce a shadow image of any defects within the bulk.	Detects voids and defects within a metal section and are best viewed as being complementary methods. Identifies most weld discontinuities including cracks, slag, lack of fusion; incomplete penetration, slag as well as corrosion and fit-up conditions.	Can detect subsurface damage; good delectability of cracks in hidden members of typical built-up sections; can be evaluated on films or digital foils; removal of paint and corrosion protection is not necessary; permanent record. Radiation is a safety hazard – requires control of nearby facility or area including lane/road closure and/or special monitoring of exposure levels and dosage to personnel. Relatively slow and expensive techniques; requires skilled operator and interpretation.		NDT
Eddy Current Testing	Inducing electromagnetic fields within a test piece and sensing the resulting electrical currents (eddy) so induced with a suitable probe or detector. A localised change in induced current flow indicates the presence of a discontinuity in the test object. The size of the discontinuity is indicated.	Detect discontinuities near the surface (i.e., cracks, inclusions, porosity). Capable of finding small discontinuities of <100 μm in highly conductive materials. Detect heat treatment variation, plating, or coating thickness.	Low cost; requires minimum surface preparation. Reliable inspections can be performed through a nonconductive coating up to thickness of ~0.4 mm. Testing can be performed very rapidly with instantaneous results. Inspection equipment is considered portable. Requires a skilled operator to calibrate and interpret indications, limited to conductive materials, some indication may be masked by part geometry due to sensitivity variations.		NDT

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Tensile Testing	Test piece of metal strained in uni-axial tension.	Measure the mechanical properties of the steel.	Accurate measurements of mechanical properties of steel including tensile strength, yield strength and ultimate strength. May cause considerable damage to the member from which the coupon is taken, extreme care must be exercised.		Material sampling
Hardness/ Rebound Testing	An indenter (hardened metal ball) is pressed into the surface of a test piece by an accurately controlled test force.	Samples can be re-used for additional testing.	Vickers test uses a pointed indenter for use on smaller or harder samples. High level of surface preparation needed to be able to measure indentation. Krautkramer Dynapocket and Equotip rebound hardness testers are portable, quick, and reliable.		NDT
Microstructure Testing	Etch a metal sample to reveal or inspect its microstructure. Cross section an iron casting to reveal its graphite microstructure.	Evaluates the microstructure of the sample to identify defects and phases.	Requires significant preparation of surface. Experienced metallurgist is required to interpret results. Only intended for the microstructure of graphite.		Material sampling
Timber Structures Deterioration					
Visual Inspection	Observe, classify, and document the appearance of distress and defects on exposed surfaces of the component.	Identifies defects on the surface only.	Simplest and least expensive method. Detects surface defects, cracking and splitting of components. Need well trained inspector; need full access to surface.	✓	NDT
Timber Drilling	A component is drilled until there is a change in resistance, indicating sound material, piping or rot.	Determine the residual amount of sound timber in a component	Can gauge piping and rot within the member, test is low cost and portable. Measurement are at a single point and rely on the inspectors knowledge in measuring change in resistance and shavings examination.	✓	Intrusive

Test Name	Principle	Application	Advantages/ Limitations	Applicable to Level 2	Test Type
Ground Penetrating Radar	Radio frequency waves from radar transmitter are directed into the material. The waves propagate through the material until a boundary of different electrical characteristic is encountered. Then part of the incident energy is reflected and the remainder travels across the boundary at a new velocity. The reflected (echo) wave is picked up by a receiver. The transducer is drawn over a surface and forms a continuous profile of the substrate below.	It can detect several parameters within timber components such as the rot, piping, and splitting, in situ density and moisture content variations.	Expensive; reliant upon operator judgement; complicated equipment setup; specialist experience in data interpretation. Relative results can be impacted by variables not of interest (moisture content, shelling). Additional tests may be needed to confirm results.		NDT
Nuclear Densometer	This test method returns a measure of soundness by determining the average density along a member.	It can detect several parameters within timber components such as the rot, piping, and splitting.	Requires calibration, mounting on a centralising jig/bracket and the need to undertake additional testing to verify results. Radiation is a safety hazard – requires control of nearby facility or area including lane/road closure and/or special monitoring of exposure levels and dosage to personnel. Relatively slow and expensive techniques; requires skilled operator and interpretation.		NDT
Resistograph	The resistograph uses a 2 mm diameter drill bit to drill the member at a constant feed rate and rotational speed. The unit measures resistance encountered as the bit advances into the timber to detect decay, voids and other irregularities in the member	It can detect several parameters within timber components such as the rot, piping, and splitting.	Provides an accurate portrayal of internal soundness but, as for conventional drilling, only provides results at discrete drilling points.		Intrusive

3.10 Inspection Frequencies and Program

Forward programming of inspections is essential to enable the efficient coordination of personnel (including specialist personnel) and equipment. To optimise the inspection program different programming techniques can be used, particularly for Level 2 inspections.

3.10.1 Level 1 – Routine Inspection

Routine maintenance/surveillance inspections are typically required at 6–12-month intervals and should be programmed alongside other inspection work required for the rest of the road network. These should be at a frequency not greater than 12months and is sufficient to confirm that the required level of serviceability and user safety of the asset is maintained.

3.10.2 Level 2 – Condition or Defect Inspection

An overall program for Level 2 inspections should be prepared for a period of up to 10 years based on the specified frequencies. The program should include details such as the name and location of the structure, date and type of last inspection along with the date and type of proposed inspections. This is particularly important when scheduled Level 3 inspections are also required for specific structures on the network.

From this, a detailed program of inspections can be produced for a particular year. This detailed inspection program must be flexible enough to accommodate availability of resources and access equipment. Environmental factors such as tides and river levels may need to be taken into account when programming inspections, as may traffic volumes. Access plays a large part in inspecting and maintaining some structures, both in cost and time. However, it is important that adequate and safe access is provided, because if access is poor the quality of the inspection will suffer. Areas with the most difficult access may be the most important to inspect and maintain. If inspection requires the use of specialist access equipment (such as an underbridge inspection unit), it may be possible to reduce hire charges by inspecting as a group all the bridges that require the equipment.

Level 2 inspection frequencies can be determined using a range of methods. Three possible methods are outlined below:

- **Cyclic inspection cycle.** This is similar to Level 1 inspections and is the simplest programming method. It includes assigning a standard inspection cycle to all bridges such as 2, 5 or 7 years. This period can be reduced for particular bridge where the bridge is known to have deteriorated, or other specific risk factors are identified.

  The advantages of this approach include:

  - simple to implement
  - all bridges are inspected at relatively short durations to allow new issues to be identified early
  - implementation is relatively straight forward allowing all bridges on a single route to be inspected at the same time.

- **Age or condition-based frequencies.** These are currently used by some jurisdictions to determine the inspection frequency.

  For age-based inspections frequencies can include:

  - up to 10 years of age – 2 yearly
  - 10 to 40 years in age – 4 yearly
  - over 40 years in age – 2 yearly.

  For condition-based inspections frequencies can include:

  - condition State 1, 2 – 4 to 7 years
  - condition State 3, 4 – 2 yearly.

The advantages of this approach include:

- a reduced inspection load, which is particularly important for jurisdictions with large portfolios or are financially constrained
- depending on the inspection increments programming is still a relatively simple activity.

- **Risk based inspection frequencies.** Risk based frequencies. Risk based frequencies consider the operating environment of the structure and the environments potential impact on the structure. Two possible models have been provided in the sections below.

Simple Risk Based Inspection Frequencies

This approach can use a simplistic matrix with only two frequencies (which aligns with current practice in some jurisdictions). An example of the simplistic approach frequency allocation is shown in Table 3.10. Several risks may exist for any structure, and all should be assessed to determine the minimum inspection frequency.

Table 3.10: Example risk based inspection frequencies

Risk	Risk tolerance	Inspection frequency
Structure length	Under 50m	4 yearly
	50m or over	2 yearly
Number of spans	Under 5	4 yearly
	6 and over	2 yearly
Proximity to marine environment	Under 1km	2 Yearly
	1km and over	4 yearly
Aggressive environment	Bridge is in corrosive atmospheric conditions	2 yearly
	Bridge is not located in corrosive atmospheric conditions	4 yearly
Constantly wet invert	Water constantly present	2 yearly
	Predominantly dry stream bed	4 yearly
Route designation	Designated over mass route	2 yearly
	Not an over mass route	4 yearly
Increasing cyclic loads	Are HCVs increasing at of greater than 5%	2 yearly
	Are HCVs increasing at less than 5%	4 yearly

Detailed Risk Assessed Inspection Frequencies

More complex approaches can be adopted that the simple method outlined above. NCHRP Report 782 - *Proposed Guideline for Reliability-Based Bridge Inspection Practices* provides a methodology for developing a reliability-based frequency for inspections. Below is a summary of this approach and full details are available in the NCHRP report.

Implementing this approach to determining inspection frequencies is likely to reduce the number of level two inspections required compared to a 24 monthly cyclic approach. However an increased resource input into planning is required. This increase is not only during the implementation phases, it also includes the ongoing assessment of inspection frequencies. However this investment would be potentially offset through early intervention is issues identified.

The process involves the owner of the assets establishing an expert panel to define and assess the durability and reliability characteristics of bridges within their jurisdiction. The panel analyses portions of the bridge inventory by bridge type to assess inspection needs by using engineering rationale, experience, and typical deterioration patterns to evaluate the reliability characteristics of bridges and the potential outcomes of damage. This is undertaken through a process that consists of three primary steps:

- **Step 1: What can go wrong, and how likely is it?** Identify possible damage modes for the elements of a selected bridge type. Considering design, loading, and condition characteristics (attributes), categorise the likelihood of serious damage occurring into one of four occurrence factors ranging from remote (very unlikely) to high (very likely).
- **Step 2: What are the consequences?** Assess the consequences, in terms of safety and serviceability, assuming the given damage modes occur. Categorise the potential consequences into one of four consequence factors ranging from low (minor effect on serviceability) through to severe (i.e., bridge collapse, loss of life).
- **Step 3: Determine the inspection interval and scope**. Use a simple 4 × 4 matrix to prioritise inspection needs and assign an inspection interval for the bridge based on the results of Steps 1 and 2. Damage modes that are likely to occur and have high consequences are prioritised over damage modes that are unlikely to occur or are of little consequence in terms of safety. An inspection procedure is developed based on the assessment of typical damage modes for the bridges being assessed.

While this methodology proposes an inspection procedure be developed based on the assessment of typical damage modes, it would be prudent at the time of inspection to complete a full level 2 inspection to enable new defects to be identified.

3.10.3 Level 3 – Specialist Inspections

Level 3 inspections are typically non-cyclic due to the nature of this inspection type, these inspections need to be specifically planned and funded. However, there are some inspections that are cyclic in nature that can be programmed utilising any of the techniques outline for Level 2 inspections. These potentially include:

- complex structures
- underwater inspections on significant bridges
- difficult access inspections
- monitoring inspections if specialist equipment is required.

3.11 Inspector Training and Certification

An important issue, which has been reported from UK research (Middleton 2004), is the unreliability of the visual inspection reports undertaken as part of the normal inspection procedures. When determining the condition of a bridge, and in particular evaluating any evidence of distress or deterioration, one relies heavily on the reports by the site inspectors who are required to report in detail on any deficiencies observed in the structure.

A study (Moorem et al. 2001) of the inspection process undertaken in the US by 49 experienced bridge inspectors, showed that there was a very significant likelihood that major defects would not be picked up. Very simple measures were identified to improve the performance of inspectors. Compulsory eye tests for all inspectors were suggested as one such measure. Other important factors influencing the performance of inspectors were psychological conditions such as a fear of heights or fear of traffic. It was also shown that the quality of the inspection was strongly correlated with the formal qualification level of the inspectors.

3.11.1 Current Practice

Current jurisdictional practices vary across Australia and New Zealand. These are summarised in Table 3.11. At present no jurisdictions have a requirement for on-going eye test, as recommended by Moore et al. Requirements for the documentation of ongoing assessments of inspectors is limited to Victoria.

Structural engineers or professional engineers are typically specified for Level 3 inspections. For some inspection types, specifically trained personal are required such as specialist construction divers for underwater inspections, or Level 1 or 2 inspectors for immediate response inspections after unplanned events such as impact damage, earthquakes, floods, and fire.

Table 3.11: Current inspector training and qualification requirements

Jurisdiction	Experience	Role	Training provider	Training requirements	Ongoing assessment
Level 1 Inspections					
Transport for NSW	Dependant on what needs to be inspected.	Work Supervisor / Inspector.	Not required.	Not required.	Not required.
NZ Transport Agency Waka Kotahi	5 years.	Maintenance of highway structures.	NZ Institute of Highway Technology.	Required for Level 1 and 2.	Not stated.
Department of Infrastructure, Planning, and Logistics, Northern Territory	Extensive practical experience.	Road and highway structure routine maintenance crew member.	ARRB and IPWEA provide Level 1 training in NT.	Level 1 training required.	Not stated.
Department of Transport and Main Roads, Queensland	2 years minimum with bridge crew.	Bridge construction/ maintenance/ rehabilitation crew member. RoadTek Employee.	RoadTek Inhouse Training.	In-house accreditations process.	Yes.
Department of Infrastructure and Transport, South Australia	Not stated.	Routine maintenance contractor.	Not stated.	ARRB's Level One Bridge Inspection" course or an interstate equivalent.	Not stated.
Department of State Growth, Tasmania	Not stated.	Routine maintenance crew member.	Not stated.	Not stated.	Not stated.
Department of Transport Planning, Victoria	Extensive practical experience in road and bridge routine maintenance.	Routine maintenance crew member.	Not stated.	Not stated.	Not stated.
Main Roads Western Australia	Not stated.	Works Supervisor; or Asset Management Officer.	MRWA, ARRB and IPWEA provide Level 1 training.	Recommended to have completed training from nominated provider.	Not stated.
Level 2 Inspections					
Transport for NSW	Accreditation required.	TfNSW trained bridge inspector.	TfNSW internal trainer.	Internal training course (over 90% of inspection done in-house).	2 yearly refresher for internal personnel and as requested for external providers.
NZ Transport Agency Waka Kotahi	At least 5 years experience of the construction, inspection and maintenance of structures or geotechnical structures.	Professional Engineer, or extensive experience.	NZIHT and IPWEA provide training.	Must have completed a NZ Transport Agency endorsed inspection training course.	Not stated.

Jurisdiction	Experience	Role	Training provider	Training requirements	Ongoing assessment
Department of Infrastructure, Planning, and Logistics, Northern Territory	Extensive experience in the inspection, construction, design, maintenance, or repair of road structures.	Not stated.	ARRB and IPWEA provide Level 2 training in NT.	Must attain accreditation through attending an NT approved accredited Level 2 training workshop.	Not stated.
Department of Transport and Main Roads, Queensland	5 years.	Construction, maintenance, rehabilitation, or inspection of road structures. RoadTek Employee.	RoadTek In-house training.	In-house accreditation process.	Yes.
Department of Infrastructure and Transport, South Australia	Demonstrated experience and capability in bridge inspection.	Professional Engineer.	ARRB and IPWEA provide Level 2 training in SA.	Prequalified in accordance with the requirements of DPTI Prequalification for Professional and Technical Services 17C811.	Not stated.
Department of State Growth, Tasmania	Not stated.	Not stated.	ARRB Level 2 course required.	Internal resources utilised. Required to attain accreditation through an approved Level 2 training provider.	Inspector performance is subject to ongoing review, including independent audits of completed inspections.
Department of Transport Planning, Victoria	Extensive practical experience in either inspection, construction, design, maintenance, or repair of road structures.	Not stated.	ARRB Level 2 course required.	Prequalified in accordance with the requirements of VicRoads Prequalification Scheme for the BI2 category. Attendance at a VicRoads Level 2 accreditation briefing and satisfactory performance in the accreditation briefing assignment.	Inspector performance is subject to ongoing review, including independent audits of completed inspections.
Main Roads Western Australia	**Concrete/Steel:** Minimum of three years' experience in Bridge design, maintenance or management. **Timber:** Minimum of three years' experience in maintenance or management in specialised crew.	**Concrete/ Steel:** Structural Engineer, Engineering Associate. **Timber:** Tradesman, Foreman, Plant Operator, Works Supervisor.	**Concrete/Steel:** MRWA or ARRB. **Timber:** Timber bridge training is undertaken through on job training in specialist crew.	**Concrete/Steel:** MRWA have developed a 2-day conversion course to enable other jurisdictional approved inspectors to undertake inspections. **Timber:** Timber bridge training is undertaken through on job training in specialist crew.	Auditing by a Senior Structures Engineer.

Jurisdiction	Experience	Role	Training provider	Training requirements	Ongoing assessment
Level 3 Inspections*					
Transport for NSW	Extensive experience specified.	Structural Engineer accompanied by TfNSW trained bridge inspector.	Not required.	University bachelor's degree in civil engineering Trained in the TfNSW Bridge Inspection Procedure.	Not required.
NZ Transport Agency Waka Kotahi	Experience of the construction, inspection and maintenance of structures or geotechnical structures.	Professional Engineer.	NZIHT and IPWEA provide training in bridge inspection.	Must have completed a NZ Transport Agency endorsed inspection training course.	Not stated.
Department of Infrastructure, Planning, and Logistics, Northern Territory	Membership of the Institution of Engineers Australia, with extensive (minimum of 5 years) and current highway structure design and construction experience and/or load testing experience if applicable.	Professional Engineer.	Not stated.	Not stated.	Not stated, peer reviews of reports maybe undertaken.
Department of Transport and Main Roads, Queensland	Must have the ability to correctly identify and interpret the severity and nature of structural and material defects, assess their criticality, and make the appropriate recommendations with respect to required action.	Professional Engineer.	Not stated.	Not stated.	Not stated.
Department of Infrastructure and Transport, South Australia	Not stated.	Not stated.	Not stated.	Not stated.	Not stated.
Department of State Growth, Tasmania	Not stated.	Professional Engineer.	Not stated.	Not stated.	Not stated.
Department of Transport Planning, Victoria	20 years experience.	Pre-qualified Proof Engineer.	Not stated.	Not stated.	Not stated.
Main Roads Western Australia	Professional Engineer.	Professional Engineer.	Not stated.	Not stated.	Auditing by a Senior Structures Engineer.

** Many States and Territories do not specify specific training requirements and ongoing assessment requirements for level 3 inspectors. For those where a professional engineer is required on going training and assessment is undertaken as part of the registration process.*

3.11.2 Bridge Inspection Training Outcomes

While some road agencies provide in-house training courses that meet their specific requirements and there are also external training organisations available, there is no Australia or New Zealand-wide recognised inspector certification. The scope therefore exists for road agencies to harmonise some aspects of the inspection process such as technical engineering aspects, supplemented with jurisdictional specific supplements on reporting, data collection, and rating criteria.

The benefits of successfully implementing a training and certification program is that it will provide for certification of inspectors in specified disciplines and will serve to impart the basic knowledge and skills necessary to accurately report on bridges for network-wide uniformity of inspection. On completion of the course, participants will have been trained to inspect bridges in accordance with the requirements of the relevant inspection standards of their road jurisdiction. National harmonisation of inspection standards would further increase the availability of certified inspectors. This could potentially be achieved through having a two-part training system. The first part would be a national accreditation and second part would be a local supplement.

3.11.3 Bridge Inspection Training Course Content

Any inspection training course should, as a minimum, be based on the relevant jurisdiction's inspection manual and should include:

- inspection policy and procedures
- safe inspection techniques and safe work method statements
- access and other equipment requirements
- data recording and coding of items
- depending on the jurisdiction the training is being provided for, the course should provide specific content on uniformity/consistency of condition rating, or identification and mapping of defects
- visual aids and material relevant to common structure forms within the network
- significance of observed defects in relation to:
  - public safety
  - durability
  - structural performance
- non-destructive testing applicable to Level 2 inspections
- reporting.

The course should include field inspection exercises coupled with assessment to demonstrate practical application of the theory.

On completion of the course the participants who meet the requirements should receive a certificate stating the level of competence attained. Ideally there will be a requirement for refresher training and reassessment at a pre-determined frequency.

3.11.4 On-going Competency Assessments

At present ongoing competency assessments are generally not required. Competency can be assessed through one or more of the following:

- implementing an auditing program including independent review of a nominated proportion of inspections undertaken. Some jurisdictions have auditing processes in place
- requiring Level 1 and 2 inspectors to undertake an inspection refresher course, if no inspections have been undertaken in the previous three years
- requiring a periodic assessment of reports on a nominated structure.

3.12 Inspection Auditing

Ensuring quality, complete, and accurate data and information is being collected is a critical factor in ensuring the success of inspection programs, particularly in relation to defining faults that require specialist inspections or significant rating changes. To achieve this an auditing program needs to be implemented.

The need for auditing was identified in a study by Moorem et al. (2001). This study of the inspection process undertaken in the US, by 49 experienced bridge inspectors, showed that there was a very significant likelihood that major defects would not be picked up and recommended that a rigorous audit system be in place to maintain the quality of the system.

An audit process should be a documented procedure that includes the following:

- identification of the number of audits to be undertaken over a defined period
- identification of the independent auditor
- desktop assessment of the inspection reports
- field validation of the inspection findings
- advising the inspector of the audit process
- advising the inspector of the audit findings.

3.13 Quality Control and Assurance

Inspector experience, qualifications and auditing are parts of the process that ensure quality in terms of the inspection and reporting.

The US Federal Highway Administration have documented a quality control and quality assurance framework within their manual (Federal Highway Administration 2012). This approach includes:

- program documentation
- quality control procedures - establishment and enforcement of procedures that are intended to maintain the quality of the inspection at or above a specific level
- quality assurance procedures – the sampling and other measures to assure the adequacy of quality control procedures in order to verify or measure the quality level of the entire bridge inspection program.

When developing system specific to bridge inspections, this should seamlessly integrate with the wider jurisdictional quality control and assurance systems and be aligned to the principles of the ISO9000 series.

3.14 Reporting

Inspection reports should be prepared following each Level 1, 2 or 3 inspections to:

- record current condition for present and future reference
- identify elements/components that are in a condition that would warrant repair/replacement
- provide the raw data for management of the total bridge asset
- provide raw data for use of other teams within the organisation.

Inspection reports should be consistent so that reports for scheduled inspections (Level 1 and Level 2) prepared by different inspectors or at different periods may be compared directly with one another. A sequence of the inspection reports prepared for a bridge over a period of years may be used to determine any changes in structural behaviour and to obtain an estimate of the rate of deterioration and consequently of the useful life remaining as well as assisting in the determination of maintenance and rehabilitation measures.

Each road jurisdiction has their specific reporting requirement within their bridge inspection manuals. To effectively record the inspection findings, the inspection report should contain some or all the following:

- written statements, which should be clear, concise and accurate - summarising the condition of each element of the structure, describing defects and if required indicating a rating against a consistent scale
- sketches detailing the nature and extent of significant defects
- photographs showing the general structure (approaches, elevations and underside views) and all significant defects.

3.14.1 Level 3 Inspection Report

Given that Level 3 inspections are detailed investigations that generally target or address a specific issue relevant to an individual structure or a class of structures, reports will tend to be more comprehensive as there is a need to include:

- background and reasons for undertaking the inspection
- the scope of works
- detailed description of the condition of those parts of the structure that have been inspected, including where appropriate, test results, photographs, and sketches
- any significant changes, e.g., works or deterioration, since the last inspection
- description of any testing undertaken, details of the information collected and an interpretation of the findings
- any information relevant to the integrity and stability of the structure
- interpretation of inspection/assessment (e.g., load rating) findings
- a management plan outlining the scope, timing, and rough order cost estimate of any proposed intervention measures such as:
  - further investigation or monitoring
  - maintenance
  - rehabilitation
  - strengthening
  - replacement
  - level of service restrictions.

4. Bridge Assessment and Load Rating

4.1 Introduction

The objectives of evaluating the load carrying capacity of a bridge, also known as rating, include:

- safeguarding road users
- ensuring that it has adequate strength to carry the load or loads required
- extending the useful life of the bridge.

According to AS 5100.7, bridge assessment is a multistage process which involves data collection of bridge construction details, current bridge condition, capacity assessment, load effects and fatigue assessment in order to determine if the bridge can safely carry traffic.

Load rating is the outcome of a bridge assessment exercise, in which the bridge load capacity is determined for a specific referenced load.

Overloading and other severe load histories, deterioration, rehabilitation and strengthening may cause changes in a bridge's load carrying capacity with time. Therefore, the load carrying capacity calculated at a particular date may not be the same as the future or past capacity, or even the same as the original design capacity. A record of the changing magnitude of the load carrying capacity during a bridge's life provides useful information about its level of service. A bridge load capacity assessment is typically considered in the following circumstances:

- the bridge is required to carry increased live or other loads
- the bridge suffered physical damage from natural or man-made actions such as vehicle overloading, vehicle impact, fire, flooding, ship impact and earthquake
- significant deterioration of the bridge components is present, e.g. by chemical or physical weathering and fatigue effects
- there are deficiencies in design and/or construction of the bridge
- rehabilitation or strengthening works have been completed on the bridge.

Bridges constructed during earlier periods may have been designed (as required) to different load standards and have used different materials or materials of different standards than recent bridges. The load carrying capacity may be affected by these factors.

The type and frequency of a load may also influence a bridge's capacity to carry that load. A bridge, at a particular date, may possess a different load carrying capacity for:

- frequent passages of vehicles within legal limits
- repeated passages of vehicles granted a period permit
- infrequent passages of vehicles granted a trip permit.

Refer to the relevant road agency's technical guidelines for specific criteria for evaluating the load carrying capacity of a bridge, such as NZTA (2016b), TMR (2013).

Such evaluation needs to be preceded by an inspection to determine any significant features that might affect the result. Inspection will normally concentrate on superstructure members as they are usually critical for live load, but the possibility of others being critical should not be neglected. Some possibilities are timber piles, foundations affected by scour, deteriorated bearings and support members badly cracked or showing reinforcement corrosion.

The evaluation process consists of determining the working load capacity of critical members for either normal live load or overload as appropriate. Comparison of the capacities with standard loading at both levels leads to parameters for posting (for live load) and rating (for overload) that are used to characterise the load capacity of the bridge.

Evaluation of the load carrying capacity of a bridge should be certified by a registered engineer.

4.2 Methodology

The following assessment methods can be used (AS 5100.7):

- a comparison check to the original design loading
- simple theoretical analysis based on the design parameters
- more sophisticated theoretical analysis using techniques such as finite element analysis or other advanced analysis methods
- analysis using the results of field investigation of material properties, bridge component dimensions, dead and live loads, foundation capacity and the like
- bridge specific live load assessment
- field or laboratory test loading, e.g. proof, static and dynamic load testing
- structural health monitoring.

Each method has differing accuracy, which should be considered in the assessment. All assumptions relevant to the assessment should be recorded. The likely behaviour of the structure should be taken into account in the analysis considering the material type and structural behaviour.

Other bridge design standards may be considered in the assessment subject to the approval of the relevant agency.

4.3 Information Required

To determine a bridge’s current load carrying capacity the following information should be collected:

- a set of as-constructed plans, showing member sizes and structural details such as connections (if unavailable, the bridge should be measured and drawings prepared)
- construction reports
- identification of all materials used in the bridge and their current strengths
- details of all repair works and modifications
- an inspection report detailing the current condition, particularly specifying any factors that may affect the load capacity.

It is also useful to have access to:

- original design plans and calculations
- previous inspection reports (if available)
- statistics of traffic using the bridge, such as number and mass.

4.4 Load Rating Process

The concept of rating is based on the limit states design principle that the assessed minimum strength capacity of the bridge shall be greater than the assessed maximum factored load applied.

The available live load capacity of the bridge shall be rated and compared with the effects of a nominated rating vehicle or loading. The rating process involves (see also AS 5100.7):

- determining the structural capacity of elements in the bridge to carry live load effects (moment, shear, etc.), taking into account the field geometry measurement if required and the bridge condition
- calculating the load effects produced in the elements by the permanent loads (such as dead loads, superimposed dead loads) and traffic loads (nominated rating vehicles, co-existing vehicles). Other transient loads such as temperature variation and wind effects may be considered
- determining the load rating factor of each element.

The element having the lowest value of load rating factor is thus the most critical in the bridge.

It should be noted that, a fatigue assessment may be required as part of the load assessment process. Refer to AS 5100.7 for further details.

A higher tier assessment may also be undertaken, if required, where the load rating factor for a specific bridge and nominated rating vehicle is assessed as being less than required. Higher tier assessment methods include:

- advanced analysis methods
- field testing (proof test and dynamic load testing)
- structural health monitoring
- reference to other recognised codes and standards subject to the approval of the relevant jurisdiction.

To facilitate rapid comparison between bridges, a standardised rating system appropriate to the road jurisdiction's needs should be adopted.

4.5 Evaluation of Structural Capacity

The ultimate limit state (ULS) capacity of concrete, structural steel, and timber components must be determined in accordance with the relevant sections of AS 5100.7. Components that do not meet the scope of the AS(AS/NZS) 5100 series and specific requirements of AS 5100.7 should have their capacities calculated using alternative procedures based on established and recognised theories, analyses, testing and engineering judgement. Similarly, the NZTA (2016b) is applicable for New Zealand jurisdictions.

The assessment of the structural capacity of a bridge must be made under the direct guidance of a professional engineer experienced in bridge design or assessment, who verifies that the actual structural condition of the bridge has been taken into account in the assessment and load rating.

The structural capacity of a bridge depends upon:

- materials and material properties
- bridge geometry
- capacity reduction factors
- bridge conditions.

4.5.1 Materials

The materials used in the bridge and their strengths must be correctly identified. Some materials may be difficult to differentiate, as, for example, wrought iron and steel. Their appearances are similar and fabrication methods may be identical. A rolled girder or a riveted fabricated girder in a bridge built within a decade or so of 1900 may be either wrought iron or steel. The only reliable identification is by microscopic examination for the slag inclusions in wrought iron.

All materials have undergone significant development of properties since their introduction to bridge building. The increase in strength of different materials with the date of construction is given in Ontario Ministry of Transportation and Communication (1983). Older steels may have lower ductility limits.

The same material may exist in different strengths, and all members in the same bridge may not necessarily be of the same strength. Fabrication and erection techniques may modify the strength or properties of the parent material at critical locations, such as at joints.

Where material properties are unknown, they can be determined using one of the following methods (AS 5100.7):

- review of other relevant plans and documents
- estimation by considering the date of bridge construction
- analysis of tests or samples obtained from the bridge or from specific bridge components
- other methods approved by the relevant agency.

The most reliable method for determining material properties for the bridge rating is to test the actual materials used in the critical elements. Some non-destructive methods, e.g. Schmidt Hammer, for the strength of concrete are available, but the most precise method is to remove specimens from the bridge for testing, although this is often impractical.

Refer to *AGBT Part 2: Materials* for detailed discussions on the properties of various materials.

4.5.2 Material Strengths

Concrete

In most reinforced concrete members, where flexure is critical, the exact concrete strength is not of real significance because both shear and moment strength are in practice governed by the reinforcement-yield value. It will therefore not often be necessary to obtain measured values. However, it may be worthwhile if any of the following are critical:

- reinforced concrete columns
- reinforced concrete arches
- reinforced concrete continuous beams, for negative moment.

For prestressed concrete members, concrete strength may be critical, but it is likely that in most cases there will be better records of design and/or test values.

Non-destructive test methods, such as ultrasonic pulse velocity and surface hardness methods, may be used to correlate the concrete strength in damaged and sound regions of a structure. If compressive strengths are estimated using non-destructive methods, calibration factors should be determined using concrete cores from the structure where appropriate, and the level of uncertainty should be accounted for in the estimate of predicted strengths.

Reinforcing Steel

Historically, steel strengths have been quite variable, so the nominal values are likely to be conservative in most cases. It is usually well worthwhile to obtain yield strengths, and this can be done from measurements of steel hardness. Various instruments are available to do this non-destructively, and it is advisable to obtain the services of a testing laboratory. It is, of course, necessary to remove some cover concrete to perform the testing.

If measured strength is exceptionally high, the requirement such as Section 6.4.4(a) of the NZTA (2016b) should be noted to guard against compressive concrete failure.

Where corrosion of the reinforcement is suspected, investigation of the reinforcement's condition should be undertaken. This will involve exposure of the reinforcement at critical sections, and at sections at which corrosion is likely to be most pronounced as evidenced by cracking and spalling of the cover concrete, to enable measurement of the remaining steel section to be undertaken.

Prestressing Steel

Accurate knowledge of the strength of prestressing steel is not usually required because the effective force in the tendons is of more significance. There is at present no practical way of measuring prestressing steel strength non-destructively, so reliance must be placed on existing records or on values from standard specifications.

Structural Steel

The statement on reinforcing steel applies essentially to structural steel.

Timber

If identification of the species of timber is required, the services of the Forest Research Institute in Rotorua or other specialists may be obtained.

Analysis of Test Results

Where strengths are measured, Table 6.2 of the NZTA (2016b) requires an adequate number of test results to give statistically reliable values. It should be noted that the reliable value increases rapidly with the number of tests.

4.5.3 Field Measurement of Geometry

When a more accurate estimate of the capacity of a bridge is justified, the structural capacity should consider the current geometry, dimensions and section properties of the bridge and its components, including the foundations. The assessment of structural resistance should allow for all geometric imperfections and eccentricities caused by inaccurate construction, damage, or any other cause.

4.5.4 Assessment of Capacity Reduction Factors

In the absence of information to the contrary, AS 5100.7 allows an assumption to be made that the bridge and its components are in their 'as-constructed' condition. Where inspection of the bridge confirms that the bridge is in sound condition, the design values for the capacity reduction factors can be used.

For the determination of appropriate capacity reduction factors, it can be assumed that the capacity reduction factors incorporate a factor of 0.95, to allow for member size and geometric deficiencies. If accurate assessments are made of member sizes and geometric deficiencies and the results included in the assessment of structural strength, the capacity reduction factor may be divided by 0.95, thereby increasing the load rating of the bridge.

Bridge foundations can be rated using a similar approach, in which actual foundation material properties are used, with capacity reduction factors being taken as the material factors specified in AS 5100.3.

Where the capacity of a bridge similar to the bridge being rated has been assessed by load testing, consideration may be given to adapting that rating. It may be necessary to use lower capacity reduction factors depending upon the level of loading used and the similarity between the two bridges. In the adaptation of the rating, the computer models that have been developed for the load-tested bridge and which have been calibrated against test results can be used.

4.5.5 Bridge Condition

The existing condition of the bridge can be assessed by a Level 2 or Level 3 inspection.

The findings from the inspection should be taken into consideration when assessing the current capacity of the components of the structure, including the foundations. No load rating can be considered valid until the inspection has been undertaken.

Structural health monitoring may be used to supplement Level 2 or Level 3 inspections.

The inspection records should be of sufficient detail to allow changes in condition to be assessed during future inspections. Refer to AS 5100.7 for the detailed requirements on the inspection records.

4.5.6 Main Member Capacity and Evaluation

The majority of bridges consist of a simple span beam system in which the critical section is likely to be at midspan. However, the possibility of an unusual arrangement of curtailed reinforcement or steel cover plates, or an unusual prestressing cable layout, could make other sections critical. In situations where shear capacity is critical, measurement of concrete strength can be worthwhile. By similar reasoning, diaphragms are not normally critical if undamaged. In order to determine the load distribution between beams, a grid analysis is usually the most appropriate method.

In a system other than simple spans, it may be more difficult to identify critical sections, and a computer grid analysis may be required for this reason as well as to determine load distribution between beams.

4.5.7 Deck Capacity and Evaluation

Concrete deck panels with all edges relatively rigidly supported and restrained by girders and adjacent deck panels resist loading primarily by membrane arch action and tend to fail in a punching shear mode.

Not all slab panels meet the requirements to enable them to be considered as acting in membrane arch action. Where conventional elastic analysis has been used to rate deck slabs, it has been found that, in the majority of reinforced concrete decks on longitudinal beams, either transverse negative reinforcement or longitudinal positive reinforcement is critical, but the possibility of other sections requiring consideration should not be forgotten.

Where deck slabs exhibit extensive cracking, the cracks should be marked and their growth monitored over time. Load testing offers a method for assessing the deterioration of the deck slab panels and of monitoring the ongoing progressive deterioration of the deck slab over time.

4.6 Determining Load Effects

Refer to AS 5100.7 for the detailed code provisions for determining load effects. The following should also be taken into consideration:

4.6.1 Dynamic Load Allowance Factors

AS 5100.7 allows the DLA to be reduced to 0.3 if the roughness of the road and bridge is controlled at an International Roughness Index (IRI) of less than 4.0 for the length of the bridge plus 400 m on each approach. It requires frequent road roughness monitoring. Also, for controlled speed situations, such as for HLP and indivisible loads, the DLA can be taken as 0.1 when the maximum speed is 10 km/h. A DLA = 0 can be applied if the maximum speed is 5 km/h.

4.6.2 Reduced Load Factors

The load factors can be modified subject to the approval of the relevant agency, where appropriate statistically significant measurement is undertaken (AS 5100.7). The modification of the load factor should take into account the actual loading, measured loading effects, actual capacity and material properties.

4.6.3 Type and Frequency of Loading

The type and frequency of loading will influence the rating, as different acceptable stress levels and different load factors may be chosen for:

- Infrequent passages of individual trip permit vehicles of above legal limit highway loading, travelling on specially issued over-mass trip permits. Note that actual check weighing is required to be certain of the vehicle's axle weights and gross weight. Even where there is good will on the part of the vehicle operator, experience has shown that large errors in weights assessed from the load manufacturers' information can often be expected.
- The load effects on the structure of specific heavy vehicles may be reduced by placing restrictions on the conditions of travel, provided that these restrictions are strictly enforced. These include speed restrictions to reduce impact, and lateral position restrictions, such as travel along the centreline, to reduce the portion of the vehicle load carried by a bridge element.
- Field testing may be used to measure the response of critical bridges under load. However, this is expensive and the decision to adopt it needs to be justified.
- Repeated passages of period permit vehicles or groups of vehicles with above legal limit highway loading, travelling under period permit. Period permit vehicles such as road trains should be compared against the normal T44 load factors and loads, but that may not be sufficient particularly for medium length continuous bridges. It is therefore necessary to look at other combinations. Each structure should be analysed and load rated for a suite of vehicles that will facilitate the assessment of period permit vehicles even though the T44 is considered to be the prime reference vehicle for rating.
- Frequent passages of vehicles, such as legal limit highway loading with no restrictions on travel. Where there is insufficient capacity for carrying the maximum legal gross or axle group load, a load limit sign should be posted on the structure that reflects this shortfall in load capacity.

4.6.4 Load History and Deterioration

Concrete

The effect of deterioration upon the load carrying capacity is largely a matter for the judgement of an experienced inspecting engineer, but, where possible, the magnitude of the deterioration should be quantified.

While cracking of structural significance may indicate overstress, most forms of concrete deterioration (spalling, scaling, efflorescence) usually are most significant in their effects on durability rather than strength.

Corrosion will reduce the area of reinforcing or prestressing strand and hence lower the strength of the member. Pitting caused by severe corrosion, nicking by mechanical damage, and fatigue may lead to eventual fracture of prestressing strand with consequent reduction in strength of the member.

Iron and Steel

In the past, iron and steel have mainly been used in thin sections with splices, stiffeners, and joint details that are liable to stress concentrations and fatigue under repeated loading.

Corrosion reduces the cross-sectional area of a member. This may not be of significance to the load carrying capacity if it is not at the controlling section. Asymmetrical corrosion may produce eccentricity effects. Severe corrosion may produce pitting, which may affect the structure's behaviour under shock loading.

Fire damage may affect the metallurgy of iron and steel, altering its structural properties.

Timber

Attack by borers, insects, etc. reduces the cross-sectional area for load carrying. Rotting reduces the strength of the timber.

Masonry

Factors that modify the load carrying capacity of masonry include the fretting of the jointing mortar, fretting or erosion of the voussoirs, cracking in the joints or blocks or deformation of the arch's shape. The modifying influence of these factors is covered in Department for Transport (2005).

4.7 Proof Loading

Proof loading is sometimes useful to verify theoretical findings, especially in cases where it is difficult to model the structure adequately for computer analysis. Criteria are laid down in NZTA (2016b) and in AS 5100.7. If proof loading is contemplated, the services of an experienced laboratory should be obtained.

4.8 Mitigating Risks Posed by Weak Bridges

Options for reducing risks posed by weak bridges, until such time as they can be strengthened or replaced, include the following:

- imposition of a speed restriction
- imposition of vehicle gross weight and/or axle weight restrictions
- limiting the number of heavy vehicles permitted on the bridge at the same time
- closing the bridge to heavy vehicles and rerouting these around a bypass, e.g. via an adjacent ford through a stream.

5. Maintenance

5.1 Introduction

When considering maintenance activities, the *International Infrastructure Management Manual* (Institute of Public Works Engineering Australasia 2015) describes two broad types, reactive maintenance and preventative maintenance.

Reactive Maintenance

Reactive maintenance variously termed 'corrective maintenance' (correcting faults and failures) or 'unplanned maintenance'. This activity reinstates service after a fault or failure by carrying out maintenance activities on failed assets. Actions may comprise repair or replacement to restore to a serviceable condition or a coat of paint to prevent further deterioration, for example. This includes routine maintenance comprising less frequent, less technically simple actions, and planned specialist operations.

Preventive Maintenance

This activity carries out maintenance tasks to prevent faults and disruption from faults, it includes 'proactive maintenance' or 'planned maintenance' activities that enhance the life of the asset, slowing down asset deterioration and delaying the time for asset renewal. Preventive maintenance is typically periodic or routine in nature, comprising frequent, technically simple, repetitive tasks. They are generally, simple maintenance tasks undertaken by road patrol or maintenance personnel, when necessary, as determined from regular inspections or as prescribed by maintenance manuals/plans.

The cost of undertaking reactive maintenance typically increases as preventive maintenance activities decrease (and vice versa). This is an important factor to take into consideration when planning (and deferring) maintenance programs.

This section discusses the objectives of undertaking maintenance.

5.1.1 Programming of Maintenance

A program of maintenance work will be based on reactive maintenance items identified in the inspections, in priority order, together with periodic preventive maintenance activities such as maintaining drainage, replacing damaged traffic barriers etc. As the amount of work will not be known until after the inspection program is complete, the program needs to be flexible. It is important to follow up the maintenance work to ensure that defects identified have been attended to.

5.1.2 Maintenance Objectives

The objectives of undertaking maintenance are to:

- safeguard the travelling public
- preserve serviceability and load carrying capacity for as long as possible
- minimise the cost of repairs caused by premature deterioration.

Maintenance involves little or no increase in the current level of service of a bridge.

Maintenance is preventative in nature. The adage ‘prevention is better than cure’ is eminently true for bridges where defects can rapidly have serious consequences if action is not taken (Organisation for Economic Cooperation and Development 1981). It is best undertaken by planning on a regular basis. This ensures that individual bridges and individual items are not overlooked. It is most effective if begun when a bridge is new and continued throughout the whole service life (Purvis & Berger 1983). Preventive maintenance applied to structures in good condition appears to be a very cost-effective strategy (Fitzpatrick, Law & Dixon 1981).

Particular maintenance tasks may have been specified by the designer, but more usually this is not the case and only the more obvious tasks will have been set out in an inspection report (Section 3.14).

Maintenance that is planned, rationalised economically with consideration of desired levels of service, and undertaken regularly leads to overall minimisation of bridge costs, except in situations where massive disruption to traffic or services may justify its deferral.

Maintenance undertaken regularly is also valuable in the early detection of defects requiring rehabilitation.

Maintenance activities may be considered to be undertaken at two levels:

- regular maintenance undertaken by road patrol or maintenance personnel and requiring only the equipment and parts normally carried in the patrol vehicle or available locally; regular maintenance should be carried out as soon as necessary, frequently if need be
- routine maintenance undertaken either by road maintenance patrols or specialist bridge maintenance gangs and often requiring extra tools, special equipment and ordering of materials; routine maintenance is usually programmed in advance from routine and detailed inspections and undertaken at approximately one to two year intervals.

Additional information on maintenance may be found in Harding, Parke and Ryall (1990) Paper Nos. 19 and 20.

5.1.3 Regular Maintenance

The need for, and timing of, many regular maintenance tasks is determined from regular inspections undertaken by road patrols or maintenance personnel.

Many regular maintenance tasks are intended to prevent retention of water and ensure its free drainage away from the bridge.

The individual regular maintenance tasks to be undertaken, when necessary, include:

- cleaning, washing, re-erecting or replacing dirty, misaligned and missing signs and delineation markers
- removing obstructions restricting road users’ clear vision
- replacing damaged and missing barrier components (both approaches and bridge)
- making good (feathering out) settlement of the running surface between the approaches and bridge
- reinstating the footway surface
- localised repair of the road surface
- cleaning all drains, side drains, channels, inlet and outlet pits, sumps, gutters and scuppers on the approaches and the bridge
- cleaning out membranes and associated drains and removal of all foreign objects from deck joints. Some joints may have membranes covered by plates. The plates must be removed, the membrane cleaned and the plate reinstalled
- tightening loose bolts in deck joints and replacing missing bolts
- cleaning vent holes where bridge superstructures are provided with them to reduce flotation forces during flood

- cleaning bearings, bearing sills and sill drains
- cleaning weepholes in abutments and wingwalls to ensure free flow of seepage water
- filling scour holes in embankments and slope protection
- removing debris, growth and silt from under the bridge and associated trash racks. This is particularly important after a large flood as trees and branches brought down in the flood get caught up in the piers and can be a severe fire risk as well as impeding future flood flows
- removing long grass and small bushes from under and adjacent to the bridge. This should be done at the start of the summer for fire protection. It can be done by controlled burning or the use of herbicides providing these are environmentally acceptable
- filling scour holes and ruts with large rocks. Take care not to damage the bridge structure when placing the rocks. See also Section 6.11.

For additional information on drainage systems see Section 6.8, and for information on regular maintenance see AASHTO (2007), and New Zealand Institute of County Engineers (1985).

5.1.4 Routine Maintenance

The need for and timing of routine maintenance tasks is determined from routine and detailed inspections.

The individual routine maintenance tasks to be undertaken, when necessary, include:

- tightening loose bolts and replacing missing bolts in barriers
- replacing corroded, damaged and missing components in barriers or painting where necessary
- resetting barriers to correct height and alignment
- stopping leaks in drainage structures, replacing if necessary
- extending and modifying drains that repeatedly require regular maintenance
- replacing leaking membranes and seals in deck joints
- replacing and repairing damaged components in deck joints to ensure integrity and free movement
- sealing of cracking in concrete elements (Sections 6.2 and 6.4.4)
- application of protective coatings (such as silane coatings – see Section 6.4.4) to concrete elements
- washing all iron and steel surfaces subject to deposits of aggressive salts and which are not washed clean by rain. The interior faces of girders of bridges in industrial areas or marine environments may be subject to deposition. Experience has shown that such deposits can promote corrosion
- cleaning all iron and steel surfaces subject to accumulation of dirt, silt, other debris and bridge droppings without damaging the protective coating. Experience has shown that such deposits can promote corrosion
- maintenance and/or touch-up painting of elements as necessary (Section 6.5.5)
- renewing as necessary the cathodic protection systems (Section 6.4.4)
- poisoning all vegetation growing in joints of masonry and cut off all vegetation that protrudes. Removal of the roots may cause extra damage to the mortar
- tightening and fixing wedges used in propping
- undertaking more extensive scour repairs such as placing mattresses and gabions if scour repeatedly occurs despite regular maintenance (Section 6.11).

For additional information on regular maintenance see AASHTO (2007), and New Zealand Institute of County Engineers (1985).

5.2 Common Defects Associated with Specific Bridge/Material Types

The focus of this section is to identify commonly observed defects in the context of specific bridge/component types as well as material type. The cause and mechanisms associated with specific material-related distress/deterioration are not discussed in depth in the following sections as these are covered in *AGBT Part 2: Materials*.

The section is intended as a general guide for the inspector or engineer dealing with a particular bridge type. Reference is made to other sections of this document and guide where applicable. It is important to recognise that, given regional variations in design, construction and maintenance practice, as well as operating environments, inspectors and engineers should familiarise themselves with jurisdiction specific manuals and publications such as VicRoads' *Road Structures Inspection Manual* (VicRoads 2014), Mandeno (2008) and Tilly et al. (2008).

It is important to understand the structural form and actions of the bridge under maintenance so that any defect detected may be correctly assessed for importance to the safety and durability of the structure as a whole. The clauses below are intended to highlight defects that are commonly known to present maintenance issues for specific bridge types. For example, the engineer should know the common defects that limit safety and durability of, say, concrete bridges, where and how to find the defect, be able to assess how serious the defect is, and know of the maintenance options for remedial work.

In extreme cases, it may be necessary to close the bridge to traffic until further information can be obtained from the site investigations. Alternatively, it may be sufficient to reduce the traffic loading and introduce emergency propping measures whilst detailed information is gathered and more permanent solutions are sought.

5.2.1 Timber Bridges

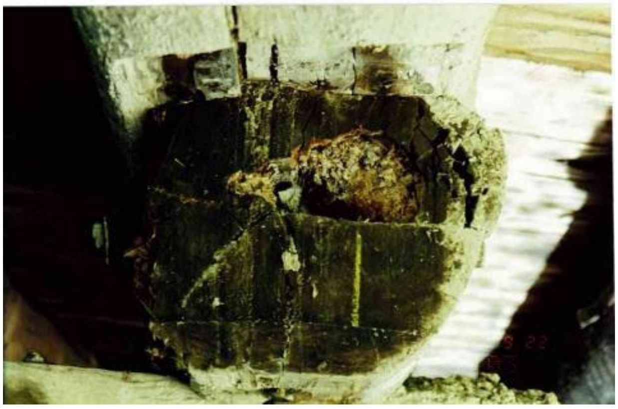
Timber members and components should be inspected for defects caused by decay (Figure 5.1), weathering, insect attack (Figure 5.2), splitting (Figure 5.3 and Figure 5.6), vehicle impact, fire damage (Figure 5.8), and proper connection to supporting members (Figure 5.4 and Figure 5.5). Look for evidence of collision damage by road and rail vehicles. Damage will be evident in the form of shattered timber.

See Section 6.6 for recommended routine and preventative maintenance of timber elements.

Timber components should be sounded with a hammer and/or probed or picked with a knife, ice pick, or prying tool to assess if the wood is sound or not. Additionally, some jurisdictions require drilling of hardwood timber components to detect any internal piping decay. Also check timber for fungal decay, insect attack, weathering, and wear.

Temporary repairs utilising timber should be identified and assessed as appropriate or in need of replacement by permanent repair.

Figure 5.1: Failed corbel from timber decay



Source: MRWA (n.d.).

Figure 5.2: Example of insect infestation in timber bridge

Source: MRWA (n.d.).

Figure 5.3: Example of split timbers in bridge

Source: MRWA (n.d.).

Figure 5.4: Example of loose joint in timber bridge

Source: MRWA (n.d.).

Figure 5.5: Example of poor timber connection in timber bridge

Source: MRWA (n.d.).

Figure 5.6: Example of split timber pile

Source: MRWA (n.d.).

Timber piles

Check for decay in the timber piles, caps, and bracing. Check particularly at ground or water level, and at joints and splices, since decay usually begins in these areas.

Specific checks and observations should be made for:

- piles – for material distress, splits, deflection, misalignment and settlement (Figure 5.6)
- splices and connections for tightness and for loose bolts
- the condition of the cap at those points where the beams bear directly upon it and at those points where the caps bear directly upon the piles. Note particularly any splitting or crushing of the timber in these areas
- pile caps for excessive deflection under passage of heavy loads
- timber piles in saltwater or marine environment to determine damage caused by marine borers and shipworms
- rotted or damaged timbers in the backwalls of end bents that function as abutments.

Timber Substructure

Timber abutments should be checked for insect and fungus attacks, as this can lead to material failure and vertical settlement.

Look at piers for signs of collision damage by vehicles or watercraft.

At timber pier protection systems, look for evidence of deterioration as a result of decay, fungal growth, vermin attack, and weathering. Also look for signs of collision damage by watercraft.

Timber Superstructure

Inspect the floor beam members along the deck bearing surface to see if the lower surface of the deck bears uniformly without crushing. For example, inspect the timber deck on a pony truss. Examine the floor beam members at their support points to see if there is adequate bearing area on the support and to see if crushing has occurred (Figure 5.7).

Fire Damage to Timber Superstructure

Fire can cause damage to destroy or weaken members. Besides checking for timber members that have been hollowed or otherwise damaged by fire, the galvanising on bolts and fittings should be checked for distortion and cracking caused by heat and firefighting. The inspection should also identify growth and debris that would fuel fires at a later time.

An example of the damage resulting from someone starting a fire in the halfcap region of a Western Australian bridge is shown in Figure 5.8 to Figure 5.11. Damage was also caused by the emergency services when they cut out a perfect wandoo stringer to access the hot spot.

Figure 5.7: Example of bridge span supported on split timber logs

Source: MRWA (n.d.).

Figure 5.8: Damage to superstructure of timber bridge by fire and wandoo stringer cut by emergency services to gain access to the hot spot

Source: MRWA (n.d.).

Figure 5.9: Fire damage in timber bridge

Source: MRWA (n.d.).

Figure 5.10: Damage caused by emergency services when they cut out a perfect wandoo stringer to access the hot spot

Source: MRWA (n.d.).

Figure 5.11: View of cut wandoo stringer

Source: MRWA (n.d.).

Figure 5.12: Incomplete deck joint

Source: MRWA (n.d.).

Timber Decks

Timber decks should be checked for looseness, dampness, decay, splitting, crushing, fastener failure, and wear. Especially close attention should be given to locations where timber decking rests on other members. These areas hold water, are frequently damp and are especially vulnerable to decay.

Timber kerbs should be inspected for decay, splitting, insect attack, weathering, proper anchorage, and proper alignment.

Rough surfaces due to wear are a frequent problem with timber wearing surfaces. Also, observe if the wearing surface is properly fastened and not rotting.

It is worthwhile checking that deck joints function properly and that parts are not missing or out of place (Figure 5.12) and inspect for signs of insect attack or fire damage.

Timber Trusses

Inspect timber members for decay, weathering, insect attack, splitting, and fire damage.

Defects such as looseness due to movement in a joint or failure to hold members firmly can result in progressive deterioration of the structure.

Misalignment, either by sag in the truss or lateral buckling in truss compression chord members or abrupt misalignment of secondary members, such as kerbs and rails, is an indicator of some problem of failing capacity and performance.

5.2.2 Reinforced Concrete Bridges

Look for signs of wear and deterioration, as these are related to durability. Visible signs are cracking, reinforcement corrosion, spalling, surface erosion, faults in drainage and leakage of water, construction defects, surface deposits (salts), and distortion of shape.

The most common problem associated with durability of concrete bridges is corrosion of the reinforcing steel. An indication of the current and future risk for corrosion can be determined from:

- a detailed visual inspection, including identification of cracks and delamination, reinforcement cover depth
- chloride content of the concrete
- carbonation depth.

Additional information is given in Section 6.4.

Reinforced Concrete Piles

Reinforced concrete piles should be checked for:

- cracked bearing seats or spalled concrete
- deteriorated concrete and cracks in pile caps
- cracked, spalled or disintegrated concrete, especially at the waterline or groundline
- verticality
- erosion or undermining of the foundation by scour (refer to as-built foundation data and streambed cross-section data)
- evidence of tilt, settlement, or misalignment.

Reinforced Concrete Spread Footings

Checks should be made to ensure that the footings are well supported and that the foundation material is solid and coherent and in good condition to bear the loads from the bridge. The footings should be checked for structural defects and any distortion or movement that may indicate settlement or instability. The footings and associated substructure should be checked to ensure that the superstructure is properly supported.

Reinforced Concrete Substructures

Reinforced concrete substructures should be checked for:

- deteriorating concrete in areas that are exposed to roadway drainage
- cracking and possible movement of the piers and abutment back wall. Check particularly the joint between the backwall and the abutment
- impact damage by vehicle or flood debris
- stone masonry for mortar cracks
- vegetation growth
- water seepage through the cracks
- weathering
- spalled (or split) blocks.

What to look for:

- Check for scour or erosion around the abutments and piers, and for evidence of any movement (rotational, lateral, or vertical).
- Measure the alignment of the abutments and piers using surveying equipment, or plumb bob and tape.
- Measure the clearance between the beam and backwall. Off-centred bearings at abutments and piers, and inadequate or abnormal clearances between beams and backwall are indications of probable movement.
- Determine whether drains and weepholes are clear and functioning properly. Seepage of water through joints and cracks may indicate accumulation of water behind the abutment.
- Report any frozen or plugged weepholes. Mounds of earth immediately adjacent to weepholes may indicate the presence of burrowing animals.
- Check bearing seats for cracking and spalling, especially near the edges. This is particularly critical where concrete beams bear directly on the abutment.
- Check bearing seats for presence of debris and standing water.

An example of scour undermining a roadway approach to a bridge is shown in Figure 5.13 to Figure 5.16. Note the abutment wall shows both a colour change where the rock protection has subsided and the bottom of the skirt wall can be seen. Also note the separation between the fill and the concrete approach kerb. The out-of-alignment of the cast-in situ kerb initially alerted the inspectors to the issue.

Figure 5.13: Scour damage to riprap protection at abutment and scour of abutment material exposing the top of the concrete piles

Note: Location – Gisborne New Zealand

Source: F McGuire (c2009).

Figure 5.14: Scour at bridge abutment leading to movement of kerb at surface

Source: MRWA (n.d.).

Figure 5.15: Kerb out of alignment due to settlement of abutment fill caused by scour at bridge abutment

Source: MRWA (n.d.).

Figure 5.16: Visible misalignment of kerb

Source: MRWA (n.d.).

Figure 5.17 to Figure 5.22 show the 'before' and 'after' repair state of a pin jointed column connection that was badly corroded.

Figure 5.17: General view of bridge with pin jointed columns exhibiting corrosion

Source: GHD and MRWA (n.d.).

Figure 5.18: Pin jointed column joint prior to repair

Source: GHD and MRWA (n.d.).

Figure 5.19: Removal of bitumen from pin jointed column prior to repair

Source: GHD and MRWA (n.d.).

Figure 5.20: Corrosion of pin joint at base of the column prior to repair

Source: GHD and MRWA (n.d.).

Figure 5.21: Pin joint after blasting to clean joint

Source: GHD and MRWA (n.d.).

Figure 5.22: Pin joint after application of primer

Source: GHD and MRWA (n.d.).

Reinforced Concrete Superstructures

Failure of materials is usually caused by:

- standing water
- poor bridge drainage
- mortar cracks
- scour.

What to look for:

- Measure the clearance between the beam and backwall. Off-centred bearings and inadequate or abnormal clearances between beams and backwall are indications of probable movement.
- Determine whether drains and weepholes are clear and functioning properly. Seepage of water through joints and cracks may indicate accumulation of water behind the abutment.
- Report any frozen or plugged weepholes. Mounds of earth immediately adjacent to weepholes may indicate the presence of burrowing animals.
- Check bearing seats for cracking and spalling, especially near the edges. This is particularly critical where concrete beams bear directly on the abutment.
- Check bearing seats for presence of debris and standing water.
- Check the bearings and expansion joints are free to move in the manner intended. Seized bearings and joints can transfer load to other bridge components not designed for such loads resulting in distress and damage.
- Check for any trapped flood debris. In the event of inundation of the superstructure by flood, silt, logs, and debris can become trapped within the structure.
- Check for impact damage by vehicles or flood debris. Figure 5.23 and Figure 5.24 show an example of severe impact damage to concrete superstructure beams. In addition to exposing the reinforcement hydraulic fluid has seeped into the damaged area. Another example of extreme impact damage is shown in Figure 5.25 to Figure 5.28. A digger on a transporter hit the Onewa Road Bridge that crosses the Auckland Southern Motorway in Auckland, New Zealand. The damage was so severe the bridge was replaced.

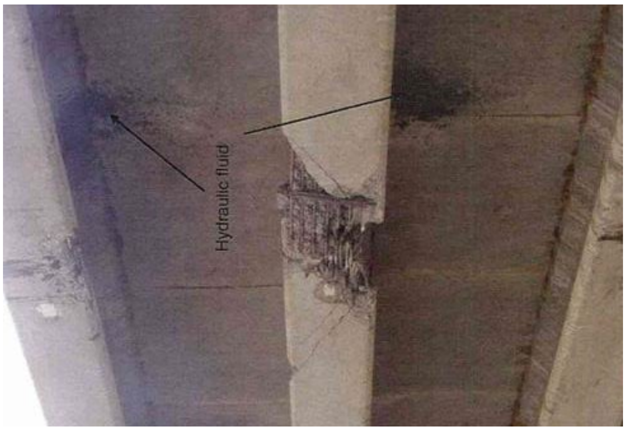
Reinforced Concrete Decks

Inspect the wearing surface and determine the type of surfacing.

Check the following:

- asphalt overlays – look for cracking, ravelling, potholing, shoving, and rutting. While these defects relate to routine road surface maintenance they can lead to structural degradation of the supporting deck, and rough riding surfaces increase dynamic loads to the bridge
- concrete overlays – look for scaling, spalling, cracking, and exposed reinforcement
- monolithic deck – look for scaling, spalling, cracking
- kerbs – look for spalling, scaling, and other forms of concrete deterioration
- deck joints – look for proper function, wear and tear, any missing parts
- concrete members – look for cracks, spalls, scaling, and efflorescence
- any exposed reinforcing steel
- proper alignment of deck members and joints
- debris accumulated on the surface or trapped to block drainage, ponding of water and blocked drainage scuppers
- vehicle impact damage to the deck, bridge barriers, handrails, deck joints, and kerbs
- any chemical spills or surfaces that may be unsafe for vehicles.

Figure 5.23: Impact damage to concrete beam



Source: MRWA (n.d.).

Figure 5.24: Impact damage to concrete beam exposing steel reinforcement

Source: MRWA (n.d.).

Figure 5.25: Impact damage to concrete exposing steel reinforcement in the bottom flange and causing extensive cracking

Source: MRWA (n.d.).

Figure 5.26: Impact damage to Onewa Road Bridge, Auckland Southern Motorway, New Zealand

Source: Opus International Consultants (n.d.).

Figure 5.27: Impact damage to the underside of Onewa Road Bridge, Auckland Southern Motorway, New Zealand

Source: Opus International Consultants (n.d.).

Figure 5.28: The digger that caused the impact damage to Onewa Road Bridge, Auckland Southern Motorway, New Zealand

Source: Opus International Consultants (n.d.).

5.2.3 Prestressed Concrete Bridge Maintenance Issues

The basic issues are similar to those for reinforced concrete (Section 5.2.2).

Post-tensioned concrete bridges are particularly vulnerable to corrosion and severe deterioration where internal grouting of tendon ducts is incomplete and moist air or water can enter the ducting system. The ingress of water and salts into tendon ducts is most likely at joints in segmental construction, other construction joints and anchorages at the ends of members.

Experience in the UK led to the requirement for existing post-tensioned concrete bridges with grouted tendon ducts to be examined in a special inspection program over a five-year period (Highways Agency UK 2008).

General Methods of Inspection for Post-tensioned Bridges

Methods of inspection for post-tensioned concrete bridges range from a visual inspection to complex non-destructive and semi-destructive methods. The methods adopted in a site investigation should commence with a simple visual examination and routine surface and material tests as part of a special inspection. Progression to the more complex methods of the special inspection may be justified if there is evidence of tendon corrosion and a risk of sudden failure in a structure. However, it should be realised that serious corrosion of the tendons can occur without any visual evidence.

An indication of general corrosion of the reinforcement in the concrete may be taken as indicative of the potential for corrosion occurring in the prestressing steel. The methods of determining corrosion risk can provide a valuable precursor to the use of other inspection techniques. In particular, high concentrations of chloride ions increase the probability of tendon corrosion.

NDT can be used to assist in the detection of voids in the tendon ducts. If no voids are found this does not preclude the possibility of corrosion occurring. However, in fully grouted ducts any corroded and broken wires will quickly re-anchor and the risk of full loss of prestress should be reduced. An assessment of the potential for a sudden mode of collapse should be undertaken, and the necessity for further investigations should be determined.

If voids are found and the conditions within the concrete are conducive to corrosion of the steel, then internal examination of the tendons should be undertaken. The method for gaining access to the tendon duct should be chosen considering the position of the duct and the degree of damage that will be caused. In all cases, drilling holes should be carried out with the agreement of the project manager, and utmost care must be taken to ensure that the tendon is not damaged.

Visual Examination of Post-tensioned Bridges

The visual examination of post-tensioned bridges should be undertaken systematically. The inspection should be carried out in such a way as to identify actual and potential areas of distress. The inspections should be undertaken by persons with experience of post-tensioned structures.

Prestressed bridges are normally designed to avoid cracks in the concrete. The development of cracks can have serious durability implications and may indicate loss of prestress. Cracks along the line of tendon ducts may be indicative of corroded and broken wires or tendons. Such cracks may be formed by the bursting forces that are generated as a broken wire slips and then re-anchors.

Signs of general corrosion on the surface of the concrete may be indicative of conditions within the concrete that are conducive to corrosion of the tendons. The presence of any water leakage through the deck should be recorded and the source located.

Void Detection

The detection of voids in post-tensioning ducts is important in isolating potential areas where corrosion of the tendon may occur. The methods of detection can be non-destructive and a guide to the use of such techniques is included in BS 1881 Part 201. Determining the position of any voids, prior to an internal examination to ascertain the condition of the tendon, should restrict the degree of damage caused to the structure. However, the only certain method of determining the tendon condition is by exposing it for visual inspection.

Examples of Vulnerable Details

The list below does not necessarily give any rating; it is up to the engineer to decide:

- segmental joints where these may be rated in order of decreasing vulnerability (in situ mortar narrow, mortar wide, match cast glued)
- types of prestressing systems
  - those with a lack of redundancy e.g. a small number of large tendons where a severe local defect might have a serious effect on strength
  - those with tendons located in the upper part of a deck where failure of the deck waterproofing may lead to corrosion
- types and locations of anchorages
  - anchorages recessed into the upper surfaces of the deck or located within joints that may have reduced protection against corrosion
- joints crossing tendons
  - where records indicate a lack of water tightness or vulnerability due to leaking deck drainage fittings or failure of deck waterproofing in the vicinity of joints
- waterproofing systems
  - where records indicate poor performance of systems (current and previous).

Prestressed Concrete Piles

The basic issues are similar to those for reinforced concrete (Section 5.2.2).

Check for:

- cracking, scaling, spalling, and abrasion
- impact damage
- the area of the waterline and pay particular attention to areas that are intermittently wet and dry.

Prestressed Concrete Substructures

The basic issues are similar to those for reinforced concrete (Section 5.2.2).

Check for:

- cracking, spalling, and abrasion.

When reporting cracks record the length, width, location, and orientation (horizontal, vertical, diagonal, etc.) of each crack. Also indicate the presence of rust stains, efflorescence, or evidence of differential movement on either side of the crack.

Prestressed Concrete Superstructures

The basic issues are similar to those for reinforced concrete (Section 5.2.2).

Check for:

- cracking, spalling and abrasion
- impact damage

Check for any settlement, distortion, misalignment or sagging of prestressed members that may be an indicator of prestress defects.

Check prestressing anchorages for cracks and rust stains that may be an indicator of defects in the prestressing system.

Prestressed Concrete Decks

The basic issues are similar to those for reinforced concrete. See Section 5.2.2.

Check for:

- cracking, spalling and abrasion
- impact damage to the deck
- items such as bridge side protection, handrails and kerbs.

Cracks may also be an indication of overloading, corrosion of the reinforcing steel, or settlement of the structure. Report each crack in terms of; the length, width, location, and orientation (horizontal, vertical, diagonal, etc.). Also, indicate the presence of rust stains, efflorescence, or evidence of differential movement on either side of the crack.

Check that deck joints and any joints in precast deck units are sound.

Check for any rust stains or water leakage near prestressing ducts that may be an indicator of corrosion of prestress reinforcement.

5.2.4 Steel Bridge Maintenance Issues

Defects in a steel bridge generally can be attributed to the environment in which the bridge exists. Defects can also arise from the load history, poor detailing, workmanship, or manufacture. Inspection should detect signs of distress including failure of protective coating, loss of section, loose or defective fastenings, fatigue, cracks in components, impact damage, structural deformation and distortion, manufacturing defects, and detailing faults.

Checks should be made for corrosion, cracks, buckles, kinks and yielding due to overstressing. Other components such as connections, cover plate ends, connection hardware, fasteners, and welds should be checked especially carefully. Loose and missing bolts and rivets should be identified. Inspection should be made under areas containing debris build-up and other damp areas because these areas are especially vulnerable to corrosion. Pins and eyebars on pinned eyebar trusses should be examined. Check pins and eyebars for corrosion and cracks. Also check the tightness of the pin nuts, etc.

Fire damage may affect the metallurgy of iron and steel, altering its structural properties.

Defects identified should be recorded, assessed, and programmed for maintenance remedial work. In the event that defects are detected in critical members or components that may result in collapse or critical safety issues, the bridge should be closed immediately.

For additional information on iron and steel maintenance issues see Mandeno (2008) and Tilly et al. (2008).

Corrosion Issues for Steel Bridges

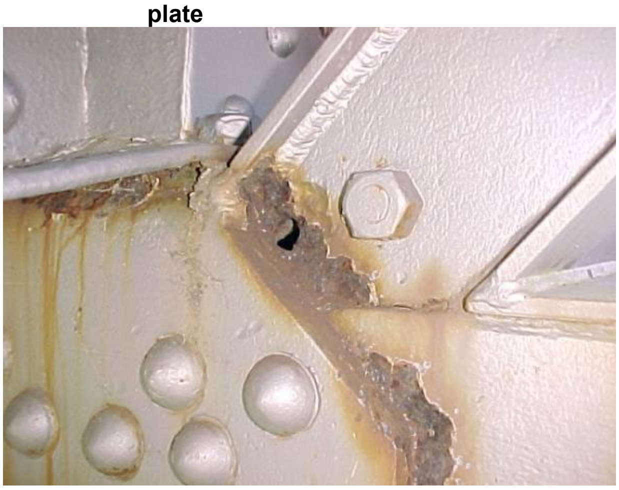
The main asset management issue for steel bridges is corrosion. For a particular type of steel bridge there are specific areas that are more prone to corrosion than other areas. For example, U-shaped bottom chords and the inside of closed box chords collect dirt and moisture, which accelerates the corrosion process (Figure 5.29).

Interfaces of steel/steel at connections (Figure 5.30 and Figure 5.31) and steel/timber on timber decks are prone to 'crevice corrosion' (Figure 5.32). With this type of corrosion the affected area may not be visible from the exterior with the result perforation of part of the member may be the first sign of a problem.

Figure 5.29: Severe corrosion of a U-shaped truss bottom chord

Source: D Carter (c2009).

Figure 5.30: Crevice corrosion in steel truss at the intersection of a diagonal with the gusset plate



Note: The only way to prevent crevice corrosion is to seal the interface. Source: D Carter (c2009).

Figure 5.31: Example of crevice corrosion at steel/steel riveted connection

Note: The increase in volume of the corrosion product has forced out the unrestrained corner. Source: D Carter (c2009).

Figure 5.32: Crevice corrosion between steel member and timber decking

Note: The corrosion is not visible. Source: D Carter (c2009).

Figure 5.31: Severe breakdown of protective coating requiring full removal by grit blasting

Source: D Carter (c2009).

Figure 5.32: Deterioration of protective coating in a moist marine environment leading to corrosion of the steel

Source: D Carter (c2009).

Locations where water constantly drops onto steel will cause deterioration of the protective coating leading to corrosion of the steel (Figure 5.33).

Figure 5.33: Corrosion of cross girders and rivet heads

Source: D Carter (c2009).

Steel members in close contact with a concrete face present a problem to apply protective coatings. Deterioration may go unnoticed for years unless inspections are rigorous.

In marine environments in areas where air flow is minimal, such as the underside of top chords of a truss, concentrations of salt may be high, leading to accelerated corrosion (Figure 5.29 to Figure 5.33).

The adage for steel bridge inspection is – 'If you can't touch it, you haven't inspected it'.

Impact

Low-level steel overpasses are susceptible to impact damage from high and wide loads.

In the instance of an impact occurring to a steel bridge it is important that a conservative approach is taken until such time as the structural implications are assessed. This may involve closing the bridge or imposing a load limit in the first instance.

When assessing impact damage it is important to inspect all members and the overall span from a global perspective and not just those where the impact occurred. Collateral damage can occur at joints, supporting members, bearings and bearing fixtures. In addition the out-of-plane displacements of members and the whole truss need to be critically reviewed. The stability of displaced members needs to be carefully considered.

A detailed structural assessment of the implications of the damage needs to be carried out to determine under what loading the bridge can operate, if at all. A detailed survey should be carried out to obtain the spatial position of all members, connections, and displacements. A structural model can then be set up to assess the structural implications of the damage.

It should be noted that permanent displacements will have resulted in cold working of the steel that will change the weldability of the material. Any repairs involving welding must be clear of areas of cold-worked steel.

Inspection

Inspection of steel bridges should be carried out by inspectors and engineers who have undergone specific steel bridge inspection training.

For a particular type of steel bridge areas susceptible to corrosion need to be identified. It is recommended that a meeting of experienced inspectors and engineers should be convened to develop the inspection plan for a particular bridge.

Experience has also shown that using two inspectors on major bridge inspections results in a more thorough inspection.

The report, *Reliability of inspection for highway bridges, Volume 1: Final Report,* includes recommendations on methods of inspection and strategies to improve their reliability (Moorem et al. 2001).

Fatigue Prone Areas

The increase in the percentage of heavy vehicles on major routes has highlighted the fact that some details on steel bridges, particularly trusses, are prone to fatigue. Instances of fatigue cracks have occurred in members where top tension flanges have been curtailed to facilitate connections.

Additional information on strengthening and rehabilitation is given in Section 6.3 and Section 6.5.

Steel Piles

Steel piles and pile bents should be inspected and checked for:

- signs of settlement, tilting
- the presence of rust, especially at the ground level line. Over water crossings, check the splash zone and the submerged part of the piles for rust
- debris around the pile bases. Debris will retain moisture and promote rust
- rotation of the steel caps due to eccentric connections
- broken bracing, connections and loose rivets or bolts.

See also Sections 6.3 and 6.5 for additional detailed information.

Steel Substructures

Steel substructures should be inspected and checked for:

- signs of settlement
- signs of member distortion, impact damage by vehicles and flood debris
- the presence of cracks
- the presence of rust
- loss of surface protection coatings
- rotation of the steel caps due to eccentric connections
- broken bracing, broken connections and loose or missing rivets or bolts
- the condition of web stiffeners.

Steel Superstructures

Steel superstructures may comprise of rolled sections, plate web girders, trough boxes, and trusses.

Steel superstructures should be inspected and checked for:

- signs of settlement
- signs of member distortion, impact damage by vehicles and flood debris (Figure 5.34 to Figure 5.37)
- the presence of cracks
- the presence of rust
- loss of surface protection coatings (Figure 5.38)
- debris deposited on steel members or trapped in expansion joints which will retain moisture and promote rust
- seized or out-of-position bearings supporting the superstructure. Non-functional bearings can transfer load to other bridge components not designed for such a load, resulting in distress and damage
- rotation of the steel caps due to eccentric connections
- broken bracing, broken connections and loose or missing rivets or bolts
- the condition of web stiffeners
- evidence of damage or loss of material properties caused by fire.

Figure 5.34 shows an example of flood debris accumulated on the bottom flange of a steel beam and around the abutment bearing.

See Figure 6.12 for possible faults in a steel beam, Figure 5.37 for an example of a timber log trapped inside the superstructure after a flood and Figure 5.38 for an example of delamination of steel beam protective coating and wasp infestation on the web.

Figure 5.34: Example of flood debris and silt accumulated on bottom flange of steel beam and around the abutment

Note: Location – Gisborne New Zealand.

Source: F McGuire (c2009).

Figure 5.35: Impact damage to steel bridge

Source: MRWA (n.d.).

Figure 5.36: Detailed view of impact damage to steel bridge

Source: MRWA (n.d.).

Figure 5.37: Example of timber log trapped inside a bridge superstructure

Note: Location Gisborne New Zealand.

Source: F McGuire (c2009).

Figure 5.38: Example of delamination of steel beam protective coating and wasp infestation on web

Note: Location Gisborne New Zealand. Source: F McGuire (c2009).

Orthotropic Steel Decks

An orthotropic bridge or orthotropic deck is one whose deck typically comprises a structural steel deck plate stiffened either longitudinally or transversely, or in both directions. This allows the deck both to directly bear vehicular loads and to contribute to the bridge structure's overall load-bearing behaviour. The orthotropic deck may be integral with or supported on a grid of deck framing members such as floor beams and girders.

As a result of inadequate knowledge about the performance characteristics, particularly in regard to fatigue and traffic loading, early designers created bridges that were too light and tended to crack in the welds under repeated use by trucks. Fatigue has been the cause of a number of orthotropic bridge failures. Research since the 1970s has resulted in criteria for more reliable design.

Delamination of asphalt and concrete wearing surfaces from the steel plates on orthotropic decks is known to be a typical fault, particularly if the steel deck plates are flexible or in the range of 10 mm to 14 mm. Modern materials and technologies, however, are providing improved solutions for driving surfaces. Thermoset resin-extended asphalt and epoxy concrete can help limit cracking and delamination. Some designs rely on the composite action of the surfacing with the deck steel plate to increase the load carrying capacity. As a consequence delamination or erosion of the surfacing from the substrate steel deck is most undesirable and should be repaired.

Inspection of orthotropic plate steel decks should aim to detect the existence of any:

- cracks and corrosion existing in the steel members and, welded components
- delamination or erosion of the driving surface from the steel deck plates
- leakage, corrosion, loss of section, and proper support.

Steel Connections – Riveted, Bolted

See Section 7 of *AGBT Part 2: Materials*.

Steel Truss Bridges

The design and maintenance practices for truss bridges have evolved over a considerable time. A number of significant bridge failures in the past gave rise to research leading to improved design and maintenance requirements for truss bridges. As a result, the topic of fracture mechanics and the effects of multiple stresses at less than yield of the materials are understood more thoroughly. The first recognition of redundant and non-redundant members was presented in the twelfth edition of the AASHTO Bridge Specifications (AASHTO 1977) but is now superseded by AASHTO (2016). The first guide specifications for fracture critical bridge members were issued by AASHTO in 1978 (AASHTO 1978).

Fracture Critical Members

After design engineers began to recognise the problems associated with multiple stresses at less than allowable values, further information was developed to assist in the design process and in evaluation of existing structures. In addition, as a result of some notable failures, it was recognised that many existing bridges may be nearing failure due to fatigue. Fracture critical (FC) members were recognised and defined as a member or component whose failure in tension would result in the collapse of a bridge. These are commonly referred to as non-redundant members. Methods were developed to help determine which structures must be further evaluated by designers for susceptibility to fatigue problems. Designers began to include Fracture control plans (FCP) in bridge design details. The most common types of FC members are tension flanges and sometimes parts of webs of flexural members such as beams and girders. Tension members of trusses, particularly eyebars, which commonly make up the lower chords of old trusses, can also be FC. Other tension members of trusses, such as diagonals, are also FC.

The following rules-of-thumb usually determine FC members:

- Two-girder bridges are defined as FC. Fracture of lower flanges in positive moment areas (mid spans) and upper flanges in negative moment areas (over supports) can be expected to lead to collapse of the structure. However, cracks over interior supports sometimes lead to subsequent higher positive stresses in the spans with no catastrophic collapse. Therefore, these FC components receive more frequent periodic in-depth inspections.
- All steel caps are defined as FC. While this statement is bold, an exception is difficult to imagine.
- Lower chords of trusses are FC. This determination is based on the fact that most truss bridges employ only two trusses and most are simple span.
- Secondary members such as diaphragms and stiffeners are not FC. They are rarely used in a manner where failure would lead to structure collapse. However, caution must be observed in evaluating certain truss members that may appear to be secondary when, in fact, their attachment to main FC members can provide a starting place for the main member failure.

Redundancy

The concept of structural redundancy is well known. Any statically indeterminate structure may be said to be redundant, to varying degrees, depending upon its supports. A two-span straight girder is redundant. However, a two-span curved girder is also redundant, but the support reactions are determinate. These definitions of redundancy are of little value to the field inspector who must make a determination of FC potential for various members in a bridge.

There are two types of redundancy that concern the FC inspector:

Load path redundancy

Superimposed traffic loads are supported directly by the deck, which in turn is supported by longitudinal stringers or beams. A bridge with a single box girder would therefore be non-redundant since a failure in the box would collapse the bridge. Likewise, a two-girder bridge is non-redundant since one girder cannot assume all of the load for which two are designed. However, it can be argued that a continuous two-girder bridge is structurally redundant since a girder failure would not cause collapse, but the structure would sag excessively. Three or more girders will usually have enough load capacity due to inherent design factors of safety to avoid collapse. The failure of one girder will immediately cause the loads to be shared by the other girders. However, the FHWA considers three-girder bridges with more than 4.6 m (15-foot) girder spacing to be FC. The strength of the deck system should be considered for this case. Some deck systems for wide beam spacings are two-way slabs and others have stringer and floor beam systems with one-way slabs. Those with two-way slabs will still have a load-path redundancy, while those with stringers and floor beams will be more unstable after failure of one girder in a three-girder system.

Internal redundancy

This term refers primarily to built-up members, such as riveted plate girders. A single plate or shape in the built-up member might fail without causing collapse. However, even members such as this must sometimes be considered non-redundant, since like two-girder structures, failure of one portion of the member can overload the remaining portions such that there is not sufficient remaining capacity to prevent total failure. Usually, if the cross-sectional area of the largest shape or plate in a built-up member is less than about 30 to 40% of the total member area, then the member may be considered to have internal redundancy.

Inspection Procedures for FC Members

Inspection procedures begin with proper advance planning. The more important planning aspects, usually based on an office review of the structural plans, are:

- Identify possible FC members.
- Note the particular members in the structure that may require special field attention such as built-up tension members composed of few individual pieces.
- Pre-plan necessary access to the members, including special equipment needs such as ladders, bucket truck, or climbing gear.
- Many FC members are a result of structures designed for urban situations with necessary complex alignment geometries. Proper inspection of these bridges may require closing a traffic lane. Safe traffic control must be coordinated in advance.
- If the structure involves a railroad, a railroad flagger must be coordinated with the railroad company.
- Identify and make available any necessary special tools and equipment that may be required in addition to the normal inspection gear. A high-pressure washer is often useful in cleaning areas where a large accumulation of debris might obscure view of FC areas. Non-destructive test equipment such as ultrasonic devices may be advantageous in some areas, particularly for inspection of box-type bent caps and pin-and-hanger connections.

The field inspection of all FC members consists of several steps. The most important step is a visual inspection. The inspector notes any:

- visual cracks and their direction and location
- evidence of rust, which may form at a working crack
- weld terminations in a tension area
- interrupted back-up-bars used for built-up-member fabrication
- arc strikes, scars from assembly cables or chains, or other physical damage
- cross-section changes that may cause a sudden increase in the stress pattern.

Fatigue and Fatigue Fracture

Members subjected to continued reversal of stress, or repeated loading such that a range of change in stress occurs, are subject to a behaviour called fatigue. Members that have a relatively constant, steady stress are not subject to fatigue. The term has been in use for almost a century and is currently defined by the American Society of Testing Materials (ASTM 2012) as 'the process of progressive localized permanent structural change occurring in a material subjected to conditions that produce fluctuating stresses and strains at some point or points and that may culminate in cracks or complete fracture after a sufficient number of fluctuations.'

Fatigue can result in:

- loss of strength
- loss of ductility
- reduced service life.

Fatigue fractures are the most difficult to predict since conditions producing them are often not clearly recognisable. Fatigue occurs at stress levels well within the elastic range, that is, less than the yield point of the steel, and is greatly influenced by minor imperfections in the structural material and by fabrication techniques.

Fatigue fracture occurs in three distinct stages:

- local changes in atomic structure, accompanied by sub-microscopic cracking
- crack growth
- sudden fracture.

Fatigue-prone Details

Fatigue fracture almost always begins at a visible discontinuity, which acts as a stress-raiser.

Typical examples are:

- design details such as holes, notches, or section changes
- flaws in the material such as inclusions or fabrication cracks
- poor welding procedures such as arc strikes
- weld terminations.

Certain structural details have been long recognised as stress-raisers and are classified as to their potential for damage. These details appear in the AASHTO (2002) and other technical publications. Most of these common details should be familiar to the fracture critical bridge inspector. Proper consideration of member detail and sizing during design will help control stress level and thus control crack growth. The stress range, or algebraic difference in the maximum and minimum stress, also becomes important. The most effective way to control cracking and eventual fracture is sensible detailing. Details such as out-of-plane bending in girder webs and certain weld configurations can cause crack propagation and fracture.

Design for fatigue also includes observing a FCP. The FCP identifies the person responsible for assigning fracture-critical designations. It establishes minimum qualification standards for welding personnel and fabrication plants. It also specifies material toughness and testing procedures. The specific members and affected sections are also identified in the FCP. During fabrication, these members should be subject to special requirements.

Fatigue failure is always an abrupt fracture, called a brittle fracture. A brittle fracture is distinguished from a ductile fracture by the absence of plastic deformation and by the direction of the failure plane, which occurs normal to the direction of applied stress. Other failure surfaces due to high stress are usually at an angle to the direction of the stress and are often accompanied by a narrowing or necking of the material. Brittle fracture failures have no narrowing or necking present.

The three main contributing factors to brittle fracture are:

- stress level
- crack size
- material toughness, sometimes called fracture toughness.

Small, even microscopic cracks can form as a result of various manufacturing and fabrication processes. Rate of propagation, or growth, of cracks also depends on the stress level and the material toughness. Material toughness is the ability of a material to resist brittle fracture. This resistance is primarily determined by chemical composition and to some extent by the manufacturing processes. Usually, higher strength steels are more susceptible to brittle fracture and have lower toughness. Toughness can be improved by techniques such as heat treatment or by quenching and tempering.

Weld Details

Inspectors concerned with FC inspections must acquaint themselves with the characteristics of good and poor structural details and be able to identify those details in the field. Welding creates the details most susceptible to fatigue and fracture. Therefore, it is imperative to recognise features prone to FC failure.

Major FC problem areas are at weld discontinuities or changes in geometry such as:

- toes of fillet welds
- weld termination points
- welds to girder tension flanges from other connections such as stiffeners or diaphragms
- ends of welded cover plates.

Welded cover plates on rolled beams were a very common detail until fatigue failures began to be recognised by bridge engineers. Whether the weld is terminated or continued around the end of the cover plate, the condition is at best questionable.

Weld attachments to a girder web or flange can reduce fatigue strength as the length of the attachment increases. Such details are commonly used to attach diaphragms and wind bracing to main structural members, either at the flange or web. Details such as run-off tabs and back-up bars may also provide possible stress riser discontinuities if not smoothed by grinding after removal.

Inspectors should familiarise themselves with acceptable and unacceptable fillet weld profiles in order to recognise potential problem areas in the field.

Fatigue in Secondary Members

Secondary members may also have fatigue problems. For instance, main girder stress reversal may induce vibrations in lateral bracing or diaphragms. In many cases the number of stress reversals in the secondary member is a magnification of those stresses in the main member. The attachment of plates to a girder web may cause out-of-plane bending in the web, a situation not usually considered by the designer.

In general, secondary members are not subject to an FC inspection. However, some secondary members, even though designed only as secondary members, such as lateral wind bracing in the lower plane of a girder system, will act as primary members. These cases generally occur in curved or heavily skewed structures. A curved bridge will have twisting or torsional effects due to the live loads that are partially resisted by the diagonal lateral wind bracing. These braces, particularly those near supports, should be inspected for possible fatigue cracks.

Proper Welding and Repair Techniques

Proper welding of structural steel members is a tedious process under the very best of conditions, which are usually found in the fabrication shop. Any field welding, whether it is a welded girder splice, retrofit detail, or repair, should be closely examined for visible problems. Many shop splices are accomplished by automatic welding machines under controlled conditions and can be smoothly ground to eliminate surface discontinuities. Field splicing operations are subject to exposure to the elements and difficulties in stabilising the pieces to be joined. In addition, the welding is usually done by hand and therefore subject to human error. Welded field splices for bridges should be constructed with supervision, careful inspection and be done by certified welders. The welded field splices should be of the same quality, as shop splices may be further inspected by radiographic (X-ray) techniques. The inspector should also be aware of problems that may arise from the use of improper field repair processes. Often a well-intentioned repair can actually make a member even more susceptible to brittle fracture.

FC Inspection Techniques

FC inspection techniques may include NDT to determine the condition of a structural member. There are several types available, including radiographic, ultrasonic, dye penetrant, and magnetic particle inspection. All are acceptable methods, but each has limitations and may not be suitable for a particular situation. One single technique may not be sufficient to assess damage, and a combination of more than one may be advisable. Usually these types of inspection are best left to personnel who have undergone the proper training.

The selection of the type of NDT method for a particular location is usually a function of the detail. For instance, potential cracks at the ends of welded cover plates are often inspected by the use of radiographic methods. Cracks in pins are best inspected by ultrasonic techniques. Subsurface defects such as inclusions may be found by magnetic field irregularities, and cracks adjacent to fillet welds at tee-joints are usually inspected by dye penetrant.

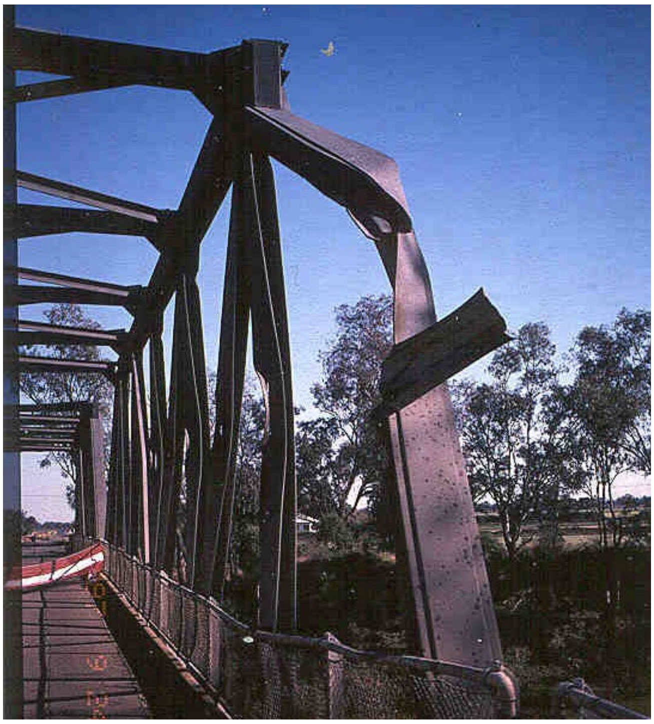
As a consequence, the inspection and maintenance of steel truss bridges is a specialised skill for which the inspector and engineer should be trained and competent.

Impact Damage

The risk of impact damage to a through-truss bridge from errant or over-height vehicles is high. Members can be severely distorted (Figure 5.39 and Figure 5.40 may require temporary support against collapse (Figure 5.41). Contingency measures to mitigate the risk include:

- installation of continuous rails along the face of the truss to prevent vehicles pocketing behind vertical and diagonal members (Figure 5.42)
- strengthening of end diagonals (referred to as principals or principal diagonals) portal bracing to resist impacts at the point of entry into the truss
- installation of over-height detection warning systems
- installation of supplementary lateral bracing on the top of the truss clear of any possible impact (Figure 5.43).

Figure 5.39: Impact damage to a steel truss diagonal member



Source: D Carter (c2009).

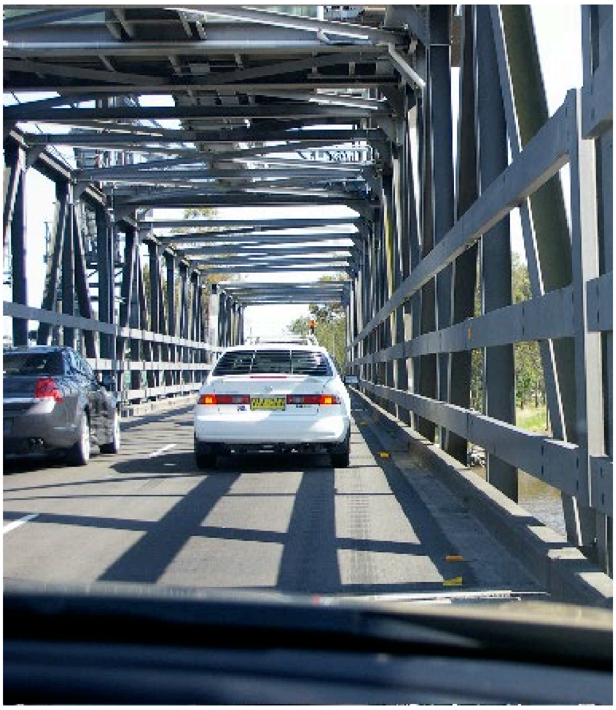
Figure 5.40: Impact damage to a steel truss vertical member

Source: D Carter (c2009).

Figure 5.41: Impact damage to a steel truss vertical member with temporary stabilising member in place

Source: D Carter (c2009).

Figure 5.42: Continuous rails installed to prevent vehicles pocketing between vertical and diagonal members



Note: Location – Pacific Highway truss bridge at Hexham, NSW. Source: D Carter (c2009).

Figure 5.43: Supplementary lateral bracing installed at the top of the truss to be clear of possible vehicle impact

Note: Location – Manning River Drive truss bridge at Taree, NSW. Source: D Carter (c2009).

Procedures have been developed to either straighten or replace damaged truss members. Heat treatment is one method that has been used (Figure 5.44 to Figure 5.46). The Bridge Section of Roads and Maritime Services (Roads and Maritime), NSW can provide information on the procedures.

Fire Issues for Steel Trusses

The implications of a fire on a steel truss are considerable and can lead to loss of structural function as the temperature of the material increases.

The damage can include distortion and buckling of members as a result of displacements that occurred when the material was heated to a temperature that caused softening (Figure 5.47 to Figure 5.49).

Figure 5.44: Heat treatment to straighten impact damaged steel truss 1

Source: Roads and Traffic Authority (RTA) NSW (n.d).

Figure 5.45: Heat treatment to straighten impact damaged steel truss 2

Note: Close-up of jig incorporating jacks to apply additional stress to assist and maintain straightening of bent steel member. Source: RTA NSW (n.d.).

Figure 5.46: Heat treatment to straighten impact damaged steel truss 3

Note: Jig attached to top truss member. Source: RTA NSW (n.d.).

Figure 5.47: Fire damaged steel truss bridge

Source: RTA NSW (n.d.).

Figure 5.48: Fire damage to steel truss – damage to joint

Source: RTA NSW (n.d.).

Figure 5.49: Fire damage to steel truss – warped and distorted members

Source: RTA NSW (n.d.).

Corrosion of Steel Trusses

Areas where water pools or dirt accumulates will inevitably lead to corrosion in the long-term.

The potential for the corrosion of steel trusses depends on a number of factors:

- maintenance of protective coatings
- exposure conditions
- moisture traps
- dirt build-up in members
- water penetration at supports and through decks
- areas where water impinges on coatings, e.g. from scupper pipes not clear of steelwork
- members with steel/steel interfaces
- hidden steel faces
- inaccessible areas.

Inspection of Steel Trusses

The inspection of wrought iron and steel trusses requires a systematic approach to ensure that problem areas are identified.

The key issue is to identify critical areas where deterioration is likely to occur. The critical areas will vary from bridge to bridge.

A detailed inspection can be broken down into the following:

- planning phase
- overall inspection – general condition of coatings, obvious areas of deterioration
- member by member inspection
- joints
- connections
- identified potential problem areas.

Access to carry out a detailed inspection requires that members be within arm's reach to ensure defects such as fatigue cracks, distortions and other forms of distress can be detected.

Fatigue-prone Areas

The increase in the percentage of heavy vehicles on major routes has highlighted the fact that some details on steel bridges, particularly trusses, are prone to fatigue. Instances of fatigue cracks have occurred in members where top tension flanges have been curtailed to facilitate connections.

Wrought Iron and Cast Iron Bridge Maintenance Issues

Wrought iron and cast iron are materials that occur in older bridges. Cast iron was often used for piles, columns and occasionally arches until the last decades of the 19th century. Wrought iron was often used for superstructures and substructures until the early decades of the 20th century. The defects that these two materials exhibit are in general very similar to those described above for steel. It should be recognised that the homogeneity and purity of the material will not be up to the standards of present-day steel so that the inspection process has to take into account a range of material variability:

- In cast iron, blow holes associated with the casting process may be observed. Cracking may be induced by shock loading or from stresses induced by the cooling of the material immediately after casting.
- The slag inclusions in wrought iron may act as starting points for corrosion, particularly in the trough or hollow members and the laps of riveted joints where water may be trapped.

Typical maintenance work may include:

- cleaning of debris from bottom chords and patch painting
- replacement of deteriorated transoms and tightening of bolts
- replacement of loose rivets in main girders and cross girders
- repair of bent components and minor steel items
- replacement of lattice arches if the load limit is required to be increased or the operating regime changed.

5.2.5 Fibre Reinforced Polymer (FRP) Bridge Maintenance Issues

A fibre reinforced polymer (FRP) is a polymer matrix that is reinforced with a fibre or other reinforcing material. FRP composites are anisotropic and hence the properties are directional. The FRP can be produced in various forms such as fabrics, pre-cured bars and plates to suit the application. High-strength fibres such as glass, kevlar, and carbon are used as the reinforcing fibre. Thin laminates of FRP composites can be externally bonded to structural members using epoxy adhesive to repair or strengthen structural members.

The treatment by FRP can significantly increase the ULS capacity of structural members. Usually there is only a small gain in the serviceability state capacity as the area of FRP is small compared to the steel reinforcement and it is operating in a similar strain environment. Also, these increases in capacity cannot be fully utilised because of the need to provide for acceptable levels of safety against the occurrence of both ULS and serviceability limit states (e.g. deflections, cracking, stress rupture, and fatigue). According to the *Guide for the Design and Construction of Externally Bonded FRP Systems for Strengthening Concrete Structures* (ACI 2008), it is recommended that the increase in load carrying capacity of reinforced concrete or prestressed concrete members strengthened with an FRP system should be limited (Nanni 2004).

In bridge applications, FRP fabrics may be adhered to beams and slabs to increase their shear and flexural capacity, and can be wrapped around columns to increase their load carrying capacity and ductility for seismic events.

See Figure 5.50 for an example of FRP with carbon fibres used to strengthen a flat bridge slab. Pre-cured FRP elements are more suited for flexural upgrade of columns, beams, and slabs.

The bond between the FRP system and the existing concrete is critical for the development of composite behaviour with the existing member. Appropriate surface preparation is essential to a successful FRP strengthening application.

It should be noted that the use of FRPs requires specialist advice to ensure the treatment proposed to strengthen the bridge component is appropriate and properly executed.

Figure 5.50: Example of FRP treatment with carbon fibre to strengthen a flat slab

Source: GHD and MRWA (n.d.).

FRP treatments themselves can lead to issues. VicRoads, for example, identified the following problem and remedial treatment issue with defective FRP treatment (VicRoads 2003). The problem arose from a carbon fibre polymer (CFP) reinforcement that had been added to concrete components. The polymer fibre/epoxy resin/concrete interface has cracked across the ends and along the sides, and the polymer began to peel at the ends. No anchorage straps were installed at the ends to prevent the peeling. There were also some minor areas of delamination along the length of the CFP where air bubbles possibly formed during installation, or where the CFP sagged before the epoxy resin set properly. The areas needed to be re-bonded to provide full strengthening again and to prevent further peeling at the ends.

The solution proposed called for:

- holes drilled at 100 mm centres to take the injecting ports
- the cracks blown clean and dried by high-pressure compressed air
- the ports fixed in place with epoxy paste
- ports cleared by poking a thin wire through them, then the crack sealed between the ports with epoxy paste, ensuring the paste penetrated the top of the crack to hold it in place during the pressure injecting
- the crack on the other side was fully sealed to prevent epoxy loss
- epoxy resin was pressure injected into the cracks to seal them and re-bond the carbon fibre strip or sheet to the base concrete again
- once the epoxy had set and cured, the injecting ports were ground off to improve the aesthetics of the member
- for peeling problems at the end of the carbon fibre strip, a cover strip was installed to aid in anchoring the ends and prevent possible future peeling.

Refer to the NZTA *Bridge Manual* (NZTA 2016b), Section 8 and AS 5100.8 for design advice.

For additional information on this subject refer to ACI (2008); FIB (2001); Nanni (2004).

5.2.6 Bridge Approach Issues

Settlement at the road approach to the bridge (Figure 5.51 and Figure 5.52) can cause a depression immediately before the bridge resulting in increased dynamic loading of the deck and supporting superstructure. Correction of the surface to produce a smooth transition can significantly reduce such vehicle impact loadings, with consequent reduced bridge stresses and benefit to the serviceable life of the structure.

Figure 5.51: Settlement at road approach

Source: MRWA (n.d.).

Figure 5.52: Transverse cracks in bridge approach

Source: MRWA (n.d.).

5.3 Special Bridge Maintenance Issues

5.3.1 Aggressive Water and Soils

Bridge members and components can be adversely affected by chemical agents contained in water and soil. The following measures may be utilised in specific circumstances:

- All iron and steel surfaces subject to deposits of aggressive salts and which are not washed clean by rain should be washed. The interior faces of girders of bridges in industrial areas or marine environments may be subject to deposition. Experience has shown that such deposits can promote corrosion.
- Coatings may be used on new concrete to provide additional protection in an aggressive environment. Coatings may also be used to seal and hide surface defects and inactive cracks on otherwise sound concrete. Specialist anti-graffiti coatings are also available.
- Defective deck joints that leak water should be repaired or replaced.
- Bridge components can corrode if subjected to moist deposits of dirt, dust, and debris. The dirt, dust, and debris that promote corrosion should be removed. If the bridge is subjected to a marine environment particular attention should be given to preventing corrosion of steel components. Nuts might be of a material incompatible with the bolts or the material being joined. This may lead to electrolytic action if not separated by a non-conductive washer.

5.3.2 Electrolytic and Galvanic Corrosion

The processes involved in electrolytic and galvanic corrosion are described in Section 8.1 of *AGBT Part 2: Materials*. The important point for the inspection engineer is to know the rate at which corrosion will occur. This depends on a wide range of environmental and material factors.

5.3.3 Uncommon Deterioration Mechanisms

Uncommon deterioration mechanisms that may be encountered include:

- delayed ettringite formation in concrete
- graphitisation of cast iron
- microbially-induced corrosion.

The processes involved in the above mechanisms are described in Sections 4.1.5, 8.5.1 and 8.5.2 respectively of *AGBT Part 2: Materials*.

Acid Sulphate Soils

Acid sulphate soils are naturally occurring soils, sediments, or organic substrates (e.g. peat) that are formed under waterlogged conditions. These soils contain iron sulphide minerals (predominantly as the mineral pyrite) or their oxidation products. In an undisturbed state below the water table, acid sulphate soils are benign. However, if the soils are drained, excavated, or exposed to air by a lowering of the water table, the sulphides will react with oxygen to form sulphuric acid.

Release of this sulphuric acid from the soil can in turn release iron, aluminium, and other heavy metals (particularly arsenic) within the soil. Once mobilised in this way, the acid and metals can create a variety of adverse impacts: killing vegetation, seeping into and acidifying groundwater and water bodies, killing fish and other aquatic organisms. Sulphuric acid produced by acid sulphate soils corrodes concrete, iron, steel and certain aluminium alloys. It has caused the weakening of concrete structures and corrosion of concrete slabs, steel fence posts, foundations of buildings and underground concrete water and sewerage pipes.

As noted in Roads and Traffic Authority (2005), ‘unless concrete structures are very dense (low porosity), acid can react with the calcium carbonate and calcium hydroxide present to form gypsum (calcium sulphate). Gypsum reacts with calcium aluminates in the concrete to form etteringite. Both gypsum and particularly etteringite formation cause an increase in the volume of the affected concrete. This results in the expansion and weakening of the concrete and its eventual exfoliation and dissolution. Etching of cement and exposure of aggregate are typical early signs of the attack of acidic effluent on concrete.’

Federal, state, territory, local authorities and industry organisations in Australia are involved in the development of legislation and policies to address the environmental issues related to acid sulphate soils. This has included the development of a national strategy for the management, identification and mapping of the extent of acid sulphate soils particularly for Australian coastal regions (ASRIS 2013, Roads and Traffic Authority 2005). Additional information on the risk management and occurrence of acid sulphate soils is available (Thomas et al. 2003).

It should be noted that the term ‘acid sulphate soils’ includes both potential acid sulphate soils and actual acid sulphate soils.

Potential acid sulphate soils are soils or sediments that contain iron sulphides and/or other sulphide minerals that have not been oxidised by exposure to air. The field pH of these soils in their undisturbed state is more than pH 4 and is commonly neutral to alkaline (pH 7 to pH 9). These soils or sediments are invariably saturated with water in their natural state. The waterlogged layer may be peat, clay, loam, silt, or sand and is usually dark grey and soft but may also be dark brown, or medium to pale grey to white.

Actual acid sulphate soils (AASS) are soils or sediments that contain iron sulphides and/or other sulfidic minerals that have previously undergone some oxidation to produce sulphuric acid. This results in existing acidity (pH <4) and often a yellow and/or red mottling (jarosite/iron oxide) in the soil profile. AASS commonly also contain residual un-oxidised iron sulphides or potential acidity as well as existing acidity.

The soils and sediments that are most prone to becoming acid sulphate soils are those which formed within the last 10 000 years, after the last major sea level rise. When the sea level rose and inundated the land, sulphate in the seawater mixed with land sediments containing iron oxides and organic matter. Under these anaerobic conditions, lithotrophic bacteria such as Thiobacillus ferro-oxidans formed iron sulphides (pyrite). Up to a point, warmer temperatures are more favourable conditions for these bacteria, creating a greater potential for formation of iron sulphides. Tropical waterlogged environments, such as mangrove swamps or estuaries, may contain higher levels of pyrite than those formed in more temperate climates.

The pyrite is stable until it is exposed to air, at which point the pyrite oxidises and produces sulphuric acid. The impacts of acid sulphate soil leachate may persist over a long time, and/or peak seasonally (after dry periods with the first rains).

The corrosion of ferrous materials and other materials in soils is presented in Elias et al. (2009).

The Roads and Maritime NSW has produced guidelines for managing acid sulphate soils (Roads and Traffic Authority 2005).

Information on the location and distribution of acid sulphate soil in Australia is available from the ASRIS website online database (ASRIS 2014), which is a major part of the Atlas of Australian Acid Sulfate Soils. Additional information on acid sulphate soil may be found (CSIRO 2003; NSW Agriculture 2000; Sammut 2000).

5.4 Impact Forces and Overloading

5.4.1 Damage caused by Impact against the Structure

Damage to the structure can be caused when:

- ships strike a glancing blow against piers, abutments and fendering
- trains are derailed or motor vehicles collide against piers and abutments
- overheight loads carried on trains or motor vehicles impact against the underside of the bridge superstructure
- over width loads or projecting parts of trains or motor vehicles, such as open doors on railway cars, having a narrow clearance to the bridge substructure, strike piers, abutments or guardrailing
- heavy floating debris carried by rapidly flowing flood waters impacts against the bridge structure
- airborne debris carried by strong winds strikes the structure.

The damage typically caused includes:

- spalling of concrete members, with or without exposure of reinforcing steel
- cracking of members, both steel and concrete, local bending or buckling of steel members
- permanent deformation of members, damage to posts, railings and parapets, rupture or fracture of members
- collapse of bridge.

5.4.2 Vehicle Overloading

The damage that can be caused by vehicle overloading includes:

- fatigue of steel members
- cracking of concrete members
- fracture of members
- collapse of bridge.

5.4.3 Other Overloading

Other causes of overloading of bridges or members include:

- build-up of flood debris caught against the structure, generating large hydraulic forces
- extremes of temperatures, either causing excessive movements until a restraint is reached or high temperature differentials within the structure
- high winds
- excessive build-up of snow
- excessive build-up of road metal on deck.

5.5 Foundation Movement

References on foundations include Hambly (1979), Terzaghi and Peck (1967), and Tomlinson (2015).

Foundation movement can be caused by consolidation or instability of the underlying strata. Variability of the underlying strata properties or thickness across the site may cause differential movements to occur. Foundation movement usually first becomes evident only when the bridge geometry changes noticeably or the substructure elements start cracking.

5.5.1 Consolidation

Consolidation can be caused by:

- Consolidation of unconsolidated materials due to the extra loads placed on them at the time of the bridge construction.
- Changed moisture content in reactive clays caused, for example, by the planting or removal of trees, installation of watering systems, leaks in drains and a rise in water level due to flooding. These clays can expand as well as shrink, thus causing upward movement of the foundation.

5.5.2 Instability

Causes of instability include:

- change in pore water pressure within the foundation strata from altered water table levels caused, for example, by bores and wells
- sliding of rock masses along fault or jointing planes
- slip circle failure of a slope such as an embankment or by adjacent excavation
- subsidence associated with mining activities or underground cavities.

5.6 Stream Forces

5.6.1 Hydraulic and Hydrostatic Forces

Additional information on waterways is available in and *AGBT Part 8: Hydraulic Design for Waterway Structures*.

The flow of water in a watercourse generates lateral pressures on elements of the bridge immersed in the flow. Normally only the piers and sometimes the abutments are immersed, but large floods may submerge the whole bridge, and the current against the superstructure may augment considerably the total lateral force acting on the bridge. Submersion may also generate a significant buoyancy force, particularly if the superstructure is of a shape that allows entrapment of air in pockets. To reduce this buoyancy, vent holes should be provided to permit entrapped air to escape to the surface.

5.6.2 Log Impact and Obstruction

The effect of flood borne debris carried along on the current is firstly to generate an impact force on the bridge if it is struck, and secondly, if the debris becomes caught on the bridge, to enlarge the frontal area presented to the current and consequently increase the lateral force.

5.6.3 Scour

Bridges often constrict the natural waterway as a result of the construction of embankments for the approaches. During major floods water velocities may be greater than those that previously occurred naturally, and this can cause major scour damage. Melville and Coleman (2000) recognise that scour is an important bridge failure problem as it can have significant adverse financial and environmental consequences for a road jurisdiction.

Bridge openings that are either misaligned with the waterway or sited on a bend in the stream may be subject to scour of the bank on the outside edge of the bend. Since the water on the outside edge of a bend speeds up, it has a greater potential to cause scour, particularly if the material is friable or softened by water.

The presence of other obstructions in the waterway also speeds up the flow, thus increasing the potential for scour.

Activities since the advent of European settlement in Australia have significantly altered the hydrological behaviour of catchments, with the result that in many cases the flow regime is now out of equilibrium with the waterway. Substantial stream degradation (scour) can occur over time, limited only by controls, either natural such as rock bars, or engineered, such as drop structures. Under these conditions bridge structures can become vulnerable as this scour progresses upstream from the next downstream control point.

5.7 Bearings

This section presents information on failure modes of common bridge bearing types and their root causes. Refer to *AGBT Part 3: Typical Superstructures, Substructures and Components* Section 14 for information on bearing function, design considerations and descriptions of typical types of bridge bearings. Refer to *AGBT Part 6: Bridge Construction*, Section 19 for more information on supply and installation, as well as construction issues of common types of bridge bearings. Refer also to Section 6.9.3 of this document, for information on the rehabilitation and strengthening treatments for these issues.

Bearings are a small percentage of the overall bridge cost, but problems arising from shortcomings in them may lead to distress not only in the bearings but in the remainder of the structure. The resultant repairs may be expensive and extremely difficult. In some situations it may be impossible to replace or reset bearings, even though provision for this operation has been required by AS 5100.4.

5.7.1 General Forms of Bearing Deterioration

The general forms of bearing deterioration include:

Corrosion of Bearings

Bearings are frequently located in confined spaces that collect dust and dirt. If located below a leaking deck joint, the resultant water trapped in the dirt may promote corrosion of the metal components. Corrosion may also be caused by galvanic action between the different metals either within the bearing or between the bearing and the rest of the structure.

Restriction of Bearing Movement

Accumulations of dirt, dust, corrosion products and debris on the bottom plate of roller or sliding bearings will restrict free movement. Drying out of the lubricant will increase the resistance of roller bearings to movement. Incorrectly installed stops, clamps to restrain movement during transport of the bearing, which are not removed after installation, excessively long holding down (HD) bolts and incorrect initial setting of the bearing may all restrict the free movement or rotation.

Deterioration of Bearing Materials

The polytetrafluoroethylene (PTFE) often used for the sliding surfaces of bearings may be scored by accumulations of dirt, dust and debris, increasing the resistance to sliding. If not adequately bonded to the bearing plates, PTFE sheet may extrude under the action of structure movement. The elastomer used in elastomeric bearings is subject to oxidation, which alters its properties leading to cracking. When used in the bearings of bridges with other than small spans, sheet lead will be extruded due to movement of the structure. Where materials in bearings are subject to excessive loading or are inadequately confined they may be subject to crushing or extrusion. For example, the elastomer in a pot bearing can be extruded if it is not adequately confined by the seal.

Deterioration of Bearing Seats

Retention of moisture, such as that from leaking joints, in accumulations of dust, dirt and debris will promote deterioration of the materials in the bearing seats. Such deterioration includes corrosion of steel, either structural steel sections or reinforcing steel in concrete. Excessive forces due to restraints of bearing movement may cause spalling in concrete and local buckling in steel members. Problems at bearing seatings are aggravated by the high-tensile splitting and spalling stresses due to the concentrated bearing loads, and relatively small tensile forces from other loads or a small extent of deterioration can cause cracking.

5.7.2 Typical Failures of Specific Bearing Types

In addition to the general forms of deterioration described in Section 5.7.1, issues related to specific bearing types have been observed as follows.

Common Failures of Metal Bearings

For metal bearings, the following forms of deterioration are common:

- corrosion of steel, particularly at the sliding interfaces leading to a frozen bearing
- cracking or spalling in the supporting concrete and loss of support under the bearing
- misalignment or excessive movement, excessive rotation of rockers
- bearings seized or frozen due to excessive debris, rust build-up, or failed lubrication system
- failure of sliding interfaces
- corroded and/or missing bolts, indentation and deformation of metal plates, failed shear keys.

Figure 5.53 shows an example of a sliding plate bearing in poor condition.

Figure 5.53: A sliding plate bearing – in poor condition

Source: TMR QLD (n.d.).

Sliding Bearings

Defects in the sliding component can include:

- tearing of the PTFE membrane that separates the two stainless steel plates, generally caused by grit or other foreign material trapped between the sliding surfaces
- scoring of the stainless steel plates, again caused by contaminants such as grit trapped between the sliding surfaces
- misalignment causing binding on the lateral restraint guides – generally only a sign of some other problem such as differential settlement of the foundations
- PTFE ‘flow’ (dimensional change in the thickness of the PTFE membrane), caused by uneven pressure – usually the result of misalignment due to differential settlement in the foundations
- seizure due to corrosion or to disintegration of the graphite sheet if there is one. This is the main fault that occurs with sliding bearings.

The structural implications of these defects are that a larger force is needed to cause sliding than the designer intended, or that the bearing seizes altogether causing undesirable stresses in both the superstructure and the substructure.

Rocker Bearings

Typical defects found during inspections are:

- excessive wear in the key and keyways or pins. This defect is often just a product of age but the condition can be aggravated by dirt and grit carried into the working parts of the bearing by water leaking through faulty deck joints
- loose bolts or cracked welds. These are often caused by excessive load brought on by partial seizure of the bearing
- corrosion caused by water and dirt build-up in the bearing region due to faulty joint and drainage systems (Figure 5.54).

Figure 5.54: Corrosion of a rocker bearing

Source: RTA NSW (n.d.).

The structural implications of the above faults are the introduction of unacceptable stresses into the main members of the structure.

Most of the faults associated with this type of bearing are related to excessive wear and corrosion. Both conditions are aggravated by accumulation of dirt and water in the bearing area. The most common cause of this condition is failure of the deck joint.

Roller Bearings

Defects often found include:

- seizure of the bearing caused by accumulated dirt, debris, corrosion, loss of lubrication or misalignment
- corrosion of the steel baseplates, rollers, bolts, pins and guide devices
- shearing of the bolts or other fixing devices.

The structural implications of the above faults are that they can introduce high stresses into the main support members of the structure and can lead to failure elsewhere.

Elastomeric Bearings

Faults which occur with elastomeric bearings are:

- shear failure in the form of delamination of the steel plates and layers of rubber, caused by excessive vertical stresses, horizontal movements or seismic forces
- rupture of the protective rubber sheathing, caused by excessive horizontal movements and/or excessive compression or rotation
- corrosion of the baseplates and/or the laminated steel plates, usually caused by leakage from defective joints

- excessive compression due to decomposition of the rubber in older models of this type of bearing
- loss of contact between the bearings and adjacent surfaces
- missing, distorted or corroded dowels and deterioration of materials
- ‘walking’ of the bearings under loaded conditions
- bearings ’overhanging’ the adjacent mortar pads or plinths.

Figure 5.55 shows an example of an elastomeric bearing with cracking and deterioration of the elastomer.

Elastomeric strips are commonly used under double hollow core units and similar forms of construction. The performance of this type of bearing is generally satisfactory but they should be inspected for excessive compression in particular.

The structural implications of the above faults are the same as described for other bearing types. They can induce unacceptable stresses into the main members of the structure and fail to support effectively the vertical loads and transmit these through to the substructure.

Shear failure in this type of bearing occurs only when the bearing design parameters for horizontal movement are exceeded. This sometimes occurs because the bearing was installed when the bridge superstructure was not midway between its fully expanded and fully contracted travel or as intended in the design. In this case it may exceed its design shear loads at one end of the thermal cycle. Shear failure may also occur in prestressed concrete bridges with continuous or interlinked spans due to insufficient allowance for long-term creep shortening.

Figure 5.55: Cracking and deterioration of elastomer

Source: RTA NSW (n.d.).

Pot Bearings

Faults that occur with this type of bearing are:

- the elastomeric pad may suffer compressive rupture at one edge if the bearing has been subjected to more than its designed rotation
- shearing of bolts caused by horizontal movements for which this type of bearing may not be designed
- corrosion of the baseplates, pots and fixing bolts, generally caused by leakage through faulty deck joints, possibly causing seizure of the bearing
- internal deterioration of the elastomer, extrusion of elastomer, leakage (Figure 5.56)
- broken seals
- excessive vertical, horizontal and/or rotational movements
- cracking or spalling of supporting members
- excessive displacement of stainless steel-faced sliding plate
- uplift/separation of the PTFE sheet and stainless steel sliding surface.

The structural implications of the fault described in the second bullet point above are excessive forces transferred into other parts of the structure and the cause is generally failure of a bearing or joint elsewhere in the structure.

Figure 5.56: Corrosion and extrusion of elastomer

Source: RTA NSW (n.d.).

Cylindrical and Spherical Bearings

Typical defects found in these bearings are similar to those found in sliding bearings:

- torn sliding membranes, generally caused by dirt entering the sliding surface area. This type of bearing is particularly prone to entrapped dirt and water if it involves a concave bottom plate
- corrosion and scoring of the stainless steel plates
- loose bolts or fixing devices
- indentation and deformation of metal plates
- sliding interface problems.

The structural implications of these faults are the same as those for most bearing defects. They will tend to restrict the design movement capacity of the bearing and transfer the resulting stresses into the main structural members of the bridge and to other bearings.

5.7.3 Root Causes of Bearing Failures

The root causes of bearing failures may come from all stages including the selection of the bearing type, design, manufacture, testing, installation, to the maintenance stage.

Selection of Right Bearing Types

All bearings have their own limits to function properly. The selection of a bearing type should ensure that the movement ranges of the bearing can accommodate the required movement ranges of the structure. Otherwise the bearing may be damaged by excessive movements. For example, elastomeric bearings should not be used for movements greater than 100 mm and rotations greater than 0.04 radians.

Design

- components designed without adequately considering critical aspects of installation, inspection and maintenance
- movements due to temperature gradients, post-tensioning, etc. not accurately addressed
- bridge geometry related failures: misalignment or improper orientation of the bearing with respect to the direction of movement.

Manufacture

- fabrication tolerance errors
- insufficient use of testing to ensure performance matches the design specification
- non-compliance materials.

Construction/installation

- inadequate or improper installation of bearings
- disintegration of poorly prepared bearing seatings, bedding anchorage, setting/releasing of transit bolts where applicable
- cleaning of epoxy mortar splashes or other deleterious materials.

Maintenance

- accumulation of detritus and water
- failure of the expansion joint system that leads to water leaking to underneath the bearings.

Other Sources

- uneven loading on bearings of skew bridges
- attack by chemicals, fire, corrosion and unforeseen events (impacts)
- excessive non-thermal induced movements of piers and abutments
- bedding mortar - cracking of mortar and gaps between the mortar and bearing.

In addition, the following causes have also been observed in practice:

- insufficient allowance for construction rotation, arising from beam hogs and camber issues, may cause 'lift off' from the bearing at one or more corners
- in-service failure is uncommon and usually relates to incorrect compound formulation, and poor manufacture controls on metal to rubber vulcanisation, causing delaminating and separation of the internal rubber from the internal steel plates
- poor internal alignment of steel plate layers which manifests as visually non-uniform bulges and ribs of rubber surface around the perimeter of the bearings.

5.8 Deck Joints

This section presents information on failure modes of common types of bridge deck joints and their root causes. Refer to *AGBT Part 3: Typical Superstructures, Substructures and Components*, Section 15 for information on deck joint function, design considerations and descriptions of typical types of bridge deck joints. Refer to *AGBT Part 6: Bridge Construction*, Section 19.3 for more information on supply and installation, as well as construction issues of common types of bridge deck joints. Refer also to Section 6.9.4 of this document, for information on the rehabilitation and strengthening treatments for these issues.

Additional references include Dahir and Mellot (1988) and Purvis and Berger (1983), Austroads (2012).

5.8.1 General Forms of Deck Joint Deterioration

The various joint components in bridge decks that typically suffer deterioration are:

- joint nosings
- joint anchorages
- seals and sealants
- cover plates and finger plates
- flexible drains.

The typical failures of these components are presented as follows.

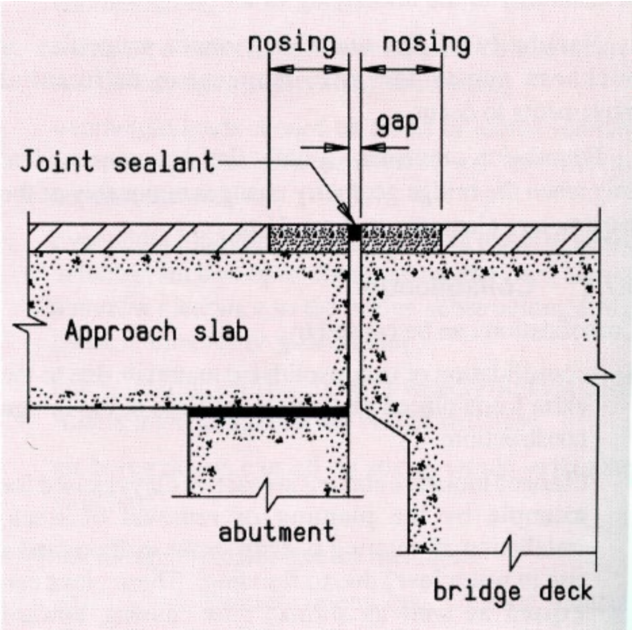
Failures in Joint Nosings

Nosings of epoxy concrete, as in Figure 5.57 are particularly susceptible to failure by de-bonding at the interface between the nosing and the deck due to the large shear stresses caused by the difference in the coefficient of thermal expansion of concrete and epoxy.

Other causes of failure from all material types include:

- unfavourable temperature, moisture or cleanliness conditions at time of installation
- impact damage by road maintenance plant
- wheel impact from traffic due to a poor vertical profile along the wheel lines.

Figure 5.57: Filled joint with nosing



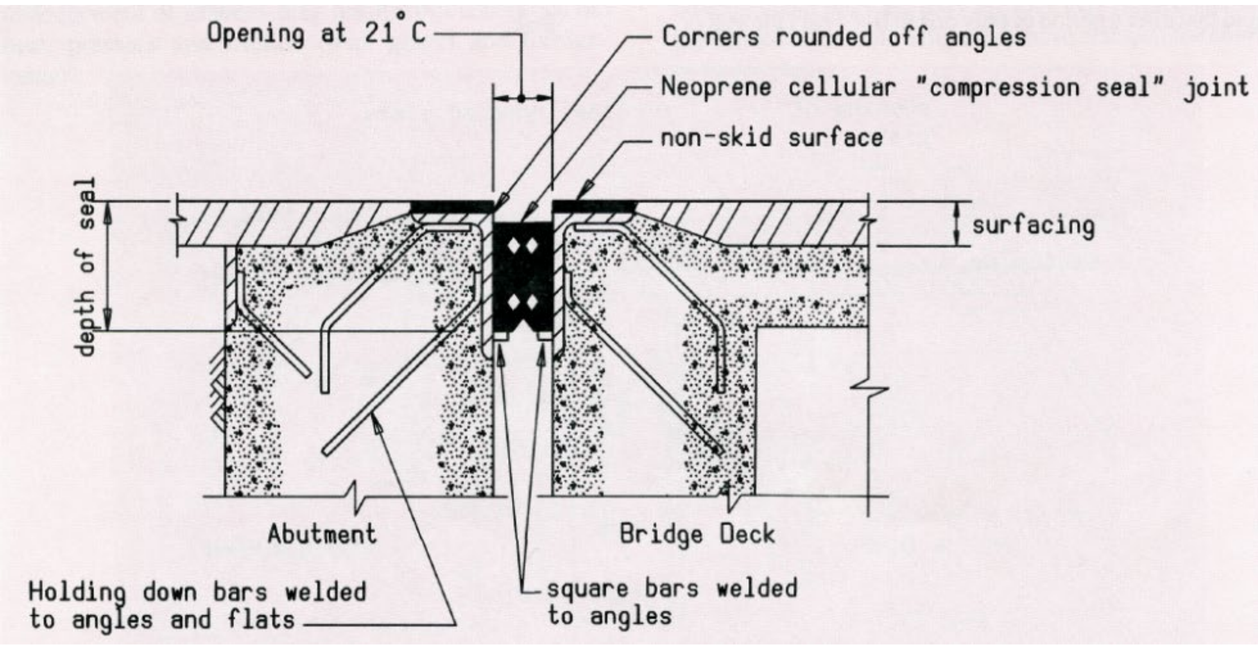
Failures of joint anchorages

Filled joints, compression seal joints, reinforced neoprene joints, sliding plate joints, finger plate joints and strip seal joints may all use steel protection angles or flats of steel or neoprene base plates held down by anchors.

Protection angles or flats and base plates are typically anchored by straps or bars welded to the angle or flat and anchored in the concrete (Figure 5.58 for example). Under the repeated impact of wheel loads, the welds suffer fatigue and crack right through.

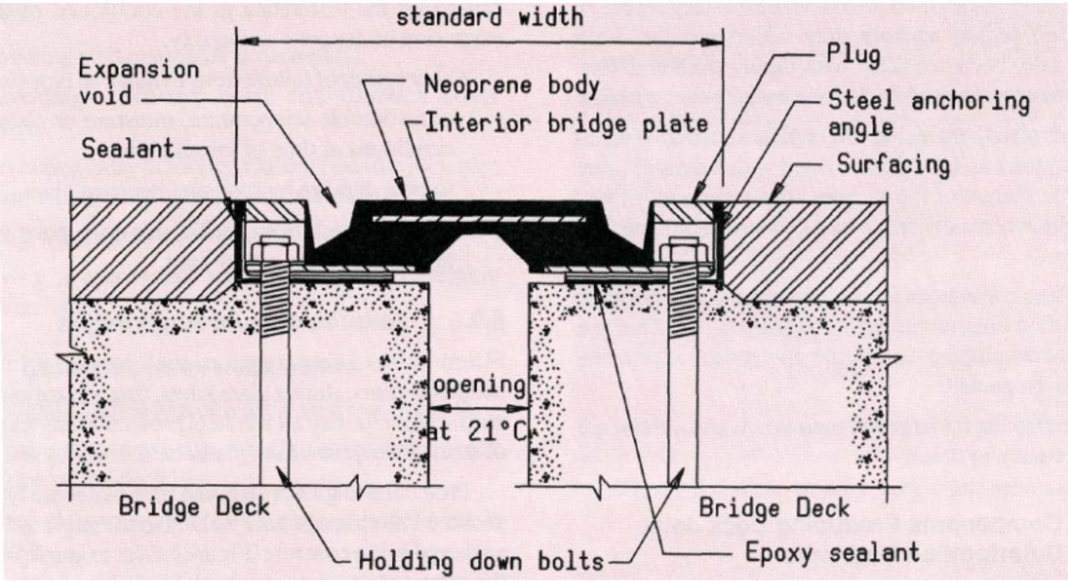
Another mode of failure is due to inadequate bond length of the anchoring straps or bars. Lack of compaction of the concrete under the angles due to entrapment of air when casting reduces the bedding and increases the forces on the straps, thus increasing the probability of failure. With the anchoring destroyed, the protection angles become loose and may even break away, threatening the safety of traffic. For this reason, flats are normally used now instead of angles.

Figure 5.58: Joint with compression seal



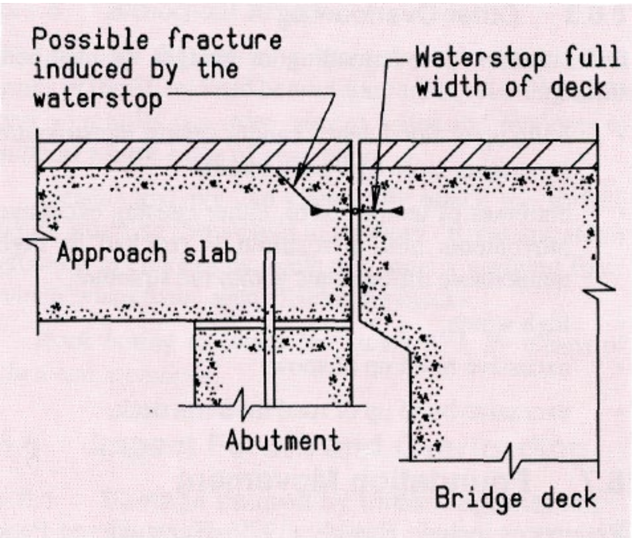
Many joints are typically anchored by HD bolts (Figure 5.59 for example). Inadequate anchorage due to an insufficient number or size of bolts or insufficient anchoring of the HD bolts will lead to failure of the anchors under the impact of wheel loads. Expanding wedge-type masonry anchors may work loose under the impact of wheel loads.

Figure 5.59: Reinforced neoprene joint



Waterstops (Figure 5.60) may induce fracture of the concrete deck from the end of the stop up to the top surface, leaving a large piece of concrete broken off at the end of the deck.

Figure 5.60: Waterstop for very small movements



Failures of Seals and Sealants

Various types of seals and sealants have been used and suffer a range of deterioration. Deterioration may be so rapid that after a period of only one to five years the seal or sealant may not perform adequately. Some typical problems include:

- Neoprene compression seals lose their initial compression over a period of time and under extreme cold may be unable to expand sufficiently to keep the joint sealed. The seal will leak, and may even fall out. Debris may also enter the gaps between the seal and the nosing, thus preventing the seal from resealing the joint when it next contracts.

- Neoprene strip seals may be punctured by debris trapped in the joint when it closes or by debris hammered by the wheels of traffic. The seal will then leak.
- Polysulphide and polyurethane sealants may be damaged by debris pressed into the soft surface by wheels. They are subjected to repeated tensile stresses that may cause the sealant to break away from one or both edges of the joint and repeated compressive stresses that may cause the sealant to be extruded from the joint. In general, polyurethane sealants have been found to perform better than poly-sulphides.

Sealants may be attacked by chemicals or grit washed off the deck or by ultraviolet light. Early deterioration or dislodgement of sealants may result from conditions of heat, pressure, and friction (with gravel and foreign matter).

Failures of Cover Plates and Finger Plates

The usual cause of looseness of cover plates and finger plates is by the failure of the anchorages. This failure may be aggravated by an initial incorrect setting of the plates resulting in high impact loadings under wheels, or by fouling with the superstructure when the bridge expands, possibly resulting in high forces.

Looseness of cover plates and finger plates may also be caused by defective or incorrectly installed bolts that hold them down to the base plate. For example, if the HD bolts are too long they may tighten against the bottom of the hole before they have clamped the cover or finger plates.

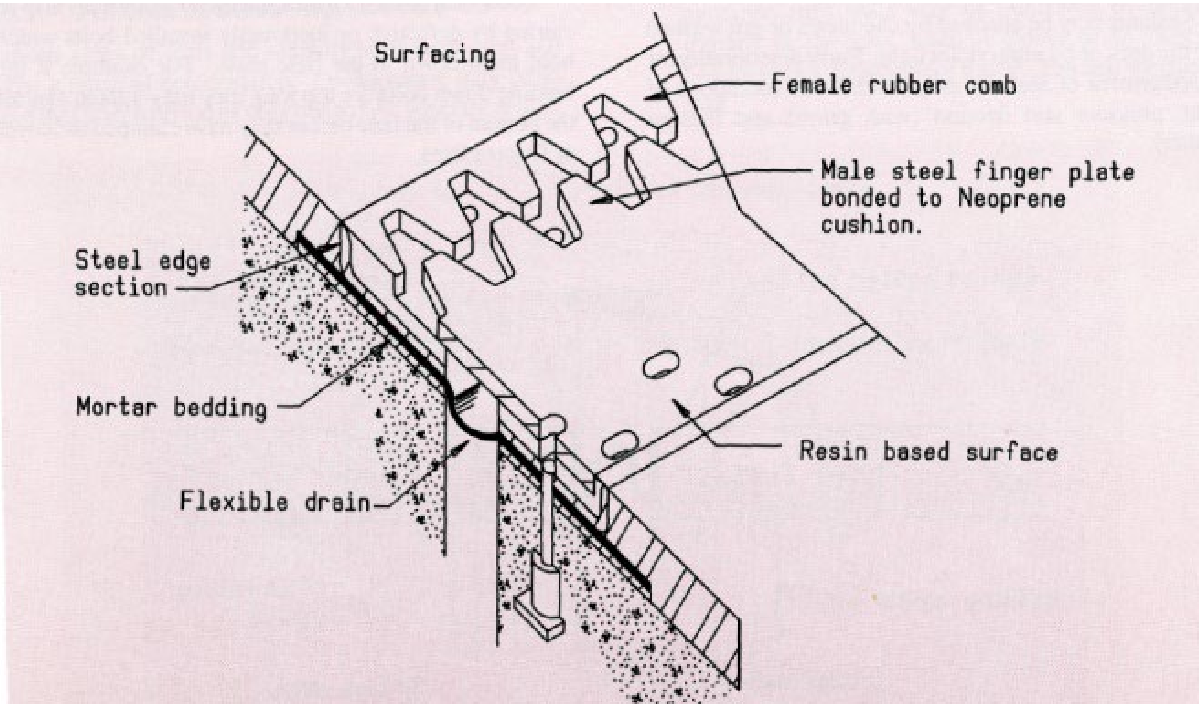
Welded cover plates may fail at the welds. Cover plates may be distorted by traffic, impact damage or fouling at full bridge expansion. Finger plates may occasionally lose a finger.

Failures of Flexible Drains

Flexible drains are frequently used under cover plates and finger plates to catch the water flowing down through the joint and lead it away (Figure 5.61 for example). As these drains are usually installed on the same grade as the crossfall, they do not have sufficient fall to clear the silt. Debris continues washing into them, fills, and blocks them. The repeated contraction and expansion of the bridge packs the debris tighter and tighter until eventually the drain ruptures, causing water leakage.

Some metal flexible drains may crack due to the fatigue caused by repeated cycles of expansion and contraction, then start leaking.

Figure 5.61: Finger plate joint bonded to neoprene cushion



5.8.2 Typical Failures of Specific Deck Joint Types

The following failures have been observed in specific joint types.

Sealant Filled Joints

Possible defects of this joint type include:

- adhesion and cohesion cracks or loss of the joint sealant and/or hose
- cracks and spalls in the adjacent deck or headers
- section delaminated from the deck
- joints impacted with debris or stones
- broken bitumen/cork filler; ripped out in chunks by traffic.

Sliding Plate Expansion Joints

The following failures have been reported:

- corrosion of the steel plate and debris collection obstructing the free movement of the superstructure
- loose and broken anchorages due to repeated impact and weathering actions.

Compression Seals

The following defects of this joint are common:

- adhesion failures from abrasion or tearing
- cracks and spalls in the adjacent deck or headers
- damaged nosings
- damaged or lost seals
- seals deteriorated due to UV exposure or hardening
- broken stud and loose nosing angles
- loss of a compression set, falling through the gap, protruding above the trafficked surface.

An example of a compression seal joint failed due to loss of adhesion is shown in Figure 5.62.

Figure 5.62: Loss of adhesion on sides of the compression seal

Source: RTA NSW (n.d.)

Asphaltic Plug Joints

Defects that occur with this type of joint include:

- De-bonding at the interfaces with the underlying deck slabs and adjacent surfacing, generally caused by the joint being required to cope with movements beyond its capacity. The inspector should check to ascertain if seizure of bearings or another fault at an adjacent expansion joint has caused restriction of movement, transferring all movement to the end where the de-bonding has occurred. Where this is not the case the most common causes of de-bonding are under-design, faulty materials proportioning, or poor surface preparation.
- Excessive depression in the wheel paths generally caused by incorrect proportioning of aggregate to matrix or by using a matrix unable to cope with high ambient temperatures. In joints skewed to the direction of traffic flow, these deficiencies may also result in ‘flow’ of the joint material along the joint under the traction forces exerted.

Plastic cracking in the joint surface – generally due to the use of a matrix composition unable to cope with low ambient temperatures or to sudden movements such as seismic events or an excessively lively superstructure. Plastic cracking may also be caused by excessive strain or fatigue from a high number of loading cycles.

The implications of the defects listed above are that excessive deformation of the surface of the joint can cause unacceptable impact loads, particularly from heavy traffic. De-bonding and deep cracking will allow water to get through to the substructure and bearing areas and will also allow progressive break-up of the joint material.

Figure 5.63: Asphaltic plug joint skewed to the traffic flow showing flow of joint material out of recess due to traction forces

Source: NZTA (2001).

Figure 5.64: A strip seal on Craig Gilbert Bridge, bolts were loosened

Source: Department of Planning Transport and Infrastructure South Australia (n.d.)

Strip Seal Joints

Typical defects of this joint type include the following:

- accumulation of debris
- loosened anchor bolts due to vibration
- elastomeric seal punctures or breaks up
- delamination and failure of Felspan joints
- breakage of the aluminium retainers and fixings.

Figure 5.64 shows an example of a damaged strip seal joint in South Australia due to loosening bolts.

Finger Joints

Typical defects of this joint type include the following:

- a section of finger joint becomes loose, due to loosening of the threaded anchors
- misalignment or broken fingers
- water leaking and debris accumulation
- corrosion of metal parts
- debonding of the upper and lower plates from the central neoprene pad
- failed or missing catch drains or membranes
- frequent maintenance required
- fatigue issues
- not compatible with seismic requirements.

An example is shown in Figure 5.65 where metal fingers bonded to an elastomer joint in Western Australia failed with debonded, missing sections and broken anchor bolts.

Figure 5.65: Failures of metal fingers bonded to an elastomer joint on Shelley Bridge No. 931 in WA

Source: MRWA (n.d.).

Modular Joints

Faults that may develop are:

- uneven friction on sliding surfaces, leading to uneven gaps between spacers
- wear on the sliding surfaces leading to noisy operation
- chips and other debris lodged above the seals
- de-bonding of seals from the spacers, leading to leakage
- fatigue cracks
- frequent maintenance required
- debris accumulation in seals
- reflective cracks in the concrete deck directly above the support boxes
- corrosion of metal components.

Proprietary Expansion Joints – Moulded Rubber

Typical defects of this joint type include the following:

- failure of anchorage systems due to repetitive live load impacts
- splitting of the seals and loosening of the retainers.

This joint was identified as having inadequate anchorages and exhibiting excessive wear of the elastomeric moulding around the retainers. In addition, the failure develops relatively rapidly after initiation.

Figure 5.66 shows an example of a Felspan joint in the ACT with damaged seal and missing sections.

Bonded Metal/Elastomer Joints

Typical defects of this joint type include the following:

- leakage
- bolt failures causing joint to lift
- bolt corrosion, missing cover pads
- wearing of rubber ribs on top surface
- breaking up of transition strips
- delamination of elastomer/metal plate interface
- glands split or pulled out of the housings
- cracked concrete or asphalt nosings.

Figure 5.67 shows an example of a deteriorated bonded metal/elastomer joint in the ACT.

Figure 5.66: A Felspan joint on Bridge 2097, damaged seal and missing sections

Source: Roads ACT (n.d.).

Figure 5.67: A deteriorated Transflex joint on Bridge 4087

Source: Roads ACT (n.d.).

5.8.3 Root Causes of Deck Joint Failures

The root causes of expansion joint failures can occur at all stages, including selection of the joint type, design, manufacture, testing, installation and maintenance. The possible causes of deck joint failures related to specific expansion joint types are discussed below for the most popular types of joints.

Compression Seal Joints

For compression seal joints, the possible causes of the failures are:

- selection of the right expansion joint type: compression seal joints should not be used for movements of more than 80 mm and for joints skewed more than 20 degrees
- deterioration of seal material causes the loss of its ability to spring back or widen when the gap opens up in the winter season after the seal has been compressed during the summer season
- frequent vibration of the structure causes the seal to protrude above the trafficked surface
- total shrinkage and movement of the bridge supports can cause the seal to fall through the gap.

Strip Seal Joints

The possible causes of failures of strip seal joints are as follows:

- the performance of the joint depends on the correct choice of seal size and seal material. The strip seal joints should only be used for movements less than 85 mm and for joints skewed less than 30 degrees
- the seals not being properly installed by a specialist installer
- poor installation is usually the main cause, such as poor or non-facilitated torqueing of the fixings or poor concrete vibration resulting in voids under the metallic side retainers
- breakage of the aluminium retainers and fixings due to poor bedding epoxy under the joints.
- vibration is typically the main cause for failure of the anchorage system
- debris entrapped on the glands may lead to tearing, puncturing or pulling out of glands under passing traffic or from joint movements. As the joint closes up, the incompressible debris wedges in the gland crevice which can cause glands to rupture. Traffic wheel loads transferred to glands through built-up debris may tear the gland or pull it out from the metal retainers (Roads and Traffic Authority 2008).

Fingerplate Joints

Root causes of problems include:

- Poor concrete vibrating practices and poor or inadequate tensioning procedures for anchor bolts are the predominant root causes of in-service failures of finger joints.
- Fingerplate joints should only be used for movements less than 200 mm unless special consideration is given to fatigue failure of the fingers. In addition, the fingers may have locking-up issues when the joint is in a fully closed position, not allowing any lateral movement.
- Unacceptable noise from a rough riding surface may occur due to anchorage problems or the vertical misalignment of the fingers. Failures of the hold-down cap screws or the base plate anchorages may cause loose plates which may be a severe hazard (Roads and Traffic Authority 2008).

Modular Joints

Critical factors affecting the performance of modular expansion joints can be classified in three categories: loading and structural response, fatigue resistance, and construction and durability. The first two categories relate to dynamic behaviour of the joint structure (Ancich & Bradford 2006). It is also mentioned by Braun (1996) that unrealistic load models and unsuitable design solutions have been used in the design. The last category may include the following causes:

- poor concrete consolidation around support boxes and edge beams. This causes differential settlement of support boxes, increasing the span of the centre beams which results in increased live load stress ranges
- accidental casting of the concrete into the support boxes preventing the thermal movement of support boxes
- reflective cracking of the concrete deck directly above the support boxes that can adversely affect the durability of the joint
- accessibility issues.

Bonded Metal/Elastomer Expansion Joints

The possible causes of failure of bonded metal/elastomer expansion joints include:

- The long-term joint movements were not fully calculated in the design, thus the joint design capacity was exceeded.
- Shallow installation depth and the joint's high stiffness may cause high horizontal forces on adjacent decks, thus causing damage on the nosings.
- The failure of the anchorage system may be caused by shearing of the bolts due to direct impact from vehicles if the panels are not protected by an impact-absorbing header material.

5.9 Barriers

The function of a barrier is to restrain pedestrians and errant vehicles from falling over the side of the bridge or approaches. To do this effectively the barrier should:

- limit deceleration to an acceptable limit
- smoothly redirect a colliding vehicle
- remain intact after a collision, and be quickly repairable
- project flush with or inside the face of any kerb, to prevent vehicles vaulting
- define the limits of the carriageway, but permit adequate visibility
- maintain compatibility between the approach barrier and bridge barrier, be effectively spliced and transitioned for continuity, and able to absorb all superstructure movements
- provide protection to vehicles and pedestrians when required to serve both.

Not all barriers used will qualify on all of the above requirements. Barriers comprise some or all of:

- metal railing
- metal posts
- concrete barriers.

5.9.1 Deterioration of Barriers

Barriers may be subject to damage by impact or inadequate provision for thermal expansion and contraction. This can cause spalling in concrete or local deformation in metal.

Concrete barriers and metal posts and rails may be subject to the same forms of deterioration as other concrete or metal components of the bridge.

Metal rails and posts may also be subject to corrosion caused by galvanic action if dissimilar metals are in contact with one another.

6. Rehabilitation and Strengthening Treatments

6.1 Introduction

This section initially discusses key basic and general considerations to be aware of in deciding on appropriate rehabilitation and strengthening treatments for:

- concrete bridges
- steel bridges
- timber bridges.

This is followed by a discussion of treatments and techniques aimed to improve material properties and/or improve the durability and function of specific bridge members critical to the overall strength, integrity, and fitness for purpose of the bridge structure. Guidance is given on the inspection and identification of defects, their interpretation, diagnosis, and repair taking into account material properties, element shape and the function the defective element plays in the bridge structure. These are discussed under the following headings:

- concrete
- steel
- reinforced and prestressed concrete
- structural steel
- timber
- approaches
- drainage system
- hardware
- foundations
- waterway
- seismic damage.

Also, there is a wealth of published material on rehabilitation and strengthening treatments (ACI 2008, Nanni 2004, Organisation for Economic Cooperation and Development 1983, and Tilly et al. 2008). It is an area that is under development as new procedures and technical advances are made. The reader is therefore recommended to refer also to other manuals published by the Australasian road jurisdictions: AS 5100, NZTA *Bridge Manual* – Section 7: Structural Strengthening (NZTA 2016b).

A design engineer should be commissioned when it is determined that a bridge or bridge element requires increased strength or capacity. Numerous design techniques are available to increase strength and capacity, but specialised knowledge is required for a successful outcome.

Examples of such techniques to strengthen or improve the ductility of concrete or steel bridge members are:

- bonded steel plates or FRP composite materials
- external prestressing
- steel sleeves or FRP composite materials to provide shear strengthening and ductility enhancement of reinforced concrete columns.

Design information is available from AS 5100 (Set), and the NZTA *Bridge Manual* (NZTA 2016b).

6.2 Concrete Bridges

Concrete bridges in general are made integral during construction and so replacement of defective elements is rarely an option.

It is practical to find and repair construction defects effectively and economically when the affected area is small. Small patches of uncompacted 'boney' concrete can be dug out, as can the occasional piece of steel that has low concrete cover and corrodes early in the structure's life.

Systematic construction failures such as listed below may lead to early corrosion over large areas of the structure:

- lack of concrete cover
- lack of concrete compaction
- lack of concrete curing
- use of an inappropriate concrete mix.

Repair of such extensive concrete defects can be more expensive than removing and replacing the entire structure. In such cases, careful regular inspection to determine when the bridge becomes structurally unsafe will be the most economical course of action.

6.3 Steel Bridges

6.3.1 Inspections

The major causes of damage in steel members are:

- corrosion, once the paint system or galvanising layer has broken down
- impact damage from vehicles, flood debris, etc.
- fatigue cracking at details that cause stress concentrations.

Most corrosion and impact damage is obvious to visual inspection. Finding fatigue cracks requires skill (knowing where to look) and specialist detection techniques.

Keeping steel clean is the best way to prevent early breakdown of protective surface coatings. Moisture and sand/silt deposited by floods, or accumulating around bearings under deck joints (due to loss of the waterproofing seal), are common causes of major corrosion in steel girders that can be prevented or reduced by regular maintenance and cleaning of debris.

Closed steel box beams are rarely completely sealed against moisture and must be able to be inspected. Suitable corrosion prevention coatings must be applied to the inside during construction, and provision for maintaining the interior coatings must be included.

6.3.2 Repairs

Corroded steel should be cleaned to bare metal and recoated with a suitable paint film. The local conditions (temperature, moisture, wind, etc.) must be considered in selecting coating systems as they may be sensitive to the environment. In general, field-applied systems have a shorter life than original paint systems applied in controlled workshop conditions. It is always more effective to repaint before the existing paint film breaks down.

Structural analysis should determine whether the corroded elements need to be cut out and repair plates welded in, or whether the remaining steel section is adequate. Corroded minor members such as bolted cross bracing may be removed and replaced more economically than being cleaned, repaired and repainted in place. Structural stability must be maintained while replacing members (usually one at a time).

Components bent by impact can be straightened with jacks. On thicker sections, controlled heating by an experienced worker may be necessary. Sections with torn elements would normally be cut out and re-welded. Experienced bridge welders with appropriate qualifications are essential to ensure repairs have the required load capacity and fatigue resistance.

Repairs of major elements may require propping or jacking of the main span, or use of a temporary bridge girder (e.g. modular emergency bridging).

Fatigue cracking early in the life of a bridge (viz. the first 30 years) would indicate the designer has miscalculated loads and number of cycles, or has used a fatigue sensitive detail. Alternatively, fabrication errors, or additional critical welds, may have been made during construction (lifting lugs, erection bracing etc.).

The implication is that all similar details are likely to suffer fatigue cracking during the life of the bridge, and inspection methods and frequency need to be adjusted.

Effective repair of fatigue cracks requires the removal of the stress concentration. Simple welding repairs may fracture in fatigue more quickly than the original due to welding stresses. Some cracks can be stopped with a hole drilled at the base of the crack. Peening of welds may increase fatigue life. Sometimes fatigue cracking can be reduced or eliminated by increasing the radius of the cut-outs or section changes causing stress concentrations. Specialist advice should be sought, and some laboratory testing of similar details may be necessary to determine effective repair methods.

6.4 Reinforced and Prestressed Concrete

6.4.1 General

The durability of a reinforced or prestressed concrete structure is defined as its ability to withstand the expected wear and deterioration throughout its intended life without the need for undue maintenance. The principal factors affecting durability include:

- attention to design details, including reinforcement layout, appropriate cover and provision for shedding of water from exposed surfaces
- good mix design
- correct construction practices, including adequate fixing of reinforcement and the placement, compaction and curing of the concrete.

Deficiencies in one or more of these factors can lead to premature deterioration of a structure. The most common form of deterioration is reinforcement corrosion.

NZS 3101 and AS 5100 (Set) provide guidelines for detailing and specifying reinforced and prestressed concrete structures to achieve the required design life. They consider the effects of concrete quality and curing, chemical content, concrete cover, alkali aggregate reaction (AAR), abrasion from traffic and freeze-thaw cycles and exposure conditions. The increased risk of reinforcement corrosion due to salt contamination is allowed for through the requirement for greater depths of cover concrete in exposed coastal environments.

The NZTA *Bridge Manual* (NZTA 2016b) requires a bridge design life of 100 years in normal circumstances (to align with AS 5100 Set). To achieve this increased life expectancy the Manual includes a modified version of NZS 3101, Table 5.5 with increased concrete cover depths.

Concrete bridge structures will deteriorate where the combination of design details and construction quality have provided insufficient durability for the environmental conditions. Where deterioration has occurred it is important to identify its nature and its cause to allow the effects on the performance of the bridge to be assessed and appropriate remedial options to be developed.

More detailed information on concrete bridge durability is given in NZTA (2001), *Bridge Inspection and Maintenance Manual,* Appendix 13.1.

6.4.2 Visible Defects

A bridge is affected by microclimates that control the nature of deterioration and hence the type of defects that develop in certain elements. The superstructure is the most exposed part and is susceptible to moisture-sensitive deterioration such as freeze-thaw, AAR and reinforcement corrosion, as well as to traffic effects such as abrasion and impact. Although less exposed, the substructure of the bridge is still susceptible to reinforcement corrosion, particularly where exposed to salt-laden winds in coastal environments, where the concrete is wetted by driving rain or water leakage through the deck as well as tidal and splash zones. Shrinkage cracking in concrete bridge decks commonly provides a passage for water leakage, and reinforcement corrosion may result. Bridge piers, piles and abutments may be exposed to salt water attack and to abrasion caused by aggregate movement in the river bed.

The defects that commonly affect concrete bridges are discussed in detail in *AGBT Part 2: Materials* and summarised below:

- cracking
- reinforcement corrosion
- spalling
- surface erosion
- drainage and leakage of water
- construction defects
- surface deposits
- failure of applied finishes
- distortion of shape.

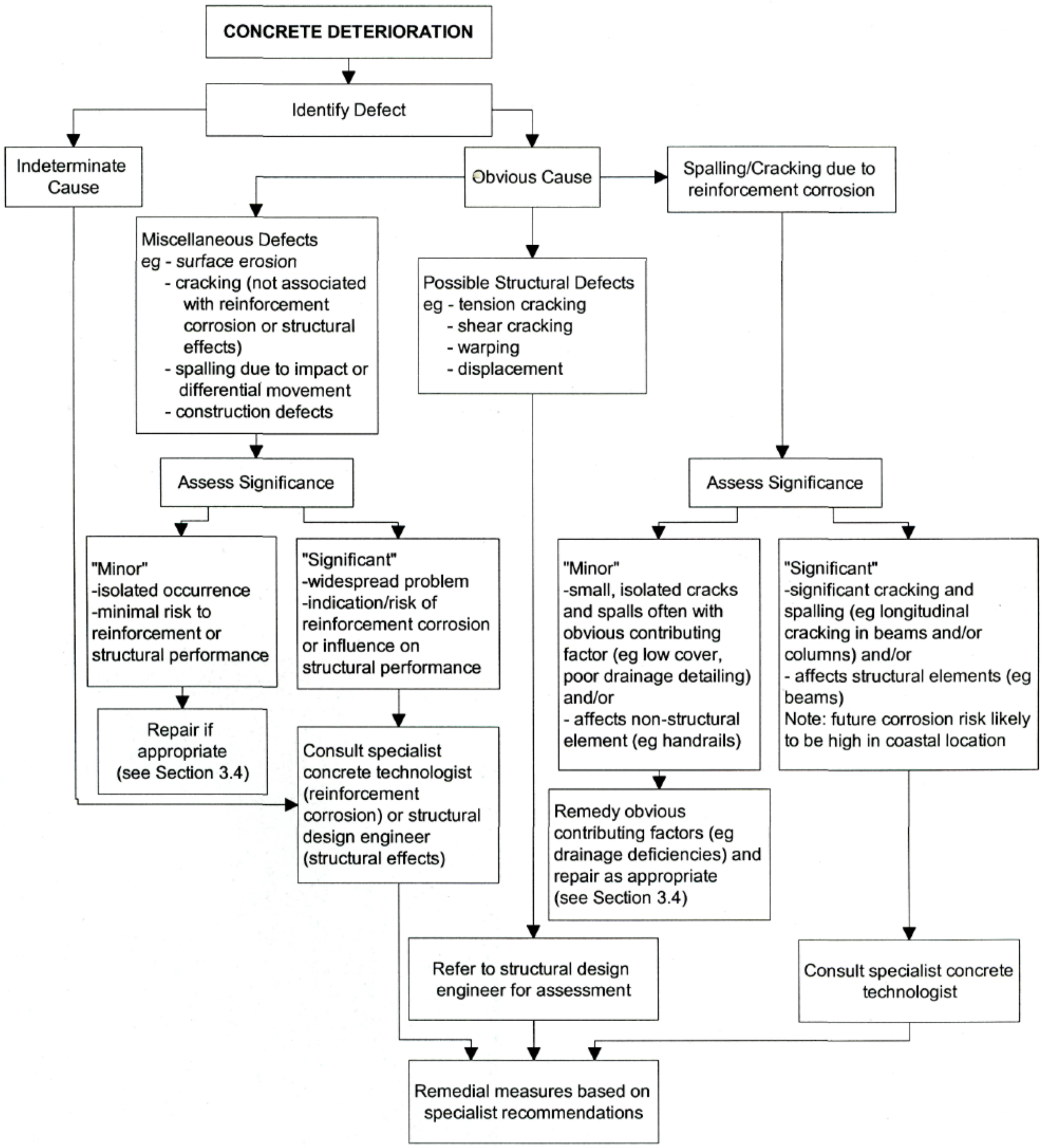
6.4.3 Inspection and Diagnosis

General

Before any repair can be contemplated, it is essential that the causes, extent and severity of the concrete deterioration are accurately diagnosed and assessed. Defects may be aesthetic (e.g. construction stains), indicating possible problems (e.g. honeycombing, efflorescence, lack of cover), non-progressive (e.g. crazing, shrinkage cracking), or progressive (e.g. reinforcement corrosion, working cracks, frost damage, alkali-aggregate reaction).

An approach to defining the significance of concrete deterioration and selecting the most appropriate option for assessment is outlined in Figure 6.1. This emphasises the need to consult experienced concrete technologists or structural design engineers to assess significant concrete defects when their cause is not obvious. A common cause of repair failure is the use of inappropriate repair methods or materials because the cause or significance of the deterioration has not been correctly assessed.

Figure 6.1: Criteria for assessment of concrete deterioration



Note: Section references refer to parent document.

Source: NZTA (2001).

Interpretation of Observations

The need for specialist input to identify the cause, extent and repair of observed defects is most commonly associated with reinforcement corrosion and defects indicative of structural distress.

Detailed visual inspection will establish the nature and extent of reinforcement corrosion, then the more specialised techniques outlined below can be used to establish the cause. Sampling to establish the deterioration mechanism will also help to identify the future corrosion risk and the most appropriate remedial options. Where significant corrosion is evident, cover concrete should be removed at critical locations and the section loss to reinforcing steel determined.

The assessment of prestressing steel corrosion is similar to the process for reinforcing steel, although specialist techniques are required to assess tendons in grouted ducts. Pre-tensioned steel is usually associated with high-quality precast concrete and so has a low risk of corrosion, although where corrosion has occurred the potential loss of prestress is a significant structural concern. Post-tensioned tendons rely on grouting for their primary protection, and this can be variable both in quality and continuity. Voids in the ducts not only increase the risk of corrosion but also reduce the degree of bonding.

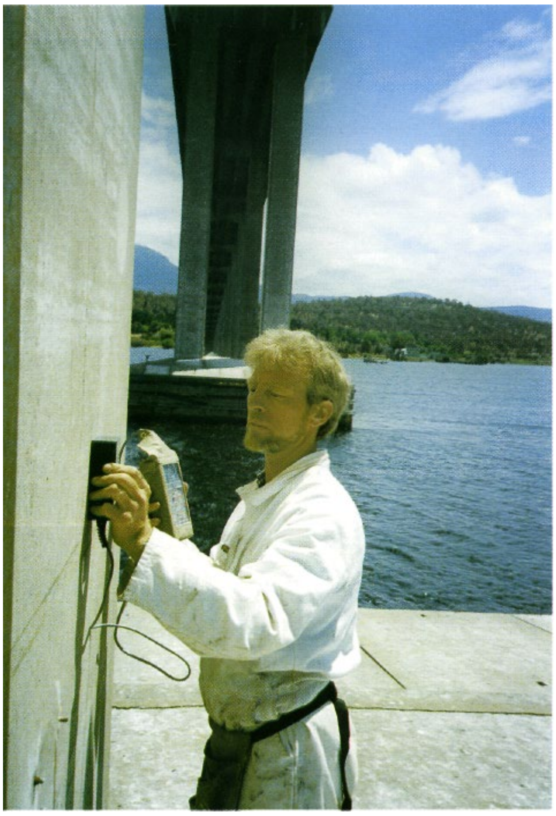
Structural implications of cracks and distortions are assessed by considering first the effect of the damage on the performance of the component, and second how the integrity of the whole structure is affected. Existing codes, structural analyses, drawings, specifications, soil investigations, construction records and previous inspection reports should be studied. Foundation movements and estimated actual loads should be compared with those assumed in the original structural analysis. For prestressed structures the conformity of structural analysis, drawings, actual stressing forces, and the effects of concrete shrinkage and creep should be checked. Environmental factors that may accelerate deterioration should also be considered.

Test Methods

Testing may be required to assess the cause and extent of deterioration. Some of the tests can be carried out by inspection personnel to assist in the initial evaluation. Other techniques should be used by specialist concrete technologists to assist in identifying the most appropriate remedial options. Available test methods are listed as follows.

- common methods:
  - hammering to detect delaminations
  - taking Schmidt hammer measurements to detect variability in concrete quality
  - core sampling to determine concrete strength and crack depth
  - crack-width monitoring
- specialist methods:
  - in situ or laboratory testing for permeability
  - microscopic analysis of cores
  - ultrasonic techniques for detecting cracks and voids
  - covermeter surveys to locate and verify the size of reinforcement, and to determine the depth of concrete cover (Figure 6.2)
  - testing for carbonation depth
  - chemical analysis of chloride ion contamination
  - half-cell potential mapping to ascertain the probability and extent of corrosion
  - measurement of electrical resistivity to indicate the likely rate of corrosion
  - strain and deflection measurements to indicate the effects of cracks
  - in situ, non-destructive measurement of reinforcing steel strength
  - proof load testing
  - detection of volume and continuity of voids in grouted ducts
  - detection of prestressing strand deterioration.

Figure 6.2: Detection of reinforcing steel using an electromagnetic covermeter



Source: NZTA (2001).

6.4.4 Repair

General

Repairs may be carried out on concrete bridges for many reasons but repairs to mitigate reinforcement corrosion damage are by far the most common. Resin injection of structurally significant cracking is also frequently carried out.

The methodology chosen to mitigate the effects of reinforcement corrosion depends on technical, economic and strategic factors. If technically appropriate, electrochemical repair methods (e.g. cathodic protection, desalination, re-alkalisation) offer a longer maintenance-free period than conventional patch repair systems. Service lives in excess of 25 years are claimed for some of the electrochemical methods, and although they have a relatively high capital cost, the extended life of the repair means the whole-of-life costs are likely to be favourable.

Conventional patch repair methods based on proprietary cementitious materials are technically well-advanced and show good performance when executed correctly. However, in some circumstances, such as on coastal bridges where the concrete is extensively contaminated with chloride ions, patch repairs are unlikely to provide long-term durability, and further cycles of repair will probably be required. In such cases an electrochemical treatment such as cathodic protection or desalination would provide a more durable repair. However, the final choice of repair method will also depend on factors such as availability of funds for repair, logistics, the bridge replacement strategy and future alterations to the bridge structure due to road widening or realignment.

If frequent repeat cycles of patch repair are expected, then the long-term structural implications of these repairs should be considered.

The following repair methods are discussed in this section:

- concrete patch repair
- concrete coatings
- electrochemical repair
- crack repair.

Concrete Patch Repair

Materials

Concrete patch repair involves reinstating cracked and/or spalled concrete with compatible proprietary cementitious materials. These products contain polymer modifiers, admixtures and fillers to improve bond, increase strength, reduce shrinkage and decrease permeability. They are entirely pre-packaged apart from the mixing water or sometimes the gauging fluid. Manufacturers guarantee the performance of their proprietary cementitious repair systems when used in accordance with their instructions and applied by approved applicators.

Some of the main factors influencing durability of repairs are the thoroughness of the preparation and cleaning of the steel (particularly the removal of salts from the surface) as described below, the quality of the adhesion of the repair material to the original concrete, and the permeability of repair material, which is influenced both by the material chosen and by the quality of its application. The use of proprietary repair systems and approved applicators should ensure that these factors are addressed.

The quality of non-proprietary repair materials batched on site will vary. They are not recommended due to the significant risk carried by the owner of the structure in the event of ongoing durability problems.

Epoxy mortar repairs are not recommended for repair of concrete damaged by reinforcement corrosion. The thermal expansion and electrical properties of epoxy resin are significantly different to concrete, and such repairs are likely to fail prematurely (Figure 6.3) as well as promote and accelerate corrosion in adjacent parts of the structure. This latter type of corrosion is known as incipient anode corrosion.

Figure 6.3: Continuing reinforcement corrosion where repaired with epoxy mortar

Source: NZTA (2001).

Cementitious repairs may also promote incipient anode corrosion due to a change in the electrochemical condition of the reinforcing steel adjacent to the repaired area. However, the effect will be much less severe than for epoxy mortars. Recent developments in repair technology have produced a method of mitigating incipient anode corrosion using sacrificial zinc anodes in conjunction with a cementitious repair system.

Extent of repairs

Concrete patch repairs are commonly carried out on a ‘measure and value’ basis. This approach involves defining the methodology and estimating the extent of repairs required before the repair contract commences. The actual extent of repair is then measured as the repair contract proceeds.

Corrosion of steel and the cracking and spalling of concrete may be relatively localised, even though the causes of the problem are quite general. Decisions must be made on the extent of repairs needed. A visual inspection will usually be adequate for deciding on the basic form of the repairs, but not necessarily on the size of the area to be treated. For all-over repairs, e.g. re-building or re-casting, obviously no further examinations are required, but if the method of repair is to be local patching, then further inspection of the surfaces will be necessary before the full extent of the repairs can be estimated.

The area for repair is normally based on the visible spalling damage plus an allowance for additional reinforcement corrosion likely to be detected during repair. Additional areas of repair may be defined where particular circumstances (e.g. low cover depths, poor quality concrete) indicate an increased corrosion risk without any visible deterioration.

Half-cell potential mapping can detect areas of concrete where the steel could be corroding but where the symptoms are not yet visible. Repairs in these areas may be required.

Structural considerations

Where reinforcement corrosion damage is severe, its effect on strength must be considered. Where it is necessary to restore both the strength and durability of an element, the following factors will need to be taken into acccount:

- Unless load is taken off the structural element before repair, for example, by jacking onto props, the repair will only contribute to the resistance of the element to additional loads.
- The ability of a repair to take load will depend not only on its compressive strength, but also on its elastic modulus and on the strength of the bond to the concrete. For example, a material with low modulus requires greater deformation to occur for a stress to develop in the repair than does a material with a higher modulus.
- Differential shrinkage and creep between the repair material and the original concrete will affect the load-bearing contribution of a repaired section. The base concrete, being old, will creep much less than the repair. This, together with any shrinkage in the repair material, will result in proportionately less load being taken by the repair, although the bond between the repair and the base will be stressed.
- Where corrosion of reinforcement is so severe as to warrant its replacement, careful consideration must be given to the replacement methodology, including, but not limited to, splice length, preparation, welding of bars etc.

Thus the way in which a repair will contribute to the restoration of structural strength requires careful assessment, taking into account loading conditions and the properties of repair materials.

Repairs almost always involve cutting behind the corroding reinforcement. The likely effect of this will need to be carefully considered before a remedial scheme is drawn up. Structural considerations may require limits to be placed on the extent and timing of the breakout work. Particular care will be needed in regions of end anchorage of reinforcement and in any work associated with prestressed concrete.

Patch repair methodology

Proprietary patch repair materials may be applied as:

- trowellable mortars
- free-flowing micro-concretes
- spray-applied mortars.

Trowellable mortars are historically the most common type of repair material (Figure 6.4) but may need to be applied in several layers in areas where a substantial depth of repair is required.

Free flowing micro-concretes are placed into preformed boxing and allow large volumes of repair to be completed in one process. These repairs are commonly used in beam soffits.

The preferred application process for spray-applied mortars is by wet spraying of a pre-mixed mortar (Figure 6.5). Dry process shotcrete (gunite) is less suitable as the quality of the applied mortar is controlled by the nozzleman. Spray-applied mortar is more common in medium to high volume of repairs.

Figure 6.4: Initial application (by hand) of a trowellable mortar

Source: NZTA (2001).

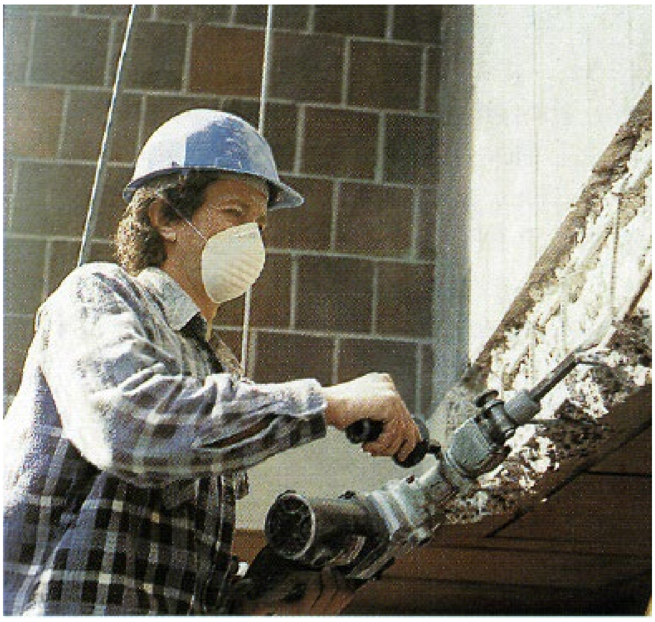
Figure 6.5: Application of sprayable mortar

Source: NZTA (2001).

Correct preparation of the area to be repaired is critical to the performance of all repairs. Recommended procedures are as follows:

- All spalling and poor-quality concrete must be removed. Either of two methods can be used. Ultra high-pressure water-blasting at pressures greater than 80 MPa (11 600 psi) has the advantage that it removes concrete and cleans reinforcing steel in one action. Water blasting may be subject to environmental constraints, particularly over waterways of environmental significance. More traditionally, pneumatic hammers have been used to excavate concrete (Figure 6.6) but they tend to fracture the surrounding concrete and a further operation is required to clean the reinforcing steel.
- Concrete should be removed from around the full circumference of the reinforcing steel and 20 mm beyond it. Removal should continue along the length of the reinforcing steel 50 mm beyond the corroded area. Reinforcing bars should be replaced where a significant percentage of the bar diameter has been removed by corrosion.
- The edge of excavations should be sawcut to a minimum depth of 10 mm. Sawcuts should be perpendicular to the surface, or angled to ‘retain’ the repair material (Figure 6.7).

Figure 6.6: Concrete excavation using a pneumatic hammer



Source: NZTA (2001).

Figure 6.7: Prepared excavation showing sawcut edges and application of reinforcement primer

Source: NZTA (2001).

- All preparation should leave a sound substrate free from dust, loose particles, and any deleterious materials.
- Where the reinforcing steel is within 10 mm of the surface, the bars may be removed (if confirmed as appropriate by an engineer), bent back into the concrete, or left in position and an additional protective layer of cementitious mortar added to the entire concrete surface.
- If concrete is removed with pneumatic hammers, abrasive blast (wet or dry) the reinforcing bars to a bright condition then thoroughly wash them with clean water.
- As soon as practical after cleaning, the reinforcement should be primed or coated to the requirements of the chosen repair system. Coating types available include cementitious slurry and zinc-rich primers.
- Reinstate the excavated area using a proprietary cementitious repair system in accordance with the manufacturer's instructions.

Concrete coatings

Coatings on concrete can fulfil three functions:

- change the appearance of the concrete
- improve surface properties of the concrete
- provide a barrier against the transmission of gases and liquids.

The principal generic types of concrete coating and their various attributes are presented in Table 6.1.

Table 6.1: Concrete coatings

Coating type	Features
Silane-siloxane	Waterproofing, chloride barriers
Silicones	Waterproofing
Stearates	Waterproofing
Epoxy resin	Versatile sealers and coatings Hard wearing, good chemical resistance, but brittle
Polyurethanes	Versatile sealers and coatings Hard wearing, good chemical and weathering resistance, flexibility and toughness
Polyester, vinyl ester, acrylate	Excellent chemical and temperature resistance. Cure at low temperatures
Acrylics	Decorative, good weathering, CO_2 and chloride barriers, crack bridging, allow vapour transfer
Vinyls and synthetic elastomers	Similar to chlorinated rubber
Chlorinated rubber	General barrier uses, weather protection, solvent sensitive
Bitumen	Low-cost waterproofing
Cementitious	Barriers against CO_2, chlorides, water, poor acid resistance

There are a myriad of different proprietary coatings available, with various claims for their performance. Caution should be exercised in the selection of coatings, and expert advice should be sought to validate claims of performance.

AS/NZS 4548 gives guidance on the selection of architectural coatings, but does not cover materials for protecting concrete in severe environments.

Coatings alter the permeability characteristics of concrete and should not be applied to structures where there is a possibility of future treatment by desalination or re-alkalisation.

Coatings are often used to complete patch repairs. Almost all repairs will produce obvious visual mismatch that can be remedied using a coating to provide textural and visual continuity. Apart from improving appearance, such coatings are normally required to provide a barrier against chlorides, carbon dioxide and water to inhibit further deterioration. Polymer modified cementitious coatings have been used to provide this type of protection and are effective at masking physical imperfections of the concrete surface.

Current technology favours a coating system consisting of a silane-siloxane penetrating primer overcoated with a clear or pigmented acrylic membrane. The silane-siloxane blocks liquid water and chloride ions but allows water vapour to pass so the concrete can dry. The acrylic membrane physically protects the silane-siloxane and provides a barrier to carbon dioxide and some protection against moisture while being permeable to water vapour. Repair material suppliers offer such coatings as a part of their patch repair systems.

Coatings may be used to seal and hide surface defects and inactive cracks on otherwise sound concrete. Coatings may also be used on new concrete to provide additional protection in an aggressive environment. Specialist anti-graffiti coatings are also available.

A coating is only as good as the substrate preparation and the application process. Care should be taken to choose a coating that is compatible with the concrete or repair material beneath it. It is important also to ensure that all properties of the proposed coating are understood. As an example, if a coating is required to enhance the appearance of a structure in a marine environment, it should also be resistant to the passage of salt-laden air.

The protective value of a coating is greatly reduced by the presence of any defects such as cracks or pinholes. Long-term durability depends on a number of factors including the chemical composition of the binder, the formulation, the total film thickness, and the application techniques.

Electrochemical Repair

Cathodic protection (CP)

Corrosion or dissolution of a metal can be prevented when its electrical potential is reduced below a certain level such that ions are prevented from leaving the metal surface. The principle of making a metal 'cathodic' relative to its surrounding material has been used for over 60 years to protect ships' hulls, marine structures, and buried pipelines. Steel is cathodically protected when it is kept 700 to 800 millivolts more negative than its surroundings by either:

- connecting it to a more electrically active or anodic material such as zinc, aluminium or magnesium in 'sacrificial anode' systems, which allow the anode to preferentially corrode
- connecting it to the negative terminal of a suitable source of DC power in an 'impressed current' system, with the positive terminal connected to a suitable anode, which might be scrap iron or a corrosion resistant material such as activated titanium.

Use of CP on above-ground reinforced concrete structures began in North America in the mid-1970s to protect bridge decks saturated by de-icing salts. Rapid development of the technique followed with a variety of anode systems being developed. After an extensive series of trials, the US FHWA concluded in 1982 that 'the only rehabilitation technique that has proved to stop corrosion in salt-contaminated bridge decks regardless of the chloride content of the concrete is cathodic protection' (FHWA 2001).

Cathodic protection of concrete is a relatively expensive remedial option. The cost is normally justified only when chloride levels in the concrete are or will be so high that conventional patch repairs will become uneconomic over the life of the structure.

Patch repair of spalled concrete is still required but only to replace the delaminated material and any non-conductive material, e.g. epoxy resin from a previous repair. The system should be designed by a consultant with experience in this type of work in accordance with a recognised standard such as the ISO 12696 or the Australian Standard AS 2832.5.

Specialist contractors are required to install a CP system. Work will include ensuring that all reinforcement is electrically continuous and installing reference electrodes, which are used to monitor the performance of the system. Activated titanium anodes may consist of rods fitted into drilled holes with a graphite-based backfill (internal type), mesh strips installed into grooves cut into the cover concrete, or mesh embedded in mortar on the concrete surface. Other types of surface anodes include thermal sprayed zinc and conductive paints.

For impressed current systems, a source of DC power will be required at a rate of about 10 mA/m^2 of steel surface to be protected. This is usually supplied at less than 24 V via a transformer and rectifier connected to mains supply. In remote areas, current may be provided from storage batteries that are recharged by solar panels or a small wind turbine.

Once installed it is important that the system is regularly monitored, because if the potentials are set too high the steel-concrete bond will be reduced, and there will be an increased risk of hydrogen embrittlement to high-strength steels.

Desalination

When a piece of concrete containing chloride ions is placed in an electrolyte between two electrodes, the negatively charged chloride ions will move towards the positive electrode (anode). If the anode is external to the concrete, and if the driving voltage is high enough, the chloride ions will leave the concrete and accumulate in the electrolyte around the anode.

In practice, desalination is performed by first removing spalling or badly cracked concrete and repairing these areas with a cementitious mortar. An anode mesh is then fixed on spacers at the concrete surface and embedded in a layer of sprayed-on cellulose fibre, or in liquid electrolyte contained in coffer tanks or percolated through layers of geotextile. The anode material itself can be an ordinary steel reinforcement mesh, or, more commonly, an electro-catalysed titanium mesh of the type normally used for cathodic protection. Desalination can take between 14 days and three months to be effective depending on the nature of the concrete and extent of chloride contamination. The anode mesh is then removed and the concrete coated. Specialist contractors are required to carry out a desalination project.

Desalination may be appropriate where chloride contamination is widespread and is the principal cause of reinforcement corrosion (e.g. substructure affected by wind borne salt attack). It may be applied to the whole structure or to the concrete elements at highest risk. The process is not appropriate where the source of chloride contamination cannot be isolated (e.g. a concrete pier in a saline estuary). In this case, cathodic protection would be more appropriate.

Desalination influences all concrete in the treated area. As a result the maintenance-free life of a desalinated structure should be greater than if repaired with a concrete patch repair system, albeit at a higher capital cost.

Re-alkalisation

Re-alkalisation involves drawing an alkaline sodium carbonate solution into the concrete from a disposable electrolytic mass on the surface. The process is driven by a voltage applied between a temporary anode embedded in the electrolytic mass (e.g. wet cellulose fibre) and the reinforcing steel.

Re-alkalisation involves the initial repair of spalled or cracked concrete, attachment of an anode mesh and sodium carbonate supply to the surface, application of an appropriate voltage between the mesh and the reinforcing steel, and removal of the electrolyte once the process has been completed. Specialist contractors are required for a re-alkalisation project. Re-alkalisation is very similar to desalination, except that a different electrolyte is used and it takes only three to seven days.

Re-alkalisation is likely to be appropriate where carbonation is the principal cause of widespread reinforcement corrosion.

The principal advantage of re-alkalisation is that it treats all cover concrete, leaving no areas of carbonated concrete. The initial capital cost of this process is likely to be higher than for a conventional patch repair, but current expectations are that no further maintenance will be required once a structure is re-alkalised, so the long-term costs are minimised.

Crack Repair

General

Cracks can either be ‘active’ or ‘inactive’ (often referred to as ‘live’ or ‘dead’), i.e. those where width varies with time or those where no further movement is likely. It is important to identify the cause and current movement of cracking because active and inactive cracks can be treated differently.

Active cracks

Once the cause of cracking has been established beyond doubt, and any possible steps have been taken to avoid further movement, it is possible to restore the structure to its original strength and durability by injecting the cracks full depth with epoxy resin specifically developed for such an application. Provided that the surfaces of the concrete in the crack are clean and sound, cracks can be successfully filled and repaired by specialised controlled pressure-injection techniques if their width is more than 0.1 mm.

The filling of cracks involves introducing the epoxy resin into the cracks to fill them completely and holding it there while it sets to a non-flowing state. Usually the cracks have to be completely sealed on all external faces to prevent the repair resin draining out.

Resin injection should be carried out by specialised contractors with experience in injection techniques. The formulator or specialist contractor must be able to demonstrate that, when the resin system proposed is injected into the cracks, dry or wet (or both), it will achieve a structural bond to the sides of the concrete at the temperature of the structure.

If it is not possible to establish and rectify the cause of the original cracking, there are two possible solutions.

The first is to cut out along the surface of the crack adjacent to it and treat it as a normal movement joint (or alternatively, cut out a normal straight movement adjacent to the crack after having repaired it by resin injection). This will involve filling with a low-modulus sealant.

The second is to inject the crack with a flexible urethane resin. The methodology and equipment used for this injection is similar to that used for epoxy resins.

Inactive cracks

The most significant inactive cracking on bridge structures is plastic cracking (both settlement and shrinkage). The cracks are generally fine and relatively straight, with individual lengths typically of up to 1 m. They should be filled with an injection resin as above or a polymer-modified cementitious slurry well worked into the crack. Early treatment is essential if contaminants are to be kept out of the crack.

6.5 Structural Steel

6.5.1 General

Most metal bridge superstructures and substructures in Australia and New Zealand are of predominantly low carbon steel (mild steel) construction, although some, often notable, structures still stand from an earlier era when wrought iron, cast steel and cast iron materials played a significant part in bridging works. More recent examples of steel bridging may incorporate high-tensile steel or stainless steel components and fastenings. For the purposes of evaluating the condition of bridges, references to steel may be considered to apply equally well to iron unless explicitly stated otherwise.

Steel bridge components usually take the form of sections hot rolled to standard sizes (or plates formed to standard dimensions. The standards used for bridge components constructed in pre-metric days are now likely to be obsolete. Steel bars, tubes, cables and castings may also have a structural function in some bridges.

Steel bridge components will generally be fastened with rivets, mild steel bolts, high-tensile steel bolts, or some patented fastening device, or may be welded. Fastenings may be designed to act in shear as individual members or may be intended to provide a clamping force across an interface to permit the generation of frictional forces between adjacent components.

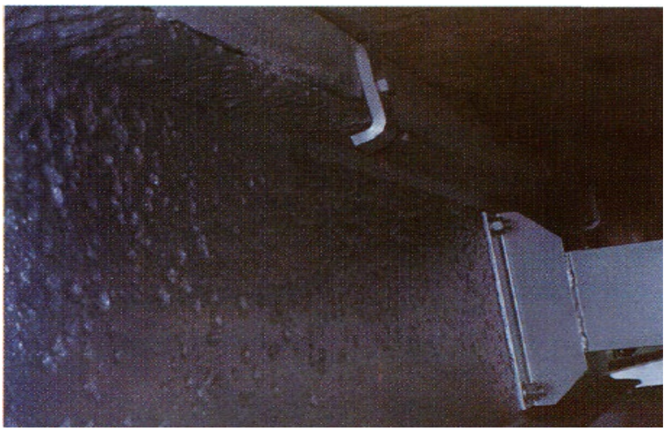
Although some use has been made of specially alloyed steels, the durability of the structural iron and steels in general use in bridging generally depends on the locality in which the structure is built and on the quality and integrity of the protective coating. Rates of corrosion of unprotected metal vary considerably from region to region, with some coastal and geothermal areas presenting very severe conditions for structural steel. Similarly, the life and performance of protective coatings vary significantly between regions. Coatings on different parts of a structure will also tend to break down at different rates.

Structural repair and maintenance (Figure 6.8 and Figure 6.9) of steelwork includes the replacement and maintenance of protective coatings (Figure 6.10 and Figure 6.11), repair of corroded members, replacement of damaged members and defective fastenings, and remedial work associated with fatigue cracking. These problems will have been identified during the inspection and evaluation process. See also Tilly et al. (2008).

Figure 6.8: Rolled steel joist span before maintenance

Source: NZTA (2001).

Figure 6.9: Rolled steel joist span after maintenance



Source: NZTA (2001).

Figure 6.10: Coating failure caused by inadequate primer

Source: NZTA (2001).

Figure 6.11: Failure of wax barrier coat

Source: NZTA (2001).

6.5.2 Defects

General

Defects in a steel bridge will generally appear as a result of the environment in which the bridge exists or as a result of a planned (or unplanned) loading history. Defects may also have been incorporated into a structure at the time of its construction through poor detailing, workmanship or manufacture.

The defects that commonly affect steel bridges are discussed in detail in *Section 8* of *AGBT Part 2: Materials* and are summarised below:

- protective coating failure
- loss of section
- loose or defective fastenings

- cracks
- weld defects
- impact damage
- deformation and distortion
- manufacturing defects
- faults in detailing.

6.5.3 Inspection

Steel bridges are inspected with the purpose of identifying any defects that may be present in the structure and to establish causes for these defects. Defects that are likely to affect the strength, safety, or serviceability of a bridge are programmed for attention as part of the remedial and maintenance work cycle.

An inspector should have a good understanding of a bridge before it is inspected. This is particularly important for complex bridges. Where appropriate and available, the following data sources should be referred to before starting an inspection:

- plans and drawings (including those showing modifications to the original structure)
- photographs (both recent and historic)
- the most recent inspection reports
- recent maintenance history
- strength and rating calculations (for both static and cyclic loading conditions if available).

A visual inspection will systematically cover the whole surface of the steel structure at close quarters paying particular attention to areas that:

- are highly stressed
- undergo significant stress reversal
- are poorly detailed
- have been observed to be defective during previous inspections.

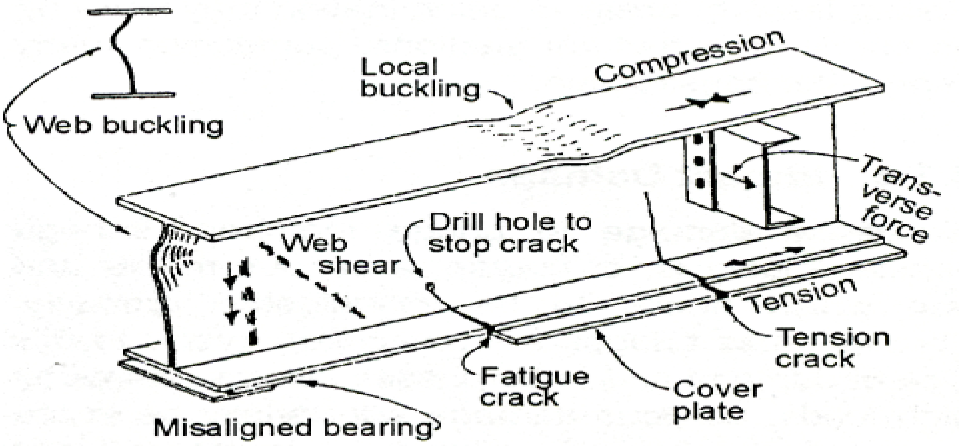
The following matters are critical to the success of a steel bridge inspection:

- Detailed notes must be taken of the condition of the protective coating on all parts of the structure using a standard method of assessment (e.g. ASTM D610 for rust ratings). Dimensions and locations of significant areas of loss of section should be noted.
- Signs of rust staining should be looked for around the heads of fasteners. This may indicate that they are loose. Confirmation can be obtained by lightly tapping the fastener with a hammer.
- Fasteners that do not conform to proper standards of installation should be noted.
- Cracks will usually show in the first instance as a trace of rust emanating from a stress raiser. The highest loaded bolt or rivet in a joint should be carefully examined in areas that are expected to be susceptible to fatigue. Particular attention should be paid to the ends and edges of welds. Secondary loading effects should be taken into account when looking for possible cracks.
- The presence of a suspected crack should be confirmed by NDT by an operator certified by the Australian Institute for NDT or New Zealand Certification Board for Inspection Personnel (CBIP) using suitable equipment. Dye penetrant and magnetic particle techniques are likely to be used in the first instance. Radiographic and ultrasonic methods may also be used for specific cases.

- Deformations and distortions will often show up as cracking or flaking paint. Any deviation can be picked up by sighting along the line of a member. Measurements of any significant deviations from the true line should be recorded.
- Probable causes of defects should be determined if possible at the time of inspection. If a cause is not immediately apparent, specialist advice may be needed.

The location and description of all defects must be methodically recorded to allow proper evaluation of their effects and subsequent monitoring or repair. Figure 6.12 shows possible faults in a steel beam.

Figure 6.12: Possible faults in a steel beam



Source: NZTA (2001).

6.5.4 Evaluation

In one way or another, all observed defects will have an effect on the strength or serviceability of the bridge. Defects that reduce the capacity or durability of the bridge or which present an immediate serviceability problem require remedial action, but others may not. The purpose of evaluation is to determine the relative significance of each defect so that the load-carrying capacity of the bridge can be reassessed and so that any remedial work required can be given proper priority. Evaluation will also assist in determining future strategies for maintenance or replacement.

The evaluation of the effect of some defects can be a complex process requiring a thorough understanding of the behaviour of the structure concerned. The interaction of primary and secondary load-carrying members, the effect of imperfectly pinned joints and the possible presence of alternative load paths need to be appreciated. A basic understanding of metal fatigue and crack mechanics is necessary to evaluate problems of this nature. The risk of failure of certain members and the consequences of such failures may need to be considered.

The allowable load factors or material stresses used to evaluate the effect of a condition are provided for in AS 5100.7 and jurisdiction specific manuals such as the NZTA *Bridge Manual* (NZTA 2016b). The frequency and type of traffic, the age and remaining life of the bridge and the size and importance of the bridge will all need to be considered in arriving at appropriate parameters. Primary load-carrying members such as main beams might need to be treated differently from secondary members such as bracing components.

Analysis of all members is required to assess the safe allowable loading for the bridge.

6.5.5 Repair of Protective Coatings

The level of maintenance required will generally be determined by the condition of the coating, but the maintenance strategy will be influenced by the ease of access. Often, removal of accumulated debris and washing of contaminants from the coating surface are all that is necessary. A regular cleaning program with minor spot painting will greatly increase the useful life of the protective coating.

To maintain in good condition a shop-applied, high-quality system it is usually more economic to carry out programmed maintenance painting than allow it to completely degrade and then attempt to replace it in situ. A field-applied coating is unlikely to give the same performance. Early and regular maintenance to touch up minor defects or upgrade areas with inadequate protection will allow any system's full potential to be achieved and is strongly recommended.

For small items (e.g. handrails and brackets) that are severely degraded, it may be more economic to remove them and hot-dip galvanise them, rather than repaint, as shown in Figure 6.13

Figure 6.13: Pitted steel posts refurbished by galvanising

Source: NZTA (2001).

Surface Preparation

After the removal of contaminants from a coating's surface by rain or maintenance washing, the next most important factor affecting the life of a protective coating is the surface condition and cleanliness at the time of its application. Where rust exists it is usually cost-effective to remove it by abrasive blast cleaning after first washing to prevent contaminants being driven into the steel surface. The use of so called 'rust-converters' is not recommended as their performance in independent tests against conventional systems has usually been disappointing.

The degree of surface cleanliness required is usually based on a Standard such as AS 1627.9, which contain descriptions and photographs of different initial surface conditions of rust (e.g. Class D for pitted steel) and corresponding descriptions and photographs for various grades of preparation using hand tools (St grades) and abrasive blast cleaning (Sa grades). Grade Sa $2^1/_2$, for example, describes a 'near-white metal' surface finish and is equivalent to an AS 1627.4 Class $2^1/_2$ or SSPC-SP 10 finish.

Appropriate safety precautions for cleaning and recoating steelwork coated with lead-based paint are given in various road jurisdiction's specifications as well as AS 4361.1.

In some situations, rust can only be removed by mechanical methods. In this case care should be exercised to ensure the surface is not burnished, which will reduce adhesion. The relevant standard is AS 1627.2 (hand and power tools).

High-pressure water-blasting is a useful method of removing aged or non-adherent coatings and contaminants, but some coatings (e.g. epoxies and urethanes) may require a light abrasion to provide a mechanical key for subsequent coats to adhere to. This is not necessary when upgrading solvent-borne coatings such as chlorinated rubbers and vinyls.

Where failure has resulted from the coating cracking at sharp edges or rough welds, these should be rounded or smoothed by grinding before re-painting. Similarly, thick edges of remaining paint layers should be feathered by hand sanding.

Coating System Selection

Many variables must be considered when selecting a maintenance coating system. Important factors are:

- compatibility with existing coating
- standard of surface preparation achievable
- climatic conditions under which re-coating will be carried out
- whether time constraints exist.

In addition, other relevant factors are:

- ease of future maintenance
- number of coats required
- appearance
- proven performance in a similar environment
- whether application is to be by unskilled labour or by a contractor with specialist equipment.

AS/NZS 2312.1 *Guide to the Protection of Structural Steel Against Atmospheric Corrosion by the Use of Protective Coatings: Paint Coatings* discusses these factors and contains a step-by-step checklist to help in the planning of a coating maintenance project. It is an essential reference for coating selection.

Pre-1970 systems based on red lead primer applied to flame-cleaned steel have been superseded by higher performance, chemically cured paints applied to abrasive blast-cleaned steel. Zinc phosphate-based primers are suitable for hand-prepared steelwork, but zinc-rich primers over blast-cleaned steel provide the best foundation to a long-life coating system.

Chlorinated rubber-based systems were often specified because of their superior resistance to moisture and chlorides. Being solvent-borne, they are also easily re-coated, over-spray problems are minimised, they can be applied at low temperatures, and they are ‘single pack’ products.

Moisture-cured urethane systems are now available. These are single-pack materials that are fast drying and are tolerant of a very wide range of climatic conditions. Because of their flexibility and compatibility with a wide range of other coatings they are often used to ‘encapsulate’ old lead-based coatings, which can cause environmental and health hazards if removed without proper containment.

Finishing or top coats are often pigmented with minute flakes of aluminium or micaceous iron oxide (MIO) to reduce breakdown of the coating (chalking) by ultraviolet rays and to give the structure a metallic appearance. MIO will also reduce the coating’s permeability and provide a key for future maintenance.

The performance of coatings that rely solely on barrier action to protect the steel (e.g. epoxy mastics) can be improved significantly by the use of a primer, which resists undercutting when the barrier is damaged or defective. Use of barrier coatings alone is often not cost-effective, especially when applied over residual rust.

The thickness and number of coats required will be determined by the severity of the environment, and the planned time to next maintenance. It is often beneficial to give the members most at risk (e.g. the bottom flanges of girders) an extra coat.

Codes of Practice such as AS/NZS 2312.1 give recommended dry film thicknesses but, for major structures, advice from a paint manufacturer or independent coatings consultant should be sought. As well as paints, the use of thermal metal spray or galvanising should be considered within the context of the structure’s total life cycle cost.

The Australian Paint Approvals Scheme provides members with lists of approved paint brands that have been found to comply with composition or durability requirements of its specifications and that are manufactured within an approved quality assurance system.

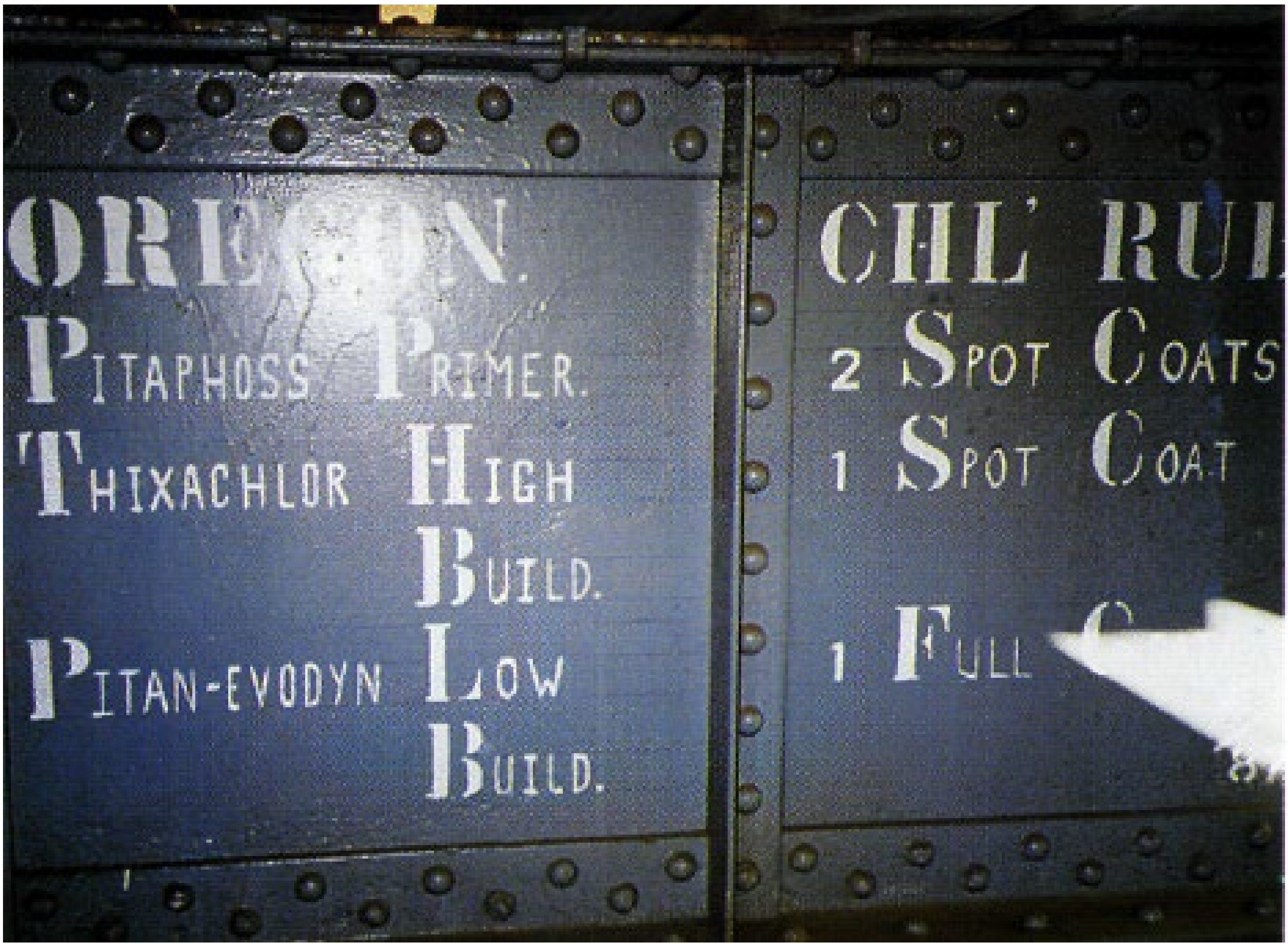
Application and Supervision

The weak link in a painted system is usually at sharp edges where it is difficult to obtain the specified coating thickness, especially when applying by spray. It is therefore recommended that before each main coat is applied, a ‘stripe coat’ be brushed onto the edges of all flanges, holes, welds, and rivets. Similarly, severely pitted surfaces or pin-holed primer coats should receive a second coat of primer.

To ensure that recommended procedures are followed, employment of an experienced third-party coatings inspector (e.g. CBIP-certified) is often a good investment. High-performance systems depend on good surface preparation and application under suitable conditions. As a minimum, daily records should be kept of application times, paint batch numbers, brands used, environmental conditions during application and dry film thickness measurements. Suitable quality control forms are published in AS 3894.10.

It is recommended that, on completion, details of the surface preparation and the paint type and thickness be stencilled onto a readily visible member for the benefit of future maintenance personnel (Figure 6.14).

Figure 6.14: Painting record



Source: NZTA (2001).

6.5.6 Repair of Defective Members

The need to repair a defective bridge component will have been established during the evaluation process (Section 6.5.4).

Because of their relative importance, differing approaches are usually taken with the repair of primary load-carrying members and secondary members.

In many instances, there is a choice of either replacing a defective component in its entirety or providing some sort of splice or strengthening plate, taking into account any introduced eccentricities. There is usually also a decision to be made as to whether the replacement or supplementary member will carry dead load as well as live loads. The final choice of method will take into account the ease of component removal and replacement, cost factors and the degree of deterioration of the component.

Provided other members can carry all the dead load as well as their share of the live load without detriment to the capacity of the structure, it will generally prove simplest to have the replacement member carry no dead load. This will require minimal temporary support and there will be no need to calculate and provide the proper degree of prestress during installation in order to allow for dead load. If, however, it is essential that the new member carries its full share of dead load, installation becomes considerably more difficult. Methods involving full temporary support for a section of the bridge, relieving frames, preheating the new component, or one of several methods of tie shortening could be required.

It may be possible to straighten bridge components that have been bent and deformed as a result of vehicle impact or some similar event to an acceptable state after heating the affected area. Nicks and gouges may be ground out to remove local stress-raisers, and cracks prepared and welded. Clearly, this sort of treatment is only possible if the resulting static and dynamic ratings of the repaired member are acceptable. Alternative methods for this type of damage include member replacement or lapping with additional components.

Loss of cross-sectional area in steel members through corrosion may need to be repaired even if the strength of the structure is not affected. For instance, corroded girder flanges may be so pitted that water is retained for long periods and corrosion remains active. An epoxy filler could be applied to a mechanically cleaned surface to improve drainage and extend the useful life of the member.

Buckling in members can often be relieved by investigating and removing the cause of the problem rather than by treating the member concerned. In some cases, however, a member may need additional stiffening or bracing, or may need to be shortened. This type of work should not be initiated without first considering the effect of a stiffer or shorter member on the remainder of the structure.

6.5.7 Repair of Defective Fastenings

Incorrectly installed high-strength friction grip bolts and fasteners can probably best be remedied by replacement with like components installed correctly (after first investigating the reason for the substandard installation).

Loose or corroded rivets may be replaced with friction grip fastenings either singly or in groups. It is probably best to replace the remainder of an entire rivet group once half of the original rivets have been replaced. New fastenings should be painted immediately after installation.

6.5.8 Treatment of Fatigue Problems

Fatigue problems are normally identified in the crack growth stage. Unrestricted, the crack is likely to continue to grow slowly until the critical crack length is reached and sudden fracture occurs. Crack growth can be slowed and sometimes stopped altogether by eliminating the small area of high local stress at the crack tip by drilling a small, smooth hole (e.g.10 mm in diameter) at or just ahead of the tip.

Investigating the reasons for the failure is essential. The general approach to overcoming this type of problem is to eliminate the stress concentrations that have given rise to the fatigue crack and continue to assist its growth, and then to look at ways to improve the situation.

It may be possible to reduce the stress in the area of the crack by introducing new load paths or removing redundant members, especially if secondary forces contribute to the stress intensity, or by re-designing a joint or connection. In conjunction with this work, peening the area of metal at the root of the crack with a pneumatic peening hammer will introduce local surface compressive forces, which are highly beneficial in slowing or arresting the progression of the crack.

Cracked welds (where the crack does not extend away from the weld into parent metal) can usually be effectively repaired by grinding out and re-welding the section of defective weld, then peening the weld until plastic deformation causes the metal to become continuously smooth. Peened indentations will be between 0.5 mm and 0.8 mm deep.

Where small cracks have initiated from rivet holes, replacing the rivets with high-strength friction-grip bolts will reduce stress concentrations and introduce compressive stresses across the joint. This method cannot be used if the crack has progressed too far from the rivet hole.

If none of these methods are appropriate, the component will need to be replaced. Suitable modifications to the original design must be made to ensure that the stress raiser that caused the problem is eliminated.

6.5.9 Preventive Maintenance

The preventive maintenance of a steel bridge starts at the design stage when proper attention should be given to the detailing of components and connections to ensure that they have adequate strength and serviceability for the structure's design life and adequate clearance for future maintenance. Provision of access to facilitate future inspections and maintenance should also be considered. Other practices that will assist in minimising maintenance of an in-service bridge include:

- proper selection of protective coating type, proper surface preparation and application over the entire coated surface
- regular washing and cleaning of protective coating surfaces
- regular clearing and cleaning of drainage ports
- Improving drainage in areas that are not adequately drained
- ensuring bearings are operating correctly
- maintaining the presence of adequate expansion gaps.

In addition, potential problem areas should be identified and appropriate action taken before structural defects become manifest. Such matters include:

- details involving abruptly curtailed cover plates on flanges or sharp re-entrant angles should be improved if they are likely to become fatigue risks
- poor welds should be ground out and replaced
- selected rivets can be replaced with high-strength friction-grip fasteners to improve the fatigue characteristics of a rivet group (e.g. the leading rivets in a joint or cover plate)
- eccentricities in joints and connections may be improved to reduced unwanted bending stresses
- the point of support of bearings may be redefined to improve eccentric movement effects.

6.6 Timber

6.6.1 General

Many timber bridges exist in Australasia. Timber bridges once formed a major part of the roading network. While many of the original timber bridges have been replaced with concrete and steel structures, timber is still a significant component of the bridge stock (Figure 6.15).

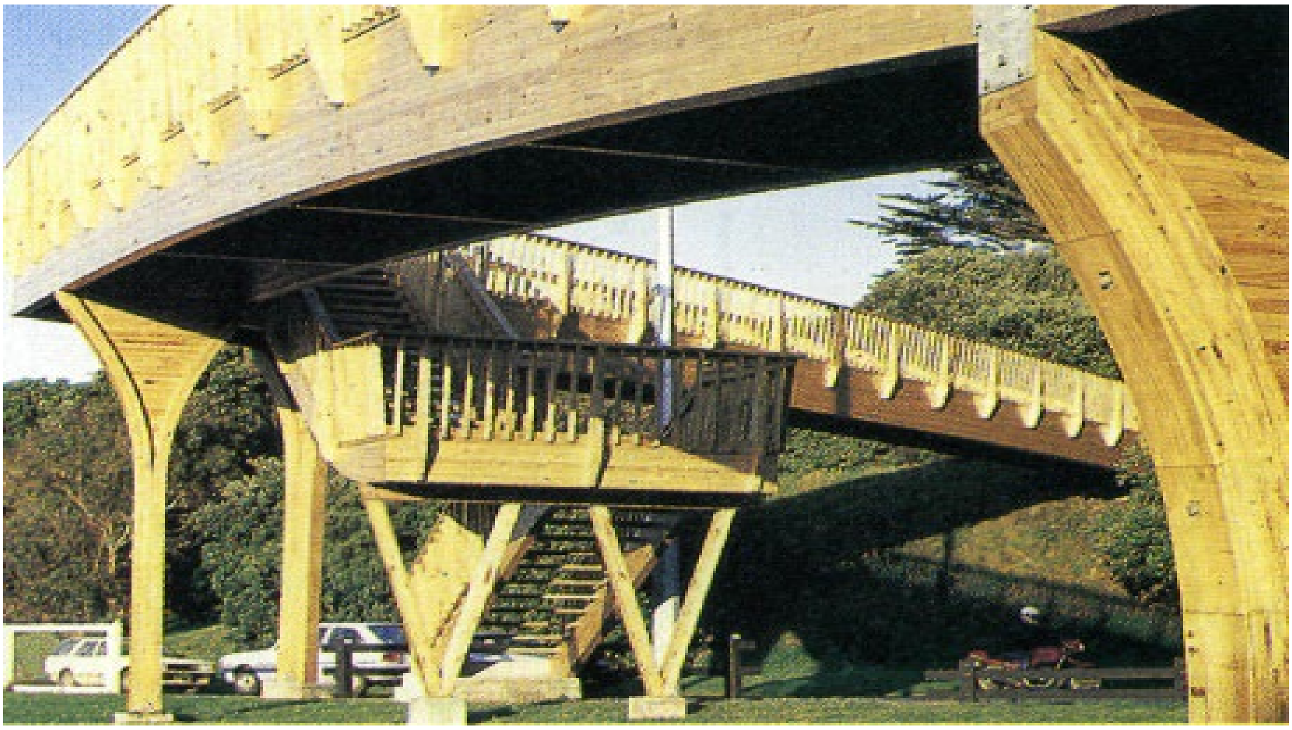
Figure 6.15: Timber truss bridge comprises timber members bolted together

Source: NZTA (2001).

Timber has been used in all parts of bridge structures and in a wide range of structural types. Structural form is largely dictated by the size of the timber members that can be cut from the natural timber source, the imagination of the designer and the skill of the builder. Timber has been used for piles, piers, abutments, beams, trusses, decking, kerbs, and rails, both in all-timber bridges and in combination with steel and concrete. A more recent trend is lamination of timber members using glue, nails, bolts, or stressing tendons to provide larger and more rigid structural units (Figure 6.16).

A New Zealand-wide survey of bridges found that timber components consistently showed the highest percentage of defects. Decay is the most serious timber defect and is the reason for most timber bridge maintenance needs.

Figure 6.16: Laminated timber bridge



Source: NZTA (2001).

Timber as a bridge-building material is not durable unless it is appropriately treated and well maintained.

Most Australian timber bridges are constructed from local hardwood. In New Zealand many are constructed from mixed Australian hardwoods and New Zealand heart native timbers. Some, however, including major structures, have preservative-treated radiata pine components. Because of their greater age, the former tend to have more defects requiring maintenance than the latter.

Some preservative treatments used before the mid-1960s involved formulations and processes now considered to be inferior. Defects, particularly in decks, can be expected in structures containing these treated timbers.

6.6.2 Defects

Many defects found in timber bridges may be explained by consideration of the material properties, coupled with installation and usage circumstances.

The defects that commonly affect timber bridges are discussed in detail in Section 9 of *AGBT Part 2: Materials* and are summarised below:

- decay
- splits, checks, shakes
- sloping grain
- knots
- accumulations
- holes
- connections and fasteners
- looseness
- misalignment
- abrasion
- vehicle impact.

6.6.3 Inspection

The aim of inspection is to identify all defects present, to establish their causes, and to evaluate their rate of advancement so that this can be followed up with an assessment of seriousness and programming for remedial action required (Figure 6.17).

Preliminary information relevant for inspection of timber bridges includes:

- age of structure
- original drawings and subsequent treatment and repairs
- prior inspection, drilling, and assessment history
- a schedule of all main timber components.

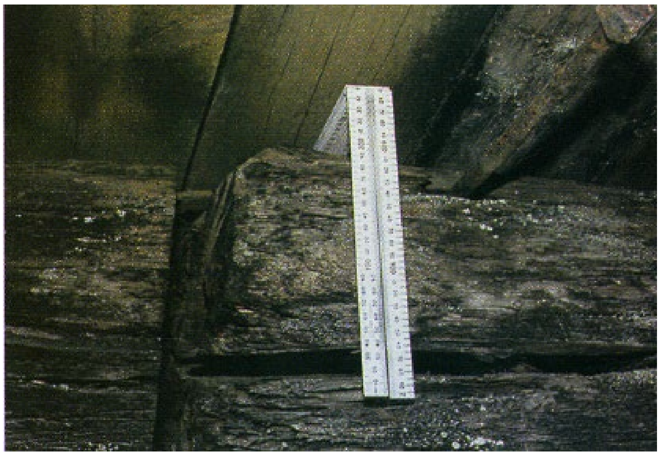
A visual inspection would include a thorough search for any of the following defects:

- accumulation of dirt, vegetation or dampness on any joints or surfaces
- drainage defects that might add to dampness of timber components
- decay particularly at joints or areas of possible dampness
- cracks or splits that might aid moisture penetration and retention in the timber (Figure 6.18)
- insect or borer infestation
- loose joints or corrosion of metal components
- movement between deck and stringers or looseness of running planks
- abrasion of the deck surface or piles
- soundness of painted or coated surfaces (Figure 6.19).

Figure 6.17: Old timber superstructure with an assemblage of beams

Source: NZTA (2001).

Figure 6.18: Large split at bearing area of beam greatly reducing load capacity in shear



Source: NZTA (2001).

Figure 6.19: Maintaining an unbroken painted coating is a problem on old timber rails

Source: NZTA (2001).

Where any watermarks, stains, or moss growth suggest possible areas of decay, particularly at joints, ends of beams and ground contact areas, such areas should be probed with a long, thin, sharp steel instrument, sounded with a hammer, and suspect areas followed up with discrete drilling or coring (Figure 6.20). The pilodyn instrument, which fires a spring-loaded pin, can be useful in obtaining a measure of resistance to penetration that may be related to density and hardness.

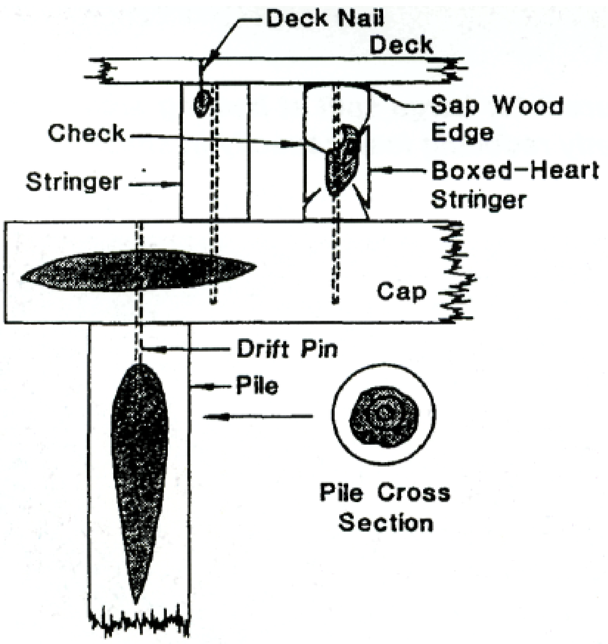
An electronic moisture meter can show which timbers are moist enough to be at risk from fungal decay. A number of other instruments and techniques are available but are more suited to research or special investigations for reasons of time and cost.

Exploratory boring is necessary to confirm the presence of decay in hardwood timber and to estimate the dimensions of sound timber that remains. Drilling with a sharp auger is the most common method, but the shavings are difficult to interpret. Methods that produce an undisturbed core from a borer or plug cutter are desirable but difficult to implement in the field. Alternatives to drilling with an auger bit include use of a Resistograph which employs a 2 mm drill bit for measuring the drilling resistance of construction timbers. The Resistograph is both a complementary tool in undertaking in-depth investigations of timber structures and as an alternative to conventional timber drilling. The minor size of the drill bit ensures the durability and structural capacity of the component is unaffected by the inspection. The Resistograph provides electronic reports of the drilling resistance which can be kept as a permanent drilling record, as opposed to traditional data recording which is dependent on interpretation of the inspector.

Irrespective of the method employed, drilling should be at locations where decay is likely to occur.

Over-drilling must be avoided and drill holes must be treated with preservative and plugged with treated dowels or plastic plugs to facilitate future inspection.

Figure 6.20: Locations where decay is likely



Source: NZTA (2001).

The objective of boring is to identify defects and measure them for assessment purposes. Measurements must be thorough and attempt to fully define all observations with a schedule covering all main members of the structure, recording location, extent, rate of change and assessed effect on performance.

6.6.4 Evaluation

Evaluation of timber structures is in most cases a complex task that, after consideration of all the best information that is available, may still involve a large degree of uncertainty and rely largely on experienced engineering judgement.

It is important that conclusions are practical and economic. Sometimes an evaluation will suggest that bridges that have been satisfactorily carrying current legal loading should be restricted to light vehicles only, despite there being no evidence of significant deterioration. Similarly, major and expensive repairs are carried out on timber bridges that in reality are in need of early renewal.

Evaluation requires the assessment of allowable timber stresses. Standard allowable design stresses are the basis for assessment, with judgement based on a number of specific member and site conditions including:

- species and grade of timber member
- size of member with allowance for defects such as splits, knots, abrasion, and decay. No effective stress capacity should be assumed in identified areas of decay
- function and importance of member either as primary member, e.g. main beam, or secondary member, e.g. deck plank, and degree of load sharing
- traffic intensity and frequency of heavy vehicles and overloads
- size and importance of the structure
- degree of risk in the event of member failure
- type of member and the reliability of stress assessment
- performance record both historic and by proof loading
- programmed intention for replacement.

It may be helpful to assess upper and lower bounds for allowable stresses and complete the evaluation for these two cases.

Figure 6.21: Timber bridge with posted load restriction

Source: NZTA (2001).

Analysis of all members is required to assess the safe allowable loading for the bridge. Procedures should follow those set out in the relevant jurisdictions manual, for example, the NZTA *Bridge Manual* (NZTA 2016b).

Old timber has a limited life. Evaluation should aim at quantifying allowable vehicle loads and lead to the determination of a maintenance strategy and the appropriate time for repair or replacement (Figure 6.22). For old timber bridges with obvious evidence of decay there is usually a problem of determining where to stop once member replacement has started. The main objective of evaluation is then to determine when total bridge replacement is necessary. Some old bridges have a heritage value and deserve preservation, possibly for pedestrian use (Figure 6.22).

Figure 6.22: Old truss bridge beyond its life limit for vehicles

Source: NZTA (2001).

6.6.5 Philosophy of Repair and Maintenance

The extent and location of defects detected during inspection will be the major influence on the repair and maintenance strategy. The success of any repair program will depend on knowledge of why the defect has occurred and the measures taken to prevent its recurrence. Experienced judgement is very important. Maintenance may be divided into three categories:

- structural repair
- rehabilitation and component replacement
- preventative maintenance.

Often old timber bridges have a history of neglect, and rehabilitation or replacement are the only viable options.

The key to a successful repair and maintenance program for hardwood bridges is management of the interaction of water with the structure. For softwood bridges, it is the use of correctly treated replacement timber. A continuing and systematic monitoring program must be instituted because any design solution hinges on an on-going commitment to inspection.

Timber bridges form a part of the historical and cultural heritage of many road jurisdictions. Some attention has been given to preserving and restoring some of the best remaining examples of older timber bridges. Obtaining suitable identical replacement timber is usually a problem and other species have been used.

See Table 6.2 for guidance on the selection of species for new timber for particular structural components.

Table 6.2: Guide to the selection of species for new timber

Structural component	Species	
	Preferred	Others suitable
Piles – unprotected in tidal waters	Turpentine (with bark left on)	-
Piles (other) Sills	Grey box Ironbark Tallow wood Wandoo Jarrah	White mahogany* WA blackbutt Yellow tingle
Girders	Grey box Grey gum Ironbark Tallow wood Wandoo Jarrah	White mahogany* Spotted gum White topped box WA blackbutt Yellow tingle
Capwales Headstocks and half caps Wales Bracing Stringers Cross girders Corbels Fender posts	Grey box Grey gum Ironbark Tallow wood Wandoo Jarrah	White mahogany* River red gum* Spotted gum White topped box WA blackbutt Yellow tingle
Decking Kerbs Bolting planks Longitudinal sheeting	Blackbutt Forest red gum Grey gum Grey box Ironbark Spotted gum Tallow wood Yellow stringybark White mahogany* Wandoo Jarrah	River red gum* Silver-top (white-top) Stringybark Tasmanian blue gum* White topped box Brush box* WA blackbutt Yellow tingle
Gravel boards Abutment sheeting Wing caps	Blackbutt Bloodwood Grey box Grey gum Ironbark Tallow wood Yellow stringybark Wandoo Jarrah	Forest red gum River red gum* White mahogany* White topped box WA blackbutt Yellow Tingle

Structural component	Species	
	Preferred	Others suitable
Handrailing Hangers (yokes or cross-pieces)	Grey box Grey gum Ironbark Tallow wood Blackbutt White mahogany* Wandoo Jarrah	River red gum* Spotted gum Silver-top (white-top) Stringybark Tasmanian blue gum* White topped box WA blackbutt Yellow tingle
Truss members (flitches, etc.)	Grey box Ironbark Tallow wood	-

*Note: Common names are used here for species; the equivalent botanical names are as AS/NZS 1148 'Timber: Nomenclature: Australian, New Zealand and Imported Species'. Note that the above list is based on Western Australian practice, and that experience elsewhere may differ. For example, NSW experience has shown that white mahogany, river red gum, brush box and Tasmanian blue gum (marked with an *) are unsuitable.*

6.6.6 Repairs – General

The general approach to the repair of timber bridges is to replace defective members. Bolted connections make this simple.

Structural repairs will depend on the cause of the defect and on the function and circumstances of the component in relation to the structure. Each case requires specific assessment and experienced judgement.

Jurisdictions with a large number of timber bridges remaining on their networks typically have standard treatments and repair details documented.

- Minor defects (such as splits and cracks) are repaired with bolts and/or steel rings.
- Timber bridges can be strengthened when a whole deck is replaced by using larger diameter girders (B class upgraded to A class) or using steel girders instead of timber.
- Decks can be strengthened and widened (by about 600 mm) by using thick (150 mm) plywood sheets such as 'Bridgewood' or equivalent. This has the added advantage of allowing a standard 'Armco' steel rail or road safety steel barrier rail to be incorporated in the repair/upgrade of the bridge. It should be ensured that widening the deck does not overload the existing components comprising the structure.
- Some agencies have placed concrete overlays on top of the timber decks. This has advantages in waterproofing the timber deck and girders, but makes subsequent maintenance of girders more difficult. This is rarely used now.
- Replacement of timber deck planks with metal trough decking (running transversely across the timber girders) was also popular in the 1970s and 1980s, but the steel troughing often suffered rapid corrosion (when filled with road base materials) and fatigue cracking failures. So, this method is no longer popular.
- 'High-tech' repair methods using fibre composites have been trialled on timber bridges overseas, often on rectangular beams or laminated veneer lumber (LVL) beams. It is difficult, or impossible, to glue fibre composites to old timber surfaces, and such techniques do not appear practical or cost-effective for repairing/strengthening of typical old Australasian bridges.
- Timber piles that rot above ground can normally have the defective section cut out and a replacement length spliced in. Care must be taken in the design and detailing of splices to ensure horizontal loads are accommodated and there are usually limitations placed on the number of piles that can be spliced in any one pile group.

- Most commonly piles rot just below the ground level where there is both air and moisture available to the active organisms:
  - Sometimes it is practical to excavate down to find a section of sound pile and replace the rotted section with reinforced concrete spliced to a sound pile above ground.
  - Alternatively a steel pile section can be driven adjacent to the defective pile by removing a few deck planks and driving from deck level. Where there are sufficient clearances, small pile frames can fit under bridges or two external piles can be driven outboard and joined by an RC beam at bed level to support interior piles.
- As many older timber bridges are on replacement programs, the required design life for repairs may be only 10 or 20 years, so durability of the repair is normally not a problem. If long-life repairs are needed, then the normal durability requirements of new construction should be applied.
- Australian hardwoods are generally durable, with certain species highly durable. Timber bridges have remained in service for over 120 years. Provided they are correctly maintained, a long service life can be expected, but some issues still require consideration:
  - Poor maintenance practice (e.g. spiking deck planks to girders, or failure to identify and treat termites) can lead to much lower service life.
  - Timber to replace girders and piles comes from old growth forests and is now scarce and expensive. As noted above, steel girders are often used to replace timber girders, as are FRP and engineered timber. Care should be taken to ensure that replacement members are of equivalent stiffness otherwise the distribution of loads within the superstructure may be altered.

6.6.7 Repairs – Decay

Three broad strategies are available for structural repairs required because of decay of timber bridges:

- replacement of decayed members
- eradication of the source of decay in affected members
- prevention of its recurrence.

In many cases a combination of at least two of these strategies will be necessary.

Replacement of Decayed Members

Replacement will be necessary when the extent of decay has reduced the residual strength of the affected member to an unacceptable level. Unfortunately, little practical information is available to correlate extent of decay with strength loss, and the decision to replace must be appraised in relation to the type and intensity of stresses imposed on that portion of the member that contains decay.

When the decayed member is removed, all adjacent timbers must be checked to ensure that decay has not spread from the defective member. It is important to remove completely any possible source of future infection. Thus, if repair involves cutting out obviously decayed sections of affected members, it is necessary to remove at least 500 mm, in the grain direction, of apparently sound wood since it is likely that hyphae (thread-like elements) of decay fungi have penetrated that far from the obviously decayed zone.

Replacement with Preservative-treated Softwood

All replacement timbers must be treated in accordance with standards accounting for the species type, moisture content and exposure to known environmental conditions. For example, NZTA specifies provisions in NZMP 3640. Any replacement timbers above deck level should be treated to Hazard Class Specification H3; deck timbers, including running planks, to Hazard Class Specification H4; and any part of the superstructure that comes in contact with soil, or which will have a permanently high moisture content, to Hazard Class Specification H5. Replacement piles in fresh water should also be treated to Hazard Class Specification H5, but in estuarine or sea water, replacement piles must be treated to Hazard Class Specification H6.

Figure 6.23: New bracing provides additional support to an old hardwood superstructure

Source: NZTA (2001).

Figure 6.24: Replacement sections of radiata pine nail-laminated deck

Source: NZTA (2001).

Replacement with Naturally Durable Hardwoods

If it is necessary to replace decayed hardwood with similar material, only new timber of Durability Class 1 or 2 should be used. It is unwise, other than for temporary repairs, to replace decayed hardwood members with those salvaged as apparently sound from other structures. Experience has shown that the residual life of such timber may be far less than anticipated at the time of salvage.

Reinforcement of Decayed Members

Occasionally it may be impractical to replace decayed members because of their location in the existing structure, but reinforcement with a parallel member or other bracing may be feasible. Because infection can spread from one member to another, untreated naturally durable hardwoods should not be used as reinforcement in close proximity to decayed members unless decay eradication procedures have been applied to the original member. Steel, preservative-treated softwoods, or concrete are preferred alternatives.

A thorough structural analysis is required to ensure the capacity of the repair and verify load distribution to members. Situations that introduce eccentric loads or tension perpendicular to the grain should be avoided.

Epoxy Repairs

Epoxy resins can be used for timber repair as bonding agents (adhesives) or grouts (fillers) in both structural and semi-structural repairs. They may be injected under pressure or manually applied as a gel or putty. Epoxies are most effective for structural repairs in dry locations when used as a bonding agent to provide shear resistance. When used as a grout for filling decay voids, it is essential that all decayed wood is removed, that moisture conditions that caused decay are rectified, and that surfaces are treated with preservative. The difficulties associated with ensuring the removal of all decayed timber from a voided area and the sealing of components to ensure grout placement has led some jurisdictions to prohibit the use of epoxy repair methods.

Eradication of Fungal Infection

Control of moisture

In the early stages of decay, all that may be necessary to arrest its development is removal of the source of moisture, allowing the timber to dry to below 25% moisture content, and prevention of re-wetting. For example, drainage patterns on approach roadways can be re-routed to channel water away from the bridge rather than onto the deck.

Cleaning dirt and debris from the deck surface, drains and other horizontal components also reduces moisture trapping.

Fumigants

Fumigants such as vapam and chloropicrin have been used successfully in the US for eradicating internal decay in Douglas fir roundwood. However, they are unlikely to be effective in hardwoods and will have only limited relevance to preservative-treated softwoods such as radiata pine where decay will be initiated at surfaces rather than internally. Fumigants do not provide long-term protection.

Diffusible fungicides

Several proprietary formulations are available. These are either gels or thick liquids based on fluorine, copper and boron salts, or fused rods of borate salts. Gels and liquids are applied to timber surfaces that are then covered for five to six weeks with an impervious wrapping to allow the chemicals to diffuse into the wood (Figure 6.25). Boron rods are inserted into holes drilled into the affected member, and the holes are then sealed.

Figure 6.25: Application of diffusible fungicide (Boracol) prior to installing concrete overlay

Source: NZTA (2001).

The principle behind both procedures is that the high wood moisture content, which has allowed initiation of decay, will act as a medium through which the fungicides diffuse and kill the decay organisms.

If there is no moisture there will be no diffusion and no decay. The fungicide will remain indefinitely and begin to work if moisture reaches the area. However, it must be noted that diffusible fungicides that are not 'fixed' into the timber will be leached out if the timber is exposed to continuing moisture penetration. Retreatment will be necessary, with frequency depending on the degree of exposure. Brush-on surface treatments have not been proven for Australian hardwoods.

6.6.8 Rehabilitation and Replacement

Many timber bridges reach an age and condition for which preventive maintenance is not appropriate and structural repairs are no longer cost-effective. If the structure is sub-standard and load restrictions apply, it is the appropriate time to review the options of major repair, rehabilitation, and replacement.

A typical case might be an old timber bridge where the running planks are loose, the deck planks below are decayed and will not hold spikes, the stringers are cracked, and zones of decay have been confirmed. Load capacity is in question and further weight restriction is not acceptable. The foundations are sound.

Options to consider are:

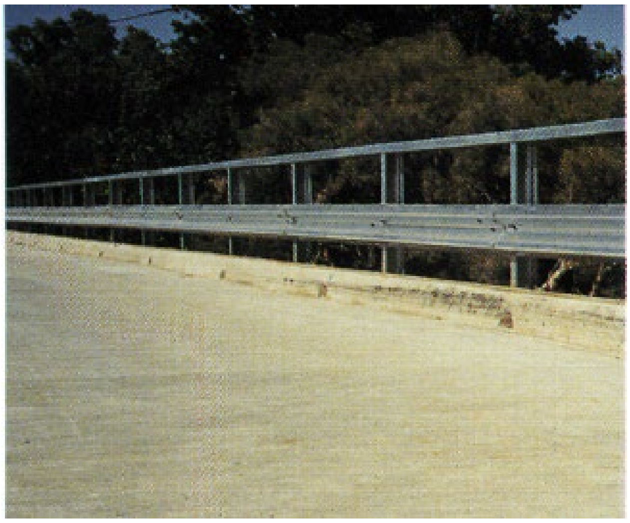
- The bridge could be propped mid-span as a temporary measure.
- The superstructure could be dismantled and sound members re-used with replacement recycled members as a short-term solution.
- The deck could be removed and additional stringers installed between existing stringers and the deck reinstated, again as a short-term solution.
- A totally new timber superstructure could be installed, using either sawn timber or laminated components.
- Other materials, such as concrete or steel, could be used either on their own or in composite action with timber. The timber deck could be designed to use a concrete deck overlay (Figure 6.26 and Figure 6.27) for good protection to the timbers below, but dead load effects must be considered.

The best solution will be based on an investigation of the economics and future expectations for the particular bridge in relation to the durability of its components and the life expectancy of the structure.

Figure 6.26: Preparation for concrete overlay

Source: NZTA (2001).

Figure 6.27: Concrete overlay complete



Source: NZTA (2001).

6.6.9 Preventive Maintenance

In addition to the general preventive maintenance requirements set out in Section 5.1, the following additional maintenance is recommended for timber elements:

Table 6.3: Routine and preventative maintenance for timber elements

Structural and deck elements	Routine maintenance (annual)	Preventative (additional to annual) maintenance (five-yearly)
Stringers/beams/ girders	Check any waterproofing. Repair as necessary. Brush coat of diffusing preservative on all repairs.	Fill or re-fill all holes with diffusing preservative. Flood all end-grain joints and splices with diffusing preservative and apply waterproofing.
Timber decking	Check for damage. Repair or replace.	Flood all exposed end-grain with diffusing preservative and apply waterproofing.
Timber kerbs	Check for damage. Repair or replace. Clear drainage scuppers and gutters. Check for loose bolts, and tighten as necessary.	Flood all joints and splices with diffusing preservative and apply waterproofing.
Timber handrail	Check for damage. Repair or replace. Check for paint damage. Repair as necessary. Check for loose bolts. Tighten as necessary.	Flood all joints with diffusing preservative. Re-paint.
Bolts	Check condition. Tighten/replace as necessary.	Fill all holes or pipings with diffusing preservative.
Truss members	As for stringers, plus check for tension failures, particularly at splice plates.	As for stringers
All timber members	Inspect for termites and decay. Treat if required.	
Corbels	Check for splitting. Bolt as necessary.	
Headstocks, capwales, halfcaps	Check for damage. Repair or replace. Check all waterproofing. Recoat as necessary after brush coating diffusing preservative on all repairs. Check for loose bolts. Tighten/replace as necessary.	Flood all end grain, joints, and splices with diffusing preservative and apply waterproofing. Use diffusible rods in exposed ends if exposure is severe.
Piles	Check for damage/degradation. Repair or replace. Check all waterproofing. Recoat as necessary after brush coating diffusing preservative on all repairs. Remove all debris. Check for scour.	Fill or-refill all prepared holes with diffusing preservative (liquid and rods, as appropriate). Flood top end-grain with diffusing preservative and apply waterproofing.
Sheeting	Check for damage. Repair or replace. Coat all end-grain with diffusing preservative & waterproof.	
Bracing	Check for damage. Repair or replace as appropriate. Check all waterproofing. Recoat as necessary, after brush coating diffusing preservative on all repairs. Check for loose bolts. Tighten/replace as necessary.	Flood all end-grain, joints, and splices with diffusing preservative and apply waterproofing.

Notes:

1 *See Glossary for definition of elements.*
2 *Barrier preservatives (such as creosote or CCA), where complete, prevent the ingress of fungi into timber. However, Australian timber bridges are typically constructed from dense hardwoods and pressure treatment, if applied, is unlikely to penetrate fully because of the refractory heartwood. Hence, all bridge timbers that have been exposed to moisture probably have some active fungal presence. At decayed locations in particular, fungi can be found up to 300 mm into apparently sound wood.*
3 *Diffusible preservatives that are brushed on to a surface may only diffuse 5 mm into hardwood and slightly more on end grain.*

Preventative maintenance of timber bridges is largely concerned with preventing or repairing the consequences of biological attack caused by fungi and borers (especially termites and marine borers).

There is a two-part strategy in minimising such attack:

- barriers to prevent ingress of new infestation and elimination of details that provide a sympathetic environment for the agents of destruction
- remedial treatment of existing infestations in situ before they can cause any further damage.

Diffusible preservatives have a useful life of at least three to five years, and a high-quality paint has a useful life of at least five to seven years. It is therefore logical to adopt a five-yearly cycle of major inspection/preservative maintenance, complemented by an annual visit for visual inspection/routine maintenance. All diffusible preservatives in holes should be renewed and holes re-capped (using treated dowel or special plastic plugs). Any further areas indicated by the inspection as needing preservative should be similarly treated.

It must be emphasised that the greatest attribute of these preservatives is their ability to diffuse through timber. It is also their main potential weakness because they are not 'fixed' into the wood structure and therefore will tend to be leached away if the medium surrounding the wood is moist or wet. The useful lifespan of the preservative is strongly affected by the severity of exposure of the member, and this must be understood when specifying a retreatment interval for the various members. These preservatives are formulated from a range of chemical compounds, which implies a range of toxicities and also variations in occupational handling safety.

Quality and price are also variable, but most have been tested and proven under Australian conditions. The choice of product for particular applications is increasingly being determined by environmental impact factors. For information on the use of diffusing preservatives see Dickinson Morris and Calver (1989), Dickinson and Murphy (1989), Edlund, Henningsson, Kaarik and Dicker (1983), Greaves (1984), and Greaves, McCarthy and Cookson (1982).

The area around all joints should be thoroughly cleaned (including paint removal where applicable) and then flooded with diffusible preservative. Where structural joints are exposed to the weather further protection in the form of fused preservative rods is advisable. Waterproofing or re-painting can be carried out about three weeks after preservative treatment.

The bridge should be cleaned of all unsound paint by an appropriate method, and all painted members (including steel truss members and navigation spans, if coating condition warrants) should be re-painted. Note that timber members with sharp edges or corners should always be rounded off prior to painting, as paint has a tendency to curl up at the edges, with a break-down of the coating emanating from the edge.

Because untreated timber has poor durability, preventive maintenance is very important to protect the timber from the elements that cause deterioration.

Decay

Decay is the most serious defect and the objective is to eliminate conditions that cause decay.

Timber bridges usually need repair because the moisture content of the wood reaches levels conducive to fungal attack. Decay becomes established because either the natural durability of the timber is insufficient for the end-use or, in a softwood member, preservative treatment is inadequate to protect the timber in the particular decay hazard environment. Replacement timbers treated to current wood preservation specifications, e.g. (NZMP 3640), should give a service life well in excess of 30 years. However, it is necessary that any timber surfaces exposed during on-site cutting receive liberal applications of an oil-based or solvent-based preservative. Creosote or copper napthenate (Metalex Green) formulations are ideal for this.

The most effective means of preventing decay is to keep the timber dry. This involves simple tasks of cleaning, draining, removal of debris and growth and applying waterproof coatings. Measures to protect against excessive moisture uptake are more applicable to untreated, naturally durable hardwoods used for repair of hardwood bridges. Protective measures are generally unnecessary for softwoods treated to hazard Class H4 Specification and above (NZMP 3640).

For timber locations that have a high decay hazard and where other preventive maintenance options are impractical, treatment with a diffusing preservative may offer a solution (Section 6.6.7).

Timber Decks

Ideally, the deck should provide a waterproof cover over the bridge beams and substructure. Many timber bridge decks comprise of transverse planks (often with running planks over) connected by spikes either directly to the beam system or via spiking planks (to minimise decay of girders). Shrinkage of timber leaves cracks that trap dirt and moisture and allow water onto the substructure. Measures are required to ensure that decks are watertight and that drainage on decks and approaches is controlled to allow run-off away from the substructure. The deck should be kept clean of loose gravel, debris, or vegetation growth that traps moisture. Trees shading and sheltering the bridge should be cut back. A bitumen or asphalt seal should be maintained wherever possible as a waterproof membrane over the bridge.

Loose timber deck members result in noise, impact loadings, and possible tyre damage. When spikes will not hold, splits or decay are usually found in the timbers below. Loose deck timbers often result from an underlying problem such as substructure decay. The cause of looseness should be identified before undertaking repairs, as more major structural repairs may be required.

Kerbs and Rails

Kerbs and rails should be of durable timber. New Zealand native timbers are known to be less durable. Maintenance of a good paint system is important. Connections, shaded and end areas require regular attention. Alternatively, treated radiata pine and more durable hardwoods may be left unpainted.

Connections

Connections of timber members are a potential area for problems. It is important to keep surfaces dry, to ensure that steel components are galvanised and that treatments of paint, grease, sealants and plugs are effective.

Organic solvent, water repellent formulations containing fungicides, or oily preservatives such as creosote should be applied in repair work in situations where two or more members overlap to form water-trapping joints. Use of materials such as mastics, ‘Malthoid’ or paint is only recommended in situations where they will not encourage water entrapment.

Holes Drilled During Inspections

All holes drilled or bored for assessing the interior condition of members should either be extended right through the member to allow drainage, or be flooded with an oil-based preservative such as creosote, or tightly plugged with a preservative treated dowel or plastic plugs.

6.7 Approaches

6.7.1 General

Bridge approaches are an integral part of the structure, and faults in the approaches can be as serious as those in the main structure.

Based on previous research, approximately 5% of New Zealand bridge failures can be attributed to partial or total loss of approach fill. While this figure will vary with the various agencies it is a significant factor.

Defects tend to fall into two categories relating either to traffic safety or to structural deterioration, but many have an effect on both of these. Either way, defects are usually very noticeable to the public and deserve prompt attention.

6.7.2 Inspection and Evaluation of Defects

The aim of inspection is to identify defects and inadequacies in the approaches, and to determine the causes and rates of deterioration. This will allow an evaluation of the significance of each and an appropriate solution to be arrived at.

Apart from having drawings of the structure and approaches, it is also useful to know the:

- age of the structure
- age of the pavement
- average annual daily traffic count
- percentage of heavy vehicles
- size of typical heavy vehicles.

The inspection should include a thorough examination of all the features discussed below.

6.7.3 Defects and their Correction

Alignment Geometry

Many bridges have substandard approach geometry. This should be considered a defect if accident records show that it seriously compromises road safety. It usually relates to lack of visibility or to slow-speed curves in a high-speed environment. Extreme vertical curvature can also increase the impact loads on the approach or structure.

Correction of road geometry is a major operation, but shape correction may be acceptable. Measures aimed at reducing speed may mitigate the effect. These could include improved road marking, marker posts and signs, including advisory speed signs.

Pavement

Pavement defects on bridge approaches are normally considered with maintenance of the rest of the highway. However, because they can affect both impact-loading on the bridge and road safety, they need to be inspected as part of the bridge inspection as well. Defects include loss of chip, flushing of excess binder, slicking as well as shear failure resulting in cracking, heaving or rutting (Figure 6.28).

Figure 6.28: Settlement of approach caused by lack of fill containment

Source: NZTA (2001).

Figure 6.29: Failure caused by seepage and poor drainage

Source: NZTA (2001).

Settlement

Settlement usually shows up as a localised depression in the pavement adjacent to the end of the bridge. The amount of settlement necessary before problems develop can be quite small. Depressions in excess of 15 to 20 mm will be detected by the road user, and when greater than 25 mm will quickly become unacceptable. Apart from the road safety aspect, the effects on the bridge will be increased impact loading and fatigue, with particularly serious consequences for the deck joint nearby. The pavement will suffer the same effects, leading to further accentuation of the depression.

Progressive settlement can often be detected in the approach traffic barrier if the pavement has been corrected by filling over the years without re-levelling the barrier rail.

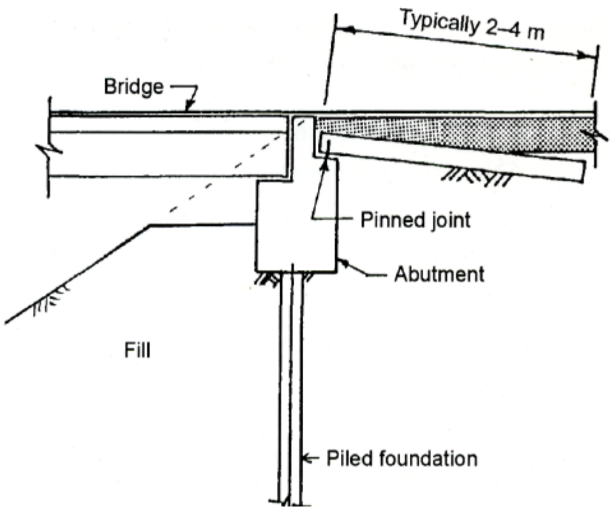
Settlement may be the result of:

- Plastic deformation of the ground underlying the fill itself if it has not been properly compacted or is of inappropriately graded material.
- Migration of fines from the fill if it is poorly graded with excess fines. This can occur by piping, after high water levels caused by flooding, or by water from defective drains. Seepage from the fill can be a warning sign of this problem (Figure 6.29).
- Movement of the abutment or wingwalls, which reduces support to the fill.
- Poorly shaped fill around and in front of a spill-through abutment (Figure 6.30).
  A horizontal berm should be formed in front of the abutment, allowing the projected fill slope to line up with the road surface immediately behind the abutment, as shown in Figure 6.31. Otherwise, material will tend to move through under the abutment and cause settlement.

Figure 6.30: Ineffective fill containment

Source: NZTA (2001).

Figure 6.31: Abutment with settlement slab and properly shaped fill



Source: NZTA (2001).

Filling the pavement depression is a short-term solution, but it is preferable to determine the cause and correct it.

If settlement is caused by plastic compression in the ground underlying the fill, a possible solution is to remove and replace the fill with lightweight material. Unsuitable fill may be replaced by correctly graded material.

If the cause is migration of fines, it may be sufficient to improve the drainage system by lining channels. Horizontal perforated drains drilled into the fill to remove groundwater may also be beneficial.

If there is insufficient fill in front of a spill-through abutment, as described above, this should be corrected if there is sufficient space to do so. Otherwise, some method of retaining the fill beneath the abutment may be sufficient, such as placing rocks. If there is no settlement slab, construction of one, such as is required by the NZTA *Bridge Manual* (2016b), will also improve the situation by preventing a depression forming immediately next to the abutment.

See also Sections 6.10, 6.11 and 6.12.

Erosion, Drainage and Slipping

Erosion of fill may be caused by scour in the waterway or defective drainage either on the approaches or the bridge. Slipping may occur due to instability in both cuts and fills on the approaches.

If erosion is caused by scour, river protection by gabions or riprap may be required, as described in Section 6.11.

If erosion is due to defective drainage, lining channels is the most effective treatment, and this could include energy dissipation measures where water has to flow down the fill slope. Diversion of water discharging from the bridge may be necessary. Regular cleaning of drains to prevent blockages will also help, and this should include flushing out of soil drains if these are present. Improved vegetation on slopes will also help control the flow. Guidance on drainage systems is provided in NRB (1997).

If slips are a frequent problem, flattening the slopes is an obvious solution, but the drainage improvement described above may be enough to correct the situation by itself.

See also Sections 6.8 and 6.11.

Traffic Barriers

Traffic barriers on approaches are most likely to be in the form of non-rigid barriers such as W-section or thrie beam barriers, although rigid concrete or steel barriers may be used in some situations.

The most obvious defects in approach barriers are collision damage. Other faults may include steelwork corrosion, loose joints, and slack cables in breakaway cable terminals. The barrier rail may become misaligned vertically due to fill settlement, or horizontally due to expansion forces if the posts lack sufficient support in the ground. All non-rigid barriers should terminate in an anchorage sufficient to resist the design force. In older installations this may be a concrete block, with the rail twisted down to meet it at ground level. This terminal does not meet current standards, which are set out in AS/NZS 3845.1 Road Safety Barrier Systems and Devices: Road Safety Barrier Systems.

A check should be made of non-rigid barriers to see whether the rail is at the correct height to operate as designed.

If the effective height has decreased because the approach has been filled to correct settlement, posts should be reset. See also Section 6.9.7.

Signs and Road Marking

- advisory speed
- load and/or speed restriction for heavy vehicles
- bridge end markers
- one-way bridge
- narrow bridge.

The condition of all these items should be checked and rectified as necessary.

6.8 Drainage System

6.8.1 General

Ineffective drainage of runoff can affect a bridge in several ways:

- flooding of the deck due to blockage of the drainage system may create a serious traffic hazard
- uncontrolled water flow over concrete or steel surfaces below deck level may lead to durability issues
- debris collects, it will retain moisture and promote corrosion
- water is discharged off the bridge other than into a proper drainage channel and may cause erosion of approaches and possibly undermining of foundations
- water trapped in blocked pipes may freeze, rupturing the pipes or cover concrete.

It is therefore essential that bridge drainage systems are regularly inspected and maintained to ensure that water is directed away from the structure in a controlled manner and as intended.

Most potential drainage problems can be eliminated by good design and correct installation. Features that affect bridge drainage are described in the following section. These descriptions can be used as a checklist during inspection to determine if any corrective action needs to be taken.

6.8.2 Drainage Features

Deck Slopes

To ensure effective drainage of the bridge deck a minimum cross-slope of 2% and a minimum longitudinal grade of 0.5% are recommended, with gutters graded at least at 1%. Care should be taken to maintain these grades during any re-surfacing operations.

Kerb Channels

Channels may become blocked with silt and sealing chip, particularly if the slopes and grades are insufficient. If the channels are not cleared regularly, plants will grow in the accumulated debris. This will exacerbate the problems of deck flooding and moisture retention in the concrete.

Drainage Inlets

Blockage of inlets with storm debris and/or rubbish is a common cause of drainage problems. Grates need to be hydraulically efficient, strong enough to support traffic, securely fixed, corrosion resistant and not present a hazard to bicycle traffic. To cope with partial blockages their inlet area should be twice the calculated area required. Regular removal of collected debris can help minimise the effects of inadequate design (Figure 6.32).

Drainage Pipes

To avoid clogging with mud and debris it is important that pipe systems have a minimum diameter of 150 mm (200 mm preferred), a minimum radius of 450 mm and are laid at an absolute minimum slope of 2% (but preferably 8%). In addition, clean-out plugs and elbows should be provided at appropriate places and be easily accessible. Open channels or troughs under expansion joints can fill rapidly if they are not regularly maintained.

Care is required when maintaining asbestos-cement pipes or channels and PVC pipes exposed to direct sunlight, as these become brittle with age.

Figure 6.32: Typical debris collecting on grate

Source: NZTA (2001).

Figure 6.33: Drainage outlet discharging water onto concrete deck soffit

Source: NZTA (2001).

Drainage Outlets

A common deficiency of drainage systems is their failure to discharge clear of the structure. This may cause staining and corrosion of steel and concrete beams or substructure (Figure 6.33). If it allows debris to build up, the situation will be aggravated by moisture retention and plant growth. If the discharge is incorrectly positioned, it may also cause the potentially serious problems of erosion of embankments or undermining of foundations. Figure 6.34 shows a simple method of avoiding drainage material discolouring adjacent girders.

Figure 6.34: Drainage outlet below adjacent steel beam soffit

Source: NZTA (2001).

Drip Grooves

Run-off should be prevented from tracking along concrete deck soffits or down the faces of girders and piers by casting grooves into the underside of edge beams or the edge of deck soffits. A drip groove can be rendered ineffective if it is too shallow, or if it is straddled by incorrectly installed baseplates for traffic barrier posts, or if it is filled up with material by insects. These problems may be aggravated on bridges with super-elevated decks, because the increased slope will require a deeper drip groove if it is to be effective.

Drainage of Voids

Box girders and other voided members should have drain holes as a precaution against build-up of condensation or leakage. Such holes may get blocked by birds or insects.

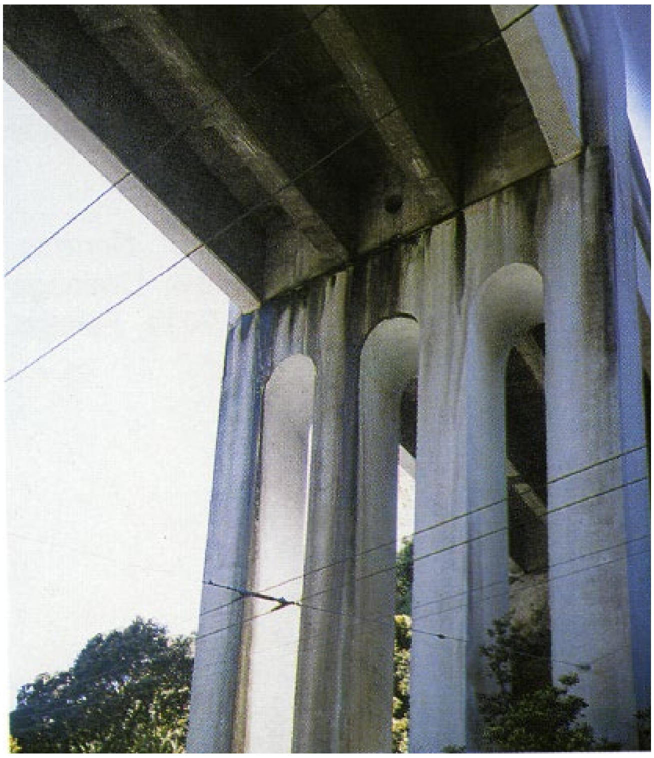
Leaking Joints

Most bridges have expansion joints to accommodate thermal or seismic movement (Section 5.8). These elements usually include provision to prevent leakage from the deck onto bearings, hinges or other substructure components. Leakage may occur from faulty installation, inadequate crossfall of collector channels, ruptured membranes, misplaced compression seals, and adhesive or sealant failure (Figure 6.35). Allowance for such failures should be made at the design stage to avoid subsequent problems with moisture retention and accumulation of debris.

Leaking Decks

Leakage may occur in the vicinity of construction joints or shrinkage cracks, especially if reflective cracking through overlying bitumen or asphaltic concrete is present (Figure 6.36).

Figure 6.35: Staining of pier caused by leaking deck joint



Source: NZTA (2001).

Figure 6.36: Leaking deck joint

Source: NZTA (2001).

6.8.3 Maintenance

The key to the success or failure of any existing drainage system is regular maintenance. The frequency of attention will depend largely on the particular environment.

Drainage Systems

Many problems can be avoided by collecting and removing debris from the kerb channel before it enters the system to cause a blockage. For example, spillages of wet concrete, grain or other granular material should be promptly removed, as should objects with potential to block pipes like cans, milkshake containers, dead birds, etc.

Where pipes are blocked through build-up of silt, high-pressure water is a commonly used cleaning aid. Where water is unable to break through, a back-flushing technique to reverse the normal direction of flow can be effective. Back-flushing with compressed air can also be used to clear badly plugged pipes, but if non-encased PVC pipes are included in the system caution is required to avoid bursting them.

Gully traps are normally cleaned by shovel or suction pumping where available. Gullies will require special attention after gritting for icy conditions.

Substructure

The area beneath an open expansion joint should be cleared of debris at the same time as the rest of the system. The same applies where a joint seal has failed and allows debris to pass through.

6.8.4 Rectification of Defects

Drainage Systems

Severe ponding problems on concrete decks caused by lack of fall may be reduced by the judicious drilling of 100 mm diameter drain holes at 2 m centres near the kerb and fitting them with droppers or a collection system where appropriate and as permitted or required by the Resource Consent. Inlets that are prone to blockage by floating debris (e.g. pine needles) can be improved by fitting a domed screen. This will allow water to continue to enter the drains under the floating material until it can be removed. Figure 6.37 shows an example of an inaccessible drainage channel, and Figure 6.38 shows a replacement system fitted with an inspection and jetting point.

Where new drains are necessary, they should be fitted with sufficient inspection points to allow for both the introduction of high-pressure cleaning water and the ejection of any debris. When retrofitting a drainage system, it is vital that any potential choke points are easily accessible so that build-up of debris can be managed.

Substructure

Where substructure concrete is water-stained, runoff should be intercepted by grooving, or by insertion of dropper pipes, which should extend below any adjacent beam. Free fall of water from a deck drain through the air should not cause an erosion problem where the fall exceeds 7.5 m, otherwise erosion protection such as riprap or paving will be necessary.

Leaking deck joints should be repaired (Section 6.9.4) and deck cracks sealed with rubberised bitumen.

Where a drip former is not working properly a possible solution is to attach a small galvanised steel or aluminium angle to the underside of the concrete to create an equivalent effect. The angle would need to be sealed against the surface.

Figure 6.37: Inaccessible drainage channel

Source: NZTA (2001).

Figure 6.38: Replacement drainage system with inspection and jetting point

Source: NZTA (2001).

6.9 Hardware

6.9.1 Definition

Hardware is the generally accepted term used to describe the various bridge components, attachments, and devices that are not regarded as main members of the deck, superstructure, substructure, or foundations.

6.9.2 Materials

The principal components of a bridge, e.g. beams, piers, abutments, may have a life exceeding 50 to 80 years with little, if any, need for maintenance.

Bridge hardware has a shorter life and requires more maintenance than the principal components. Accordingly, the design and maintenance of bridge hardware requires careful consideration of initial product selection, maintenance programs and provision for replacement.

Bridge hardware components use a wide range of materials, all with a definite service life. The most common are:

- steel
- plastics – in particular PTFE (poly-tetra-fluoroethylene)
- rubber
- concrete
- epoxy-mortar
- timber
- adhesives.

Function, required life, loading and maintenance requirements of the various components need to be considered at the design stage and when selecting replacements, particularly when considering the type of bearings and joints to be used. Maintenance of bearings and joints can cause major disruption to traffic. They can be costly to replace and they are the components most prone to the debilitating effects of overloads and excessive thermal, shrinkage and creep movement.

All the materials listed above are subject to attack from the external environment.

Steel and timber can be protected by a suitable thickness of good quality protective coating.

Durability of reinforced concrete can be enhanced by suitable coatings, but primary protection of reinforcement comes from dense, impermeable cover concrete (see *AGBT Part 2: Materials*).

There are many types of plastic, some of which are subject to chemical attack, others to radiation such as ultraviolet rays from the sunlight. Most plastics also deform under sustained loads and high temperature.

Rubber may be natural or synthetic (e.g. chloroprene). Both have been used, although the current preference is for natural rubber in bearings. Natural rubber stiffens with age and requires specific additives to counteract chemical and radiation effects.

Adhesives, used to bond stainless steel or PTFE sliding surfaces to steel backing plates, or steel to rubber in older elastomeric bearings, are also susceptible to ageing and embrittlement. Users should ensure that the correct type is chosen for each application and that the manufacturers' instructions are followed.

Any proposal to replace or modify any bridge hardware element should be submitted to an experienced bridge designer to ensure that the design philosophy and behaviour of the structure are not compromised.

The materials used in the repair of various components should as far as possible be fully compatible with the materials of the original component.

High-pressure water blasters can cause as many problems as they solve. Real care needs to be exercised around bearings and joints when using such equipment so that water or dirt is not forced into inappropriate places and elastomeric materials are not removed.

6.9.3 Bearings

This section presents information on the rehabilitation and strengthening treatments for the failure modes of common bridge bearing types. Refer to *AGBT Part 3: Typical Superstructures, Substructures and Components*, Section 14 for information on bearing function, design considerations and descriptions of typical types of bridge bearings. Refer to *AGBT Part 6: Bridge Construction*, Section 19.2 for more information on supply and installation, as well as construction issues of common types of bridge bearings. Refer also to Section 5.7 of this document for the typical failures related to specific bearing types.

Bearings transmit superstructure loads to the substructure. They also provide for longitudinal movements and rotation due to live load deflection, expansion and contraction, and small seismic movements. They are vitally important to the efficient functioning of the structure. If they are not kept in good working order, stresses may be induced into the structure that can substantially shorten its service life.

In many bridges, bearings, particularly elastomeric bearings, are not fixed positively to the structure, but depend on friction to prevent progressive displacement.

It is desirable that friction effects or the fixing of the bearing components to both the substructure and the superstructure prevent the components parting company or 'walking' in a seismic event.

All bearings need to be protected from sand blasting and grinding operations, and care needs to be exercised when high-pressure water blasting not to drive grit and other contaminants into places where they may cause surface damage.

Sliding Bearings

The following points should be considered in the maintenance and repair of sliding bearings:

- If PTFE 'flow' has occurred, the probable cause will be differential settlement or rotation causing uneven loading on the membrane. Rotational effects will be remedied when the bearing is reset.

- Where the PTFE membrane has been deformed or ruptured, the bearing will require full rehabilitation. It is generally inappropriate to attempt this work on site. It will, therefore, be necessary to either replace the bearing with a spare unit or use a temporary bearing while the rehabilitation is carried out under factory conditions. In most cases the stainless steel sliding surfaces will need to be re-polished or, if badly scored, replaced.
- Where misalignment has caused bending or cracking of the side guides, these will require straightening and/or re-welding.
- If HD bolts or slide guides are bent or fractured, the cause of the problem should be investigated. Differential settlement or seizure at another point in the structure may have caused the problem. In this case treat the cause first and then repair the bearing fault.
- Usually the defects are caused by water or wind-borne grit entering the sliding surface area. The future performance of the bearing may be enhanced by the installation of a protective shroud. When installing a shroud, take care that it is able to cope with the full travel of the bearing.
- Where only minor corrosion of the base plate and/or top plate has occurred, and the sliding surfaces are still satisfactory, wire brushing to remove the corrosion product and the application of high-quality protective coatings can be done on site. Under no circumstances should grinding wheels or other grit-producing methods be used to remove the corrosion products, as inevitably some grit will enter the sliding surfaces.
- Bearings with metal sliding surfaces should be lubricated.

Rocker Bearings

The following points should be considered in the maintenance and repair of rocker bearings:

- When effecting repairs, any defects in the joint should also be addressed, and the time between the two exercises should be minimised. If the wear in keys, keyways and pins is not too severe, the bearing should have all corrosion product removed and be treated with a high-quality protective coating.
- If the bearing is so badly worn or corroded as to require replacement, replacement with a more modern type of bearing such as a low-profile elastomeric pad should be considered.
- Any damaged bolts should be replaced and any loose bolts re-tightened to the recommended torque.
- All bearings should be lubricated.

Roller Bearings

The following points should be considered in the maintenance and repair of roller bearings:

- Regular attention to lubrication is essential.
- If minor corrosion has occurred, wire brushing and new protective coating is sufficient.
- If seizure has occurred, dismantling in a workshop will be required. If undamaged, the bearing may be re-used.

If the bearing components are severely corroded or flats are worn on the rollers, the unit should be replaced. In this case consideration should be given to replacing it with a more modern unit. If the horizontal movement is large, a sliding bearing could be used, preferably with a low-profile elastomeric pad mounted under the sliding bearing base plate to accommodate rotation. If the horizontal movement is small, an elastomeric pad bearing may satisfy both horizontal and rotational movement demands.

Elastomeric Bearings

The following points should be considered in the maintenance and repair of elastomeric pad and sliding bearings:

- Where any significant damage such as delamination, rupture of rubber cover layer, or perishing of the rubber has occurred for whatever reason, the only remedy available is replacement of the pad.
- Where the base plates or top plates have corroded, the corrosion product should be removed by wire brushing, the parts given a high-quality protective coating of paint and the cause of the defect remedied by repair of the defective deck joint.

Pot Bearings

The following points should be considered in the maintenance and repair of pot bearings:

- Loose or damaged bolts should be re-tightened or replaced as necessary.
- Compressive rupture of the elastomer is generally the result of excessive rotation. Before replacing the pad, the cause of this phenomenon should be investigated. It may be the result of differential settlement or rotation of the foundations. If so, and if it is reasonable to assume differential settlement or rotation has now stabilised, the superstructure should be jacked back up to its original height and the new bearing installed on a raised pad, making suitable provision for rotations that may have occurred.
- Corrosion of the base plates, bolts, and pots is generally the result of water leakage through faulty joints. The bearing should be removed, the metal components sand blasted, then painted with a high-quality protective coating. Once the bearing is repaired the faulty deck joint causing the problem should be repaired without delay.

Spherical Bearings

The following points should be considered in the maintenance and repair of spherical bearings:

- Damaged units should be replaced by a more satisfactory bearing such as a low profile elastomeric pad. If the decision is made to repair the original unit, it must be done under factory conditions.
- Any loose bolts should be re-tightened or replaced as necessary.
- Where no significant damage is apparent, the bearing performance and durability can be enhanced by application of a protective shroud to prevent water and wind-borne dirt and grit gaining access to the sliding surface area. Future inspection access must be considered in the application of a shroud.

6.9.4 Deck Joints

This section presents information on the rehabilitation and strengthening treatments for the failure modes of common bridge deck joints. Refer to *AGBT Part 3: Typical Superstructures, Substructures and Components* Section 15 for information on deck joint function, design considerations and descriptions of typical types of bridge deck joints. Refer to *AGBT Part 6: Bridge Construction*, Section 19.3 for more information on supply and installation, as well as construction issues of common types of bridge deck joints. Refer also to Section 5.8 of this document, for information on failure modes of common types of bridge deck joints and their root causes.

The performance requirement of the deck joint area is dynamically very different from the rest of the bridge deck.

The type of joint used depends on the range of movement and on current practice at the time the bridge was built.

Many older bridges have simple open joints at every support, resulting in minimal movement per joint. These bridges generally do not give joint problems. Bridges of a later era have fewer joints with more movement per joint and often give rise to maintenance problems. The current preference is to eliminate joints if possible and absorb the movements by compression and relaxation of the approach filling.

Open joints allow water and debris to fall through the joint and can be a potential source of deterioration due to corrosion of affected bridge elements. For this reason current practice favours the use of sealed expansion joints. Guidelines for the use and selection of sealed expansion joints are included in Highways Agency UK (1994a and 1994b).

Features and potential problems common to different joint types are described below. This section also includes more specific comments on various joint types. It is recognised that there are many variations on all these, and the diagrams should be taken as indicative only.

General Problems

Many joints have steel angle nosings protecting the edges of the concrete at the gap. A common problem is loosening of the angles caused by breaking up of the concrete beneath them. This is generally due to insufficient compaction of the concrete under the angle and the presence of air trapped there during concreting. Movement leads to failure of the welds connecting the anchoring bars. When replacing such angles, bleed holes should be drilled in the horizontal legs to release the air during concreting. Sometimes angles or plates are held down by bolts, and these frequently need tightening or replacement.

Many joints are susceptible to becoming clogged by debris lodged in the movement gap, and require regular cleaning out. If debris is allowed to build up, movement can be inhibited and superstructure compression forces can cause problems to bearings and substructure elsewhere. The material might also be forced down into the joint seal, thus rupturing it and causing leakage.

Leakage of water and debris through a joint can cause corrosion of steel parts of bearings, HD bolts and linkage bolts, as well as structural steel main members. The effect on concrete parts can be just as severe, as the water may promote corrosion of reinforcement, as well as unsightly staining of visible surfaces. Corrosion effects are made worse if debris is allowed to build-up because it retains the moisture.

Vertical misalignment of the two sides of a joint may lead to damage to the joint itself, which must be repaired. This fault may also indicate a failure in the bearings, which must also be corrected.

Many types of joints incorporate directly or indirectly a nosing of ordinary concrete or an epoxy mortar. Epoxy mortars are relatively cheap but often have a higher compressive strength than the adjacent concrete, and a different expansion coefficient. For these reasons they tend to crack perpendicular to the joint and de-bond, particularly where the traffic loads are greatest and in the region of shrinkage control cracks. A life of 5 to 10 years appears to be usual in a highly trafficked situation, but a longer life has been observed in less-trafficked situations.

Deck joints incorporating steel components at the road surface often appear as a depression in the surface, because commonly used asphaltic binders do not bond satisfactorily to the steel. The joint becomes progressively depressed relative to the running surface on each side as this is built up with successive re-sealing. Impact loads are increased, rideability is decreased, and the joint becomes increasingly noisy. This problem appears to arise because the asphaltic binder used on the deck and on the adjacent roadway is also used on the steel components.

Research shows that the magnitude of creep and shrinkage shortening has often been underestimated at the design stage, with the consequence that the deck joint provided cannot sustain the movements. The rotational movement of the span ends, induced by traffic loading, has also often been overlooked in the design and similarly contributes to deterioration or failure of the deck joint.

Poor geometry of the approach roading often leads to high-impact loading at bridge abutment deck joints. A smooth surface profile with the joint finished to the appropriate level and grade is important. Technical literature for the particular joint type should be referred to, but in general the seal element should be set just slightly below the level of the adjacent road surface.

Information on the rehabilitation and strengthening treatments of popular joint types are presented below.

Open Joints

During inspection, all the relevant faults should be checked for. The channel, if any, should be checked for blockage and the underlying substructure should be checked for build-up of debris.

The following points should be considered in the maintenance and repair of open joints:

- Maintenance largely consists of clearing debris carried through the joint onto the channel or the substructure, especially around bearings.
- Where possible, install a collector drain to carry water clear of the substructure. In many cases this is not possible because of lack of access.
- If the nosings are faulty and need to be broken out, the joint should be replaced with a more suitable type providing a fully sealed joint. Options available would probably include elastomeric in metal runner joints or asphaltic plug joints.

Sealant Filled Joints

During inspection, the relevant faults should be checked for, but the principal failure in this joint is de-bonding. This is generally due to movement of the structure in excess of the capacity of the joint, or to improper surface preparation, but may also be due to chemical incompatibility with the joint nosing material. Embrittlement and failure of joint materials may also occur.

The following points should be considered in the maintenance and repair of sealant filled joints:

- Where movement has exceeded the capacity of the sealant material, consideration should be given to the use of a different joint system. Compression seals and asphaltic plug joints are potential retrofit options.
- In replacing the joint material, take note of the most recent developments in sealant materials. Some systems are now available that have greatly enhanced elastic properties and improved priming coats that will ensure better bond characteristics.
- Adhesion is a key factor when reinstalling sealants and critical to this is adequate preparation of the joint surfaces. It is essential to remove all traces of the previous joint material or bitumen, as most sealants are sensitive to these materials.
- When replacing the sealant, ensure that the surface is kept low enough that tyres do not touch the surface of the compound.
- Where a layer of asphaltic concrete is placed over a sealant-filled joint, reflective cracking can be minimised by ensuring that the number of joints is such that the movement accommodated at each joint is small. A narrow strip of suitable membrane should be laid over each joint prior to asphaltic concrete surfacing so that the surfacing is de-bonded over a finite length to provide for elastic deformation of the asphalt under the deck movements. Regular maintenance should keep the joint clear of chips or other debris.

Sliding Plate Joints

During inspection, all faults should be checked for, in particular:

- loose bolts, studs or broken welds that allow the sliding plate to rattle, creating a very noisy joint
- vertical misalignment of the sliding plate and the supporting steelwork on the opposing face. This also promotes failure of the fixing bolts or welds
- restricted movement caused by sealing chip, particularly if the re-sealed surfaces intrude into the region of the joint.

The following points should be considered in the maintenance and repair of sliding plate joints:

- Other than the addition of a sealing strip, where this is not already present, little can be done to enhance the performance of this type of joint.
- If there is any vertical misalignment caused by differential movement between spans, the anchoring mechanism holding the sliding plate to the slab will inevitably fail. This may be caused by span end rotation and/or compression in the bearings under live load. Before repairing such a fault, it is necessary to correct the underlying cause of the differential movement.
- Regular maintenance should keep the space next to the edge of the sliding plate clear of sealing chips or other debris.

Compression Seals

Inspectors should check for de-bonding of the seal, which can occur due to excessive opening of the gap, or if the seal is set too high so that tyres can tear it or load it either directly or through debris lodged in the gap. The seal element should be recessed 3 to 6 mm below the road surface to prevent contact with passing tyres, but shallow enough so that debris does not accumulate.

The following points should be considered in the maintenance and repair of compression seals:

- As for strip seal joints, frequent cleaning out of chips and debris from above the seal will prevent traffic loads being transmitted to it.
- If replacement of the seal is required, care should be taken to use a rubber section appropriate to the movement at the joint and to strictly follow manufacturers' recommendations for application of the adhesive. It is advisable to use the same make and model as specified in the original design, otherwise the seal will probably stand proud of the road surface.
- If the angle nosing on a joint of this type needs replacing, it is essential to drill 10 mm bleed holes in the upper surface of the angle to ensure air is released from the replacement concrete.
- When retrofitting other joint types (e.g. sealant filled joints) with a compression seal all traces of bitumen or existing sealants must be removed from the nosing to ensure adhesion.
- The condition of the metal protection armour should be periodically inspected for corrosion if the protective coating fails and the protection renewed if required. Slippery surfaces shall be made skid-resistant, as appropriate to the site. Clear the joint gap of debris, and push protruding seals back.
- As waterproofing of compression seal joints is not continuous, a complete water drainage system should be provided.

The performance of the joint depends largely on the quality of installation and the correct choice of the seal size and seal material. The following recommendations should be considered (Roads and Traffic Authority 2008):

- The size of the joint gap opening and bridge temperature should be measured to estimate the probable gap width at the installation between the armouring before ordering the seal to ensure the correct size and depth for the seal.
- The joint should not be used in decks with greater than 20 degrees skew.

- Compression seals should be proportioned in a working range of 40% to 80% of uncompressed width in accordance with the manufacturer's specifications to ensure that the seal remains in compression during its service life.
- The seals should be set below the deck level and at a uniform depth without excessive longitudinal stretching to prevent protrusion above the roadway surface when fully compressed.
- The seals manufactured from ozone-sensitive neoprene compositions should not be used as they may cause compression set after a few years of service.
- AS 5100.4 requires that metal protection armouring shall be provided for this joint type. For older bridges where it is not possible, this type of joint can be installed by forming the gap narrower than the design width and saw cutting with a diamond blade immediately prior to seal installation to ensure correct and uniform width for installation.
- Strip seal joints with wedge shaped metal retainers are preferred over compression seal joints as concrete compaction under the steel protection angles is problematic.

In addition, special types of compression seal joints have been available on the local market which are designed based on a combination of compression joint and epoxy-bonded rubber seal technologies. The seal is bonded into the gap walls, which may be steel, concrete, polymer modified concrete or aluminium members, thus providing waterproofing and prevention of dislodgement of the seal.

Asphaltic Plug Joints

The following points should be considered in the maintenance and repair of asphaltic plug joints:

- Where the joint has failed by de-bonding or plastic cracking, all movements acting on the joint should be investigated and the suitability of this type of joint for the imposed movements confirmed.
- Where the type of joint is unsuited to the situation, the only permanent remedy is total removal of the joint and replacement with new material better designed to cope with the particular needs at that site. Worn joints can be renovated by specialist suppliers by removing the surface and replacing it to the required profile.
- A more difficult problem to overcome in retrofitting this type of joint to existing bridges is providing sufficient width and depth of joint to cope with the expected movements. Most existing deck and diaphragm configurations restrict both width and depth availability because relatively thin decks and thin surfacing are used in this country.
- Where deck capacity allows, application of an asphaltic concrete overlay on the deck will increase the depth available for the joint.
- The failure of some joints of this type is undoubtedly due to the adoption of minimal width and thickness configurations. The joint configuration needs to be critically considered.

Strip Seal Joints

During inspection, all the faults described in Section 5.8 should be checked for. If the joint has epoxy nosings, a check should be made for any cracking or de-bonding, as this will eventually lead to spalling and break-up of the material.

The following points should be considered in the maintenance and repair of strip seal joints:

- Frequent cleaning of the joint is of vital importance to keep it in good operating condition.
- A further aid in preventive maintenance is to avoid chip seals in the immediate vicinity of the joints. Asphaltic concrete is much less prone to releasing aggregate than surface chip seals.

- Where damage has already occurred to the membrane, the joint or membrane must be replaced. If the membrane is held in place by bolted proprietary pads, or steel nosings with shaped anchorage recesses, this is a reasonably simple task and causes only minor delays to traffic. But where the sealing membrane needs to be broken out, a new seal installed and new epoxy nosings cast, traffic will be disrupted and the completed repair requires protection by a steel cover plate until curing is complete. The re-casting task can only be carried out in fine weather and when temperatures are above 10 °C.
- Failed glands can be patched using a new length of gland if partial repairs are necessary.
- Problems to the performance of joints can be caused by build-up of debris at shoulders and lightly trafficked areas, which can be reduced by scuppers (Roads and Traffic Authority 2008).

When installing the new membrane care should be taken to ensure that:

- sufficient hog or sag is built into the free section of the membrane to accommodate maximum contraction of the structure during cold weather
- sufficient gap is left between the edges of the nosings to allow expansion during hot weather
- replacement membranes are chosen on the basis of manufacturer's recommendations to cater for the required movements.

Installation of a strip seal below a sliding plate or finger joint may require a multiple ripple seal (i.e. a membrane that is corrugated to allow for more movement) in order to accommodate the larger movement.

The following measures are recommendations from various manufacturers and are required (Roads and Traffic Authority 2008):

- Even though manufacturers claim large movement capacity for strip seal joints, the maximum allowable movement is 85 mm to be compliant with AS 5100.4.
- The flush internal gland-type is preferred over a draped, single-layer gland-type to avoid debris collection.
- Bent or mitred retainers should be used at kerb and traffic barrier upturns.
- Anchorages should be designed to resist all static and dynamic loads and should be thoroughly bonded to the concrete. The metal retainers should have regularly spaced vent holes to allow air to escape, unless venting is provided by the anchor bolt holes.
- Appropriate torque should be applied to tighten the anchor bolts during installation. The value of appropriate torque should be specified by the manufacturer and supported with experimental data to ensure an axial tension of 65% ULS load in the anchor.
- The appropriate number and size of anchor bolts should be specified.
- Pre-tensioned bolts should be considered.
- Failed joints shall be replaced with new strip seal joints.

Finger Joints

Inspectors should look for the following defects:

- Bent, broken, or misaligned fingers are generally a result of entrapped debris caught in the finger slots or, in the case of misalignment, differential settlement of bearings or failure of HD bolts. Before effecting any repairs, the causes 0of the fault should be investigated and remedied.
- Loose, noisy sliding plates indicate slackness or a failure in the bolts or other fixing device.
- The membrane seal sometimes mounted below the finger plates may be leaking or blocked with debris.

The implications of water leakage through faulty joints and joint structure are described in Section 5.8.

Where no waterproofing element is provided under the joint, the joint's overall performance can be greatly enhanced by adding a sealing element. Where such an element is present, but has failed through rupture or de-bonding, it should be replaced. Alternatively, it may be possible to install a drainage channel.

The following guidelines for maintenance of fingerplate joints are proposed (Roads and Traffic Authority 2008):

- Regular inspection and monitoring of these joints is required for public safety.
- Drainage troughs should be cleaned at least once a year, or more often as required.
- Damaged fingers should be repaired.
- The joint should be kept free of corrosion.
- Any loose nuts or screws should be investigated and remedial action taken urgently, as loose nuts or screws will quickly result in fatigue failure of the anchor bolts and uplift of the fingerplates under traffic.

Recommendations on the design of this joint type include (Roads and Traffic Authority 2008):

- A minimum permanent opening of the gap should be specified to prevent the joint from closing up at high temperatures.
- Fingerplates should have adequate stiffness to prevent excessive vibration and have sufficient flexural capacity to prevent bending and fatigue issues.
- Fingers should be aligned in the direction of movement to avoid exerting excessive forces on opposing fingers.
- Anchorages shall have sufficient tensile and shear strength to resist loads from heavy traffic including impact. To prevent fatigue failures of the anchorages sufficient bolt tension is required such that the load in the bolt does not change under the design ULS traffic load.
- Long debonded anchor bolts to avoid the use of base plates should be considered in new designs.
- Stainless steel drainage troughs with crossfalls of at least 8% should be provided to prevent water and debris accumulation. Stainless steel should be used for bolts, nuts and washers.
- In new long bridges with significant creep and shrinkage, resetting of fingerplates should be considered by providing extra bolt holes in the plates.

In addition, modern fingerplate joints are available on the market with special features that help to address the causes of the joint failures. The pre-tensioned bolts create a permanent compression stress between the joint and the structure, thus providing good resistance against vibrations and fatigue effects. The pre-tensioned fingers combined with a flexible and shock-absorbing design, help to protect the bridge structure underneath from fatigue-related problems and improve the capacity for accommodating different deflection, rotation or settlement across the joint.

Modular Joints

The following points should be considered in the maintenance and repair of modular joints:

- if there is uneven wear or friction or noise from sliding surfaces, the sliding medium should be replaced
- regular maintenance should keep the slots above the seals free of debris.

The following maintenance measures are suggested:

- Inspection should be frequent (once a year) to check the replaceable components such as springs and buffers.
- The joint shall be maintained in accordance with the maintenance manual supplied with the joint. The required inspection and maintenance schedule for the joint, together with work procedures required carrying out repairs and/or replacement of each component of the joint should be followed accordingly.

The environmental noise problem associated with modular expansion joints can be reduced by providing noise abatement equipment. It is recommended that all abutments underneath the joints shall have sufficient space within the abutment cavity if the installation of a Helmholtz absorber for noise abatement is required after post-commissioning noise measurements identify an actual or likely noise nuisance. Sufficient space shall also be provided for inspection and maintenance of the joint (Roads and Traffic Authority 2004).

Proprietary Expansion Joints – Moulded Rubber

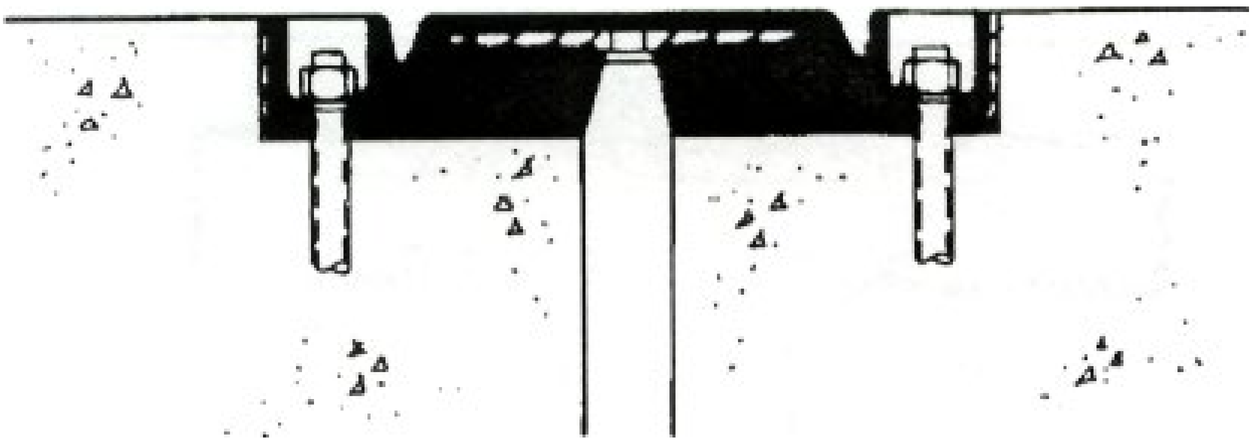
There are two basic variations of this joint type. A reinforced elastomeric plank joint is where the gap is bridged by a reinforced elastomeric plank (Figure 6.39). An elastomeric sheet seal is where the joint is bridged by a thin elastomeric sheet anchored on either side of the gap by reinforced elastomeric block nosings.

Elastomeric plank joints in particular are very stiff and substantial forces on their anchorages must inevitably develop. The failure and loosening of hold-down bolts can be related to this stiffness. The joints rely heavily on correct installation. If the joints are installed slightly above carriageway level then problems such as joint element wear, poor rideability, and excessive noise generation can be encountered.

Other faults that may develop are:

- failure of nosings (usually interlinked with the loosening of HD bolts)
- chips and other debris lodged in the shaped grooves in the rubber
- de-bonding of the rubber and steel plates
- leakage under the rubber cushion between units butted together
- loss of plugs protecting the hold-down bolts.

Figure 6.39: Typical reinforced elastomer joint



Source: NZTA (2001).

The following points should be considered in the maintenance and repair of reinforced elastomer joints:

- Bolts should be checked for tightness as a specific task in the maintenance program.
- If the rubber and internal plates have been de-bonded, the cushion will need to be replaced.
- Regular maintenance should keep the grooves in the rubber free of debris.

__Bonded Metal/Elastomer Joints__

The following points should be considered in the maintenance, repair, design and installation of bonded metal/elastomer joints:

- Anchorage bolts should be re-tightened periodically to compensate for creep of the elastomer and other loosening effects.
- This joint should be used with caution due to its high replacement cost and variable field performance.
- Selection of the correct size and correct installation play an important role in ensuring a good performance.
- The bridge temperature at the time of installation should be taken into account when calculating the joint gap. If the joint is installed at too cold a temperature, the joint may buckle up in the middle and result in damage by traffic in hot weather. If setting the joint in hot weather, excessive stretching in cold weather later may damage the elastomer or the anchorages.

It may be necessary to use adhesive sealants and to jack segmental panels together during installation to minimise leakage.

6.9.5 Holding Down Bolts

Holding down (HD) bolts in this context are used to fix the superstructure to the substructure. They are generally located within the diaphragm on concrete structures and through the beam flanges on steel structures.

In older bridges, HD bolts provided horizontal restraint against earthquake movement as well as vertical restraint, but in current designs the two functions are usually separated.

In concrete structures they often pass through ducts that will allow for some horizontal movement in the longitudinal direction.

It should be noted that current design criteria do not require the superstructure to be specifically held down, except where the dead load is small and likely to be reversed, for example, at a joint between two cantilevers. Where there is no HD device, horizontal restraint is achieved by some other means such as shear keys.

Faults that occur with HD bolts are:

- loose nuts caused by repetitive movement cycles of the superstructure
- bending and shear failures caused by excessive horizontal movements
- corrosion, generally caused by leakage from defective deck joints or exposure in an aggressive environment
- cracking of superstructure or substructure concrete caused by excessive horizontal movements or jamming of the HD bolts in their movement slots.

The structural implications of these faults are that the superstructure may not be effectively tied down to the substructure, and horizontal restraint may not be effective. In the event of a significant seismic shake, a span could be pulled off the supporting pier or abutment.

The following points should be considered in the maintenance and repair of HD bolts:

- Where HD bolts on a concrete bridge have bent, it is often the result of post-tensioning of the superstructure after the bolts have been installed. In this case, where insufficient allowance for movement has been made in the HD bolt ducts passing through the diaphragm, high stresses can be induced in the diaphragm area. The only effective remedy for this condition is to cut the existing HD bolts and drill in new bolts from the deck level. The new bolts should be grouted into the abutment or pier caps with 'non-shrink' grout and tightened down into recesses cut in the deck.
- Sufficient compressible filler should be placed above the bolts to allow for bearing compression and secondary concrete poured in the recesses to protect the bolt heads. Where the pier or abutment cap concrete has cracked in the embedment region of the bolt, the concrete should be broken out and patch repaired. If the cracking is not too severe, the cracks should be repaired by epoxy injection.
- Corrosion of exposed sections of the bolts is generally the result of leakage of deck drainage through defective joints. The exposed section of the bolt should be cleaned of all corrosion product and protected by coatings of paint. The defective joint should be repaired without delay.

6.9.6 Seismic Restraint Devices

Seismic restraint devices are used to resist larger movements from earthquake events. They may include such items as span linkage bolts to control longitudinal movements, keys and/or cleats to resist lateral movement and in older bridges HD bolts to resist movements in all directions.

Where bolts are used to link spans and to tie end spans to the abutment backwalls, the bolts generally pass through the diaphragms, are fitted with shock-absorbing rubber pads and are fitted in such a way that the independent components of the structure can move up to 100 mm before the device applies restraint.

Where bolted cleats are used to resist lateral horizontal movements, they are generally bolted to the substructure, and rubber pads ensure the forces are evenly distributed.

In some concrete structures lateral restraint is achieved by casting keyways in the substructure and keys as an integral part of the superstructure, or by upstands on the substructure bearing against the outer beams.

Faults that can occur with these devices include:

- Span linkage bolts installed too tightly to allow for normal thermal movements in the superstructure. This defect can transfer unacceptable loads to bearings and HD bolts. Where the bolts should have allowed for large relative movement, a significant seismic event could lead to a span coming off its support seating elsewhere in the structure. Alternatively, the linkage bolts may have been designed to restrict relative movement to a magnitude that can be:
- accommodated by the bearings and expansion joints
- misalignment causing seizure of the device, preventing normal thermal movement. This is often a sign of other problems such as differential settlement of the foundations or bearing failure
- corrosion of metal parts, generally caused by leakage through faulty joints or exposure in an aggressive environment
- cracked or spalled concrete keys and keyways, generally caused by misalignment faults.
  The underlying cause of the misalignment must be investigated
- bent cleats and sheared bolts, which may also be the result of misalignment due to differential settlement of foundations or bearing failure.

Most of the above faults will prevent the various components of the structure from moving as designed, causing excessive stress that will result in damage to other members of the bridge.

The following points should be considered in the maintenance and repair of seismic restraint devices:

- Where span linkage bolts have been installed too tightly, it is generally sufficient to slacken them to allow the required movement. However, it is prudent to investigate if this condition has already led to excessive strain in other areas. The linkage may have been designed to restrict relative movement to a magnitude that can be accommodated by the bearings and expansion joints. It is therefore advisable to check with the design office before adjusting linkages.
- If superstructure is misaligned, check that the seismic device is not cracked or bent and that the fixing devices are not damaged. If damage has occurred, replacement rather than repair is usually recommended as these devices are designed to cope with severe stresses during a significant seismic event and any weakening can detract from their ability to restrain the structure during such an event.
- Galvanised hardware within a kilometre of the sea should be washed down with clean water from time to time to prevent build-up of salts. In severe environments galvanised steelwork should be given some protection compatible with the associated elastomeric pads.

6.9.7 Pedestrian and Traffic Barriers

Pedestrian barriers (handrails) provide security for pedestrians using the structure and are not designed to withstand vehicular impact. Pedestrians may be protected from vehicles by a traffic barrier separating the carriageway from the footway. Where no separate barrier is provided, vehicles are generally confined to the carriageway by a kerb and a combined pedestrian/traffic barrier at the outer edge of the footpath.

Traffic barriers are designed to confine traffic to the bridge carriageway. A system commonly used is the W-section guardrail mounted on steel posts on the bridge deck and bolted to wooden posts on the approaches. Other types of non-rigid barrier (e.g. thrie beam) and rigid barrier (solid concrete or steel post and rail) are also used.

The design principle for W-section guardrail is that, under an extreme vehicle impact, the HD bolts fail or the wooden posts break off, allowing the rail to deflect outwards and resist the forces by ribbon tension. For other types of non-rigid barrier, the supporting posts bend under impact to allow ribbon tension to be developed. The height of the W-section guardrail and other non-rigid barriers relative to the road surface must be in accordance with the road jurisdiction's standards. If it is too low, vehicles may overturn, rather than being re-directed along the rail. For the same reason all W-section guardrail post packing-out blocks must be in place so that when a post bends under impact the rail will be raised rather than lowered. The W-section guardrail or other non-rigid barrier must be effectively anchored at each end in order to develop tension. Rigid barriers do not deflect under vehicle impact.

Defects that can occur with non-rigid barriers include:

- rot in timber handrails caused by loss of protective coatings and fungi attack
- corrosion of steel components, including bolts on both handrails and traffic barriers, caused by loss of protective coatings
- loosening of components by excessive movement, impact or vandalism
- misalignment, often leading to shear failures caused by failure of bearings or differential settlement of foundations (Section 6.7.3).
- impact damage.

These defects place road users and pedestrians in danger and repairs should be carried out without delay.

The following points should be considered in the maintenance and repair of handrails and traffic barriers:

Timber Handrails

Where damaged by impact, rot, or severe weathering, they should be replaced.

Where the damage is restricted to weathering of protective coatings, the timber should be sanded back and a full protective coating system applied from primer through to topcoats.

Any corroded straps or bolts should be replaced or cleaned of all corrosion products and an appropriate protective coating applied.

Steel Handrails

Damage generally consists of loss of protective coatings with subsequent corrosion, and/or loose fixing devices. All loose paint and rust should be removed by grit blasting and a full good quality protective coating system applied.

Damaged bolts or other fixing devices should be replaced.

Non-rigid Traffic Barriers

In most instances damage will be from vehicular impact. The damaged rail sections should be replaced using new bolts in both the splice areas and the fixing system to the posts. Bent posts should be replaced using new HD bolts. For W-section guardrail, these bolts have a special necked section designed to fail at a specific loading. If the barrier rail has suffered a severe impact, they may have been weakened and should not be re-used.

Non-rigid barriers are usually hot dip galvanised. If this sacrificial coating is defective the rail should be replaced and the defective section refurbished and kept for future replacement tasks.

Rigid Traffic Barriers

For rigid concrete barriers, damage from vehicular impact may occur to the face of the barrier. Severe damage should be repaired as for other concrete elements. For rigid steel barriers, damage from vehicular impact may occur to post and rail elements and connections. Severely damaged elements should be replaced using new bolts. Corrosion protection may also be damaged by vehicle impact and should be repaired.

6.9.8 Services Supports and Access Attachments

Services supports are generally brackets carrying service lines in ducts on the outer sections of deck soffits. Where service lines are located beneath the cover slabs on footways or inside box girders, the supports may be saddles mounted on the floor of the box girder or the deck surface beneath footways. Usually the brackets will be of steel, bolted to the underside of the deck.

While the items are technically the responsibility of the service utility concerned, the bridge owner has an interest in ensuring that they are kept in good condition.

Access attachments can range from full walkways in larger structures to eyebolts embedded in the deck soffit on smaller structures.

Faults that can often be found with these items are:

- corrosion of steel brackets caused by loss of protective coatings
- loose bolts, generally as a result of inadequately applied torque when installed
- misalignment of the service line due to damage to footways caused by differential settlement or abnormal movements in the structure.

The structural implications of these faults are less important than the danger to people using the areas beneath the bridge or the walkways on it. In addition, failure of the service the bridge is carrying can cause considerable disruption to users.

The following points should be considered in the maintenance and repair of services supports and access attachments:

- missing or loose bolts should be replaced and tightened as necessary
- loss of protective coatings on the brackets may require the brackets to be removed, wire-brushed to remove corrosion products and re-coated.

6.10 Foundations

6.10.1 General

This section considers only the effects of external forces on foundations. The effects of deterioration of materials are dealt with in the relevant sections on those materials.

6.10.2 Settlement

Detecting settlement of foundations is not difficult because it will normally be self-evident. The superstructure will be out of line and expansion joints may have opened or closed. If differential settlement has occurred at a support, the superstructure may also be warped, and the pier or abutment may show other distress such as cracking, particularly in piles.

If settlement has been observed, it is desirable to determine whether it is continuing and at what rate. Precision levelling over a period may be required for this.

Some causes of settlement are:

- Compressible layer – Settlement is often caused by a compressible layer such as clay or peat beneath the foundation or adjacent embankment. Increase in pressure in this layer due to the weight of the bridge and approaches causes consolidation of the material and settlement of the structure. If this is suspected, it can be verified by drilling and determination of the material properties.
- Downdrag – If settlement due to a compressible layer occurs as described above, and if a pile group penetrates the compressible layer, the settlement will induce downdrag forces on the lengths of piles above the layer. On end-bearing piles this may be sufficient to cause crushing of the base material or buckling of the piles. On friction piles the forces may lead to settlement by additional penetration.
- Scour or bed degradation – Scour during flood events can induce settlement by reducing the foundation embedment for a short time. Bed degradation over a period of years may have had the same effect. In either case, the overall safety factor of the foundation is likely to be unacceptably low.

Scour to a critical level during major floods may be suspected even though there may be no visible permanent effects. Verification of this by direct measurements during floods is difficult, as this involves being there at the right time with the necessary equipment, but there are methods of installing instrumentation to record scour automatically.

A frequent problem where critical scour is suspected is that the exact pile lengths may not be known, so that even if scour can be measured, the margin of safety is not known. Some success has been achieved in deducing pile lengths by measuring the ground resistivity between the pile and an electrode at various levels in a hole drilled alongside it. Knowing the pile length and characteristics of the bed material, the safety margin can then be estimated.

If a pier or abutment has settled, it is unlikely that it can be re-levelled, but the superstructure can probably be restored by jacking up and packing under the bearings. Before this is done, the settlement should be halted, and obviously it is necessary to determine why it is happening. The usual remedy is to underpin the structure by driving extra piles and enlarging the pile cap to accommodate them.

See also Sections 6.7, 6.11 and 6.12.

6.10.3 Pile Deformation

Abutment piles may deform because of horizontal movement accompanying loading of the ground beneath an embankment. In an extreme case this can result in plastic hinging in the piles. Such movement is most likely where there is a single row of piles in a line. It will probably show up as displacement and rotation of the abutment, and closing up of the expansion joint.

If piles are seriously deformed by ground movements, further movement can be prevented by tying back to a deadman anchor or by underpinning using raking piles, but it is unlikely the structure can be restored to its original position. It may also be possible to reduce the load from the embankment by replacing some of the fill with lightweight material. If this is placed behind the abutment backwall, and a settlement slab is placed above it, there may be some minor relaxation of the displaced abutment back towards its original position.

See also Section 6.12.

6.10.4 Abrasion of Piles or Cylinders

Foundations in gravel-bedded rivers with high-velocity flow may be subject to abrasion at bed level and below. Detection may require divers or remote TV, but should be carried out during a detailed inspection. If abrasion is significant, the effective cover remaining should be verified by measurement or by use of a covermeter.

A remedy for abrasion (Figure 6.40) is to jacket the piles with reinforced concrete. There are proprietary sleeves available to use as formwork, and the recommended method of placing the concrete is to use pre-placed aggregate and then grout it.

Figure 6.40: Abrasion of concrete pile cap

Source: NZTA (2001).

6.11 Waterways

6.11.1 General

Refer to the *AGBT Part 8: Hydraulic Design of Waterway Structures* for more detailed information on waterway related issues.

6.11.2 Waterway Problems

General

Many of the problems found in bridge waterways may be explained by consideration of the natural behaviour of rivers and the impact of man-induced changes.

Scour

Scour is the most serious problem (greater than 60% of reported failures can be attributed to it), and may be aggravated by another type of problem, e.g. debris or alignment, and may cause other problems, e.g. undermining or bank erosion.

Aggradation

Changes in alignment, upstream constrictions, or the inability of the downstream section to transport material away are the main causes of aggradation problems. Transient aggradation can occur with the transport of material, such as from a landslide.

Aggradation may cause general instability of the river downstream.

Piping

Piping occurs where there is inadequate protection against the migration of fines. It can cause subsidence or settlement of the structure, approach fill, fill containment works, or batter protection works.

Alignment Changes

These occur naturally due to meander migration and upstream aggradation, or they may be man-induced. They can cause significant increases in scour and place different parts of the structure at risk.

Blockage

Channel blockage can occur as a result of accumulation of debris or aggradation of the stream bed. This can cause or exacerbate other problems, e.g. scour or flooding.

Flooding

Flooding occurs when the structure waterway is incapable of accommodating the flow passing through it. This can lead to approach inundation, debris damage to the superstructure, and adverse effects to adjacent property.

Undermining

This is the progression of scour under the structure or protection foundation. It may also occur as a result of piping.

Swing Fences

Swing fences fixed upstream or close to a structure can generate significant changes in the waterway, exacerbating blockage and scour problems.

6.11.3 Inspection

Purpose

Inspections provide a visual assessment of the condition of the structure, approach batters, protection work, and the waterway. They also allow the causes of problems and the rate of change to be evaluated so that the seriousness can be assessed and appropriate remedial action programmed.

Procedure

The inspection should look for changes in the following parameters in addition to reporting the problems:

- land development or other changes (e.g. erosion) in the catchment area
- bed level at the structure as well as upstream and downstream
- alignment
- debris and vegetation
- bank erosion and the performance of bank protection and fill containment works
- maximum water level during recent floods, both against the structure and upstream where backwater effects occur. If the discharge or return period of the event is known, that should also be reported.

It is important for inspectors to record the likely causes or true nature of any problem to ensure the correct evaluation is made. To assist this it is suggested that full photographic sequences be taken of both the upstream and downstream channels. As a minimum, a photograph of the structure's elevation should be taken at each inspection to show the waterway cross-section.

Waterway inspections should extend a distance of at least three channel widths both upstream and downstream to look for changes in bed, alignment, and bank stability.

When observing depth of scour following a flood event it is important to be aware that scour holes tend to refill with sediment as the water level subsides especially in the coarser bed material sizes.

Where significant bed degradation or scour is taking place and there is uncertainty on the founding depth of piles or spread footings, then inspection should include the determination of these depths. This information must be known to allow a true evaluation of the risk to the structure.

Techniques using ground radar or cored holes are available for determining the extent of foundations (Section 6.10.2).

In some situations underwater inspections will be necessary.

Records

It is important that good records be kept to:

- see how the river is changing with time
- assist with evaluation of condition
- ensure planned action is appropriate, consistent and that it is carried out.

The records should include:

- original design drawings
- a chronological list of significant events such as modifications to the structure or waterway and changes within the catchment
- inspection reports whether they be routine or periodic (flood-related)
- channel profiles and cross-sections, taken periodically as appropriate
- photographs
- maximum water levels and their dates.

6.11.4 Evaluation

Waterway problems have accounted for many Australasian bridge failures. Approximately 75% of reported bridge failures in New Zealand can be attributed to waterway problems. It is therefore important that the evaluation be undertaken with due care and consideration and with the best available information. In general, evaluations rely on experienced engineering judgement and may still involve a large degree of uncertainty. In some cases it may be necessary to call on specialist assistance to make the evaluation. It is important that conclusions are practical and economic.

The inspection report should suggest the level of evaluation required to assess the security of the structure, approach and protection works.

Inspection should identify changes that have occurred or problems that are evident in the waterway or in the structure. The photographs obtained should be compared to those from the previous inspection to confirm the nature and degree of change. Should the change be considered significant then the observations should be compared to the original design assumptions. This will enable the engineer to assess the nature and degree of change to predict future waterway changes, and thereby to assess the degree of risk to the structure, approach, and protection works. Note that some problems are transient in nature and therefore make risk assessment difficult.

From this assessment of risk or impact on the structure, the appropriate course of action is determined. This might involve maintenance, securing, or structural modifications (e.g. underpinning or increasing the waterway area) as described in Section 6.11.5.

6.11.5 Maintenance and Durability Enhancement

General

The waterway of a river crossing requires an ongoing maintenance commitment in much the same way as a bridge structure.

The maintenance required can vary from the routine, to major design and construction work that may be required to prolong the life of the structure. A major problem in the waterway, such as severe degradation or channel movement, may require significant investigation, analysis and design effort to identify the cause of the problem and find an economic solution.

The extent and nature of the problems identified during the inspection and the subsequent evaluation will influence the repair and/or maintenance strategy. The most economic strategy will optimise cost and structure risk for the remaining life of the structure.

No significant work should be undertaken without the approval of the river controlling agency.

Routine Maintenance

The channel should be kept clear so that the water will be allowed to flow freely.

Remove logs, trees or other debris from the waterway and upstream and downstream channel before they can alter the course of the river and exacerbate scouring and undermining. Flow disturbances occurring up to three meanders above a bridge site can affect alignment at the bridge site. Inspections, using divers where warranted, should be made as soon as possible after floods and log jams to ensure that all debris has been removed.

Monitor the condition of existing bank or bed protection regularly and carry out maintenance as necessary. This could include, for example, the topping up of slumped riprap protection, repair of groynes or gabions, or the thinning and inter-planting of willows.

Deposition of alluvial material (aggradation) can reduce the available waterway area. If aggradation reduces the flood clearance to less than what is acceptable, then as a short-term expedient the material deposited at or near the crossing should be removed. Since this has not dealt with the cause of the aggradation, ongoing monitoring should be undertaken to check the rate of change. If the risks to the structure are high, then a detailed investigation should be initiated.

Gravel bars can deflect the water from its normal channel to cause erosion of a bank or scouring of a foundation. Removal of gravel bars can reduce such risks.

Bank and Abutment Protection

Purpose

Since bridgework generally costs more per unit length than the approach embankments, maximum economy is usually achieved by minimising the waterway. Thus a moderately constricted waterway with bank protection and/or training works will, in many cases, have been the optimum solution available to the agency that installed the crossing. In a number of these crossings the abutment or approach fill will also need to be properly protected to ensure secure batters.

The bank protection and training works are themselves structural elements that, to a greater or lesser extent, form obstructions to the flow and are therefore subject to attack. Their ongoing purpose is to:

- stabilise the river banks and channel
- protect road approaches from stream attack
- constrict the waterway to economise on bridge length
- align the flow to minimise afflux, scour and the trapping of debris
- direct the flow parallel to the piers to minimise local scour.

In addition to these purposes, protection works in the vicinity of structures generally need to accommodate run-off from deck drains or approach side drains.

Ongoing maintenance of the bank protection and river training works is essential to ensure the crossing as a whole continues to give good service.

Types

Commonly used types of flexible protection include:

- riprap (Figure 6.41)
- gabion baskets or gabion mattresses (stone filled wire baskets)
- concrete-filled bags or geotextile mattresses
- concrete blocks or slabs – loose or articulated
- vegetation – such as willows.

Occasionally rigid types of protection are used, including:

- poured concrete slabs or walls
- soil cement
- asphaltic concrete
- timber walls.

Figure 6.41: Moonshine Bridge, Hutt River illustrates the use of riprap to protect an abutment

Source: NZTA (2001).

Maintenance of Bank Protection

Bank protection can fail because of uplift, undermining, outflanking, overtopping, or washout.

Uplift can occur if the structural elements are too light to resist the uplift forces resulting from the high-velocity flow. Uplift can only be prevented by ensuring that the elements are large and heavy enough to withstand all the forces that the protection work may be subjected to from all different angles of attack.

Scour is liable to occur at the following locations near banks:

- alongside banks, in parallel flow
- around the ends of any projecting banks, where spiral currents occur caused by oblique flow
- under banks, by direct attack.

Methods of avoiding undermining include:

- continue the protection down to or below scour level
- drive a 'cut-off-wall' from the toe of the protection down to a basement level
- lay a flexible apron (also called a launching apron) horizontally on the bed at the foot of the protection. As the scour develops the material will settle and cover the side of the scour hole on a natural slope
- pave the entire bed. This is generally only economical for relatively small streams or where bed control is also required.

Care must be taken not to promote scour or progressive failure at the upstream and downstream ends of the protection. This can be achieved by turning the protection into the bank where it cannot be outflanked.

The protection should normally extend above design water level to avoid damage from overtopping.

To avoid washout or piping of the underlying soil, natural rock/gravel filters or synthetic filter fabric should be used under the protection layer.

Training Works

Training works are often employed to favourably align the flow with the bridge opening. Straightening, shortening of the flow line and increased velocities normally result. To offset the increased velocities and to maintain bank stability, some degree of stabilisation or channelling is often required.

The principal types of training works at bridge sites are:

- guide banks (or spur dikes) – built to direct the flow smoothly through the opening and to minimise the scour depth at the river crossing
- groynes (or spurs) – constructed roughly perpendicular to the river bank
- stopbanks – to prevent flooding beyond a chosen zone
- diversions – man-made channels.

6.12 Seismic Damage

6.12.1 General

This section is intended to provide a strategy for undertaking bridge inspections following a significant earthquake and to give guidance on where to look for, and how to evaluate, damage to typical highway bridges. It is based on NZTA practice and other road jurisdictions will have their specific requirements based on their risk assessment for such an event. However, the principles adopted are applicable to all road jurisdictions.

It does not cover very large or special structures. Obvious problems such as fallen spans or other extreme damage, where the severity of the damage is self-evident, are not discussed. It does not cover damage repair, and it strongly recommends that design advice be obtained before repairs are carried out.

Emergency work to open the bridge or to clear roads or waterways below may sometimes commence very shortly after the earthquake. Even if total demolition is going to be the final fate of the bridge it is desirable to investigate, record and photograph the damage before the evidence is destroyed. It is preferable that this work be done by an experienced bridge designer who has experience in forensic investigation of seismic damage and who is not involved in the emergency work. If there is an authorised Earthquake Engineering reconnaissance team in the area they will be able to assist and should be given access to the site.

6.12.2 Objectives

The overall objectives of seismic damage assessment are to:

- minimise loss of life
- minimise the economic loss to the region.

6.12.3 Strategy and Inspection Levels

To ensure that a route or network is safe for the public, bridges should be inspected following an earthquake of sufficient intensity to cause concern about the possibility of damage. All bridges within an area subjected to VI magnitude intensity (Modified Mercali Intensity Scale) shaking or greater should be inspected. Two levels of inspection are appropriate:

- a preliminary safety check, conducted immediately following the earthquake to check for safety for immediate use and for obvious damage
- a detailed structural check, which may or may not be required, and which would be conducted at some later time.

For example, in most areas of New Zealand, seismic screening of the state highway bridges will have already identified those bridges likely to be most vulnerable to damage with potential to cause loss of life.

6.12.4 Preliminary Safety Check

The following order of priority should be considered for the preliminary safety check of the bridges:

- inspect first those bridges known to be most vulnerable with potential for loss of life (e.g. as identified by the seismic screening), giving priority to those carrying the highest traffic volumes
- inspect all other bridges along the primary routes required for the passage of emergency vehicles concerned with the saving of life and property
- inspect all other bridges along other primary routes required for the distribution of essential supplies and restoration of essential services
- inspect all remaining bridges.

Each bridge should be examined quickly but with sufficient care to identify problems that could lead to collapse and compromise public safety.

Fortunately, serious damage can often be detected at road level. Nevertheless, the underside of the deck and the substructure should also be briefly examined.

The bridge and approach embankments should be observed for signs of settlement, which is a very common form of earthquake damage (Sections 6.7 and 6.10). Vertical settlement may not be a cause for immediate concern other than from the road safety aspect. However, it is likely to have increased the loading on abutment piles through down-drag. If it is also associated with horizontal movement of the ground, especially towards the bridge, the abutment may have been moved as well. This has a consequence for foundations, bearings, and expansion joints. Such ground movement may extend to the adjacent pier as well, with similar consequences.

Settlement of approaches may have damaged services in the ground. If it causes water to leak, further damage from washouts may later eventuate.

At the bridge, the lines of the handrails, kerbs, and centreline markings should be checked for horizontal and vertical discontinuities, as these will be quick indicators of problems below. Differential settlement between piers and abutments, of any one support relative to the others, may indicate serious damage to substructure members. Also, it will alter the stress distribution in continuous superstructures, which could lead to overstressing and damage at some sections.

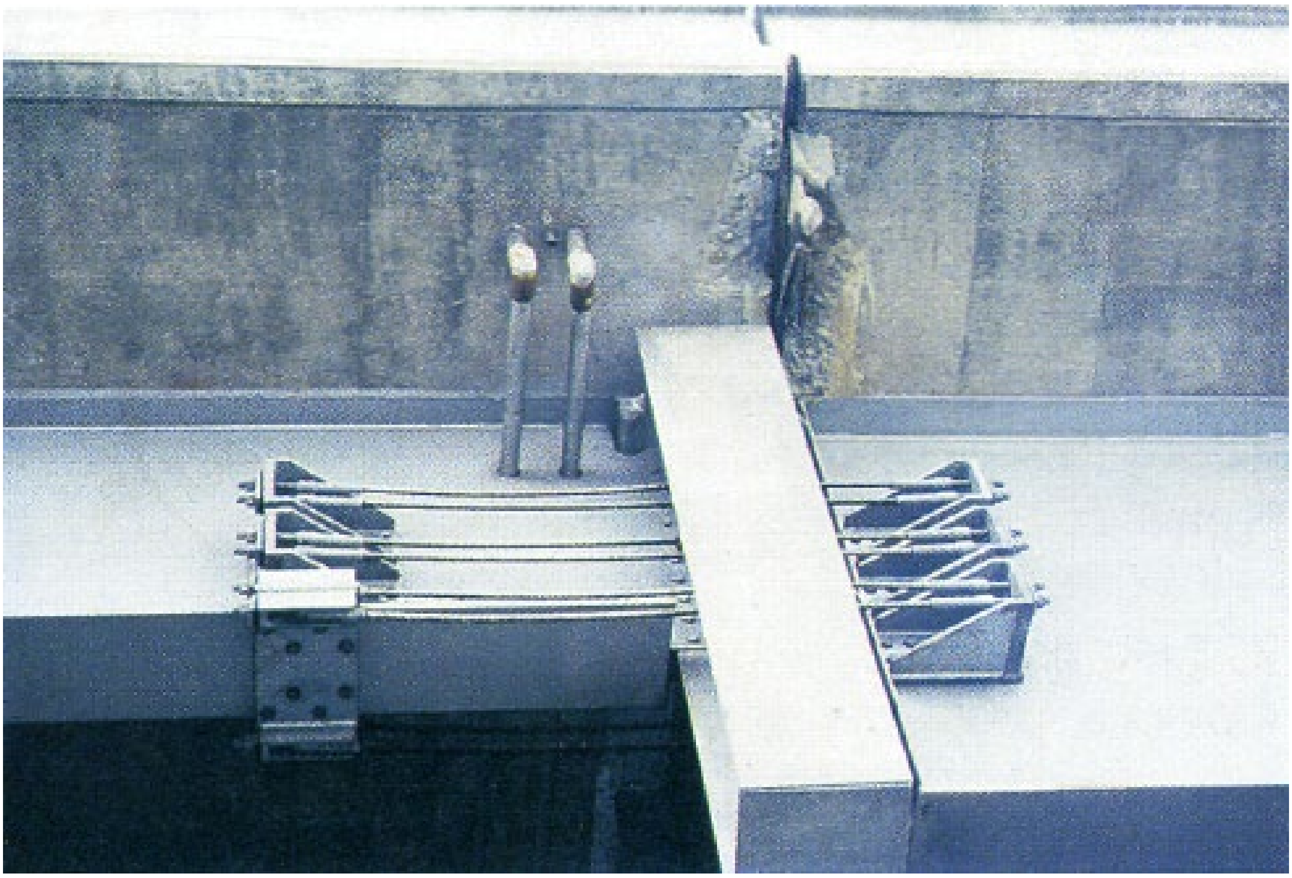
Other indicators of problems that can be seen at deck level are:

- evidence of excessive movement of expansion joints during the earthquake (Figure 6.42)
- expansion joints closed up
- knock-off devices at abutment backwalls displaced backwards and/or upwards by impact
- spalling of kerbs and decks either side of expansion joints
- buckling of handrails or traffic barriers.

A visual inspection below the bridge should reveal obvious damage such as:

- bearing failures or bearings having 'walked'
- linkage and shear key failures
- cracking or spalling and yielding of abutments and piers
- movement of abutments and piers
- damage to exposed piles.

Figure 6.42: Spalling damage at expansion joint



Source: NZTA (2001).

If damage is found, several courses of action can be taken. The bridge may be:

- left open to the public unrestricted, but noted for a detailed structural check (Section 6.12.5) at a later date
- left open to the public but with restricted speeds and/or axle loads
- left open only for emergency vehicles
- closed until temporary repairs are completed or until shoring has been installed
- closed indefinitely.

In deciding on what course of action to take, the engineer should take account of the risk of the bridge collapsing against the consequences of placing restrictions on it. That in turn will depend on the importance of the route and the alternatives available. The likely effect of aftershocks should also be considered.

It should be remembered that a bridge can sustain a great deal of superficial damage including loss of cover concrete without the vertical load-carrying capacity being affected too greatly.

Lastly, any damage discovered should be recorded, photographed, and confirmed as recent and likely to have been caused by the earthquake.

6.12.5 Detailed Structural Check

Approach Embankments

High approach embankments on soft ground are notorious for settling or slumping in earthquakes. If settlement is associated with failure of underlying soils, especially liquefaction, then soil flow through the abutment is likely to have occurred.

If lateral displacement has occurred it can be detected by such evidence as:

- heave at the toe of the embankment
- longitudinal cracking of the approach road surface
- movement of the abutment
- sand volcanoes and/or ground cracking on the flat ground.

Soil flow through the abutments will increase the lateral load on piles, and it may have caused damage to them that can only be seen by excavation. If the rotational and lateral movements of the abutment can be quantified it may be possible to carry out a back analysis, which gives an indication of the risk of pile damage. Abutments on raked pile groups cannot sustain much movement without damage.

Flow through the abutments can occur even where there are no approach embankments but where the natural ground is weak. There are many recorded cases of river banks moving closer together in earthquakes (usually but not necessarily associated with liquefaction).

See also Sections 6.7 and 6.10.

Apart from the problems with moving ground discussed above, abutments are vulnerable in other ways. Particularly with more recently designed bridges, the superstructures are often designed to act independently of abutments in an earthquake, and large relative movements between the abutment and the superstructure can take place. If the intensity was sufficient to cause yielding in the substructure this relative movement could be greater than that provided for normal service. In such cases damage is likely to include:

- hammering and concrete spalling
- failure of expansion joints or at least of their seals
- knock-off devices displaced backward and/or upwards

- linkage hardware distressed
- shear keys damaged.

Such damage is considered to be acceptable, repairable and not necessarily cause for closure.

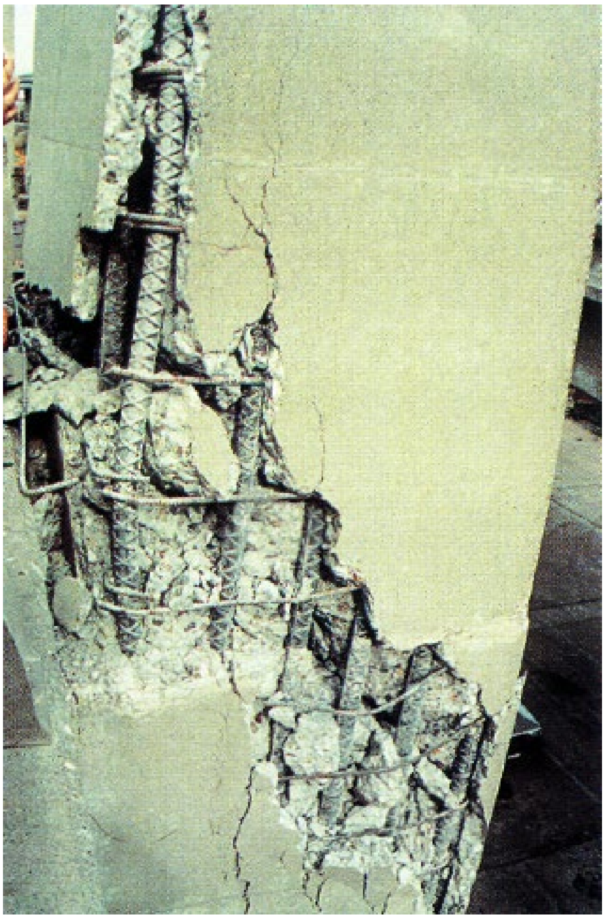
Where the abutment is independent of longitudinal superstructure movements but provides lateral restraint, the mechanism for providing that restraint should be checked, i.e. check for damage to shear keys, linkage devices, mechanical dampers, etc.

Piers

Cracking of piers to a greater or lesser extent should be noted and considered on a case-by-case basis for sealing, repair, encasing or replacing.

The piers are the visible part of the substructure and ideally are the locations chosen by designers for the development of plastic hinges in the design intensity earthquakes.

Figure 6.43: Ineffective confinement in a column



Source: NZTA (2001).

For modern bridges at least, spalling of cover concrete at the top or bottom of piers may indicate that the bridge has been subject to a design intensity earthquake and has yielded according to prediction (Figure 6.43).

If the reinforcing cage is largely intact and the core concrete properly confined, the vertical load capacity is likely to be adequate. Repairs will be required but it may be possible to carry them out with the bridge in service.

Piers should be checked for verticality. If they have moved out of plumb significantly during the earthquake the reason should be determined. If it is caused by yielding of the pier and displacement of the superstructure, then it may be desirable to straighten them before repairs to the yield zones are attempted. However, if piers are out of plumb and the superstructure is not displaced, there is a strong inference of foundation displacement.

Foundations

The foundations are the hidden part of the substructure and ideally, in modern (post-1972) bridges, have been designed with sufficient overstrength to ensure they remain elastic up to and after the bridge structure has started to form a collapse mechanism.

However, that design approach will not have been applied in older bridges, and even in modern bridges that ideal is not always achieved and piles can yield (Figure 6.44).

Figure 6.44: Movement of piles in the ground

Source: NZTA (2001).

Telltale signs are:

- large relative movements between piles and the soils
- relative displacement between pile caps
- piers out of plumb.

From each of these signs a deflection of the top of the pile can be measured and a back analysis using a range of soil parameters can be used to indicate the likelihood of pile yielding. The final check is to dig them out and examine them, but usually that is not easy to do.

Vertical displacement of piles does not always mean that pile yielding has taken place, particularly if there are no raked piles in the group. Furthermore, temporary loss of bearing capacity (e.g. from liquefaction) does not necessarily mean that capacity is lost permanently.

Vertical displacement can have serious consequences for the superstructure and particularly so for indeterminate (continuous) spans. The effect should be assessed.

For simply supported spans, the rotational capacity of bearings may have been exceeded because of the settlement.

For short deep continuous spans, distress could have been caused by small settlements that are only detected by taking levels. This should be considered as part of a thorough investigation.

See also Section 6.10.

Linkage Devices and Shear Keys

Linkage devices and shear keys are installed in bridges to limit relative movements between adjacent spans, and between spans and their supports. If they have been worked very hard by the earthquake they may have been damaged and require replacement.

To check the linkages properly it may be necessary to remove some for close inspection.

See also Sections 6.9.6 and 6.9.8.

Expansion Joints

Expansion joints that have operated beyond their design range may be damaged and require repair or replacement. Many joints have very little capacity for lateral displacement and damage can be caused by very small movement.

See also Section 6.9.4.

Holding Down (HD) Bolts

HD bolts that have been subject to large transverse shears are likely to have yielded. That does not necessarily mean that replacement is required. Recent design practice is to provide no HD bolts at all in most cases. Each case should be assessed on its merit.

See also Section 6.9.5.

Bearings

Bearings are vulnerable in earthquakes, particularly if they are also required to carry transverse shear:

- elastomeric bearings
  - Bearings that have been deflected beyond their design shear may be ruptured internally, but usually damage manifests itself at the surface and can be seen. Deep lead-rubber bearings are used for base isolation and can be subject to large lateral deflections. They should be carefully checked after an earthquake. If there is any suspicion of damage it may be necessary to remove one and check its capacity for continued performance.
  - Bearings not positively restrained in position may ‘walk’. There is a recorded case of a lead-rubber bearing escaping from its keeper ring during an earthquake.
- sliding bearings
  - If the design capacity for sliding has been exceeded, damage is likely.

- pot stay bearings
  - If a pot stay bearing has failed in shear it will be obvious.

See also Section 6.9.3.

6.12.6 Investigation Report

The results of the investigation should be recorded in a report that includes the:

- assessment of the ground acceleration at the site
- assessment of how the loads were transmitted to and from the ground and their magnitude (i.e. trace the load paths)
- recording of damage and permanent deformations, including photographs.

One of the potential difficulties of investigating earthquake damage is confirming that all the damage observed occurred during the earthquake. If investigation is sufficiently soon after the event it is easier to differentiate between new and old damage, particularly where previous inspection records are able to demonstrate the extent of any pre-existing damage.

7. Heritage Bridges

7.1 Introduction

This section is intended to complement Austroads *Guide to Asset Management*, which deals with the asset management aspects of bridge management. The focus of this section is on how to best manage the structural maintenance and management of existing heritage bridges. See Appendix B.5 for illustrative examples of heritage bridges.

Bridges that are identified as heritage items need to be managed in a manner that conserves their heritage significance over time. The best way of doing this is to keep a bridge in service performing as close to its original configuration as possible. Recognising and understanding the significance of individual heritage bridges is the first step towards their proper care and management.

The significance of each heritage item is determined in accordance with the criteria described in Appendix B.3.

A conservation management plan (CMP) is a fundamental tool in managing heritage bridges. The preparation of such plans is described in Section 7.9.

Where a CMP has been prepared for a bridge, the bridge should be managed in accordance with this plan. The plan will contain details of how different elements of the bridge are to be addressed in any work required on the bridge. Some works will be approved by virtue of accepta0nce of a CMP and can be undertaken without further approval, while any work not covered will require approval before it can be started. The CMP is likely to define a bridge curtilage to ensure that heritage significance is retained, and works may thus involve more than the bridge itself. Management of the bridge curtilage is likely to involve the local government agency in its planning role.

A strategy for implementing the works should be prepared in conjunction with the CMP. Those processes that require approval can be easily identified through the strategy.

Where no CMP exists for a bridge that is listed as a heritage structure, and work is required, such a plan should be drawn up before the work commences. If the works are urgent they should be referred for approval before work commences. Approvals may be required from within the asset owner's agency, from local government and from state and national heritage bodies.

7.2 Works on Bridges

Some works required on a bridge will fall within the scope of the approved works in the endorsed CMP, and these can be implemented without further approval. Documentation of these works with referral to the CMP is required as a record of work and conformance with the plan.

Work considered as routine maintenance should be defined in the CMP for the individual bridge. Items such as cleaning of gutters and scuppers, repair of barriers, resealing of pavements and reinstatement of scour should be identified as routine maintenance, where there is no effect on the heritage value of the bridge, for simplicity and efficiency in executing the work.

Paints and protective coatings should be sympathetic to the original colour scheme for the bridge or similar bridges of the era. It should be noted that bridge colour schemes used in the past were not necessarily the same as those used for buildings. The use of 'heritage' building colour schemes when choosing a colour scheme for a bridge repainting is not recommended. Rather, attempts should be made to establish the original colour scheme(s) present on individual bridges when first opened. The range of colours for high-performance protective coatings is also likely to be limited.

Work such as strengthening, widening, barrier replacements and installation of services on a bridge may have an impact on the heritage value of the structure and should be referred to relevant bodies for approval if not specifically covered in the CMP.

Services need to be given particular consideration because of their potential to impact upon the appearance and operation of a bridge and risks associated with failures of pipes carrying fluids or gases. In Australia, the Commonwealth's telecommunications legislation gives significant powers to carriers and limits the time available to respond to any requests for installation of services. Where services are permitted, there should be an emphasis on minimising their visual impact.

Bridges that are bypassed or that are no longer required for their current use may be adapted for some other function. An example would be a road bridge being adapted to serve as a pedestrian bridge.

Any work or changes to the structure are required to comply with the CMP and necessary approvals obtained where work or changes are not already approved in the plan.

7.3 Approval Processes

Approvals will be required for all works not covered by an endorsed CMP.

Approval processes will differ between agencies and jurisdictions and may involve the agency's designated heritage officer or relevant project manager for internal approvals and for submission to the relevant state and national heritage bodies and planning authorities. It will be necessary for bridge owners to ascertain the approval processes that apply to their particular jurisdiction.

Works requiring approval cannot commence until this is obtained.

Applications for approval should describe the proposed works with associated reasons in enough detail for an assessment to be made of their impact on the heritage value of the structure.

The design of works should endeavour to minimise the impact on the heritage value of a bridge. Early liaison with heritage authorities will facilitate the development of an acceptable solution.

7.4 Bridge Records

The recording of bridge data involves documentation of its form and condition at various times and any works that may be undertaken on it. The condition and form of a bridge at a point in time may be recorded using survey, drafting and photographic techniques. Stereo photogrammetry with digital models may be a useful tool for recording, for accurate measurement to assist with structural modelling and for monitoring any permanent deformations and deterioration. Photogrammetric models may be used as a main way of preserving the bridge information and yet avoiding the preservation of the real bridge and avoiding costs (Figure 7.1).

Figure 7.1: Photogrammetric model of Richmond Bridge

All work undertaken on a bridge, other than basic routine maintenance such as deck cleaning, should be recorded.

Records need to be made in sufficient detail to enable subsequent readers to understand exactly what works have been undertaken and what effect these may have had on the heritage value of the bridge. These records form part of the historical record of the bridge, which includes any changes since its construction, and should be managed in accordance with that level of importance in mind. Too often important records are lost, left languishing in a pile of filing, or are destroyed because the information contained, particularly that relating to construction and early years of a bridge's life, is considered to be no longer important for the management of the bridge. All files relating to existing bridges should be marked 'not to be destroyed' or similar. Archival records of original bridge designs and any subsequent alterations are also important in identifying a sequence of events shaping a bridge's life. Archival records are also important for determining the appropriate load capacity of a structure.

Where disposal of a heritage bridge is being considered, a heritage record of the structure should be prepared before disposal whether by transfer of ownership or by demolition. Records should be transferred with transfer of ownership so the new owner can maintain the history of the structure.

Records of demolished bridges should be archived for future reference.

Recording requirements may be defined in heritage and archival legislation relevant to a particular jurisdiction.

7.5 Interpretation

Providing opportunities for interpretation and appreciation of the significance of a heritage bridge by members of the public is a desirable aspect of managing bridge heritage.

Well thought out access for members of the public, including parking provision, will facilitate the appreciation of heritage bridges. Such access needs to recognise that the bridges will generally be working structures, often carrying vehicles, at some height above the surrounding ground and public safety is an important consideration. Adverse impacts of facilitating access, including vandalism, also need to be considered.

Where public access is available, interpretation panels may be provided. The Institution of Engineers, Australia has an established plaquing program for significant items of engineering heritage, including bridges. In NSW, the Roads and Maritime and the NRMA also undertook a plaquing program for Australia's bicentennial in 1988. Consideration needs to be given to the form and location of interpretive material on or adjacent to the bridge site so that it does not detract from the bridge itself. Signs or plaques adjacent to, rather than attached to the bridge fabric, are preferable.

Brochures are another option to be considered in providing interpretation. Brochures may be specific to a structure or developed as part of a bridge heritage experience as in Victoria. In NSW, the Roads and Maritime has produced a pamphlet, in the form of a driving tour, on the historic bridges of the Hunter Regions.

7.6 Disposal of Heritage Bridges

Bridges that are no longer required by an agency through processes such as bypassing with a road of improved standard may be disposed of by either adapting them to some other use, providing access on a road of lesser importance that may be managed by another agency, or demolition. It would be very rare to have a bridge that is considered so significant that it has to be retained, and yet used for no purpose (such as pedestrian access or to carry services). For example, the deBurgh truss bridge over the Lane Cove River that carried a Water Board pipeline after it was decommissioned as a road bridge was no longer used until it was destroyed by bushfire in 1994.

Adaptation for an alternative use may require works to be carried out on the bridge, and the CMP will be used for the control of this work. If the works are significant, the CMP may need to be altered to reflect the changes.

Bypassed bridges need to be assessed for future maintenance required to maintain public safety and for their heritage value. If the bridge is assessed as being of significant heritage value then the agency may be obliged to maintain the structure to preserve this value.

Bypassed bridges that are still the property of the agency need to be maintained in a safe condition if access can be gained to the bridge. The extent of barriers that are installed to prevent access, and the legal obligations if people gain access, needs to be determined by the agency.

From a heritage perspective, an important aspect of the heritage significance of a bridge is lost when it is bypassed, that is, its association with a particular transport network, its development and growth. However, some bridges are so significant in their own right that they should be retained even if no longer operating. It is preferable for a bridge to have some purpose, allowing public access, rather than remaining unused. It is hard to justify the expenditure of public money on a structure that the public cannot use or appreciate. Some alternative use should be found that should not be at the expense of its heritage significance, or the item should be demolished and its parts salvaged for the maintenance of other bridges of its type (if possible).

If a bypassed bridge is assessed as not being of significant heritage value there is still a cost to the agency if it is maintained, and demolition of the bridge may be an option. Approval has to be obtained for demolition of any heritage bridge. Note that demolition is the least preferred, but sometimes the only practical option.

A bridge owner would need a strong case to demonstrate that there was no prudent and feasible alternative to demolition. Disposal processes may be governed by heritage legislation.

7.7 Funding

Owners of heritage bridges will commonly be working in an environment of limited funding and a range of competing objectives such as safety, transport efficiency, minimisation of life cycle costs and environmental responsibility, including heritage aspects.

Alternative sources of funding may be available for the heritage aspects. In Australia, options include Environment Australia's *Cultural Heritage Projects Program*.

7.8 Operational Issues

7.8.1 General

Heritage bridges will commonly have been designed and built at a time when loadings were lower or less well understood and when society was less litigious. Bridge managers thus need to consider a number of developing and emerging issues in managing bridge heritage, including the changing legal framework.

7.8.2 Bridge Loadings

The principal purpose of bridges managed by road jurisdictions is to carry vehicular traffic. Permissible masses and volumes of vehicles, particularly heavy vehicles, are continuing to increase and are generally well in excess of those envisaged at the time of design and construction of the bridges. In the latter part of the 20th century, vehicle loadings increased at a rate of approximately 10% per decade. The new Austroads traffic loading standard provides for further substantial increases in design vehicle loadings over the next century. Bridge managers are thus faced with a continuing challenge of managing bridges to perform their primary role while conserving their heritage value. Bridges may need to be assessed for their ability to carry existing or proposed loads, load limiting options may need to be considered, and strengthening or adaptation undertaken as appropriate.

Processes for determining other applied loadings, such as flooding, have also changed.

Assessment and strengthening should consider the range of loads that may be applied, including vehicular loads, dynamic loads, acceleration and braking, temperature effects, stream and wind loads, parapet and barrier loads as well as earthquake loading.

7.8.3 Risk Management

As indicated, the manager of a heritage bridge needs to balance a range of aspects including:

- the safety of the travelling public, including the ability of the bridge to carry temporary and future vehicular loadings
- differences between geometric, structural and other standards of the bridge and those embodied in contemporary standards
- the interests of other stakeholders
- conservation of heritage and other values, such as tourism and identification with the bridge
- the legal context, including roads and heritage legislation and principles of feasance and negligence
- limited resources, particularly funding, in the context of managing a larger stock of other road and bridge assets.

The key responsibility of a bridge asset manager is ensuring the provision of a bridge that satisfies current traffic requirements and which functions in a safe manner. The focus of a heritage practitioner's perspective is ensuring that items of cultural heritage are managed in respect of their assessed level of significance and in accordance with legislation and guidelines. When these two responsibilities are in harmony with each other, the balancing of these issues is relatively easy. However, when these two issues are in conflict, the balancing of the various issues becomes more difficult. This is why an understanding of exactly what makes a heritage item significant, and the level of that significance, is the key to managing that item. For a bridge of not very great heritage significance, the operational disadvantages may outweigh the benefit to the community of retaining that item. On the other hand, a very highly significant bridge may warrant retention at the expense of improved load ratings or geometric standards on a route, or may justify the construction of an adjacent or alternative bridge, rather than the demolition of the existing one.

The first step toward managing heritage bridges is understanding their significance, at the very least as individual structures, but more preferably as a group or groups of bridges. Once this has been completed, then heritage significance can be incorporated as a factor into planning, asset management and risk management processes, rather than being considered when a project to upgrade or replace a heritage bridge is well progressed. When this significance is understood, the best tool for helping to maintain that significance within the practical considerations of day-to-day activities is a CMP.

A structured risk management approach, such as that outlined in the Australian and New Zealand risk management standard AS/NZS ISO 31000, may be of assistance in establishing an appropriate balance.

7.9 CMPs

7.9.1 General

A CMP is a document in which the heritage significance of an item or place is defined, and the management policies that are appropriate to enable the significance of the place or item to be retained in its future use or development are stated in accordance with the Burra Charter. Kerr (2000) (Appendix B.1) *The Conservation Plan* is a valuable document in understanding how to translate the principles of the Burra Charter into a CMP. In New Zealand, the comparable document is *Preparing Conservation Plans* by Bowron and Harris (2000) (Appendix B.1).

In the case of surviving heritage bridges, the majority are still in use, even if the mode of transport they serve has changed over time. The need to prepare a CMP for a heritage bridge is centred upon establishing its heritage significance (if this has not already been done), how this significance is expressed and how best to manage this significance within the operational, legal, financial and other constraints faced by bridge owners.

7.9.2 Process

Trigger for CMPs

As shown Figure 7.2 there are three possible triggers for the preparation of a CMP for a heritage bridge.

A proactive approach may be used as part of the management of a larger heritage bridge asset. In New South Wales, the Roads and Maritime has a policy of preparing a CMP for each of its bridges assessed as being of state or regional heritage significance. The Tasmanian Department of State Growth is similarly progressively preparing CMPs for its significant heritage bridges. In an ideal world, an agency would prepare CMPs for every heritage item under its ownership or control. However, for agencies with large numbers of heritage items, this is not feasible. Thus the preparation and endorsement of CMPs for an owner's most significant items should be the first priority and a minimum goal.

The need to undertake significant maintenance or other works, such as strengthening for increased vehicle loadings, may also provide the trigger for the preparation of a CMP for a bridge.

Many communities identify strongly with heritage bridges in their locality. Heritage bodies, such as the National Trust and heritage branches of the Australian Institution of Engineers, may also have a strong interest. CMPs may be prepared as a result of strong interest expressed by local communities and heritage bodies so that the heritage significance can be better understood and conservation policies developed.

Skills Required for Various Elements

Study teams put together for the preparation of any CMP must be multidisciplinary in nature in order for the team to have the necessary skills and knowledge to cover the range of issues that a CMP is expected to address. Ideally this should be a combination of expertise in engineering and heritage management, with further specialisation required for particular projects. A team may include engineers with expertise in the relevant materials and structural forms, conservation architects, industrial architects, industrial archaeologists and other heritage professionals. The services of an archaeologist may be required when dealing with a site likely to contain subsurface archaeological remains.

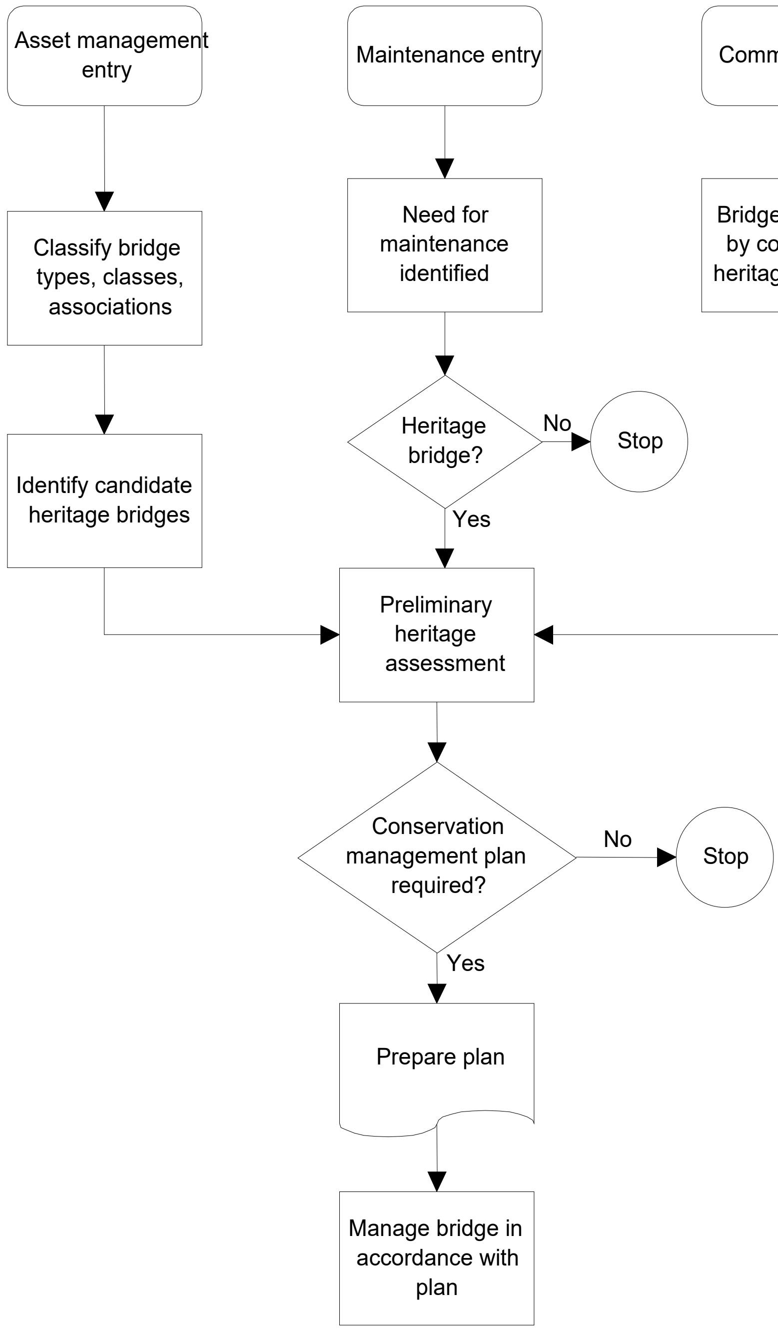
Consultation

Authorities consulted during the preparation of a CMP can be divided into two groups, statutory and non-statutory. Statutory groups are usually government agencies that deal with non-indigenous heritage. Agencies administering indigenous heritage may become involved if indigenous heritage sites are known to occur in the vicinity of the bridge. Local government involvement may also be required, particularly if the bridge is listed on a local government planning instrument such as a heritage schedule of a council, state heritage body or the Register of the National Estate (O'Connor 1986, O'Conner 1997 in Appendix B.4) for the criteria that was used in the process to identify bridges for that Register. Consultation with a federal heritage body, such as the Australian Heritage Commission, may also be required and is mandatory when a bridge is entered on the Register of the National Estate or where there is Commonwealth ownership or funding.

Non-statutory groups who may have an interest in the item and who may wish to contribute information to the CMP are, for example, local historical societies or other community-based heritage groups, the National Trust of Australia or New Zealand, the Institution of Engineers Australia, the Institute of Professional Engineers New Zealand or the Royal Australian Institute of Architects. Contact with formal organisations is usually achieved by writing letters but for consultation with the general community letter box drops, public meetings, features (in the form of a press release) or advertisements in the local paper, may be more appropriate means of reaching the target audience.

As stated above, if it is known or suspected that indigenous sites are also present in the immediate vicinity of a bridge, it is also advisable to contact the local Aboriginal Land Council, Tribal Council or other indigenous community group(s) in order to obtain their opinions concerning a bridge or its environs, and particularly on whether the bridge or bridge site have any particular associations with that community that need to be considered when addressing the social significance of the item.

Figure 7.2: Management process flowchart



Approvals

The management policies developed in a CMP may have implications for the owner of the heritage bridge, for adjoining property owners and for the relevant local government agency, with the setting of a heritage bridge often an integral part of its heritage significance.

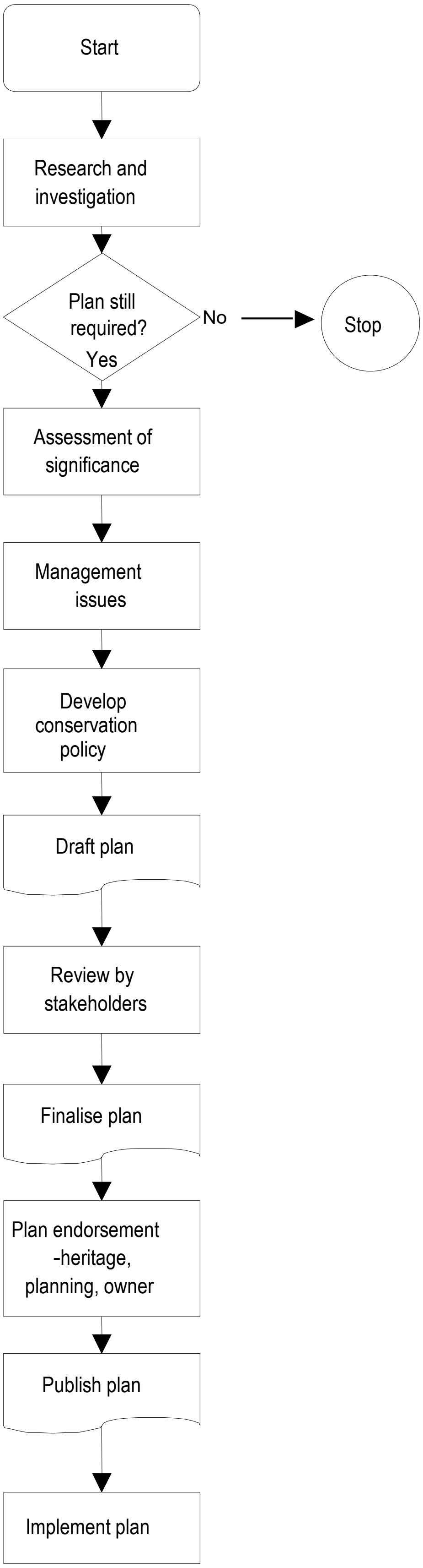
Prior to implementation, a CMP may require the endorsement of:

- the bridge owner
- the relevant local government agency
- State heritage councils
- Australian Heritage Commission
- New Zealand Historic Places Trust (Pouhere Taonga).

7.9.3 Elements of the Plan

The steps involved in the preparation of a CMP are shown in Figure 7.3.

Figure 7.3: Typical process for developing and implementing CMPs



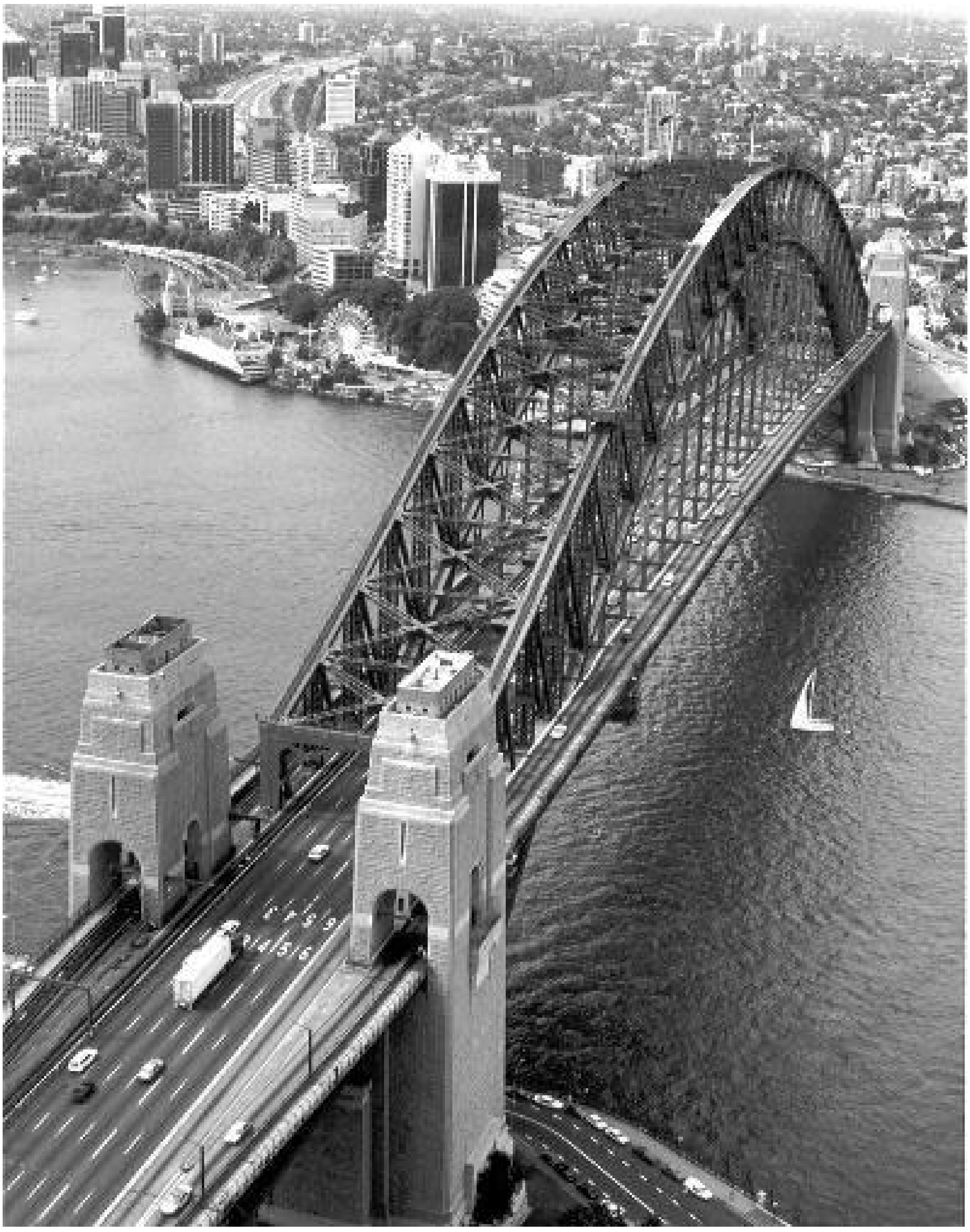
Introduction

The introduction should include a brief background to the item, its heritage significance, why the CMP was commissioned, and the scope of the plan.

Historical outline/development of structure

This section should describe the history of the bridge, the evolution of its design concept and construction techniques, and the history of the site upon which it is built. These elements are essential in understanding the historical significance of the item in its setting. For example, the Sydney Harbour Bridge (Figure 7.4) has significance in being one of the world's largest steel arch bridges, but the bridge also has significance to the history of the development of Sydney, as its construction allowed the opening up of the land to the north of the harbour, and it provided a vital land transport link across the harbour.

Figure 7.4: Sydney Harbour Bridge



Physical analysis

A detailed discussion of the elements that comprise the bridge is required. The form and fabric of each element of the bridge, and the function of that element, should be discussed, but in a manner that is accessible to the non-specialist. The condition of the bridge, and any changes that have been made to its design, should be addressed in this section. It is helpful if the contribution of the individual bridge elements to the heritage significance of the whole item is assessed at this stage, as this is an important aid in determining whether proposed changes are likely to have a significant impact upon the significance of the bridge. For example, in a timber truss bridge, the truss spans are likely to be the most significant elements of the bridge, while the abutments are likely to be of much less significance, and thus less integral to maintaining the significance of the bridge as a whole. As a broad rule of thumb, less intervention or modification would be considered allowable to the most significant bridge elements than to those that are either commonplace or heavily modified.

Fabric surveys

Fabric surveys involve observations and assessments of the materials from which a bridge is constructed, including relevant history and condition, and the form of the structure to develop recommendations for conservation works.

The fabric survey needs to distinguish between defects and weaknesses, where defects can be repaired but weaknesses need to be managed. It is a consequence of this assessment that all actions on the bridge should be evaluated for their effect on the determined weaknesses. For example, many old stone bridges have their stones bedded in site soil or a weak mix of quicklime and soil that is readily washed out by water passing through the structure. It is essential that weatherproofing pointing on these bridges is kept in good condition and that stormwater runoff not be allowed to enter the fabric.

Structural engineering assessment

In many cases, heritage bridges will be subject to significantly higher loads than those for which they were designed if, in fact, a basis for design can be determined. It is also likely that structural detailing or even the form of design, such as for arch bridges, is not addressed within contemporary design and assessment codes.

There will be a need to review the capacity of a bridge to carry the loads to which it may be subjected to determine whether it can carry those loads without restriction or whether limitations on usage or strengthening will be required.

There may be a need to use fundamental engineering principles, specialist analysis techniques and software, and bridge instrumentation and testing as part of the assessment process. Assessment, rather than design, philosophies should be used.

Hydraulic

Hydraulic assessment methodologies and the data available to support them will generally have changed significantly since a heritage bridge was designed and constructed. Assessment may involve the use of *Australian Rainfall and Runoff: a Guide to Flood Estimation* (Ball et al. 2016) or comparable publications and modelling software. As with the structural assessment, there will be a need to develop strategies to address any identified deficiencies.

Traffic

As with structural and hydraulic aspects, it is likely that traffic-related standards will have changed since a bridge was designed and constructed. Aspects such as horizontal and vertical geometry and lane and footway widths will need to be assessed against contemporary standards and management strategies developed. Annual Average Daily Traffic figures (if known) should be included to give an indication of whether traffic volumes are increasing, decreasing or stable, and whether provision needs to be made for greater traffic demand in the future.

Statement of significance

The significance of the bridge has either been established prior to preparation of the CMP (and most likely has been the trigger for the production of the CMP), or otherwise takes place during the course of CMP preparation. Existing statements of significance should be reviewed as part of the preparation of a CMP.

Criteria for establishing the nature and degree of significance of a heritage bridge are described in Appendix B.3. Often heritage items are considered to be significant because they are the sole example of a class of item. However, consideration needs to be given to the relative significance of one example of a class of items of which there are numerous examples. In this instance it is essential to determine whether the bridge is a good example to demonstrate the general nature of this class of bridges. Another consideration is that of rarity, particularly if a bridge is a rare example of a group of bridges that were once common. It is possible for a bridge to be assessed as being both representative and rare.

Management issues

Management issues that may arise from the CMP investigation, and which can be addressed in the recommendations, generally pertain to changes that need to be made to the item in order for it to continue in a safe operating condition, or addressing unsympathetic changes that may have been made to the item in the past. In addition, the ongoing maintenance of the item needs to be addressed and whether any routine practices enhance, or detract from the significance of the bridge.

Bridge curtilage

A bridge curtilage, which may include previous crossings, should be established to limit unsympathetic development within the immediate environs of a heritage bridge. In some instances, listing upon a heritage register may be contingent upon a curtilage for the item being established. The curtilage should not just encompass the immediate environs of the bridge, but should also include associated features such as the remains, or location of earlier bridges or river crossings, historic plantings associated with the structure and any rest or recreational facilities (such as picnic areas or swimming holes) associated with the bridge site. Should the bridge be located within a built-up environment, or one which is likely to move toward a higher density of settlement than previously, recommendations should also be made to governing, and if necessary restricting unsympathetic developments within the vicinity, if not within the curtilage, of the bridge. For example, if a clear view from a viewing area such as a tourist lookout or park toward a heritage bridge is a distinct feature of the visual environment of a place, then measures should be put in place to ensure that these elements are not destroyed by unsympathetic development in the area.

Legislative and planning issues

The CMP needs to cover the legislative constraints that apply to the conduct of work on, or the making of alterations to, the bridge. Typically these are state or territory heritage legislative constraints that apply because the bridge is a ‘relic’ (i.e. in excess of 50 years old) or is listed on a state heritage register. In some jurisdictions, there are also specific requirements that apply to heritage items under the care or control of state or territory government agencies. In addition, a bridge may also be listed on a local government planning schedule, such as a local or regional environmental plan, which could impose certain requirements such as the need to obtain development consent prior to undertaking any alterations to the item.

A distinction needs to be made in the CMP between routine maintenance works, which relate either to 'housekeeping' of the bridge (painting, cleaning) or replacement of like fabric with like (particularly with timber bridges), and alterations, such as upgrading of the road surface, safety barriers, signs and falling object prevention measures.

The CMP should outline routine maintenance practices and, if detrimental to the heritage significance of the bridge, recommend new practices or alterations to existing practice that will eliminate or reduce this. The CMP should attempt to cover alteration works that are planned or are likely to be required on the item in the near future, in order that the potential impact of the works on the bridge's heritage significance can be assessed, and modifications suggested, if possible. The rationale behind this approach is also that the proposed alterations can be approved at the same time as the CMP is endorsed by either the owner or the relevant state heritage body. This can circumvent the need to apply for separate permission to undertake these works at a later date. However, it is absolutely vital that the CMP not be written with a specific maintenance or use outcome in view, otherwise there is the danger of the document being written to lend weight to a particular proposed outcome. CMPs are meant to be best practice management documents and can achieve this only through retaining impartiality. As soon as a CMP is prepared in an attempt to justify a particular outcome, the value of the plan to act as a guide in future management of the item is compromised or lost.

Conservation policies

The development of conservation policy in the CMP stems from the assessment of the item's heritage significance and provides guiding principles to be followed to ensure the integrity of the key elements that contribute the bridge's significance are not lost or reduced during maintenance or modification works. The conservation policies should be more guiding than prescriptive statements unless there are specific matters to be covered. However, the policies must be written in clear, unambiguous language, in order that no confusion is caused by policies that are able to be interpreted in several ways.

References

Sources of material used in the preparation of the CMP should be acknowledged. The inclusion of references will assist those involved in the management of or interested in the bridge in the future.

Appendices

Appendices may include supporting documentation for the CMP including photographs, drawings, other documents and detailed reports, such as fabric surveys, structural assessment and hydraulic analysis.

7.9.4 Review

CMPs need to be reviewed and evaluated to reflect factors such as the execution of conservation works and changes in legislation or use.

It is recommended that plans be reviewed every 5 to 10 years. If a plan is endorsed by a state or territory heritage council or similar body, there may be a clause in the endorsement that the endorsement is valid for only a given period of time, usually five years, after which the CMP is considered no longer valid, unless a revised version has been submitted to that body for another endorsement.

7.10 Further References for Heritage Bridges

Refer to Appendix B.1 for publications that are commended to readers to assist with developing their understanding and management of bridge heritage. Many of these publications have been used in the preparation of this Guide.

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Appendix A Example of Economic Evaluation Procedure

A.1 Bridge Management Decision Making

The selection of the most effective management strategy for each bridge should be based on an economic analysis of feasible options to determine which will provide the desired level of service at the lowest overall cost over a specified period known as the lifecycle of the bridge.

A.2 The Life Span

The life span of a bridge is the period for which it is to be kept in service. At the time of design, a life span may have been specified (e.g. 120 years) implying that all elements were designed to last this period, or an initial commitment made to provide permanent ongoing maintenance procedures to guarantee this life.

Any nomination of a life span implies a terminal date towards which life cycle costing may be directed. This is very useful if the structure will no longer be required at that time but can lead to bad decisions if the structure continues to be needed after the arbitrarily nominated life span. This is important because most jurisdictions now have many bridges that have aged to a critical stage of their nominal lives structurally but which still have a long life requirement. If replacement at the end of the original design life is blindly considered to be the only option, many more economical alternatives may be overlooked.

A.2.1 Range of Management Options

The following options are available when considering a deteriorating structure:

- Specific operations – based on periodically revised estimates of expected life span, specific maintenance, rehabilitation and enhancement works are undertaken as needed. Often a number of technically viable options may be feasible.
- Replacement – where severe structural deterioration has occurred and especially where other factors such as poor alignment, susceptibility to flooding, etc. require very expensive rehabilitation, replacement will be the most economically viable option.
- Delayed replacement – in this case it will have been decided that the structure would no longer be needed after a certain date. All that is required is that inspections be carried out with sufficient regularity to ensure its fitness for any purpose for which it is used prior to closure and demolition.

A.2.2 Life Cycle Costing and Replacement

The concept of life cycle costing attempts to ensure that all appropriate future implications are taken into account when deciding present strategies. It is necessary to have a nominated life span in making these decisions, but it must be remembered that this relates to a period for which the bridge will be needed, rather than a plan to actually replace the bridge at about that time. The nominated 'life' is therefore a notional target date up to which the bridge must remain viable.

Many bridges now 30 to 50 years old will receive work that will reset their expected life span to another 30 to 50, or even more, years.

An inspection should be undertaken to review the nominated life span before each rehabilitation and strengthening action is undertaken.

There is a range of options with different costs, serviceability and expected life span implications. Life cycle costing relates the cost of each option to the overall serviceability and required longevity of the structure.

Weyers et al. (1984) and Palmer and Wyche (1990) have proposed a simple and effective method of analysis for comparing the life cycle costs of optional management strategies. For each strategy, a program of activities is drawn up in which the timing and cost of each future action is nominated. The future cost of each action is then converted to its present worth value by multiplying by the appropriate present worth factor (*PWF*), as in Equation 1.

$$PWF(n) = \frac{1}{(1+i)^{n}} \quad (1)$$

where

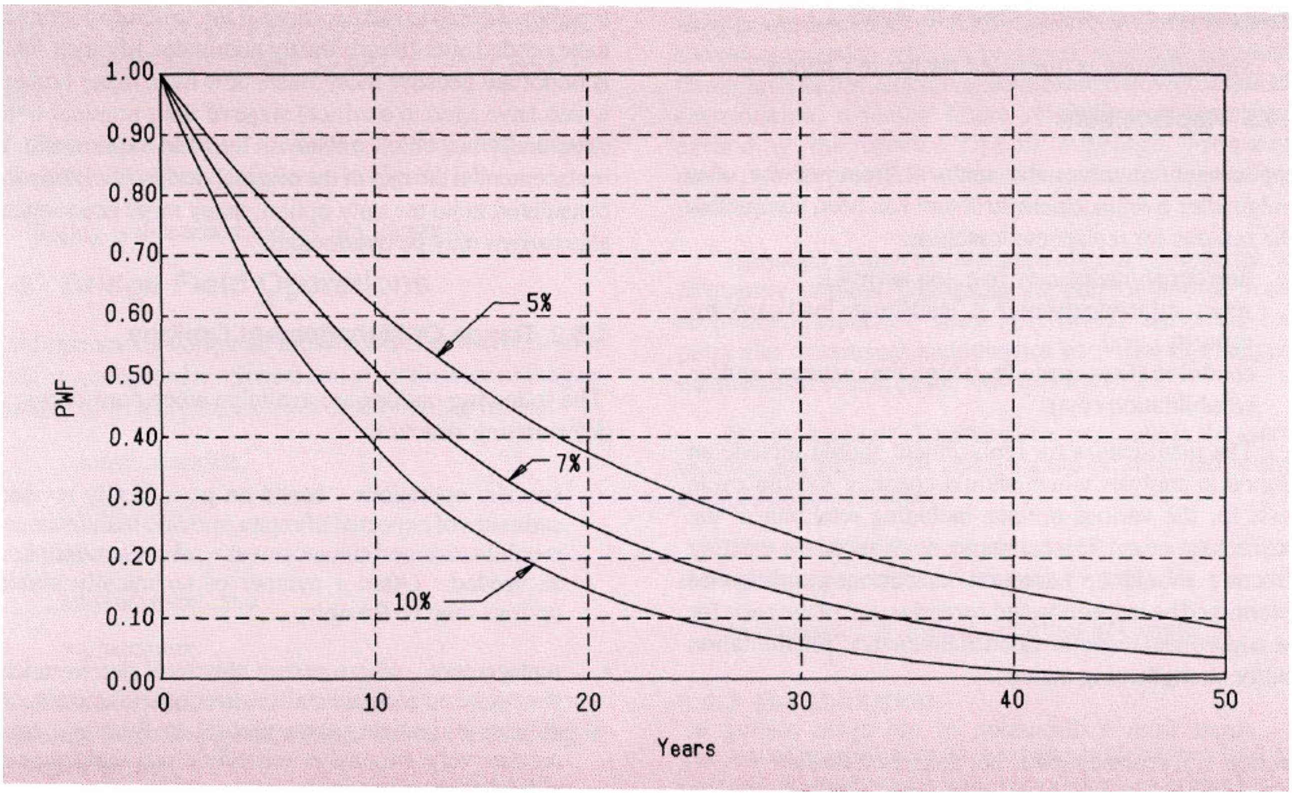
i = the real interest rate per annum

n = the time to the future action, in years

$PWF(n)$ = the present worth factor of a future action undertaken at time n.

The present worth factor is shown in Figure A 1.

Figure A 1: Present worth factor



The sum of all these present worth values is then converted to an equivalent annual uniform cost over the expected life of the bridge by multiplying by the capital recovery factor *CRF*, as in Equation 2.

$$CRF(N) = \frac{i(1+i)^N}{(1+i)^N - 1} \qquad 2$$

where

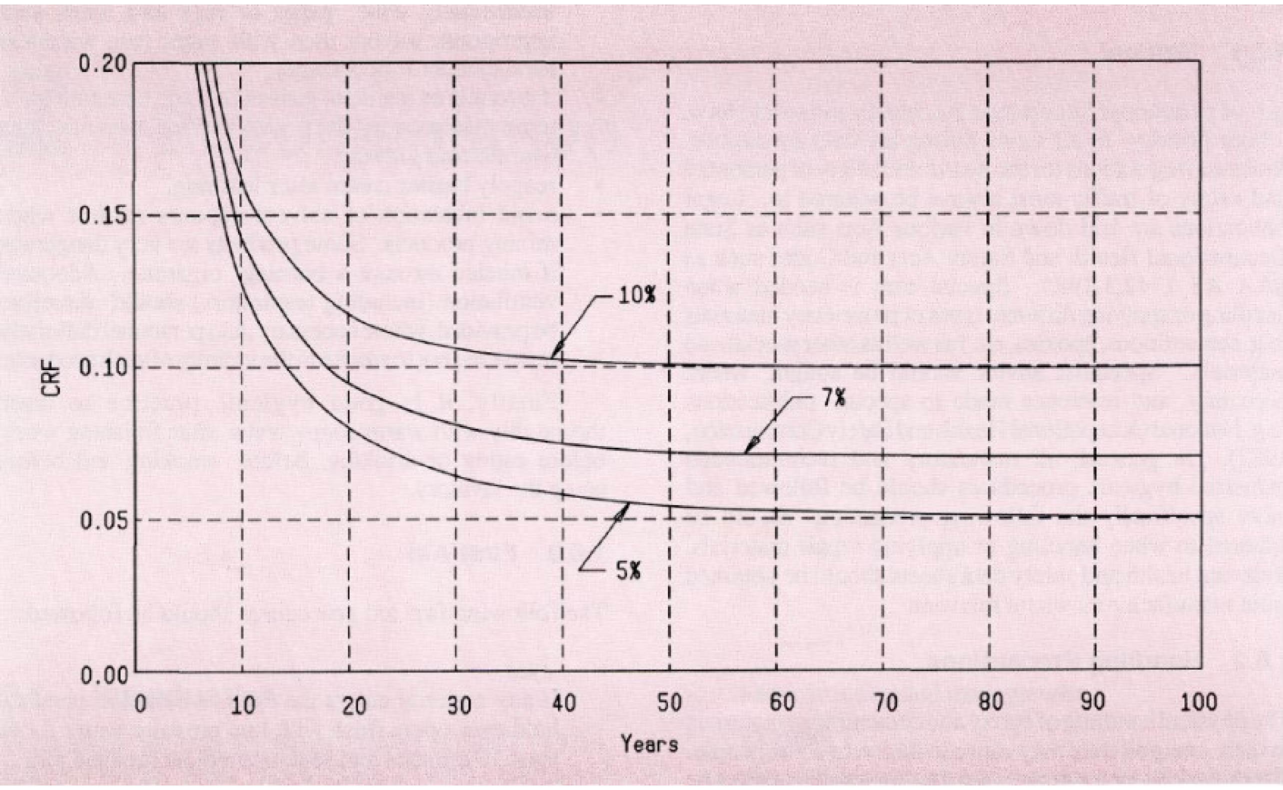
i = the real interest rate per annum

N = the expected life span of the bridge, in years

$CRF(N)$ = the capital recovery factor for the expected life span (N).

The capital recovery factor is shown in Figure A 2.

Figure A 2: Capital recovery factor



Finally, the annual cost of maintenance must be added (Equation 3).

$$CRF(N) * \sum_{k-1}^{1} \{(cost\ of\ action\ k) * PWF(n_k)\} + annual\ maintenance\ cost \qquad 3$$

The option with the lowest total annual cost is the most economic. Some aspects to be considered in the analysis include:

- Real rate of interest – any extra funds spent now have a real cost of interest, which is the difference between the inflation rate (of construction costs) and general interest rates.
  Typically, a figure of about 7% may be used with 5% and 10% also analysed to check sensitivity to interest rate variation. The economic analysis is sensitive to the real rate of interest used with high rates favouring options with low initial cost but high costs incurred later.

- Maintenance – this is necessary for the continued serviceability of any structure but certain specific actions can reduce maintenance costs. For example, construction of a concrete overlay on a timber bridge deck saves patching the road surface and clearing of deck drainage.
- Rehabilitation and enhancement may bring other benefits beyond the narrow scope of the life cycle cost analysis. For example, the removal of a load limit, a smoother running surface and an increase in width all result in benefits to road users.

To illustrate life cycle costing, consider a 15-year-old timber bridge having five spans of 6 m and a width of 7 m. A detailed inspection showed that the piles on two piers were rotted at ground level and needed to be repaired the following year, while some of the timber corbels were split and needed banding as soon as possible. The inspection also showed that the timber deck was reasonably sound but would benefit from a concrete deck overlay in five years time.

The estimated cost of each activity (in current values) is shown in Table A 1 together with the present worth factor (assuming a real interest rate of 7%) and the present worth of costs. The total present worth of costs is $57 250.

Table A 1: Example of present worth of a series of future maintenance expenditures

Maintenance activity	Time years	Estimated cost	PWF	Present worth $
Repair corbels	0	3 000	1.000	3 000
Repair piles	1	10 000	0.935	9 350
Concrete overlay	5	63 000	0.713	44 900
Total present worth				57 250

The expected life span after the concrete deck overlay is 20 years, making a total expected life span of 25 years. The capital recovery factor for 25 years, at a real interest rate of 7% per annum, is 0.08581. The equivalent annual uniform cost is therefore:

- $57 250 x 0.08581 = $4900 per year.

To this must be added an estimated annual maintenance cost of $700, making a total cost of:

- $4900 + $700 = $5600 per year.

This may be compared with replacement by concrete box culverts, at an estimated construction cost of $105 000, with no expected rehabilitation actions over an expected life span of 70 years and requiring an estimated annual maintenance of $200.

The total present worth of costs is simply the construction cost of $105 000. The equivalent annual uniform cost is:

- $105 000 x 0.07062 = $7400

and the total annual cost:

- $7400 + $200 = $7600

This simple approach neglects salvage costs and increasing annual maintenance costs with deterioration. These costs are included in Weyers et al. (1984).

Appendix B Heritage Bridges

B.1 References/Bibliography for Heritage Bridges

The following publications are commended to readers to assist with developing their understanding of bridge heritage and the management of bridge heritage. Many of the publications have been used in the preparation of this Guide.

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B.2 The Burra Charter and AHC Assessment Criteria

The Australian ICMOS Charter for the Conservation of Places of Cultural Significance (The Burra Charter)

Preamble

Having regard to the International Charter for the Conservation and Restoration of Monuments and Sites (Venice 1966), and the resolutions of the 5th General Assembly of the International Council on Monuments and Sites (ICOMOS) (Moscow 1978), the following Charter was adopted by Australia ICOMOS on 19 August 1979 at Burra, South Australia. Revisions were adopted on 23 February 1981, 23 April 1988 and 26 November 1999.

Who is the Charter for?

The Charter sets a standard of practice for those who provide advice, make decisions about, or undertake works to places of cultural significance, including owners, managers and custodians

Using the Charter

The Charter should be read as a whole. Many articles are interdependent. Articles in the Conservation Principles section are often further developed in the Conservation Processes and Practice sections. Headings have been included for ease of reading but do not form part of the Charter.

The Charter is self-contained, but aspects of its use are further explained in the following Australia ICOMOS documents:

- Guidelines to the Burra Charter: Cultural Significance
- Guidelines to the Burra Charter: Conservation Policy
- Guidelines to the Burra Charter: Procedures for Undertaking Studies and Reports.

Code on the ethics of Coexistence in Conserving Significant Places.

What places does the Charter apply to?

The Charter can be applied to all types of places of cultural significance including natural, indigenous and historic places with cultural values.

The standards of other organisations may also be relevant. These include the Australian Natural Heritage Charter and the Draft Guidelines for the Protection, Management and Use of Aboriginal and Torres Strait Islander Cultural Heritage Places.

Why conserve?

Places of cultural significance enrich people's lives, often providing a deep and inspirational sense of connection to community and landscape, to the past and to lived experiences. They are historical records that are important as tangible expressions of Australian identity and experience. Places of cultural significance reflect the diversity of our communities, telling us about who we are and the past that has formed us and the Australian landscape. They are irreplaceable and precious.

These places of cultural significance must be conserved for present and future generations.

The Burra Charter advocates a cautious approach to change: do as much as necessary to care for the place and to make it useable, but otherwise change it as little as possible so that its cultural significance is retained.

Articles	Explanatory Notes
Definitions	
Article 1. For the purpose of this Charter:	
1.1 *Place* means site, area, land, landscape, building or other work, group of buildings or other works, and may include components, contents, spaces and views.	The concept of place should be broadly interpreted. The elements described in Article 1.1 may include memorials, trees, gardens, parks, places of historical events, urban areas, towns, industrial places, archaeological sites and spiritual and religious places.
1.2 *Cultural* significance means aesthetic, historic, scientific or social value for past, present or future generations.	The term cultural significance is synonymous with heritage significance and cultural heritage value.
Cultural significance is embodied in the *place* itself, its *fabric, setting, use, associations, meanings*, records, *related places* and *related objects*.	Cultural significance may change as a result of the continuing history of the place.
Places may have a range of values for different individuals or groups	Understanding of cultural significance may change as a result of new information.
1.3 *Fabric* means all the physical material of the *place*.	Fabric includes building interiors and sub-surface remains, as well as excavated material. Fabric may define spaces and these may be important elements of the significance of the place.
1.4 *Conservation* means all the processes of looking after a place so as to retain its *cultural significance*	
1.5 *Maintenance* means the continuous protective care of the *fabric* and setting of a place, and is to be distinguished from repair. Repair involves *restoration or reconstruction*.	The distinction referred to for example in relation to roof gutters, are: • maintenance – regular inspection and cleaning of gutters • repair involving restoration – returning of dislodged gutters • repair involving reconstruction – replacing decayed gutters.
1.6 *Preservation* means maintaining the *fabric* of a *place* in its existing state and retarding deterioration.	It is recognised that all places and their components change over time at varying rates.
1.7 *Restoration* means returning the existing *fabric* of a *place* to a known earlier state by removing accretions or by reassembling existing components without the introduction of new material.	

Articles	Explanatory Notes
1.8 *Reconstruction* means returning a *place* to a known earlier state and is distinguished from *restoration* by the introduction of new material into the fabric.	New material may include recycled material salvaged from other places. This should not be to the detriment of any place of cultural significance.
1.9 *Adaptation* means modifying a *place* to suit the existing *use* or a proposed use.	
1.10 *Use* means the functions of a place, as well as the activities and practices that may occur at the place.	
1.11 *Compatible use* means a *use* which respects the *cultural significance* of a *place.* Such a use involves no, or minimal, impact on cultural significance.	
1.12 *Setting* means the area around a *place*, which may include the visual catchment.	
1.13 *Related place* means a *place* that contributes to the *cultural significance* of another place.	
1.14 *Related object* means an object that contributes to the *cultural significance* of a *place* but is not at the place.	
1.15 *Associations* mean the special connections that exist between people and a *place*.	Associations may include social or spiritual values and cultural responsibilities for a place.
1.16 *Meanings* denote what a *place* signifies, indicates, evokes or expresses.	Meanings generally relate to intangible aspects such as symbolic qualities and memories.
1.17 *Interpretation* means all the ways of presenting the *cultural significance* of a *place*.	Interpretation may be a combination of the treatment of the fabric (e.g. maintenance, restoration, reconstruction), the use of and activities at the place; and the use of introduced explanatory material.

Conservation Principles

Article 2. Conservation and management

2.1 Places of cultural significance should be conserved.

2.2 The aim of conservation is to retain the cultural significance of a place.

2.3 Conservation is an integral part of good management of places of cultural significance.

Articles	Explanatory Notes
2.4 Places of cultural significance should be safeguarded and not put at risk or left in a vulnerable state.	
Article 3. Cautious approach	
3.1 *Conservation* is based on respect for the existing *fabric, use, associations* and *meanings.* It requires a cautious approach of changing as much as necessary but as little as possible.	The traces of additions, alterations and earlier treatments to the fabric of a place are evidence of its history and uses which may be part of its significance. Conservation action should assist and not impede their understanding.
3.2 Changes to a *place* should not distort the physical or other evidence it provides, nor be based on conjecture.	
Article 4. Knowledge, skills and techniques	
4.1 *Conservation* should make use of all the knowledge, skills and disciplines which can contribute to the study and care of the *place.*	
4.2 Traditional techniques and materials are preferred for the *conservation* of significant *fabric*. In some circumstances modern techniques and materials which offer substantial conservation benefits may be appropriate.	The use of modern materials and techniques must be supported by firm scientific evidence or by a body of experience.
Article 5. Values	
5.1 *Conservation* of a *place* should identify and take into consideration all aspects of cultural and natural significance without unwarranted emphasis on any one value at the expense of others.	Conservation of places with natural significance is explained in the Australian Natural Heritage Charter. This Charter defines natural significance to mean the importance of ecosystems, biological diversity and geodiversity for their existence value, or for present or future generations in terms of their scientific, social, aesthetic and life-support value.
5.2 Relative degrees of *cultural significance* may lead to different *conservation* actions at a place.	A cautious approach is needed, as understanding of cultural significance may change. The article should not be used to justify actions which do not retain cultural significance.
Article 6. Burra Charter Process	
6.1 The *cultural significance* of a *place* and other issues affecting its future are best understood by a sequence of collecting and analysing information before making decisions. Understanding cultural significance comes first, then development of policy and finally management of the place in accordance with the policy.	The Burra Charter process, or sequence of investigations, decisions and actions, is illustrated in the accompanying flowchart.

Articles	Explanatory Notes
6.2 The policy for managing a *place* must be based on an understanding of its *cultural significance*.	
6.3 Policy development should also include consideration of other factors affecting the future of a *place* such as the owner's needs, resources, external constraints and its physical condition.	
Article 7. Use	
7.1 Where the *use* of a *place* is of *cultural significance* it should be retained.	
7.2 A *place* should have a *compatible use*.	The policy should identify a use or combination of uses or constraints on uses that retain the cultural significance of the place. New use of a place should involve minimal change, to significant fabric and use; should respect associations and meanings; and where appropriate should provide for continuation of practices which contribute to the cultural significance of the place.
Article 8. Setting	
Conservation requires the retention of an appropriate visual *setting* and other relationships that contribute to *the cultural significance* of the *place*.	Aspects of the visual setting may include use, siting, bulk, form, scale, character, colour, texture and materials.
New construction, demolition, intrusions or other changes which would adversely affect the setting or relationships are not appropriate.	Other relationships, such as historical connections, may contribute to interpretation, appreciation, enjoyment or experience of the place.
Article 9. Location	
9.1 The physical location of a *place* is part of *its cultural significance*. A building, work or other component of a place should remain in its historical location. Relocation is generally unacceptable unless this is the sole practical means of ensuring its survival.	
9.2 Some buildings, works or other components of places were designed to be readily removable or already have a history of relocation. Provided such buildings, works or other components do not have significant links with their present location, removal may be appropriate.	

Articles	Explanatory Notes
9.3 If any building, work or other component is moved, it should be moved to an appropriate location and given an appropriate *use*. Such action should not be to the detriment of any *place* of *cultural significance*.	
Article 10. Contents	
Contents, fixtures and objects that contribute to *the cultural significance* of a *place* should be retained at that place. Their removal is unacceptable unless it is: the sole means of ensuring their security and *preservation*; on a temporary basis for treatment or exhibition; for cultural reasons; for health and safety; or to protect the place. Such contents, fixtures and objects should be returned where circumstances permit and it is culturally appropriate.	
Article 11. Related places and objects	
The contribution that *related places* and *related objects* make to the *cultural significance* of the *place* should be retained.	
Article 12. Participation	
Conservation, interpretation and management of a *place* should provide for the participation of people for whom the place has special *associations* and *meanings*, or who have social, spiritual or other cultural responsibilities for the place.	
Article 13. Co-existence of cultural values	
Co-existence of cultural values should be recognised, respected and encouraged, especially in cases where they conflict.	For some places, conflicting cultural values may affect policy development and management decisions. In this article, the term cultural values refers to those beliefs which are important to a cultural group, including but not limited to political, religious, spiritual and moral beliefs. This is broader than values associated with cultural significance.
Conservation Processes	
Article 14. Conservation processes	
Conservation may, according to circumstance, include the processes of: retention or reintroduction of a *use*; retention of *associations* and *meanings*; *maintenance, preservation, restoration, reconstruction, adaptation* and *interpretation*; and will commonly include a combination of more than one of these.	There may be circumstances where no action is required to achieve conservation.

Articles	Explanatory Notes
Article 15. Change	
15.1 Change may be necessary to retain *cultural significance*, but is undesirable where it reduces cultural significance. The amount of change to a *place* should be guided by the *cultural significance* of the place and its appropriate *interpretation*.	When change is being considered, a range of options should be explored to seek the option which minimises the reduction of cultural significance.
15.2 Changes that reduce *cultural significance* should be reversible, and be reversed when circumstances permit.	Reversible changes should be considered temporary. Non-reversible change should only be used as a last resort and should not prevent future conservation action.
15.3 Demolition of significant *fabric* of a *place* is generally not acceptable. However, in some circumstances minor demolition may be appropriate as part of *conservation*. Removed significant fabric should be reinstated when circumstances permit.	
15.4 The contributions of all aspects of *cultural significance* of a *place* should be respected. If a place *includes fabric, uses, associations* or *meanings* of different periods, or different aspects of cultural significance, emphasising or interpreting one period or aspect at the expense of another can only be justified when what is left out, removed or diminished is of slight cultural significance and that which is emphasises or interpreted is of much greater cultural significance.	
Article 16. Maintenance	
Maintenance is fundamental to *conservation* and should be undertaken where *fabric* is of *cultural significance* and its *maintenance* is necessary to retain *that cultural significance.*	

Articles	Explanatory Notes
Article 17. Preservation	
Preservation is appropriate where the existing *fabric* or its condition constitutes evidence of *cultural significance*, or where insufficient evidence is available to allow other *conservation* processes to be carried out.	Preservation protects fabric without obscuring the evidence of its construction and use. The process should always be applied: • where the evidence of the fabric is of such significance that it should not be altered; • where insufficient investigation has been carried out to permit policy decisions to be taken in accord with Articles 26 to 28. New work (e.g. stabilisation) may be carried out in association with preservation when its purpose is the physical protection of the fabric and when it is consistent with Article 22.
Article 18. Restoration and reconstruction	
Restoration and *reconstruction* should reveal culturally significant aspects of the *place*.	
Article 19. Restoration	
Restoration is appropriate only if there is sufficient evidence of an earlier state of the *fabric*.	
Article 20. Reconstruction	
20.1 *Reconstruction* is appropriate only where a *place* is incomplete through damage or alteration, and only where there is sufficient evidence to reproduce an earlier state of the *fabric*. In rare cases, reconstruction may also be appropriate as part of a *use* or practice that retains the *cultural significance* of the place.	
20.2 *Reconstruction* should be identifiable on close inspection or through additional *interpretation*.	
Article 21. Adaptation	
21.1 *Adaptation* is acceptable only where the adaptation has minimal impact on the *cultural significance* of the *place*.	Adaptation may involve the introduction of new services, or a new use, or changes to safeguard the place.
21.2 *Adaptation* should involve minimal change to significant fabric, achieved only after considering alternatives.	

Articles	Explanatory Notes
Article 22. New work	
22.1 New work such as additions to the *place* may be acceptable where it does not distort or obscure the *cultural significance* of the place, or detract from its *interpretation* and appreciation.	New work may be sympathetic if its siting, bulk, form, scale, character, colour, texture and material are similar to the existing fabric, but imitation should be avoided.
22.2 New work should be readily identifiable as such.	
Article 23. Conserving use	
Continuing, modifying or reinstating a significant *use* may be appropriate and preferred forms of *conservation*.	These may require changes to significant *fabric* but they should be minimised. In some cases, continuing a significant use or practice may involve substantial new work.
Article 24. Retaining associations and meanings	
24.1 Significant *associations* between people and a *place* should be respected, retained and not obscured. Opportunities for the *interpretation*, commemoration and celebration of these associations should be investigated and implemented.	For many places associations will be linked to use.
24.2 Significant meanings, including spiritual values, of a place should be respected. Opportunities for the continuation or revival of these meanings should be investigated and implemented.	
Article 25. Interpretation	
The *cultural significance* of many *places* is not readily apparent, and should be explained by *interpretation*. Interpretation should enhance understanding and enjoyment, and be culturally appropriate.	
Conservation Practice	
Article 26. Applying the Burra Charter process	
26.1 Work on a place should be preceded by studies to understand the place which should include analysis of physical, documentary, oral and other evidence, drawing on appropriate knowledge, skills and disciplines.	The results of studies should be up to date, regularly reviewed and revised as necessary.

Articles	Explanatory Notes
26.2 Written statements of *cultural significance* and policy for the *place* should be prepared, justified and accompanied by supporting evidence. The statements of significance and policy should be incorporated into a management plan for the place.	Statements of significance and policy should be kept up to date by regular review and revision as necessary. The management plan may deal with other matters related to the management of the place.
26.3 Groups and individuals with *associations* with a *place* as well as those involved in its management should be provided with opportunities to contribute to and participate understanding the *cultural significance* of the place. Where appropriate they should also have opportunities to participate in its *conservation* and management.	
Article 27. Managing change	
27.1 The impact of proposed changes on *the cultural significance* of a *place* should be analysed with reference to the statement of significance and the policy for managing the place. It may be necessary to modify proposed changes following analysis to better retain cultural significance.	
27.2 Existing *fabric, use, associations* and *meanings* should be adequately recorded before any changes are made to the *place*.	
Article 28. Disturbance of fabric	
28.1 Disturbance of significant *fabric* for study, or to obtain evidence, should be minimised. Study of a *place* by any disturbance of the fabric, including archaeological excavation, should only be undertaken to provide data essential for decisions on the *conservation* of the place, or to obtain important evidence about to be lost or made inaccessible.	
28.2 Investigation of a *place* that requires disturbance of the *fabric*, apart from that necessary to make decisions, may be appropriate provided that it is consistent with the policy for the place. Such investigation should be based on important research questions which have potential to substantially add to knowledge, which cannot be answered in other ways and which minimises disturbance of significant fabric.	

Articles	Explanatory Notes
Article 29. Responsibility for decisions	
The organisations and individuals responsible for management decisions should be named and specific responsibility taken for each such decision.	
Article 30. Direction, supervision and implementation	
Competent direction and supervision should be maintained at all stages, and any changes should be implemented by people with appropriate knowledge and skills.	
Article 31. Documenting evidence and decisions	
A log of new evidence and additional decisions should be kept.	
Article 32. Records	
32.1 The records associated with the *conservation* of a *place* should be placed in a permanent archive and made publicly available, subject to requirements of security and privacy, and where this is culturally appropriate.	
32.2 Records about the history of a *place* should be protected and made publicly available, subject to requirements of security and privacy, and where this is culturally appropriate.	
Article 33. Removed fabric	
Significant *fabric* that has been removed from a *place* including contents, fixtures and objects, should be catalogued, and protected in accordance with its *cultural significance*.	
Where possible and culturally appropriate, removed significant fabric including contents, fixtures and objects, should be kept at the place.	
Article 34. Resources	
Adequate resources should be provided for *conservation*.	The best conservation often involves the least work and can be inexpensive.
Words in italics are defined in Article 1.	

B.3 Australian Heritage Commission Significance Assessment Criteria

Criterion A: Its importance in the course, or pattern, of Australia's natural or cultural history

A.1 Importance in the evolution of Australian flora, fauna, landscapes or climate.

A.2 Importance in maintaining existing processes or natural systems at the regional or national scale.

A.3 Importance in exhibiting unusual richness or diversity of flora, fauna, landscapes or cultural features.

A.4 Importance for their association with events, developments or cultural phases which have had a significant role in the human occupation and evolution of the nation, State, region or community.

Criterion B: Its possession of uncommon, rare or endangered aspects of Australia's natural or cultural history

B.1 Importance for rare, endangered or uncommon flora, fauna, communities, ecosystems, natural landscapes or phenomena, or as wilderness.

B.2 Importance in demonstrating a distinctive way of life, custom, process, land use, function or design no longer practised, in danger of being lost, or of exceptional interest.

Criterion C: Its potential to yield information that will contribute to an understanding of Australia's natural or cultural history

C.1 Importance for information contributing to wider understanding of Australian natural history, by virtue of their use as research sites, teaching sites, type localities, reference or benchmark sites.

C.2 Importance for information contributing to a wider understanding of the history of human occupation of Australia.

Criterion D: Its importance in demonstrating the principal characteristics of:

(i) A class of Australia's natural or cultural places; or
(ii) A class of Australia's natural or cultural environments.

D.1 Importance in demonstrating the principal characteristics of the range of landscapes, environments, ecosystems, the attributes of which identify them as being characteristic of their class.

D.2 Importance in demonstrating the principal characteristics of the range of human activities in the Australian environment (including way of life, philosophy, custom, process, land use, function, design or technique).

Criterion E: Its importance in exhibiting particular aesthetic characteristics valued by a community or cultural group

E.1 Importance for a community for aesthetic characteristics held in high esteem or otherwise valued by the community.

Criterion F: Its importance in demonstrating a high degree of creative or technical achievement at a particular period

F.1 Importance for their technical, creative, design or artistic excellence, innovation or achievement.

Criterion G: Its strong or special associations with a particular community or cultural group for social, cultural or spiritual reasons

G.1 Importance as places highly valued by a community for reasons of religious, spiritual, cultural, educational or social associations.

Criterion H: Its special association with the life or works of a person, or group of persons, of importance in Australia's natural or cultural history

H.1 Importance for their close associations with individuals whose activities have been significant within the history of the nation, state or region.

B.4 Identification of Bridges for the Register of the National Estate

The following notes have been provided by Colin O'Connor on the process that was used to identify bridges for the Register of the National Estate. His report to the Australian Heritage Commission was reprinted as Research Report CE69 of the Department of Civil Engineering of the University of Queensland.

Colin O'Connor is Emeritus Professor of Civil of Engineering at the University of Queensland, where he joined the staff in 1954 and was Professor of Civil Engineering from 1970 until his retirement in 1989. He has published reports, technical articles and three major books on bridges. The 1983 *Register of Australian Historic Bridges* was published in association with the Institution of Engineers, Australia.

The selection of bridges was aided by reference to graphs of maximum span/date and sum of spans/date for 11 bridge types. On these graphs were superimposed lines to form boundaries for six age/size categories. These graphs with the superimposed boundary lines and the governing numerical parameters were presented in Report CE69, together with a list of the 250 selected bridges.

Although these graphs were extended in some cases forward to the year 1940, most of the data used in these studies was earlier than circa 1930.

More recent data was used to extend the graphs for five bridge types in a paper presented by O'Connor to the 1997 Austroads Bridge Conference in Sydney (O'Connor 1997) (Appendix B.1). Reference should be made to the conference proceedings for the graphs. The extended graphs may be used to test the validity of the boundaries for application to more recent bridges recognising that:

- some bridge types have tended to fall out of use
- the new data is selective, in that it is essentially for larger bridges, whereas earlier data covered a full range of bridge sizes.

In the graphs, the upper inclined line defines a bridge that is large for its size. Below this is a parallel inclined line; the space between these lines defines a bridge with considerable size for its age.

The upper line is defined by its vertical ordinate at year 1900 and its slope, defined as the 'doubling period' or the period for the ruling dimension to double. In this regard, it should be remembered that all vertical dimensions are plotted on a logarithmic scale.

The lower line has vertical ordinates equal to a specified constant multiplier of the ordinates of the upper line.

In the context of this guide, the major question to assess is whether or not the upper inclined line represents a valid lower bound for bridges that are large for their age.

For metal trusses, these boundary lines still appear to be reasonable.

For metal girders, the bound lines are at the same slope as for metal trusses with a doubling period of 50 years. There has only been a very small rate of increase in the maximum span for Australian bridges of this type. Larger spans have however been achieved for bridges overseas. For example, the Zoo Bridge at Cologne, completed in 1966, has tubular box girders with a maximum span of 259 m. The Sara River Bridge in Yugoslavia, built at the same time, had steel plate girder spans with about the same maximum. For the year 1966, the upper line that defines the age/size category has a vertical ordinate or maximum span, equal to:

60×2^a, where a = (years after 1900) ÷ 50 years

This maximum span is 150 m, well below 259 m.

On the other hand, recent Australian metal girder bridges have sums of spans that relate well to the boundaries of the age/size categories.

On the whole, O'Connor is inclined not to change the boundaries.

For concrete girders, present age/size boundaries both for maximum span and sum of spans agree well with more recent data. The same is the case for the concrete arch.

The spans of Australian suspension and cable-stayed bridges are small by world standards. The maximum span at year 1900 was 75 m with a doubling period of 50 years. The vertical ordinate of 280 m at 1995 may be compared with the maximum span of the Anzac (Glebe Island) Bridge of 364 m. A line passing through 75 m at 1900 and 364 m at 1995 would have a doubling period of 41 years. Alternatively, a doubling period of 40 years would give a span of 389 m at 1995.

The author is not inclined to change the boundaries as the present lines would correctly classify Batman, Westgate and Glebe Island Bridges as large for their age.

It is concluded that the original 1985 definitions of age/size categories are still reasonable and useful in classifying bridges on the basis of their date and size. It must be remembered, however, that this is only one factor in the assessment of the significance of heritage bridges.

B.5 Illustrative Examples of Heritage Bridges

The table below provides a list of selected major heritage bridges in Australia.

Table B.1: Selected major heritage bridges in Australia

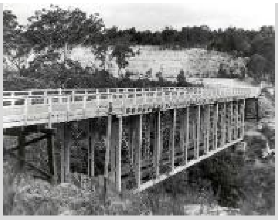
Bridge	Year	Description	Picture
Iron Bridge, Coalbrookdale Britain	1779	Iron Bridge crosses the Severn River in Coalbrookdale, Shropshire in Western England. It was designed by Thomas Pritchard and built by iron founders John Wilkinson and Abraham Darby between 1777 and 1779. It was the first iron bridge to be built in the world. The bridge spans 100 ft and has five arch ribs, each cast in two halves. It was closed to vehicular traffic in 1934 and listed as an 'Ancient Monument'.	
Richmond Bridge, Tasmania	1823	Richmond Bridge is Australia's oldest extant bridge. Its foundation stone was laid on 11 December 1823 and the bridge was opened to traffic in 1825. It is a six span sandstone arch and, with a maximum span of 8.5 m, was Australia's longest span bridge until 1836. The bridge was an integral part of the development of Tasmania's east coast and of the Richmond township. It was convict built and has associations with a number of notable persons from Tasmania's early history.	

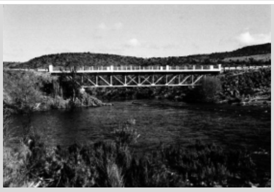
Bridge	Year	Description	Picture
Lapstone Hill Bridge, New South Wales	1833	The bridge is a masonry arch and the oldest remaining bridge on the Australian mainland. Although strengthened, at the time of this publication it was still in use. David Lennox, a pioneering bridge builder, was responsible for many early bridges in New South Wales. The bridge was built with the help of 20 convicts and completed in less than 12 months.	
Lansdowne Bridge, New South Wales	1836	The bridge at Liverpool was designed and built by David Lennox using convict labour and at the time of this publication it was still in use. Lennox had to divide his time between the construction of this bridge and the Lapstone Hill Bridge.	
Ross Bridge, Tasmania	1836	Ross Bridge was designed by John Lee Archer and built by convict labour under the supervision of two convicts, James Colbeck and Daniel Herbert. It has three spans of 8.9 m and is carefully detailed with stone cutwaters, parapets and curved stairs to the river bank at each corner. Its real distinction comes from its carvings, with the face of each rib made with 31 carved pieces of stone. Many of the carvings are symbolic or iconic. Others are representational with depictions including people, animals, rural objects and musical instruments.	
Red Bridge, Tasmania	1838	Red Bridge was built by convict labour between 1836 and 1838. It has three 7.6 m spans and, in plan, is an elongated H-shape with four long river training walls. Its name derives from the approximately 1.5 million red bricks used in its construction. It is the oldest substantially brick arch in Australia and has been an integral part of Tasmanian transport development. The bridge is located on the National Highway and has recently been strengthened to carry contemporary vehicle loadings.	
Spiky Bridge, Tasmania	C1843	Spiky Bridge was built by convict labour. It comprises a small culvert with substantial buttresses. The bridge played an important role in the development of the road system on Tasmania's east coast and has strong historical and social associations with the community. Its name derives from an unusual arrangement of stones on the top of the parapet.	
Blackman River Bridge, Tasmania	C1847	The bridge has stone piers and abutments with a timber superstructure. While the timber elements have been replaced as part on the bridge's ongoing maintenance, the form of construction has been maintained, and the structure may be Australia's oldest timber girder bridge.	

Bridge	Year	Description	Picture
McMillans Bridge, Victoria	1856	The bridge was built originally in 1856 with a timber truss superstructure. The timber superstructure was replaced in 1888-9 with two double-intersection wrought iron deck trusses and a longitudinal timber deck. The single 29 m span is of unusually light construction. The abutments are among the earliest surviving examples of Victorian bridge technology. The bridge is also one of the earliest surviving examples of a composite road bridge combining masonry abutments, wrought iron truss girders and timber deck. The bridge is associated with significant engineers in Charles Rowland, Charles Wilson and Professor WC Kernot. The design of the masonry abutments possibly reflects the style of Telford, Brunel and other early 19th century British engineers.	
Corowa Bridge, New South Wales	1861	This wrought iron bridge was built in 1892 through funding from a local businessman J Foord and was subsequently named after him. Corowa is famous for its role in hosting the conferences, which were instrumental in the development of Australia's Federation.	
Knapsack Gully Viaduct, New South Wales	1867	The bridge was built originally by Whitton in 1867 as part of the Lapstone Zigzag for the Western Railway. It was taken over as a road bridge for the Western Highway in 1926 and widened in 1939. The masonry arch construction has two spans of 6.1 m and five of 15.2 m. It has a maximum pier height of 40 m.	
Ellerslie Bridge, Victoria	1867	The bridge is Victoria's second oldest positively dated timber-beam road bridge and one of only two bridges known to have resisted Victoria's epic floods of 1870. It is the largest and most original example of an important class of colonial-era composite timber beam and masonry bridges. The bridge has a mixture of foundation types and incorporates the best surviving example of timber trestle piers over rare stone sub-piers.	
Mia Mia (Redesdale) Iron Girder Bridge, Victoria	1868	The bridge was built in 1867–68 and is a wrought iron lattice girder through truss road bridge on masonry abutments with divided lanes and a longitudinal timber deck. There are three trusses, each of which spans 45.7 m. The bridge was constructed from parts of a larger iron bridge being imported from England in 1859 to bridge the Yarra River. The ship caught fire and was scuttled in Hobsons Bay but the parts were subsequently salvaged. The site is claimed to be adjacent to the scene of a notable battle between local Aboriginal people and early European settlers. It is the second oldest metal truss bridge in the state and the oldest in rural Victoria. The three above deck trusses linked and stabilised by arches over a dual carriageway are unique in Victoria	
Maley Bridge, Western Australia	1864	The bridge was built by convict labour under the supervision of the Royal Engineers using jarrah timbers from the Perth escarpment. It has a sawn timber superstructure on limestone masonry piers and abutments. There are five spans with an overall length of 26.9 m. The bridge was restored by MRWA during the 1980s.	

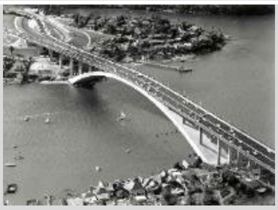
Bridge	Year	Description	Picture
Glenmona Bridge, Victoria	1871	The Glenmona (Old Bung Bung) Bridge is a continuous three span wrought iron lattice girder deck truss road bridge on bluestone masonry abutments and piers. It has an overall length of 46.6 m. It is an historic crossing place on a major early pastoral and mail route and subsequently on the main route between South Australia and the Mount Alexander gold diggings. The bridge was a prototype for a series of locally constructed wrought iron lattice girder bridges and represents the stage in Victorian iron bridge evolution that displaced the laminated timber arch bridge design.	
Alligator Creek Bridge, Queensland	1873	This truss bridge over Alligator Creek on the Bruce Highway just north of Rockhampton was constructed in 1873 and decommissioned in 1960. The main span of 40.5 m was a Finke truss with the timber girders forming the top compression chord. Vertical timber posts projected down from the girders. The bottom ends of these posts were supported on wrought iron bars anchored into the compression chord at the piers and suspended below the deck.	
Waianakarua River North Branch Bridge, New Zealand	1874	Located on SH l in North Otago this is a masonry arch structure built in 1874 and designed by John Turnbull Thomson, Provincial Engineer of Otago. It has two 18 m spans of limestone with skew arches and exhibits superb craftsmanship in the masonry work. This is dressed ashlar with the voussoirs having a reticulated finish. There is a vertical curve in the roadway and balustrades giving an enhanced visual appeal but needing some lowering and widening to meet present day standards on a major arterial highway. This is the finest masonry bridge in the country and carries a Category 1 heritage status.	
Kooringa Bridge, South Australia	1879	The bridge is a two span wrought iron bowstring girder bridge at Burra in the mid-north of the state. The girder is a cross between an arch and a truss, with the truss coming to a point at its ends. It is one of few examples of this type of structure.	
Murray Bridge, South Australia	1879	This wrought iron truss bridge was the first permanent crossing of the River Murray in South Australia providing a link with the eastern states. The 603 m long bridge, which consists of five main spans and 23 approach spans, carried rail traffic from 1886 to 1925. The town of Murray Bridge takes its name from the bridge.	
Maclean’s Bridge, Queensland	1880	Maclean’s Bridge over the Logan River on the Brisbane to Beaudesert Road was a cable stayed timber bridge built in 1880 and replaced in 1935. Wrought iron rods supported the main spans which had lengths of 18 m, 30 m and 18 m.	
Monkerai Bridge, New South Wales	1882	The bridge is the one of the oldest timber truss bridges in the state. Its most interesting feature is the flat slope of the end diagonals and the increased depth of the upper chord at the mid-span. The bridge has three 21.3 m truss spans and three additional timber girder spans.	

Bridge	Year	Description	Picture
Rakaia Gorge No 1 Bridge, New Zealand	1882	This is a most unusual bridge completed in 1882 on SH 72 in mid Canterbury, unique in New Zealand and possibly further afield. It has a 55 m deck truss without a bottom chord and was incorrectly believed to be a Bollman truss, a patented US design of Wendell Bollman and used mainly for railways. The Rakaia River bridge has a top chord that is really a wrought iron plate girder with long raking members connected to inverted struts. There is some resemblance to the Whipple and the Fink trusses. It has a timber deck and balustrades. Designed in 1877, there were many delays before its completion. A proposed railway allowed for in the design never eventuated. At the time of this publication it was still in use. It has a Category 1 NZHPT rating as a heritage bridge based on its unique technology and striking setting in the gorge.	
Dickabram Bridge, Queensland	1885	This bridge over the Mary River near Miva has three truss spans with high timber trestle approaches. It carried both road and rail traffic and was opened in 1885.	
Hackney Bridge, South Australia	1885	The bridge is a single 38 m span wrought iron parabolic arch truss on masonry abutments. It replaced the first crossing of the Torrens River constructed in 1844. The sawn timber deck was replaced by a concrete deck in 1936. The bridge still carries current traffic loads.	
Bungambra-watha Creek Bridge, New South Wales	1887	The bridge is one of only a few metal arch bridges in New South Wales. It is located near the botanical gardens in Albury and has a wrought iron deck supported on five three-pinned arches. It spans 13.7 m and was designed by McDonald.	
Annan River Bridge, Queensland	1888	The Annan River Bridge, constructed in 1888, is one of Queensland's oldest structures still in service. It comprises 22 spans of 15.3 m and consists of a timber deck resting on five wrought iron plate girders. Wrought iron headstocks are supported on segmentally constructed cast iron piles. The pile sections are attached by way of a bolted splice connection.	
Cowra Bridge, New South Wales	1893	The bridge had three 48.4 m spans, another four 27.4 m truss spans and six timber girder spans when originally opened. The four smaller truss spans on the ends appear to have been replaced. It was one of the largest timber bridges built. Known as a composite bridge due to the varied construction materials used, it had a steel bowstring span, wooden decking, and iron and steel pillars. The bridge was replaced with a concrete structure in the late 1970s. One of the main truss spans was removed and re-erected as a heritage display in a nearby park.	
Lamington Bridge, Queensland	1896	The Lamington Bridge across the Mary River south of Maryborough was one of the world's first major concrete bridges. It is a continuous structure of eleven 16.6 m spans reinforced by railway lines. It was opened to traffic in October 1896.	

Bridge	Year	Description	Picture
Hampden Bridge, New South Wales	1898	The bridge is one of three suspension bridges in New South Wales and is built from local timber, wrought iron, steel and castellated sandstone. Constructed over the Kangaroo River, it has a clear span of 77 m suspended by towers at each end. At the time of publication it was still in use.	
Clifden Bridge, Waiau River, New Zealand	1899	This Southland bridge on SH 96 opened in 1899. It was designed by CH Howorth, Southland County Engineer on behalf of the Public Works Department. The span is 111.5 m (366ft), the towers are concrete and the stiffening truss is timber. When a replacement reinforced concrete girder bridge was built in 1978 the old bridge was offered to the New Zealand Historic Places Trust and accepted as a fine example of a suspension bridge that were very common in New Zealand up to the 1930s. The Trust saw it as important to ensure some worthy examples are retained. It has the added merit of being a World War I Memorial in having a marble tablet recording the names of the fallen attached to one of the towers. The bridge, restricted to foot traffic, has the Trust's Category 1 status.	
Dunmore Bridge, New South Wales	1899	The bridge over the Patterson River at Woodville has a lift span of 17.8 m and 38 m timber truss approaches. A hand-operated winch drives a vertical shaft that runs up the pier to a longitudinal shaft geared to the insides of the rims of the main sheaves. The ropes and counterweights have now been removed.	
De Burgh's Bridge, New South Wales	1899	The largest timber truss bridge constructed in Australia was DeBurgh's Bridge over the upper Lane Cove River. Although replaced in 1967, it stood next to the new bridge until fire destroyed it in 1994.	
Horseshoe Bridge, Western Australia	1903	The bridge is a steel and brick arch over the railway in Perth City. It is curved to minimise the severity of the vertical alignment. The central section of the bridge over the railway comprises six steel beam spans with an overall length of 66.3 m. The two curved and ramped approaches are each of 12 brick arch spans approximately 66 m long.	
Grafton Bridge, Auckland New Zealand	1910	This fine bridge spanning Grafton Gully in the heart of the city was opened in April 1910. It demonstrates the progressive attitude of the local body when civic pride was at stake. The designer was RF Moore, Chief Engineer of The Ferro-Concrete Company of Australasia, the firm that carried out the construction. When built its 97.6 m (320 ft) reinforced concrete arch span was the largest in the world. There are reinforced concrete girder spans and a maximum height of 44.8 m (147 ft). In recent years considerable upgrading has been necessary but it remains a superb structure of its time. It has NZHPT Category 1 status, City Council heritage protection, and an IPENZ 'Engineering to 1990' plaque being one of the country's outstanding heritage monuments.	
River Somme Bridge, South Australia	1911	The bridge at Keyneton is a two-span under-strutted timber girder bridge. It was the last timber bridge to carry road traffic in South Australia and was bypassed in 1986.	

Bridge	Year	Description	Picture
Bridge of Remembrance, Avon River, Christchurch, New Zealand	1924	This impressive bridge is the result of a competition for a World War I Public Memorial. The successful designers were Gummer and Prouse, a firm of architects based in Auckland and Wellington using the idea of a Bridge of Remembrance. The 15 m segmental arch spans the river at a 30° angle with a vertical element of the tall arch over the roadway at the east end. There are smaller flanking arches for pedestrians with carved stone lions couchant on top. The structure is faced in Tasmanian stone with marble on one side having tablets commemorating battles. Although reinforced concrete bridges had been acceptable as expressing the nature of the material, undoubtedly the public would expect a more conventional treatment for such an important monument. In more recent times a separate vehicular bridge has been built and the conversion to a pedestrian use has embodied the architect’s concept of a bridge for people to stroll over and use at their leisure. It has a Category 1 status in the NZHPT Register and City Council heritage protection.	
Sydney Harbour Bridge, New South Wales	1932	The Sydney Harbour Bridge replaced a ferry service thus opening up the northern suburbs of Sydney. It has become known around the world as one of Australia’s greatest landmarks. It is also justifiably referred to as the largest steel arch span of its type in the world.	
Little Crystal Creek Bridge, Queensland	1932	This concrete masonry arch structure, located near Paluma in the north eastern part of the state was constructed in 1932. The bridge comprises a single 18.3 m span and is the only one of its type still in service in Queensland.	
Vincents Rivulet Bridge, Tasmania	1933	The bridge is of composite steel and concrete construction with a single span of 10 m. It was designed by Allan Knight in association with Professor Allan Burn of the University of Tasmania to prove the concept of composite action and provided the basis for construction of a number of similar bridges in other parts of the state. It is the first composite steel and concrete bridge in Tasmania and one of the first in the world.	
Nive River Bridge, Tasmania	1933	The bridge is the oldest all welded steel truss in Australia. It was important in opening the road link to the state’s west coast. Its setting is distinctive and the bridge is highly visible because of the grade and alignment of the road approaches.	

Bridge	Year	Description	Picture
McKillops Bridge, Victoria	1936	McKillops Bridge is a 255 m long road and livestock bridge across the Snowy River. It consists of welded steel trusses seated on tall one-piece reinforced concrete piers and supporting an elaborate timber stockbridge superstructure. The bridge was built by the Country Roads Board between 1931 and 1936 during two stages, with the height being increased after the original superstructure was washed away in record floods of January 1934. The site was a stock crossing for almost a century before the bridge was built. The bridge was an early example of electric arc welding. It is of aesthetic significance for its impressively long and handsome timber stockbridge superstructure, combined with modern bridge technology of grand proportions, viewed against a panoramic backdrop of mountain forest and alpine river gorge.	
Fremantle Traffic Bridge, Western Australia	1939	The bridge is a significant timber structure that is very prominent in the vicinity of Fremantle. It is the first Australian bridge seen by many migrants over many years. The bridge has 26 spans with an overall length of 205.5 m. It is the second bridge on the site, with the original bridge being built in 1864 by James Manning of the Royal Engineers using convict labour.	
Ferguson's Bridge, Victoria	1939	Ferguson's Bridge is a four span composite timber and steel road bridge with a rare super-elevated curved timber deck 56 m long. It is representative of the post-depression/pre-World War II era of Victorian transport history when rapidly increasing numbers of motor vehicles forced road agencies to abandon traditional bridge designs more suited to horse-drawn vehicles. It is associated with an early Campaspe River crossing place adjacent to a squatting run in the earliest days of white settlement.	
Birkenhead Bridge, South Australia	1940	This double-leaf bascule bridge was designed by local engineers RH Chapman and JA Fargher who both made significant contributions to rail transport in South Australia. The bridge provided a permanent link across the Port River at Port Adelaide. The bridge is still in use carrying current traffic loads.	
Karangarua River Bridge, New Zealand	1941	Located on SH 6, 19 km south-west of Fox Glacier this superb bridge was designed in 1938 and completed in 1941. An all steel structure, except for the concrete deck, it has a span of 130 m (420 ft) and a plate girder in lieu of the stiffening truss. It is a most impressive bridge aesthetically, being a worthy example of Public Works Department bridge design and execution. Although not currently classified by NZHPT this bridge certainly merits top rating.	
Floating Arch Bridge, Tasmania	1943	The bridge provided the first fixed crossing of the Derwent River near Hobart. The design was developed to cater for the width of the river, the depth to sound foundations and current and tidal forces from the river. It was built as a horizontal three pinned arch and of cellular reinforced concrete construction. The bridge was superseded by Tasman Bridge in 1964 because of storm damage, corrosion, and congestion arising from growth on Hobart's eastern shore and the delays arising from the lift span used for navigation.	
Perth Causeway Bridge, Western Australia	1952	The bridge is an early Western Australian example of a steel composite bridge. It is built on the site of the 1844 crossing, which was the first bridged crossing of the Swan River. There are 13 spans with an overall length of 226 m. The bridge is the larger of a two-bridge causeway that incorporates Heirisson Island on the eastern approaches to Perth. The structure is now infamous for its occurrence of alkali-aggregate reaction in the pier concrete.	

Bridge	Year	Description	Picture
Distillery Bridge, South Australia	1958	The bridge has seven 15.2 m spans and crosses Salt Creek at Renmark. It is the first prestressed concrete beam bridge built in South Australia.	
Narrows Bridge, Western Australia	1959	The structure is a precast segmental prestressed concrete bridge of significant size for its construction era and founded on very innovative 'Gambia' piles. External prestressing was used to allow lighter I-section deck beams. It has five spans and an overall length of 335 m. The bridge was recognised by the Institution of Engineers, Australia in 1999 as a National Engineering Landmark.	
Gladesville Bridge, New South Wales	1964	The bridge is the longest single span concrete arch in the world. It weighs 78 000 tons or double the weight of all the steel in the Sydney Harbour Bridge. It has a span length of 304 m and the crown of the arch is 60 m above water level.	
Tasman Bridge, Tasmania	1964	Tasman Bridge was first opened to traffic in 1964 and is the largest bridge in the state. It is perhaps best known from the collision on 5 January 1975 involving the *ss Lake Illawarra* that resulted in the collapse of three spans, the loss of 12 lives and major disruption to the lives of the citizens of Hobart until it was reopened in October 1977.	
Mooney Mooney Creek Bridge, New South Wales	1986	The bridge provided a great technological leap forward with a pair of closely spaced prestressed concrete balanced cantilever girders. The deck is 76 m above water level, compared with the 56 m of the Sydney Harbour Bridge.	
Anzac Bridge, New South Wales	1995	A defining landmark, this bridge is the longest cable-stayed bridge in Australia. It has a main span of 364 m and provided an innovative solution to solving traffic flow in Sydney. Importantly, its design was in response to the community's needs and requirements.	

Glossary

Bridge terminology

The terminology used to label bridge elements (also referred to as components) varies from jurisdiction to jurisdiction and region to region in accordance with local practice. This is particularly so for timber bridges.

Road jurisdiction bridge inspection manuals provide a complete list of common component definitions used within each jurisdiction and should be referenced as appropriate.

A summary of the more common terminology adopted for bridges in general is shown in Figure 1.1, for timber bridges in Figure 1.2, and for masonry bridges in Figure 1.3.

The following terms are defined as they have specific meaning in the context of this document.

Term	Definition
Alkali-aggregate reactivity (reaction)	A chemical reaction in concrete between alkalis from Portland cement and siliceous minerals in some aggregates. Deleterious expansion of the concrete may occur.
Approaches (bridge)	A relatively short length of carriageway leading up to a bridge, including embankment, pavement and safety barriers.
Approach slab	A reinforced concrete slab supported on the abutment curtain wall and the approach fill.
Asset inventory	A list of assets considered worthy of identification as discrete assets, with information such as location, design standard, construction date, maintenance history, configuration, condition, and technical details.
Asset management	A systematic process of effectively maintaining, upgrading and operating assets, combining engineering principles with sound business practice and economic rationale, and providing the tools to facilitate a more organised and flexible approach to making decisions necessary to deliver optimal community benefits. Also see total asset management (TAM).
Asset management strategy	A corporate statement of a medium to long-term goal for the comprehensive maintenance of a group of assets (e.g. road network), supported by processes for the development and implementation of plans and programs for asset creation, operation, maintenance, rehabilitation/replacement, disposal and performance monitoring to ensure that the desired levels of service and other operational objectives are achieved at a minimum cost.
Ballast wall	A narrow reinforced concrete wall, forming part of the abutment to prevent the earth fill reaching the abutment sill and bearings.
Beam	A load bearing member which supports the deck of a bridge.
Bearing (structural)	A device for transmitting horizontal and vertical forces of a bridge superstructure to the supporting structure.
Bearing capacity	The maximum average contact pressure between the foundation and the soil which will not produce shear failure in the soil.
Benefit-cost analysis	see economic costs and benefits
Benefit-cost ratio (BCR)	The ratio of the discounted benefits over the life of a project to the discounted capital costs, or the project's discounted total agency costs.
Box culvert	A culvert of rectangular cross-section.
Bridge	A structure designed to carry a road or path over an obstacle by spanning it.
Bridge management system	A systematic approach to maintenance inspection, planning, budgeting and work, usually supported by software to assist in organising and analysing data on bridge inventory and condition as well as maintenance activities (e.g. type, costs, productivity, location, history, etc.).
Clearance height	The minimum height that a structure should be designed to accommodate the design vehicle (maximum allowable vehicle heights are generally 0.2 m less than this height).
Cross beam (cross girder)	Transverse beam, used to support longitudinal members, such as girders, planks, stringers, etc.

Term	Definition
Crosshead	Part of a pier or abutment spanning between columns or piles that support the superstructure.
Culvert	One or more adjacent pipes or enclosed channels for conveying surface water or a stream below formation level.
Curtain wall	A narrow reinforced concrete wall, forming part of the abutment to prevent the earth fill reaching the abutment sill and bearings.
Curtilage	An area of land attached to a bridge and forming one enclosure with it.
Damage (bridges)	The sudden worsening of the condition of a bridge, its elements and component materials due to the effect of a sudden event such as a fire, flood, impact, accident or vandalism.
Defect	Visible evidence of an undesirable condition in the pavement affecting serviceability, structural capacity or appearance.
Delineator	A device, other than a simple post, used to indicate the edge of the formation, carriageway or traffic lane.
Designated level of service	The level of service for a given type or class of road within the designated service requirements based on road users needs for access and levels of pavement condition appropriate to the designated function of the pavement.
Design life	The period during which the performance of an asset is expected to remain at the intended service level without replacement, refurbishment or significant maintenance. Also see economic life, technical life and useful life.
Design load	The load distribution, or combination of loads, for which a structure is designed.
Design standard	Identifies particular standards used in the design, e.g. standard lane width.
Deterioration (bridges)	The gradual worsening of the condition of a bridge, its elements and component materials due to the effects of traffic and other loadings, the action of the environment on the structure and/or the actions of the constituents of component materials over a period of time.
Differential settlement	The uneven sinking of the foundations of a structure.
Dowel	A short, straight, plain reinforcing bar embedded into two adjacent blocks or slabs of concrete to permit relative horizontal movement in the direction of the dowel and hence prevent relative vertical movement between adjacent blocks.
Economic analysis	Means of analysing investment or policy decisions by comparing the benefits and costs of such decisions as far as practicable in monetary terms. Future costs and benefits are discounted to represent present day values
Economic costs and benefits	Total costs and benefits to the community irrespective of whether actual financial flows are involved. Economic costs and benefits should exclude transfer payments (such as taxes and subsidies) and these are removed from market prices to reflect resource costs. All costs and benefits are analysed in real terms.
Economic life	The lesser of the following: the estimated number of years from the post-construction opening to traffic or rehabilitation/replacement condition, which gives the minimum possible life-cycle cost, in present value terms that account for the operational costs of maintenance, rehabilitation and road user costs to achieve a designated level of service the estimated number of years from the post-construction opening to traffic or rehabilitation/replacement condition to reach a predetermined threshold value of a range of distresses beyond which the pavement is no longer acceptable for use (i.e. the estimated service life).
End beam	A transverse stiffening member girder webs or inside a hollow pier (see diaphragm).
End of deck	The back face of the curtain wall or back wall (if the wall extends to the design surface level) or the end of the superstructure (when the approach slab sits above the curtain wall).
Enhancement (bridges) Strengthening and upgrading	The improvement of the condition of a bridge above its design or initially planned level of service. Forms of enhancement include strengthening, widening, lengthening, raising and improved safety such as better barriers.
Expansion joint	A joint provided to allow relative movement to occur and to prevent the build-up of stresses due to expansion. May be of various types depending on the movement to be accommodated.

Term	Definition
Fender wall	A narrow reinforced concrete wall, forming part of the abutment to prevent the earth fill reaching the abutment sill and bearings.
Foundation	The soil or rock upon which a structure rests.
Furniture	The equipment such as sign posts, median kerbs, lighting poles, etc., which is installed to make the road network work more effectively.
Gabion	A wire basket filled with stones used to retain earth or to control scour. A certain type is termed a rock-fill wire mattress.
Hairline crack	Bridge: a crack less than 0.05 mm in width. Road: an irregularly running, thin, narrow, crevice or fissure at the surface of a concrete or clay product, not penetrating deeply.
Infrared thermography	Equipment that uses a special type of camera to measure variations in heat emitted by a structure. Near-surface defects and other mechanisms, for example moisture infiltration, can potentially be detected using these methods based on their influence on surface temperature.
Inspection (bridges)	The process of utilising various techniques to closely examine a bridge and determine and maintain records of its current physical and structural condition. Four levels of inspection are specified in increasing detail and decreasing frequency.
Intervention level	The value (extent and condition) of a condition parameter, which triggers either maintenance investigation or maintenance activity. An intervention level will identify a defect as either acceptable or unacceptable. The latter will require further consideration of the defect in relation to its location with respect to the asset, safety issues, the possibility of continuing deterioration and increased repair cost and the economics of not undertaking repairs.
Invert	The lowest portion of the internal surface of a drain or culvert.
Level of service (LOS)	A qualitative index for ranking operating conditions on roads as well as pedestrian and cycling facilities based on factors such as speed, flow rate, travel time, freedom to manoeuvre, interruptions, comfort, safety, and convenience.
Maintenance (routine maintenance, periodic maintenance)	All actions necessary for retaining an asset as near as practicable to its original condition, or to reduce its rate of deterioration.
Maintenance intervention level (intervention level)	The value of a condition parameter which triggers maintenance investigation or activity.
Mortar	A mixture of cement and/or lime and sand with water. A mixture of fine aggregate with an epoxy compound or other binder.
Network level	A type of road condition survey or data analysis where the main purpose is to monitor network performance or assist with network asset management decisions, as distinct from project decisions.
Periodic maintenance	Maintenance treatments conducted at fairly regular intervals of longer than one year. For bridges, replacement of joint seals is one example.
Photogrammetry	Imaging used to model buildings, engineering structures, vehicles, forensic and accident scenes. Typically, a camera is hand held or set on a tripod close to the subject and a 3D model or a drawing is produced. Common points are identified on each image and a line of sight can be constructed from the camera location to these points on the object.
Pier	An intermediate support in a bridge having more than one span. Part of the substructure supporting the superstructure and transferring the loads to the foundations.
Pile	A slender member driven, jetted, screwed, or formed in the ground to resist loads or thrust.
Pile cap	A structural member designed to connect and distribute loads from the above structure to a group of piles.
Post-tensioning	A method of prestressing in which tendons are tensioned after the concrete has hardened.
Predictive maintenance (RMS ITS)	A maintenance strategy based on measuring the condition of equipment in order to assess whether it will fail during a given future period. Then planning or taking appropriate action to avoid the consequences of that failure before it occurs.

Term	Definition
Prestressed concrete	Concrete in which prior to complete loading, effective internal compressive stresses are induced deliberately, usually by means of tensioned steel, to reduce or eliminate tensile stresses when loaded.
Pretensioning	A method of prestressing in which tendons are tensioned before the concrete is placed (e.g. Super T girders, deck units).
Preventative maintenance (RMS ITS)	A maintenance activity before or at an early stage in the development of one or more defects, aimed at preventing occurrence or progression of the defect(s), usually undertaken on a proactive rather than reactive basis.
Proactive maintenance (RMS ITS)	Refers to any maintenance tasks used to predict or prevent equipment failures.
Project level	A type of road condition survey or data analysis where the main purpose is to assist with decisions about proposals for a specific project on a short length of road, as distinct from network decisions.
Rating (bridge)	The process of evaluating the load carrying capacity of a bridge, accounting for its current condition and material properties. The rating process is part of the considerations of the appropriateness of rehabilitation and strengthening. Refer to AS 5100.7 and NZTA Bridge Manual Section 6 (NZTA 2016b). The assessed load capacity in accordance with the process outlined above
Redundancy (RMS ITS)	The existence of more than one means for accomplishing a given function within a system. Each means of accomplishment of the function need not necessarily be identical.
Reflection cracking	Surface cracking resulting from movement associated with cracks or joints in an underlying pavement layer.
Rehabilitation (bridge)	The actions necessary to restore a bridge to its originally intended level of service in order to retain it in service for as long as possible. It is characterised by major repairs which are remedial in nature, are costly and less frequent than those undertaken for maintenance.
Reinforced concrete	Concrete strengthened within its mass by steel bars, or mesh or steel fibres.
Reinforced soil wall (RSW)	A method of constructing retaining walls in which fill is retained by vertical steel or concrete units anchored by friction into the fill by means of galvanised steel strips.
Reinforcement	Bars, or mesh, usually of steel, embedded in concrete, masonry, or brickwork, for the purpose of resisting particular stresses, e.g. tensile, temperature related, etc.
Remaining life (remaining service life)	The period, under current or stated use (e.g. traffic volume, type and growth), during which the asset condition is expected to remain within stated limits, provided that appropriate routine and preventive (periodic) maintenance are carried out.
Retaining wall	A wall constructed to resist lateral pressure from the adjoining ground or to maintain in position a mass of earth.
Revetment	A facing of stone or other material laid on a sloping face of earth to maintain the slope in position or to protect it from erosion.
Riprap	Medium to large size rock protection, against scour, applied (usually by dumping) to the face of an embankment.
Risk	The potential realisation of the unwanted or adverse consequences of an event from which there is no prospect of gain.
Risk acceptance	An informed and formal decision to accept the consequences and the likelihood of a particular risk.
Risk analysis	A systematic use of available information to determine how often specified events might occur and the magnitude of their consequences.
Risk assessment	The processes of reaching a decision or recommendation on whether risks are tolerable and present risk control measures are adequate, and if not, whether alternative risk control measures are justified or will be implemented.
Risk management	The process of planning, organising, directing and controlling the resources and activities of an organisation in order to minimise the adverse effects of accidental loss to that organisation at least possible cost.

Term	Definition
Risk mitigation	A selective application of appropriate techniques and management principles to reduce either likelihood of an occurrence or its consequences, or both.
Risk rating	An estimate of risk assigned to an individual site or activity.
Routine maintenance	Small mainly reactive works which are normally anticipated within a budget timeframe, but their precise nature, location and timing are not known in advance. Routine maintenance mainly consists of minor activities planned on a short term basis, usually about two weeks or less. More extensive maintenance tasks undertaken either by road maintenance or specialist bridge maintenance gangs, when necessary, as determined from routine inspections, detailed inspections or special inspections, possibly using special equipment or materials specially ordered. Usually the work will be programmed.
Safety barrier	A physical barrier separating roadside hazards or opposing traffic and the travelled way, designed to resist penetration by an out-of-control vehicle and as far as practicable, to stop or redirect colliding vehicles.
Scabbing	Loss of patches of aggregate from a seal.
Scheduled maintenance (RMS ITS)	Preventative maintenance performed at prescribed or scheduled points in a system's operational life.
Scour monitoring	Use of sensors and mechanical set-ups to monitor the loss of material adjacent to piers, piles and abutments during flood events. A typical fixed scour monitoring system includes sensors, data loggers and a data transfer system. The sensors may be connected to the data logger(s) by a wired or wireless connection.
Settlement	A downward movement of the soil or of the structure it supports.
Sill beam	Part of a pier or abutment spanning between columns or piles that support the superstructure.
Skew angle	The angle between a line at right angles to the control line/setting-out line and the pier or abutment.
Span	The distance between the centres of adjacent supports of a bridge, beam or truss. The superstructure of a bridge between two adjacent supports.
Subsidence	A downward movement resulting from displacement of the underlying material.
Substructure	In a bridge, the piers and abutments (including wing walls) which support the superstructure.
Superstructure	That part of a bridge structure which is supported by the piers and abutments.
T-beam	A load-bearing structure (usually reinforced concrete) with a T-shaped cross-section.
U-beam	A load-bearing structure (usually reinforced concrete) with an inverted U-shaped cross-section.
Useful life (RMS ITS)	May be expressed as either: The period over which a depreciable asset is expected to be used; or The number of production or similar units (i.e. intervals, cycles) that is expected to be obtained from the asset. Also see economic life, design life and technical life.
Waterway	A channel or stream. The area available for water to pass through or under a structure.
Waterway area	The area of the cross-section of the stream at right angles to the direction of the flow, up to the assumed flood level.
Wing wall	The extension of an abutment wall as in a bridge, or of an end wall in a culvert, used for retaining the side slopes of the earth filling.

Guide to Bridge Technology consists of eight parts. It provides guidance to bridge owners and agencies on issues related to design procurement, vehicle and pedestrian accessibility and bridge maintenance and management practices, including the use and application of Australian and New Zealand bridge design standards.

Guide to Bridge Technology Part 7: Maintenance and Management of Existing Bridges discusses the structural management and maintenance of existing bridges including procedures at a network level and project level of management. It provides documentation of practices relevant to inspection, recording, reporting, evaluation of bridge condition and fitness for purpose, monitoring and appropriate technical treatments. Strategies for the control of heavy loads and the preservation of the physical bridge asset are also discussed.

Guide to Bridge Technology Part 7

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